

Raspberry Pi Internet Radio

Constructors Manual



A comprehensive guide to building Internet radios using the Raspberry Pi and
MPD (Music Player Daemon)

Bob Rathbone Computer Consultancy

www.bobrathbone.com

13th of July 2026

Version 8.3

Contents

Chapter 1 - Introduction	4
Chapter 2 – Hardware components overview	16
Chapter 3 – Wiring and Hardware	47
Chapter 4 - Construction details	62
Chapter 5 – System Software Installation.....	82
Chapter 6 - Installing the radio Software	109
Chapter 7 Configuring additional radio components	143
Chapter 8 – Operation	178
Chapter 9 -Troubleshooting	217
Licences and disclaimer	237
Glossary.....	239
Appendix A – Wiring diagrams and lists	241
Appendix B – Using a battery pack	247
Appendix C – HiFiBerry Device Tree Overlays.....	248
Appendix D - Audio streaming codec notes.....	249
Appendix E Radio program and configuration files	250
Index.....	252

About this manual

This manual should be used to install the RPi Internet Radio software for version 8.3 onwards. There is also a **Raspberry Pi Internet Radio Beginners Guide** available on the Bob Rathbone Web site or on **GitHub** which shows a [single](#) simple design without all of the options shown in this manual.



Version 8.2 onwards of the radio software was introduced to run on the **Trixie** Operating System released on the 2nd October 2025. Limited support is also provided on Bookworm.



As software suppliers are now rapidly abandoning 32-bit systems, all support for 32-bit systems has ended from version onwards, although most things will still work on a 32-bit system.

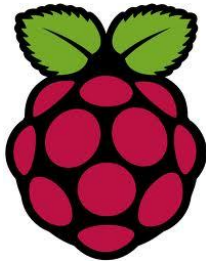
Quick Links

Topic	Page(s)	Topic	Page(s)
Adafruit	66	PiFace CAD	73
Airplay	190,162	Radio software installation	109
Buttons (Switches)	49	Radio software operation	178
Bluetooth speakers	162	Recording facility (liquidsoap)	166,209
Configuration files - Editing	167	RSS feed	170
DAC sound cards	26	Rotary encoders	50
Glossary	239	Shoutcast	200
I2C backpacks	67	Spotify	177,208,230
IQaudio Cosmic Controller	67	Speech facility (Espeak)	165
Interface boards	72	Switches (Buttons)	49
Install Icecast streaming	163	HDMI and Touch-screens	37
IR sensors	63	Tutorials and documents	145
LCD displays	163	USB Sound card	131
Media	70,171	Vintage radios	7
Menus	180	Wake up button	69
Network drives (NAS)	43,51	Web interface	152
Maintaining the Wi-Fi network	211	Wi-Fi configuration	102
OLED displays	195	Wiring	47
Pimoroni products	204		



Please note that there is also a full document index on page 252 at the end of this document.

Chapter 1 - Introduction



This manual describes how to create one of the most popular Internet Radios using the Raspberry Pi educational computer. This manual provides a detailed overview of construction and software installation.

The source and basic construction details are available from the following Web site:
https://bobrathbone.com/raspberrypi/pi_internet_radio.html or at
<https://github.com/bobrathbone/piradio6>

The main features of the Raspberry Pi internet radio are:



Raspberry Pi Internet Radio

Turn your Raspberry Pi into an Internet Radio using a variety of designs as shown in this manual. It runs on both 32 and 64-bit systems. It also runs on an X-Windows or Wayland Desktop.



Media Player

Play your favourite MP3 tracks from a USB stick, SD card or from a NAS (Network Attached Storage).



Artwork display

Displays artwork (if available) from both Radio Stations and Media (mp3 etc.) tracks.

Note: The radio station and associated album artwork is provided from the **Discogs** music database. See <https://www.discogs.com>



Airplay Receiver

This design allows the Raspberry Pi to act as an Airplay receiver. Music tracks can be played from your Apple or Android mobile phone or tablet.



Spotify Receiver

Turn your Raspberry Pi into a Spotify Receiver. This requires a Spotify Premium account.



RSS Feed Reader

This software also allows you to read any configured RSS feed. For example, your favourite news feed.



Record Radio Streams to disc

Record any Radio streams using the **liquidsoap** utility to record any streams that are using the **Shoutcast** or **Icecast** formats whilst also listening to a different radio station.

Note: **liquidsoap** only works on **64-bit OS systems** and not on **32-bit** systems.



Bluetooth speaker/headphone support

This manual contains instructions how to run the radio software with Bluetooth speakers or headphones. Bluetooth versions 1.x through 5.x supported depending upon the Raspberry Pi model used.



Digital Clock

The Internet Radio software displays as standard a digital clock and date with alarm and snooze functions.



Shoutcast

Shoutcast radio is a streaming audio which is used by some 50,000 Internet Radio stations across the internet. This radio software, using a Web interface, allows multiple playlist creation from Shoutcast radio stations by genre or country which can then be selected through the radio or Web interface menus.



Icecast

Icecast is free streaming software which supports a variety of streams such as MP3 and OGG. Icecast can optionally be installed on the Raspberry Pi and allows the currently playing station or track to be streamed around the local network or out to the Internet (Legal and Copyright issues apply).



User Interface and displays

This design caters for a number of user interfaces such as push-buttons, rotary encoders or touch screens. Also, a number of displays such as 2 and 4-line LCDs, OLED displays, TFT or full graphical displays such as HDMI and touch-screen are supported. Examples are shown later on in this manual.



Web interface

The radio software includes an optional Web interface (radioweb) powered by **Apache 2**, **MySQL** and **O!MPD**. This allows stations and playlists to be selected via Web pages on your PC, Mobile Telephone or tablet.



The **eSpeak** engine is a small, lightweight text-to-speech (TTS) program that supports a large number of languages. It is used with the radio software to assist blind or visually impaired users by “speaking” the menus out loud. It can also be used with radios without a screen to navigate the menus.



Language files enable the messages to be displayed in the user’s own language, such as French, Dutch or Spanish. The user can easily create their own language file if required.



Highly configurable allowing a broad range of displays such as two or four-line LCDs, TFTs, graphic screens or HDMI monitors to be selected at installation time. Likewise, the user input devices such as rotary encoders or buttons etc. An audio configuration script selects a range of audio devices such as DACs or Bluetooth speakers.



Comprehensive menu driven installation and maintenance. All main software components and hardware options are all menu selectable. Underlying installation and configuration scripts carry out all the usual installation and maintenance tasks.



Support for English, Russian/Cyrillic and Western European character set, dependent upon LCD capabilities. Both native and Romanized (Conversion to Latin) characters supported.

REMEMBER TO HAVE FUN DOING THIS PROJECT!



Figure 1 Fun radio using an old toaster (Courtesy Robert Knight)



This manual is continually being updated. Please check to see if there is a more up to date manual before commencing any work.

Is the size of this manual worrying you then, try starting with *Raspberry Pi Internet Radio, A Beginners Guide* at:

<https://bobrathbone.com/raspberrypi/documents/RPi%20Radio%20beginners%20guide.pdf>

Examples

This design caters for both the complete novice and more advanced constructors. Do not be put off by the size of this manual as it shows a lot of different designs. Simply read through the following examples and decide which one is the best for you. Some examples are shown in the following pages. This manual is designed to provide inspiration for your own ideas and unique solution to building an Internet Radio using the Raspberry Pi.

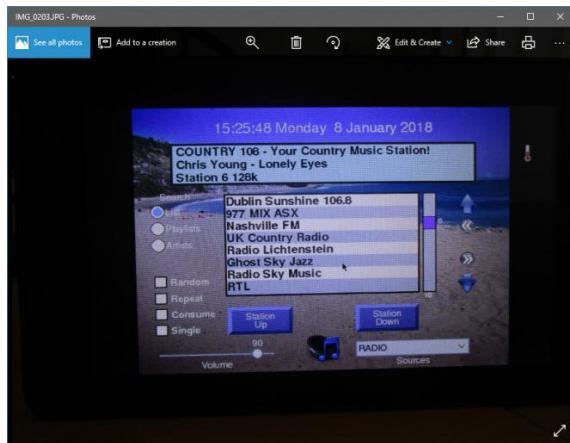


Figure 2 Raspberry pi 7-inch touchscreen radio



Figure 3 HDMI Television running the radio



Figure 4 Vintage tuning touch-screen radio

The Radio software supports the Raspberry Pi 7-inch touch screen. Using the graphic version of this software, the radio can be operated using the touch screen or a mouse and HDMI screen or TV with HDMI. A keyboard can also be attached and used to operate the radio. The touch screen version supports the same functionality as the LCD versions of the radio except for timer and alarm functions. The touch screen can also be configured to use either rotary encoders or buttons.

The HDMI/Touch screen version of the radio can also be configured to run in a window on the Raspberry Pi desktop. Here it is running on the HDMI input of a typical flat-screen television. It can also be configured to use an IR remote control using a FLIRC USB IR detector.

As an alternative to the above design this touch-screen radio is made to look like a vintage radio with a tuning dial. The green slider marks the currently playing station. When a station name is touched on the screen then the slider jumps to that position and plays the selected radio station. The design supports multiple pages of radio stations which can be scrolled left or right. The volume control slider is at the bottom of the screen. This version currently only plays radio stations and not media or Airplay. The touchscreen can also be configured to use either rotary encoders or buttons.



Figure 5 Adafruit 3.5-inch TFT

This example shows an Adafruit 3.5-inch TFT (Thin Film Transistor) touch-screen running the graphical version of the software (Version 6.7 onwards). This is the smallest screen that is currently supported.



Figure 6 Osoyoo 3.5" SPI touch screen

Osoyoo (Pinetree Electronics Ltd.) are a Canadian electronics company who provide a range of well-supported add-ons for Raspberry Pi including this 3.5" SPI touch-screen.

The also provide a variety Raspberry Pi miniature displays using HDMI, DSI and SPI technology, ranging from 2.8" to 10.1".

See <https://osoyoo.com>



Figure 7 Osoyoo 5" Capacitive touchscreen

Another excellent product from **Osoyoo** is the 5-inch DSI (capacitive touchscreen). The main advantage of the 5-inch screen is that it is big enough to display both the full-feature radio (**gradio**) and the Vintage radio simulator (**vgradio**). See **Osoyoo** tutorials on how to install this product.

<https://osoyoo.com/2019/09/20/instruction-for-raspberry-pi-5-dsi-touch-screen/>

The photograph on the left doesn't do it justice as the display is crystal clear.



Example of the PI internet radio using an Adafruit RGB-backlit LCD plate for Raspberry Pi from AdaFruit industries. It has five push buttons and is one of the easiest options to construct. If you want to build this into a case then don't use the buttons supplied with the kit but use external buttons.

Figure 8 Radio using the Adafruit LCD plate



Figure 9 Lego Internet Radio

Example of a fun radio built using this design and Lego from Alan Broad (United Kingdom). It uses a 4x20 line LCD.

This really puts the fun back into computing.



Figure 10 Pi radio using rotary encoders

The rotary encoder switch version of the radio consists of a Raspberry PI connected to an Adafruit 20-character x 4-line RGB LCD display housed. It is all housed in a LogiLink PC speaker set with two rotary encoders. The rotary encoders also have push buttons (Push the knob in). The left one is the *Mute* switch and the right one is the *Menu* switch. The blue glow in the sub-woofer opening comes from a bright blue LED.



Figure 11 Old Zenith radio using rotary encoders

Example of the PI radio from Jon Jenkins built into an old Zenith valve radio case. The two original controls have been replaced by two rotary encoders. The old valve radio inside has been completely removed and replaced with the Raspberry PI and radio components. The LCD display has been built into the top so as not to spoil the original face of the radio. This is a fine example of what can be done with this project.



Figure 12 Transparent Perspex Radio



Figure 13 Perspex radio rear view

The above example built by the author has a transparent perspex front and back panel. It uses a Raspberry Pi with a HiFiBerry sound card and a Velleman 30 watt amplifier.

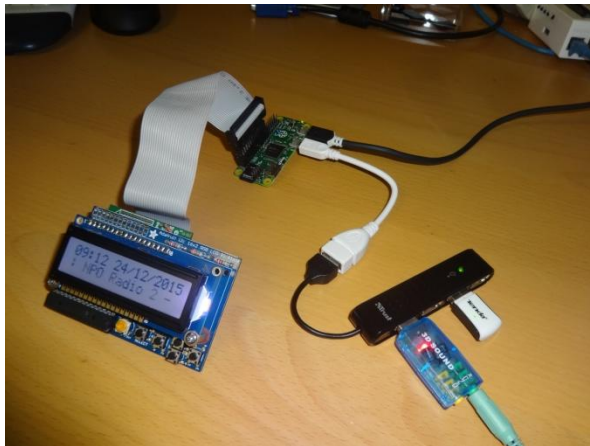


Figure 14 The Radio running on a Pi Zero

This is an example of the radio running on a Raspberry Pi Zero. In this example it uses a micro to standard USB adaptor to connect a simple USB hub. A USB sound dongle and Tenda wireless adapter are plugged into the USB hub. A USB to Ethernet adapter can also be used in place of the wireless adapter. The display used is the Adafruit LCD plate. Also note that the Pi Zero comes with an unpopulated 40 pin GPIO interface. You need to either directly connect directly wires to the GPIO interface (Not advised as they keep falling off) or solder onto a 40-pin male extender header (Advised).



Figure 15 Boom Box radio front view



Figure 16 Boom Box Radio rear view

This beautiful radio is a fine example built by the author. It is using a Raspberry PI model 2B and rotary encoders with inbuilt push button. The display is a 4 x 20 LCD. The sound system is a Velleman 30-Watt amplifier (bottom right) and two 5 ¼ inch 50-watt speakers. It has an IR sensor (Left speaker on the right side) and an activity LED (between the two knobs).

Below is a Raspberry Pi radio built into a old wine box. It uses a 2x8 character LCD and rotary encoders. The amplifier and loud speakers are from a set of old PC speakers.



Figure 17 Raspberry Pi Wine Box radio



Figure 18 Wine box internet radio internal view



Figure 19 Very small radio using the Cosmic Controller

Here is a really cute radio made using the IQaudIO Cosmic Controller and Olimex 128x64 pixel OLED display. The Cosmic Controller provides an excellent solution where space is limited or you simply want a very small radio. The audio output is on the rear of the case.

Note: The Cosmic Controller has been discontinued and is difficult to find but you can still build the radio with separate components.

Below is a fascinating use of both modern and bygone era technology. The radio shown below was created by Broesel from Austria. In this design a Raspberry Pi has been used with the software described in this manual. However, the audio amplifier has been constructed with an ECL84 vacuum valve. The ECL84 valve provides a two-stage mono audio amplifier driving an elliptical wide frequency response loud speaker.



Figure 20 RPi radio with two-stage valve amplifier

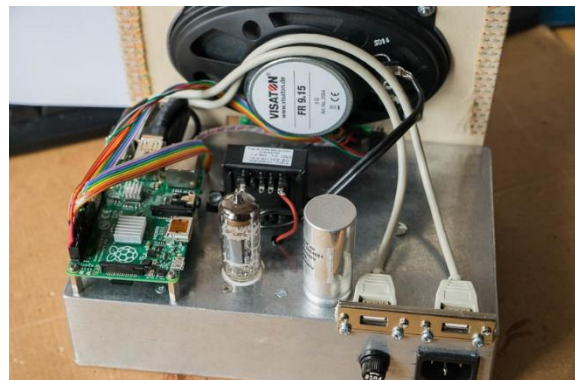


Figure 21 The RPi valve radio chassis view



Figure 22 Pimoroni Pirate Radio

The radio software supports the Pimoroni Pirate radio with pHat BEAT.

The pHAT BEAT gives high-quality, digital, amplified, stereo or mono audio and 16 RGB LEDs, in two rows of 8, which are ideal as a VU (Volume Unit) indicator, and 6 buttons to control the radio (Five on the left and one at the top). See:

<https://shop.pimoroni.com/products/pirate-radio-pi-zero-w-project-kit>

It can also be ordered with an additional speaker.



Figure 23 PiFace CAD Radio with IR Remote Control

The radio supports the PiFace Control and Display (CAD) board. This is obsolete hardware and has limited support and can only be installed on the **Bullseye** Operating System.

The PiFace CAD has five push buttons, a reset button (not currently used). Although it has an Infra Red (IR) sensor for a remote control this does not work with latest IR remote control software and is therefore not available. The PiFace CAD uses the Serial Peripheral Bus interface (SPI) on the Raspberry Pi.

Note: It is only supported on **Bullseye OS**.

Below is an example of a Raspberry Pi Internet radio, made by the author, running the graphical version of the radio which gives a vintage look-and-feel to the final radio. Two vintage Bakelite knobs help complete the vintage appearance.



Figure 24 Vintage look-and-feel Internet radio

The Raspberry Pi model 3B outputs to a Pimoroni 3W stereo D-Class amplifier driving two 3-inch loudspeakers. It is housed in a wooden case painted black with two vintage radio Bakelite knobs.



Figure 25 Vintage look-and-feel - rear view

Vintage Radio Conversion

This version of the software allows for the program to be configured without a screen for use with a vintage radio as shown below:



Figure 26 Philips BX490A (1949) Vintage Internet Radio

The radio is a Philips BX490A manufactured in the Netherlands in 1949. The purpose of this design is retain as much of the original look and feel of a vintage radio which has been converted to run as an Internet radio. It does not have any LCD display. In the above example the following controls are used:

- Far left switch - Simple tone control
- Middle left switch - Volume and mute switch
- Middle right switch – Radio channel (Tuner) or media track selection
- Far right switch – Menu switch (8 positions)
- Push button on right side (Not shown) - Standard menu switch

At the top left the so-called magic eye tuning indicator has been replaced with a Red,Green,Blue status LED. In the above picture the LED is glowing green (Normal operation). This window also contains the IR sensor and activity LED for a remote control. If the radio is busy (loading stations for example) it glows blue. For an error or shutdown the LED glows RED. The IR remote control also flashes red to indicate IR remote control activity.

The software allows espeak to be configured to ‘speak’ station and search information etc.

The details on how to construct a similar project is contained in the following documents:

<http://www.bobrathbone.com/raspberrypi/documents/Raspberry%20PI%20Vintage%20Radio.pdf>

<http://www.bobrathbone.com/raspberrypi/documents/Raspberry%20PI%20Vintage%20Radio%20Operating%20Instructions.pdf>

The illustration below shows a French Radio Schneider Frères Rondo from the 1950’s which has been converted to an internet radio by Franz-Josef Haffner, from Germany.



Figure 27 Vintage radio using a touch screen

What makes this project also very interesting is that he has removed all of the RF section of the valve radio leaving only the audio amplifier and power supply. This is an excellent example of combining old and new technology to extend the life of these increasingly rare radios.

See the Vintage Radio supplent for further details using the following link:

<https://bobrathbone.com/raspberrypi/documents/Raspberry%20PI%20Vintage%20Radio.pdf>



Figure 28 Vintage Radio with AM transmitter

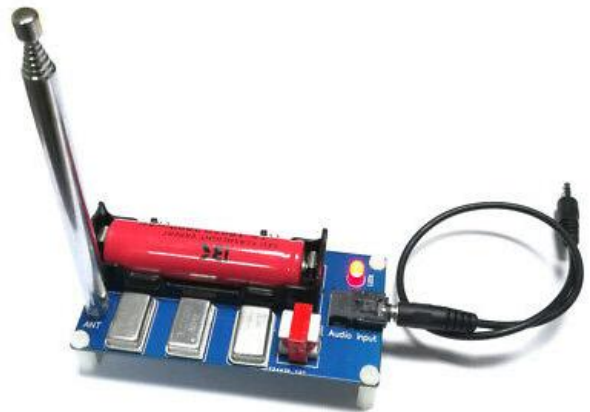


Figure 29 Lusya DIY 3-channel AM transmitter

Another way to connect to a vintage radio (if it is working) is to use a Raspberry Pi and a low power AM transmitter. In the above illustration, the box on the right contains a Pi Zero W running Bookworm OS and the radio software with a tuning knob. The audio comes out of a USB DAC which connects to the Lusya 3-channel AM transmitter which transmits on the 300 Meter wavelength (AM waveband) to a vintage radio. In this example the radio is the British made **Bush DAC 90A** medium/longwave receiver.

Details on construction can be found in the following document:

<https://bobrathbone.com/raspberrypi/documents/Raspberry%20PI%20Vintage%20Radio.pdf>

Chapter 2 – Hardware components overview

Contents chapter 2	Page
Principle components	17
Raspberry Pi computer	17
LCD Displays	20
Using DAC Sound Cards	26
Pimoroni Products	34
Audio options for Raspberry Pi 5	35
HDMI and Touch-screens	37
Radio variants	42
Connecting the LCD display	43
Housing the radio	43
Building in an IR sensor and remote control	44

Principle components

The principal hardware required to build the radio consists of the following components:

- Current versions of the Raspberry Pi computer (Version 1 boards no longer supported)
- Push buttons or rotary encoders for the user interface
- A display such as a HD44780U LCD, OLED or a Raspberry Pi touch-screen display
- Optional Digital to Analogue Converter (DAC) Sound card
- Optional InfraRed (IR remote control)

Raspberry Pi computer

The **Raspberry Pi** is a credit-card-sized single-board computer developed in the United Kingdom by the [Raspberry Pi Foundation](https://www.raspberrypi.com) originally with the intention of promoting the teaching of basic computer science in schools. It has however become immensely popular with hobbyists and engineers. The official site for the Raspberry Pi Web site is <https://www.raspberrypi.com>

More information on the Raspberry Pi computer may be found here:

http://en.wikipedia.org/wiki/Raspberry_Pi

If you are new to the Raspberry Pi try the following “Getting started” guide at:

<https://www.raspberrypi.com/documentation>

You will also find the Raspberry Pi Help Forums and other useful documentation on the above page.

Raspberry Pi model 4B

The Raspberry Pi model 4B was released in June 2019.

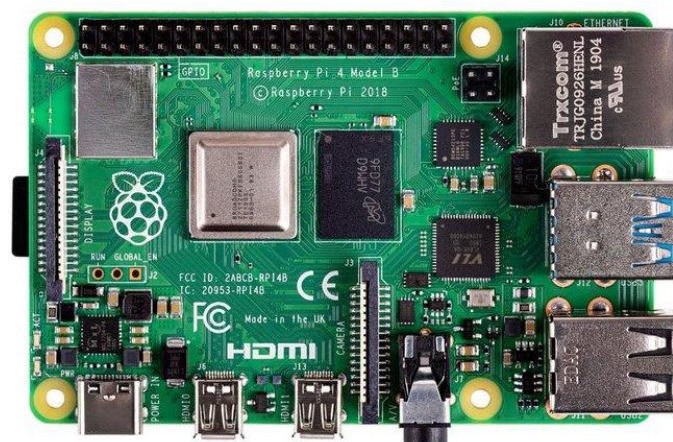


Figure 30 Raspberry Pi Model 4B Computer



Note: The Raspberry Pi 4B requires **Raspberry Pi Bullseye**, **Bookworm** or **Trixie** SD card. It will not work with earlier OS versions. However, the recommended OS for the Radio software is **Bookworm**.



As of the 2nd October the Raspberry Pi Foundation have introduced a new version of the operating system called **Trixie**. Version **8.1** and earlier of the software does not work on **Trixie**. You must use **Bookworm 64-Bit Desktop** with version **8.3** of the radio software.



Figure 31 USB-C plug

The power supply on the model 4B uses a USB-C connector. The usual micro-USB power supply will not fit used for other Raspberry Pi models will not fit. The USB-C specification allows the cable to be inserted either way around. When purchasing the model 4B also purchase the official model 4B power supply which is 5 Volt 3 Amps.



Figure 32 Raspberry Pi MicroHDMI cable

The model 4B also requires an official Raspberry Pi Micro-HDMI cable for each of the micro-HDMI ports (This is not the same as the Pi Zero HDMI adapter). Order one or two of these adaptors if using an HDMI or TV screen.

Audio and video output jack

Earlier versions of the Raspberry Pi have a separate audio output jack and composite video output. Later versions (3 and 4) of the Raspberry Pi have a new AV (Audio/Video) port which combines the audio and video signals in a single jack. Instead of using a standard composite cable, this new connector requires a 4 pole 3.5mm AV cable. To complicate matters: not all of these cables are the same! However, existing audio jack plugs are compatible with the new AV connector.

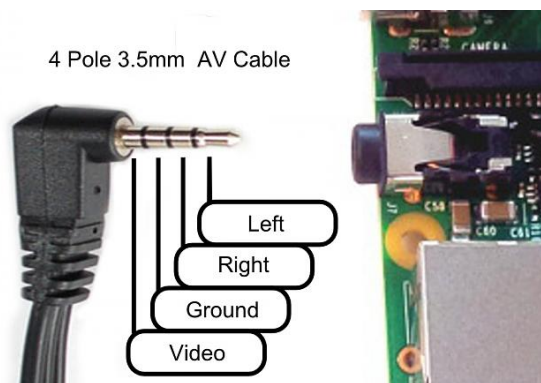


Figure 33 Raspberry Pi B+ AV cable

When choosing a cable, seek an **iPod 4 pole AV** cable. This will however result in the left and right audio channels being reversed but otherwise provides the proper connections. Using other cables, such as a camcorder cable will be hit or miss. Typically, camcorder cables have the wrong pin connections for Video and Ground. This change also can cause some issues with shared grounding with audio speakers. If separate audio and composite AV connector is required, these can be split apart using the same jack inputs as for the model A and B.

Raspberry Pi Model 5

The Raspberry Pi model 5 was introduced in September 2023 and is approximately 2.5 times faster than its predecessor; the model 4B.

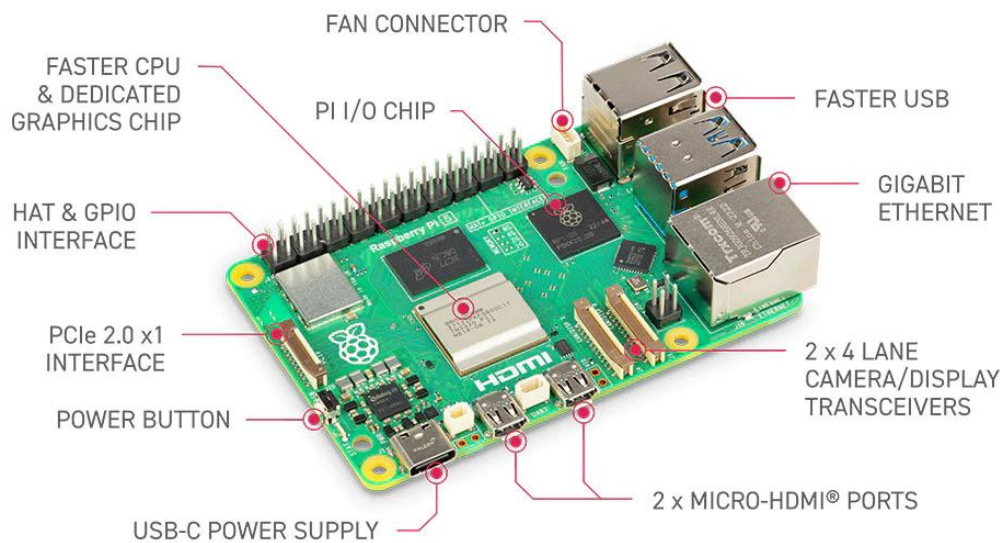


Figure 34 Raspberry Pi Model 5

Note: There is no audio or composite video output jack! See *Audio options for Raspberry Pi 5* on page 35.

Raspberry Pi Zero and Pi Zero W

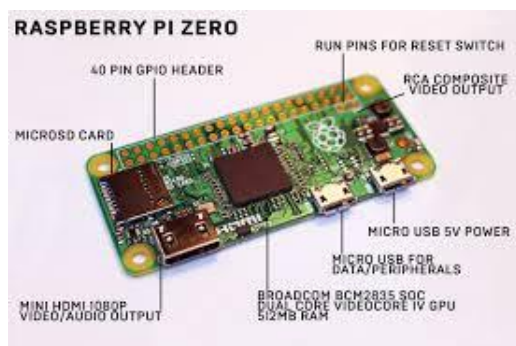


Figure 35 Raspberry Pi Zero

On the original Pi Zero, network connection is only possible with either a USB to Ethernet adapter or a Wi-Fi Dongle. Note that the USB is a Micro USB and will need a micro-USB to standard USB adapter. The Pi Zero W has onboard Wi-Fi and Bluetooth and is a better choice for the radio.



Figure 36 USB Ethernet adapter

In October 2021 the 1GHz quad core Pi Zero 2 W was added to the list of Raspberry Pi's. It appears unable to run **Trixie OS**. Use **Bookworm OS** which is available for download at <https://www.raspberrypi.com/software/operating-systems/>

The Raspberry PI Zero does not have an onboard audio output jack. Sound must be played through either the HDMI port or a USB sound dongle (See Figure 14) or one of the Pi Zero DAC boards available from manufacturers such as **IQaudio**, **HiFiBerry**, **Pimoroni** or **JustBoom**. Alternatively, use Bluetooth speakers.

One useful and simplest audio device is the **Pimoroni Audio DAC Shim** for the Pi Zero/W. Because the Audio DAC SHIM adds no extra bulk to the Raspberry Pi Zero it allows the GPIO header to be accessed as normal, for example other HATs can be fitted on top of it. It pushes over the GPIO header and doesn't require any soldering. It is a very simple way to provide audio output to the Raspberry Pi Zero/W. See *Pimoroni Audio DAC Shim for Pi Zero* on page 31

Raspberry Pi 400/500

In November 2020 the Raspberry Pi foundation released the model 400. This is housed in a keyboard case with all connectors including a 40-pin GPIO header at the back of the unit. Most notably it does not have a 3.5mm audio Jack. If connected to a HDMI monitor or TV then sound can be played through the HDMI interfaced speakers on the monitor/TV. Alternatively, a USB DAC or Bluetooth speakers can be used. As the RPi 400 has a 40-pin GPIO header it can, in theory, be connected to a suitable DAC via a ribbon cable. The Raspberry Pi 500 is untested by the author but should work without problem using Bookworm or later.



Figure 37 Raspberry Pi 400 running the graphical radio

LCD Displays

The HD44780U LCD display



Figure 38 The HD44780U LCD display

The HD44780U LCD interface is an industry standard interface for a variety of LCD displays. These can come in various sizes but the two lines by 2x16 or 4x20 character displays are the most popular. Most of these modules compatible with the Hitachi HD44780U LCD controller so there is a wide choice of these displays.



Figure 39 OLED 4 x20 LCD display

The latest displays use **OLED character** displays (Organic Light Emitting Diode) and give very good results and are becoming more popular when compared to traditional LCD displays.

See <https://en.wikipedia.org/wiki/OLED>

For pin-out details see LCD pin outs on page 50.



Also see *Configuring Russian/Cyrillic text* on page 175.

Midas LCD displays with VEE



Figure 40 Midas LCD display with VEE

Some LCDs from Midas are compatible with the HD44780U except for pin 15 (VEE) which outputs a negative voltage. For pin-out details see LCD pin outs on page 50.

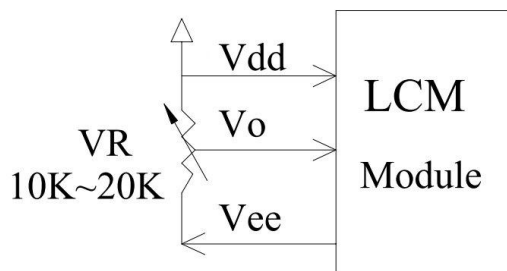


Figure 41 HD44780 potentiometer wiring

The diagram on the left shows the wiring for the 10K contrast potentiometer.

Pin	Name	Description
2	VDD	+5 Volt
3	VO	Contrast adjustment
15	VEE	Negative voltage output



Do not connect pin 15(VEE) to the +5V supply. It will damage the LCD and possibly the Raspberry Pi.



WARNING: DO NOT CONNECT AN I2C BACKPACK TO THIS TYPE OF DEVICE AS BACKPACKS CONNECT +5V ONTO PIN 15 AND WILL DAMAGE BOTH THE BACKPACK AND THE LCD.



The term **LCD** is used throughout this manual to mean both traditional **LCDs** and **OLED character** displays which are gradually replacing **LCDs**.

Midas Character OLED with MC0100 Controller

So-called Character OLEDs are gradually replacing LCDs. Midas market a wide range of such displays.

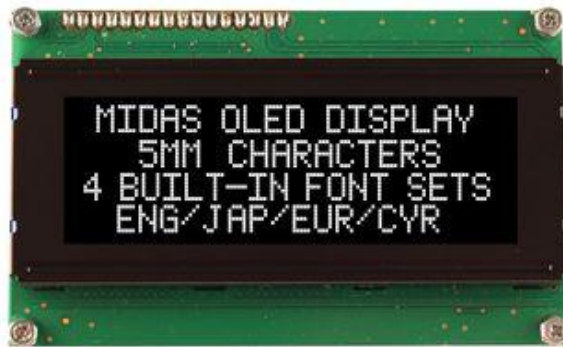


Figure 42 Midas character OLED

These displays use the **MC0100** controller which is largely compatible with the Hitachi **HD44780U** controller.

This controller can support various built-in font sets such as English, Western European, Japanese and Russian.

Since OLEDs generate their own illumination, they do not need pins 15 and 16 connected for backlighting.

Winstar 40-character LCDs



Figure 43 40-character 2-line LCD

Version 8.0 onwards supports Winstar 2-line 40-character display (Model No. WH4002A). These displays show the station/artist/track information slightly differently than other displays used in this design. The top line displays the time and date along with the station or artist details. The second line displays the station information (if supplied) or track being played. Sometimes if a station name is too long to display the date as well as the station/track information then only the station/artist will be displayed on line 1.

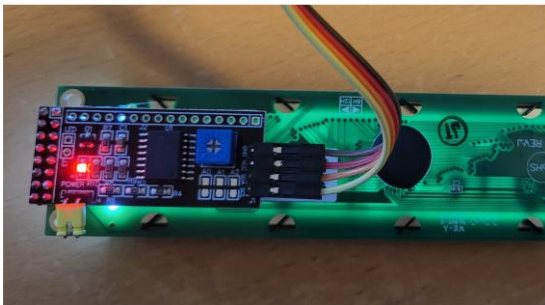


Figure 44 2x40 character LCD with I2C backpack

These displays have a 16-DIL connector and can be driven either directly from the GPIO pins or using a compatible PCF8574 I2C LCD backpack with a 16-pin DIL connector (2x8 pins) as shown on the left. This is the recommended configuration for this type of display.



Note: There are also similar 40-character displays available on the market which use **Vacuum Fluorescent Display (VFD)** technology. These were once commonly used on consumer electronics equipment such as video cassette recorders, car radios, and microwave ovens. This type of display is not currently supported and is unlikely to be in the future.

Please also note that **4-line 40-character** LCDs are not supported as this doesn't make sense as there is insufficient information available from the music stream to warrant using a display capable of displaying 160 characters.

Olimex OLED 128x64 pixel screen



The Olimex 128x64 pixel OLED is a low cost, low power, high contrast LCD display with a UEXT connector. It is controlled via the I2C or SPI interface. The power supply required is only in the range of 1 uA in sleep mode, 200 uA in operating mode and 7mA in display ON mode. View area is 21 x 11 mm. It is particularly useful where space is very limited.

See:

<https://www.olimex.com/Products/Modules/LCD/MOD-OLED-128x64/open-source-hardware>



Note: The Olimex OLED screen is not a particularly fast device when compared with say an LCD or graphics screen. However, its biggest advantage is its size.

Sitronix SSD1306 128x64 pixel OLED



Figure 45 Sitronix SSD1306 128x64 pixel monochrome OLED

The Sitronix SSD1306 128x64 pixel 0.96-inch OLED is display marketed under various names (Such as Makerhawk) has just four connections for the I2C interface.

The wiring from left to right is:

1. VCC +5 volts – GPIO header pin 2
2. GND (0 volts) – GPIO header pin 6
3. SCL I2C Clock – GPIO 3 (pin 5)
4. SDA I2C Data – GPIO 2 (pin 3)

The I2C interface hex address for this device **0x3C**. It uses the **ssd1306_class.py** driver.



Note: It is not currently possible to flip the display up-side down due to limitations of the SSD1306 driver software. Use the LUMA.SSD1306 driver instead which can be flipped.

SSH1106 1.3-inch I2C monochrome OLED



Figure 46 1.3-inch SH1106 I2C OLED

This 1.3-inch monochrome 132x64-pixel display connects via the I2C interface. It uses the SH1106 display chip.

The wiring from left to right is:

1. VCC +5 volts – GPIO header pin 2
2. GND (0 volts) – GPIO header pin 6
3. SCL I2C Clock – GPIO 3 (pin 5)
4. SDA I2C Data – GPIO 2 (pin 3)

The I2C interface hex address for this device **0x3C**.



Figure 47 SH1106 0.91" 128x32-pixel TFT

These OLEDs also come in in 128 x 32-pixel sizes from various manufacturers and can also be used with the Luma driver. The wiring from left to right in the example on the left is:

1. SDA I2C Data – GPIO 2 (pin 3)
2. SCL I2C Clock – GPIO 3 (pin 5)
3. Reset pin – Not used
4. GND (0 volts) – GPIO header pin 6
5. 3Vo – +3.3V output. Not used
6. VIN +5 volts – GPIO header pin 2

These OLEDs can also use the Luma driver and usually have an I2C address of 0x3C.

Note: These displays can be difficult to read due to their very small size.

SH1106 128x32 pixel OLED monochrome OLED

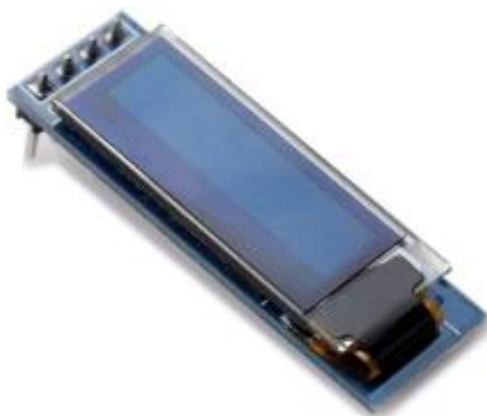


Figure 48 - 0.91-Inch 128x32 OLED

These 0.91 x 0.38-inch 128x32 pixel monochrome OLED display modules connect via the I2C Serial Interface. They are marketed under various names such as **WayinTop** or **AzDelivery**. This is the smallest display supported by this project.

Because of its size, the text can be hard to read. Also, one peculiarity is that the PIL driver software recognises the display as being 128x64 pixels and not 128x32. It is most useful where there is little space available for a display.



Note: OLED devices are very slow when compared to character LCDs or Graphics (touch) screens. Their main advantage is their size.

Grove LCD RGB Backlight



Figure 49 Grove I2C LCD RGB Backlight

The **Grove JHD1313 LCD RGB** backlight is a 2x16 character LCD with an RGB backlight. Version 4 and lower of the Grove LCD RGB LCD uses two controllers.

AIP31068L – I2C controller for the LCD

PCA9632DP1 – I2C RGB backlight driver

Version 5 onwards uses the **SGM31323** controller for the backlight.

The interface is the 2-wire I2C interface plus +5V power supply. The backlight can be set to any colour and is configurable in the radio software.

RPi 1.3" 128x64 OLED (SH1106) with joystick



Figure 50 RPi 1.3" 128x64 OLED with joystick

This device uses a 1.3-inch 128x64 pixel OLED display driven by an SH1106 chip and the SPI interface.

It comes with a 5-button joystick (on the left) and three push buttons (on the right). It is designed to fit on a 40-pi Raspberry Pi Zero-W but can also fit on a standard Raspberry Pi such as a 3B or 4B. If fitted to a Pi Zero Audio, audio output can be provided using a USB to jack-plug audio adaptor or Bluetooth speakers.



Note: This device comes wired to use an SPI interface. It can be modified to use the I2C interface using the I2C address **0x03**. In such a case use the LUMA-SH1106 I2C driver (See *Converting the 1.3" OLED pHat to use the I2C interface* on page 244).



Please note that the Pimoroni Audio DAC shim is not compatible with devices using SPI. This is because the OLED driver uses GPIO 25 which is also the MUTE pin on the Audio DAC shim. It Audio shim can only be used if the OLED with joystick is re-wired to use the I2C interface.

The following table shows the functions of the various switches and the GPIO pins that they are using. The Joystick and button wiring cannot be physically rewired however the Radio software needs to know what the wiring of each switch is and this is configured using the **configure_radio.sh** program.

Table 1 - 1.3" OLED with joystick button designations

Joystick	Buttons	Radio Function	GPIO Pin
Up		Channel Up	6
Down		Channel Down	19
Left		Volume Down	5
Right		Volume Up	26
Push		Menu switch	13
	KEY1	Mute Switch	21
	KEY2	Not used	n/a
	KEY3	Not used	n/a

Waveshare 2.42 and 1.5-Inch SPI interface OLEDs



Figure 51 Waveshare 1.42-inch SPI OLED Interface

Some OLEDs are rather on the small side and can be difficult to read at times. Waveshare make 2.42-inch OLED display that solves this problem. It has 5-wire SPI interface and the 2.42-inch interface gives good results. They also have a 1.5-inch display.

Wiring details are described in Appendix A.8 *Waveshare 1.42" and 1.5" SPI OLED display wiring* on page 246.

Using DAC Sound Cards

The sound output of the on-board audio jack on the Raspberry Pi is known to be limited. Using a DAC (Digital Audio Converter) will give much better quality and output. Several types are available. The one you choose depends upon your requirements. These DAC cards use PCM (Pulse Code Modulation) technology and the Raspberry Pi i2s interface.

The following sound cards and adapters are supported in this project:

Manufacturer	Section	Page
HiFiBerry	HiFiBerry DAC	27
IQaudIO	IQaudIO DAC sound products	28
Justboom	JustBoom DAC products	29
Various	PCM5102A DAC Sound Card	30
Pimoroni	Pimoroni audio DACs	30
Adafruit	Adafruit speaker bonnet	32
Allo	Allo DAC products	32
Waveshare	Waveshare WM8960 Audio Hi-Fi Sound Card	33
Rpi DAC	Raspberry Pi badged sound cards	33

If you are going to use an external amplifier then almost any DAC will do. If you want a complete solution, a DAC card with an in-built amplifier (typically 3W up to 25W) is a good choice. The output from these in-built amplifiers is usually so-called class D-type amplification. Another consideration is the required connection to the amplifier copper, S/PDIF or optical audio connection (Toslink).

If price is a big consideration there are a number of very reasonably priced DACs which emulate some of the better-known ones and use the same device driver software as the one, they are emulating.

For more information on DACs see https://en.wikipedia.org/wiki/Digital-to-analog_converter

HiFiBerry DAC

This version supports the HiFiBerry DAC from HiFiBerry. See <https://www.hifiberry.com>. There is a comprehensive range of DACs available from this manufacturer. A few are shown below:

1. HiFiBerry DAC+ Light/Light – Entry level and standard solution
2. HiFiBerry Digi+ Light – Optical (TOSLink) and RCA connectors
3. HiFiBerry AMP+ –Standard DAC with a 25W D-Class amplifier
4. HiFiBerry DAC+ Zero Form Factor
5. HiFiBerry 3W Miniamp Zero Form Factor

More products will be shown on their Web Site. Also, a list of **HiFiBerry** product software drivers (DT overlays) will be found in *Appendix C – HiFiBerry Device Tree Overlays* on page 248.

The HiFiBerry DAC Plus

The HiFiBerry DAC PLUS uses the 40-pin connector and has an unpopulated 40-pin header to extend the GPIO pins on the HiFiBerry DAC to use with other cards.



Figure 52 HiFiBerry DAC Plus

The DAC plus uses the 40-pin connector on new Raspberry Pis. A 40-pin dual in line male header is required (purchase separately).

Solder the 40-pin male header into the component side of the HiFiBerry DAC as shown Figure 53 below. The Radio controls and LCD screen for example are then connected on this this header.



Figure 53 HiFiBerry mounted on the Raspberry Pi

The A+/B+/Pi2 uses the following pins supporting PCM.

Pin 12 GPIO18 PCM_CLK
Pin 35 GPIO19 PCM_FS
Pin 38 GPIO20 PCM_DIN
Pin 40 GPIO21 PCM_DOUT
(All set to mode ALT0)

Pin 12 (GPIO) conflicts with the down switch on the radio. Wire the down switch to GPIO 10 (Pin 19) and configure the **down_switch=10** parameter in **/etc/radiod.conf** or by running the **configure_radio.py** program.



Note: All settings in the **/etc/radiod.conf** file use GPIO numbers and NOT physical pin numbers. So, in the above example **down_switch** is GPIO 10 (Physical pin 19).



The Pimoroni pHat is compatible with HiFiBerry DAC (Not DAC+) and uses the same Device Tree (DT) overlay.

IQaudio DAC sound products

IQaudio DAC also have a comprehensive range of products. Again, these provide excellent results. These cards fit within the Pi's form factor and provide additional full access to the Pi's 40way I/O signals allowing easy addition of IR sensors, Rotary Encoder or i2c devices (such as OLED screens) etc. See <http://iqaudio.co.uk>

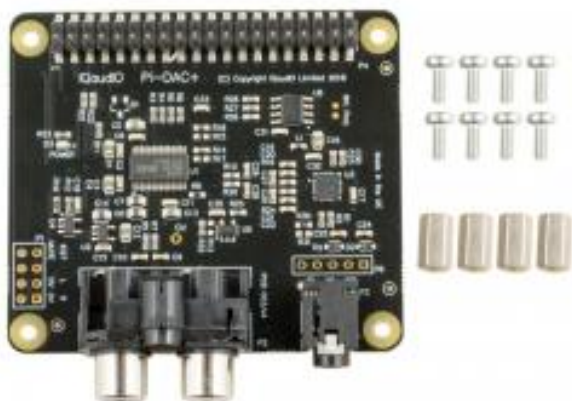


Figure 54 IQaudio DAC plus

DAC for the Raspberry Pi. This latest revision of the IQaudio Pi-DAC PRO and is pre-programmed for auto detection. Line out: 2x Phono/RCA
Headphone: 3.5mm socket.



Figure 55 IQaudio Pi-DigiAMP+

This provides the DAC+ along with a 35W amplifier which fits the. Raspberry Pi A+/B+/RPi2/3/3B+. This card requires supports up to 24v power supply and delivers the full 2.5amp to the Pi.

JustBoom DAC products

The construction using **JustBoom** products is similar to other sound cards. The radio must be wired as shown in *Table 4 Radio and DAC devices 40-pin wiring*. The `/etc/radiod.conf` configuration file must also be configured to support these devices by running the audio configuration program.

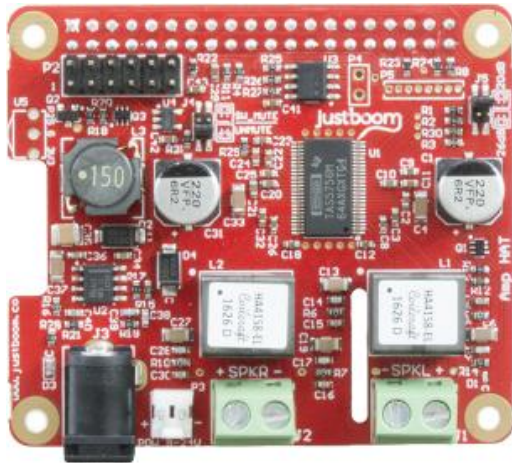


Figure 56 JustBoom Amp HAT



Figure 57 JustBoom Amp Zero pHAT

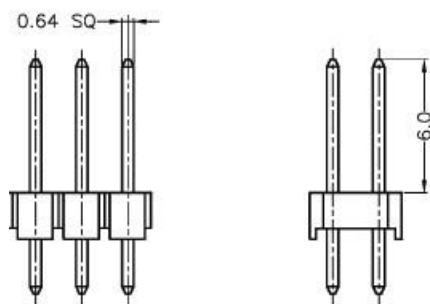


Figure 58 JustBoom Zero stacker requirements



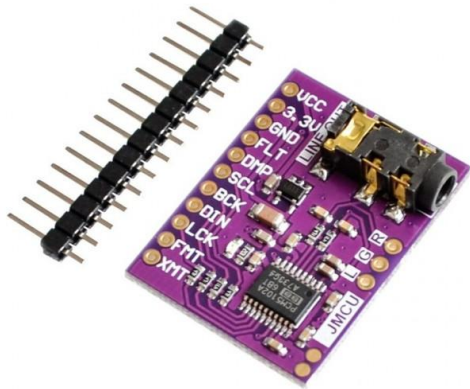
Figure 59 Using the 40-pin stacker

The **JustBoom** Zero boards are used with stackers or installed directly on the Raspberry Pi Zero. Some stackers and some 2x20 male headers on the market though are too thin or too short to provide good contact with the board. Use stackers that the pins are squared and are at least 0.6mm in width. If you are soldering the 2x20 male header on the Raspberry Pi Zero make sure that the pins are 0.6mm in width and 6mm in usable height.

Plug a suitable stacker onto the Raspberry Pi Zero. Plug the JustBoom Zero board on top of the stacker so that the pins protrude through the JustBoom Zero board.

Plug the radio interface card or ribbon cable (not shown) on top of these protruding pins.

PCM5102A DAC Sound Card



DAC	Function
FLT	Filter Select. Normal latency (Low) / Low latency (High)
DMP	De-emphasis control for 44.1kHz sampling rate: Off (Low) / On (High)
FMT	Audio format selection: I2S (Low) / Left justified (High)

There are a number of inexpensive DACs which use the PCM5102A chip and have a 3.5mm Stereo Jack 24. These give a very reasonable 24-bit sound. Despite the fact that they claim to be a pHat they do not plug directly into the Raspberry Pi. Use either jumper wires or build your own interface board. These boards use the HiFiBerry DAC device driver and the I2S dtoverlay. Wiring as below:

DAC	RPi	Pin
VCC	5V	2
3.3V	N.C.	-
GND	GND	6
FLT	N.C.	-
DMP	N.C.	-
SCL	N.C.	-
BCK	GPIO18	12
DIN	GPIO21	40
LCK	GPIO19	35
FMT	N.C.	-
XMT (1)	3.3V	1

N.C. Not Connected. (1) Connect via 10K resistor.

Pimoroni audio DACs

Pimoroni pHat DAC

The Pimoroni pHAT DAC provides an affordable high-quality DAC for the Raspberry Pi. The 3.5mm stereo jack comes soldered onto the board already. Though designed to match the format of the Raspberry Pi Zero it is compatible with all 40-pin GPIO Raspberry Pi variants.

Features:

- 24-bit audio at 192KHz
- Line out stereo jack
- pHAT format board
- Uses the PCM5102A DAC to work with the Raspberry Pi I2S interface

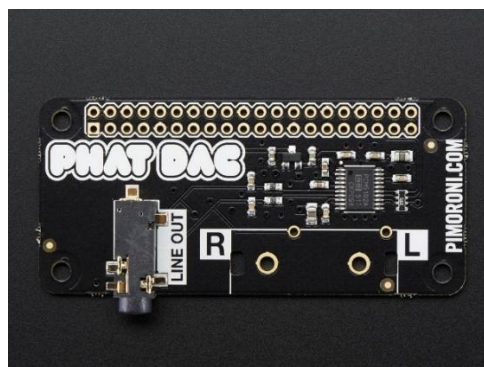


Figure 60 Pimoroni pHat DAC



Note 1: Do not use the 40 female header that comes with the board but use a 40-pin extender so that other cards can be used on top of it.



Note 2: The Pimoroni pHat is not completely compatible with the HiFiBerry DAC although it uses the same software driver. In particular to use the Alsa sound mixer a package called **pulseaudio** is required. However, **pulseaudio** is not compatible with several of the features of this package such as the **espeak** speech package. The Pimoroni pHat DAC will run fine without any mixer controls.

Pimoroni Audio DAC Shim for Pi Zero



Please note that the Pimoroni Audio DAC shim is not compatible with *RPi 1.3" 128x64 OLED (SH1106) with joystick* (See page 25). This is because the OLED driver uses GPIO 25 which is also the MUTE pin on the Audio DAC shim. It Audio shim can only be used if the OLED with joystick is re-wired to use the I2C interface (See *Converting the 1.3" OLED pHat to use the I2C interface* on page 244).

Pimoroni now supply a DAC which uses PCM5100A DAC chip. It has a friction fit header which slips pushes down over the Pi Zero 40-pin GPIO header and doesn't require any soldering and is removable.

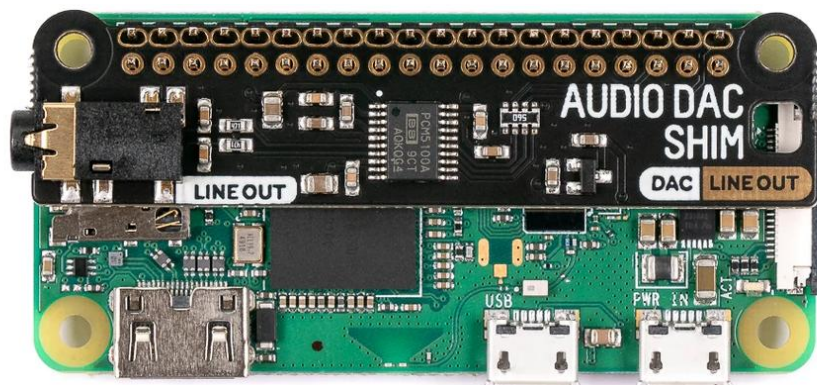


Figure 61 Pimoroni Audio DAC Shim for Pi Zero/W

The above figure shows a Pimoroni Audio DAC shim fitted to a Pi Zero W. The PCM5100A DAC chip takes high quality digital audio from the Pi Zero and pipes out crisp, line-level 24-bit / 192KHz stereo audio through the 3.5mm jack.

Because the Audio DAC SHIM adds no extra bulk to the Raspberry Pi Zero it allows the GPIO header to be accessed as normal, for example other HATs can be fitted on top of it. It should also fit easily inside any standard case. It's a simple way to add an audio output to the Pi Zero. Pimoroni even claim that it can also be fitted to a Raspberry Pi 400.

It uses the same driver as HiFiBerry DAC (Not DAC Plus or Digi). This can be configured during installation of the radio software.

Alternatively, it can be manually configured using by adding the following lines to the [All] section of **/boot/firmware/config.txt** (for **Bookworm**):

```
dtoverlay=hifiberry-dac
gpio=25=op,dh
```

If you're using a Pi that has an audio jack you might also need to disable onboard audio by either adding a # to the beginning of the following line or setting **on** to **off**.

```
#dtparam=audio=on
```

Adafruit speaker bonnet



Figure 62 Adafruit Speaker Bonnet

The Adafruit speaker bonnet is primarily designed for the Raspberry Pi Zero format but can be used on any Raspberry Pi.

It consists of a stereo 3W amplifier connected to the I2S interface of the Raspberry Pi via a 40-pin DIL female header.

It can be purchased with two miniature speakers or it may be used with any small 4 or 8-Ohm speakers.

Allo DAC products

Allo is a manufacturer of very high-quality audiophile sound products. A good example is the **Allo Piano 2.1 HiFi DAC (EU)** with **woofer output** as shown below.

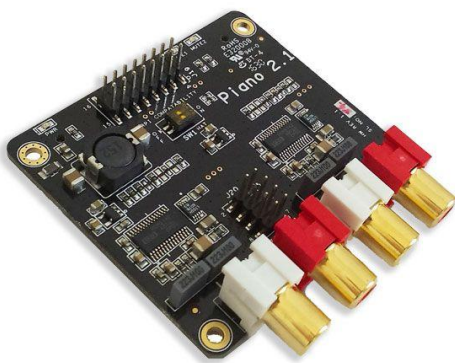


Figure 63 Allo Piano 2.1 HiFi DAC with woofer

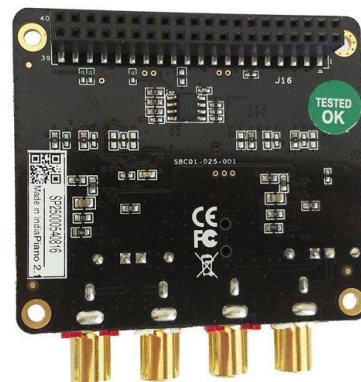


Figure 64 Allo Piano 2.1 HiFi DAC (Rear)

The **Allo Piano** has two outputs, one for normal sound ranges and one for a sound woofer. The board comes pre-programmed as 2.1 for the Raspberry Pi. The subwoofer has 2 outputs (Stereo left and right) but is mono only. However, it uses a second I2S channel on GPIO5 and mute signal on

GPIO6 both used to control the woofer amplifier. This conflicts with the LCD signals **lcd_data4** and **lcd_data5** pins.



Allo products currently need to be specially configured to correct this conflict. See *Configuring Allo Sound Cards* on page 134.

Further information about Allo products can be found at <https://www.allo.com/sparky/index-dac.html>

Waveshare WM8960 Audio Hi-Fi Sound Card

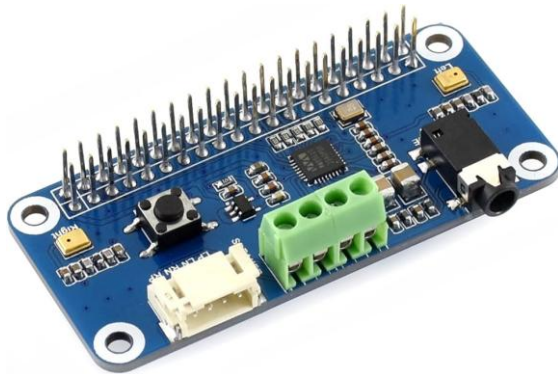


Figure 65 Waveshare WM8960 Sound Card

The Waveshare WM8960 Audio Hi-Fi Sound Card integrates WM8960 low power stereo CODEC, which communicates via I2S interface. It also has two high-quality MEMS silicon Microphones (Not used in this project) mounted on the left and right of the board. Onboard standard 3.5mm earphone jack, play music via external earphones. Onboard dual-channel speaker interface, directly drives speakers. **Note: Waveshare** only support for this card on **Bookworm** or **Trixie**.



These cards are often sold with a pair of speakers which plug directly into the card. The particular speakers shown are 5W enclosed speakers.

For more information about this sound card see <https://www.waveshare.com/wm8960-audio-hat.htm>

This card has special installation and uninstallation scripts. See *Installing the Waveshare WM8960 DAC* on page 136.

Raspberry Pi Audio Cards



Figure 66 Raspberry Pi DAC Pro

Raspberry Pi DAC Pro

The **Raspberry Pi DAC Pro** is the highest-fidelity audio HAT. With the Texas Instruments PCM5242, the DAC Pro provides an outstanding signal-to-noise ratio (SNR) and supports balanced/differential output in parallel to phono/RCA line-level output. It also includes a dedicated headphone amplifier. This card is a rebadged **IQaudio DAC Pro**.



Figure 67 Raspberry Pi DigiAMP+

Raspberry Pi DigiAMP+

The **Raspberry Pi DigiAMP+** is a high-performance audio HAT. With its on-board Texas Instruments TAS5756M stereo amplifier, it delivers a direct connection to passive stereo speakers at up to 35 Watts per channel with variable output, and is ideal for use in Raspberry Pi-based Hi-Fi systems. Powered by an external 12–24V DC power supply, it connects directly to Raspberry Pi's GPIO header, providing power to the Raspberry Pi itself. This card is a re-badged **IQaudIO DigiAMP+**.

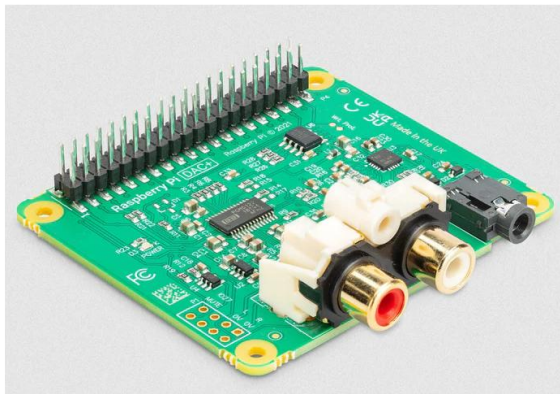


Figure 68 Raspberry Pi DAC+

Raspberry Pi DAC+

Rebadged from the **IQaudIO DAC+**, the Raspberry Pi DAC+ is a low-cost audio output HAT for the Raspberry Pi (any 40-pin model), supporting 24-bit 192kHz high-resolution digital audio. It uses a Texas Instruments PCM5122 DAC to deliver stereo analogue audio to a pair of phono connectors, and also provides a dedicated headphone amplifier. The DAC+ is compatible with any 40-pin Raspberry Pi and requires no soldering. It uses a Texas Instruments PCM5242 DAC for even higher signal-to-noise ratio.

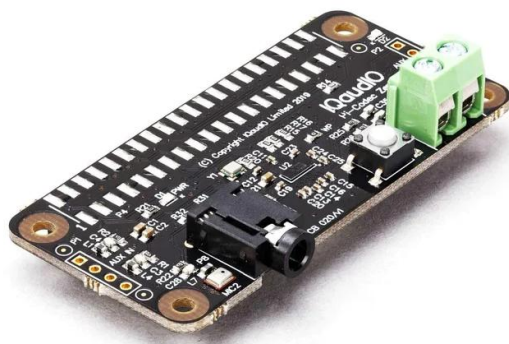


Figure 69 Raspberry Pi Codec Zero

Raspberry Pi Codec Zero (Mono)

The Codec Zero is a Raspberry Pi Zero-sized audio HAT that delivers bi-directional digital audio signals (I2S) between a Raspberry Pi and the Codec Zero's on-board Dialog Semiconductor DA7212 codec, and allows you to use a variety of input and output devices. With features such as programmable green and red LEDs, plus a push button for user input, Codec Zero is suitable for projects requiring a microphone input. It supports a 1.2W 8Ω mono speaker.

Pimoroni Products

Pimoroni are a UK based company who produce electronic products for both Raspberry Pi and Arduino. They make a range of all-in-one audio boards for Raspberry Pi, with high-quality digital audio. Their Website is at <https://shop.pimoroni.com/>

Pimoroni Pirate Radio



Figure 70 Pimoroni Pirate Radio - Rear view

The illustration on the left shows the rear of Pimoroni Pirate radio. The amplifier consists of dual I2S DAC/amplifiers for stereo audio (MAX98357A) at 3 Watts per channel.

The Pirate radio comes as a kit (Soldering skills required). The Pimoroni software is disabled and the software from this project used instead. Note: It does not have a screen.

Pimoroni Pirate Audio



Figure 71 Pimoroni Pirate Audio

No soldering skills are required to construct this project when using a Pimoroni Pirate Audio range of products and a Raspberry Pi Zero with a pre-soldered 40-pin header.

You need the **Bookworm OS** and not **Trixie**.

Bookworm can be downloaded at:

<https://www.raspberrypi.com/software/operating-systems/>

It comes with a 240x240 pixel colour 1.3-inch IPS (In-plane switching) display which gives a very good viewing angle. The display is driven by an ST7789 controller.

There are four variants of the Pimoroni Audio but they all use the same DAC and display software:

1. Pirate Audio Speaker - MAX98357A DAC with mini 1W / 8Ω speaker
2. Pirate Audio Line-out - PCM5100A DAC chip with 3.5mm output stereo jack
3. Pirate 3W Stereo Amp - MAX98357A DAC with mini 3W amplifier output
4. Pirate 3W Headphone Amp - PCM5100A DAC driving a PAM8908 headphone amplifier



Note: The software for Pimoroni Pirate Audio is very basic at the moment in particular the menu selection function. It is hoped to improve this at a later date.

Audio options for Raspberry Pi 5

The **Raspberry Pi 5** does not have an on-board audio jack. This means that it requires one of the following three options for audio output:

1. A half size audio DAC such as an Adafruit I2S DAC bonnet
2. A USB Bluetooth audio dongle
3. Bluetooth speakers or headset

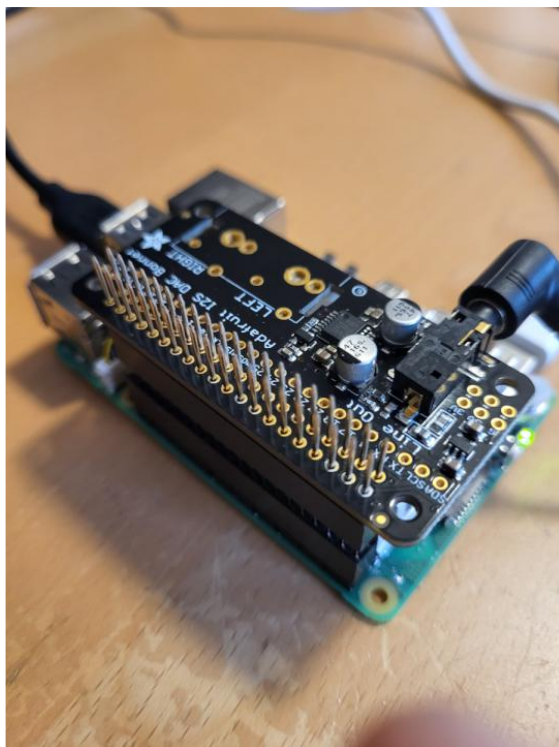


Figure 72 Raspberry Pi with a Adafruit DAC



Figure 73 Raspberry Pi 5 DAC cooling fan clearance

Figure 72 above shows an Adafruit I2S DAC bonnet mounted on a 40-pin header extender. This is needed to give adequate clearance for the cooling fan (if fitted) as shown in the second Figure 73.



Note: It is probably advisable to fit a half-size DAC (Pi Zero format) rather than a full-size DAC to improve cooling fan airflow. A full-size DAC Hat may still allow for adequate cooling but no testing has been done on this.

HDMI and Touch-screens

Types of monitors and touchscreens

This section is only relevant if configuring an HDMI or touchscreen interface. If using an LCD or OLED display then skip this section. There are three types of monitors or touch screens used in this project as shown below.



Figure 74 HDMI monitors/touchscreens

HDMI monitors/touchscreens

They use an HDMI (High-Definition Multimedia Interface) connection like you will usually find on your TV or computer monitor. The latest Raspberry Pi's have two miniature HDMI connections.

They connect to the Raspberry Pi using an HDMI cable and are easy to install (plug and play) and use the standard X-Windows or Wayland interface.



Figure 75 DSI monitors and touchscreens

DSI (Display Serial Interface) touchscreens.

Connection is provided by a thin ribbon cable connecting from the **DSI** interface on the Raspberry Pi as shown in the illustration on the left.

These types of displays or touchscreens use fairly simple installation procedures which can vary between manufacturers.



Figure 76 SPI interface touchscreens

SPI interface touch screens

These displays/touchscreens use the SPI (Serial Peripheral Interface) and either plug directly into the GPIO header on the Raspberry Pi or via jumper wires. Installation procedures can be complex and unfortunately these displays are often poorly supported by their manufacturers who seem to be struggling to get these screens to work with the **Trixie OS**.

They are also much slower than other types of display and are very sluggish on earlier types of Raspberry Pi's (Model 3B for example).



Note: There are currently serious problems with small 2.8" touch-screens under both **Bookworm** and **Trixie OS**. Do not use! Contact bob@bobrathbone.com if you wish to use these. Also see *Touch screen installation* on page 41.

Adafruit 2.8 and 3.5-inch TFT touch-screens

The radio software supports the Adafruit 3.5-inch TFT touch screen (480x320). The small size can make the controls difficult to use but it will still work. It is best to use a touch-screen stylus.



Figure 77 Adafruit 3.5-inch TFT touchscreen

Waveshare 2.8 and 3.5-inch TFT touch-screens

Waveshare provide a 2.8 and 3.5-inch TFT touch screen (480x320 and 720x480 pixels respectively). These have 26-pin DIL connector which plugs in directly to the GPIO header.



Please note that at the time of writing this device currently only works with the Bookworm 64-bit Operating System. A new version for Trixie OS is being worked upon. Also, there is a problem with touch-screen operation. Also note that screens with less than 480x320 pixels are no longer supported (2.8" screens for instance).



Figure 78 Waveshare 2.8-inch TFT touch screen (unsupported)

The above screen shows a 2.8-inch Waveshare touch screen (Courtesy - Recep A. Güleş). Note the revised layout due to its size when compared to the 7-inch touch screens. Unfortunately 2.8" screens are not supported in this version. This may change in the future.



Note: This software has only been tested with the Raspberry Pi 7-inch touch screen, the Waveshare 3.5-inch screen and the Adafruit 3.5-inch TFT touch-screen . Smaller than 7-inch screens may prove difficult to operate. The following resolutions are supported: 800x480, 720x480, 480x320 or 1024x600 pixels. Screens smaller than 480x320 pixels are not supported in this version including 2.8" screens.

Raspberry Pi 7-inch touch screen

As an alternative to building a radio using limited LCD screens it is possible to build a radio using the Raspberry Pi 7-inch or 3.5-inch touch screen or any other HDMI screen (touch-screen or otherwise). If the screen does not have touch capability, then it is possible to use it with a mouse or keyboard or both. Also, the touch screen can be used in conjunction with rotary encoders or push buttons.



Figure 79 Raspberry Pi 3 with 7-inch touch screen

There is a very good setup guide for the Raspberry Pi touch-screen at:

<https://www.modmypi.com/blog/raspberry-pi-7-touch-screen-assembly-guide>

Connection via ribbon cable

Smaller touch screens usually plug directly into the Raspberry Pi GPIO header, however that may not be convenient especially if you want to build it all into a case. Fortunately, it is possible to purchase a 40-way DIL male to female ribbon cable which allows the display to be panel mounted.

The following illustration shows TFT screen on the connectors side. If it is required the display can be put on the GPIO side but will be displayed upside down. Luckily it is easy to flip the display upside down as shown in the installation instructions.



Figure 80 TFT connected by a 40-pin male/female ribbon cable

MHS 3.5-inch RPi Display

The MHS 3.5-inch RPi Display appears to be a badged version of the same hardware as the Waveshare 3.5" TFT. The installation software is also very similar.



Figure 81 MHS-3.5-inch RPi Display (Courtesy Brent Fraser)

More information the MHS 3.5-inch RPi Display will be found at:
http://www.lcdwiki.com/MHS-3.5inch_RPi_Display

Touch screen installation

Installation of touch-screens can be complex and there are so many available. Also, manufacturers are struggling to keep up with constant changes to the Operating System. It is impossible for the **Bob Rathbone Computer Consultancy** to provide support if your touch display doesn't work properly. You need to get the **Raspberry Pi Desktop** running on your device before you attempt installation of the **radio** software. Below are the links to the touch screens mentioned here.

Manufacturer	Model	Link
Waveshare	2.8-inch	https://www.waveshare.com/wiki/2.8inch_Capacitive_Touch_LCD
"	3.5-inch (C)	https://www.waveshare.com/wiki/3.5inch_RPi_LCD_(C)
Adafruit	2.8-inch	https://learn.adafruit.com/adafruit-pitft-28-inch-resistive-touchscreen-display-raspberry-pi/overview
"	3.5-inch capacitive	https://cdn-learning.adafruit.com/downloads/pdf/adafruit-pitft-3-dot-5-touch-screen-for-raspberry-pi.pdf
MHS	3.5-inch	http://www.lcdwiki.com/MHS-3.5inch_RPi_Display
Osoyoo	3.5-inch	https://osoyoo.com/2022/12/15/osoyoo-3-5-inch-spi-touch-display-for-rpi/
"	5-inch	https://osoyoo.com/2019/09/20/instruction-for-raspberry-pi-5-dsi-touch-screen/

For others go to the manufacturer's site for information and support.

Radio variants

Before starting you need to make a choice which type of radio you are going to build. There are several combinations of user interface and display type which can be constructed as shown in the following tables.

Table 2 Display Type options

Display Type	
1	Two-line 8-character LCD
2	Two-line 16-character LCD
3	Four-line 16-character LCD
4	Four-line 20-character LCD
5	Two-line 40-character LCD
6	Adafruit 2x16 RGB Plate (I2C)
7	Raspberry Pi 7-inch touch screen
8	Olimex 128 by 64-pixel OLED
9	Adafruit 3.5-inch TFT touch screen
10	No display (Vintage radio design)
11	Pirate Radio (No display)
12	Waveshare 1.8/2.3" touchscreen
13	Grove I2C 2-line 16-char. LCD
14	OLED displays supported by LUMA (SH1106, SSD1306 etc.)
15	SH1106 128x64 1.3" OLED (SPI interface)
16	Waveshare 1.5 and 2.42-inch OLED with SPI interface

Table 3 User interface options

User interface	
1	Five or six push buttons
2	Two rotary encoders with push buttons
3	Adafruit RGB plate with push buttons
4	Raspberry Pi 7-inch touch-screen
5	Adafruit 2.8" or 3.5" TFT
6	Mouse (HDMI/Touchscreen only)
7	Keyboard (HDMI/Touchscreen only)
8	IQAUDIO Cosmic controller
9	Pirate Radio – 6 push-buttons
10	IR remote control – LCDs, OLEDs & TFTs
11	FLIRC IR remote control – Windows only
12	Waveshare 1.8/2.3" touchscreen
13	pHat with 5-position joystick and 3-buttons
14	Web interface using !OMPD
15	MPD command line client interface (mpc)

Any type of **HD44780U** LCD display can be used with any user interface except the FLIRC IR remote control. The HD44780U can either be connected directly to the GPIO pins or via a so-called I2C (also known as IIC) backpack. The **Adafruit RGB plate** has a two-line 16-character display and comes with five inbuilt pushbuttons.

It also has its own I2C interface using the MCP23017 chip so it does not require a separate I2C backpack.

The **PiFace CAD** comes with a two-line 16-character display and comes with six inbuilt pushbuttons. It uses the SPI interface from Motorola. The touch screens can be used with or without rotary encoders or push buttons. The touch screen variants can also use a mouse and keyboard.

It is a simple choice of which display (two or four lines, 8,16 or 20-characters LCDs or a touch screen or HDMI screen, OLED display or Pirate radio) and whether to use rotary encoders or push-button switches as the user interface. The rotary encoder options give the most natural feel for the radio as most conventional radios use knobs to control the volume and station tuning. The keyboard interface, whilst supported on the touch-screen versions, is a very limited option.

There is a configuration program called **configure_radio.sh** which configures the choice of display and user interface required. It can be safely re-run at any time. The vintage radio software (Display option 10) specifically intended for converting an old radio to an Internet radio whilst retaining the original look and feel of the radio. It has no LCD display.

The four lines LCD can display more information than two-line versions.



Note: The touch screen software (**gradio.py** or **vgradio.py**) cannot be run at the same time as the LCD version of the radio software (**radiod.py**). It is a case of using one or the other. It is however possible to switch between **gradio.py** or **vgradio.py** programs during operation.

Connecting the LCD display

There are two ways of wiring up the display:

- Directly connect the LCD to the GPIO pins. This uses six GPIO pins.
- Connection via an I2C backpack or inbuilt I2C interface. These use the two-pin I2C interface

The first choice uses more wiring but is the cheapest option. The second choice uses an I2C backpack which is an extra component to be purchased. However, I2C backpacks are reasonably cheap.

Housing the radio

This manual describes a couple of ways of housing the radio. A few ideas are below:

- A custom-built case as shown in this manual
- Old plastic boxes or food containers
- Construct a case using Lego
- Use a pair of speaker housings that have enough room
- Install in an old vintage radio (really cool)
- Use an old wooden wine box
- Use an old video recorder, CD player or desktop set
- Buy a PC speaker set with enough room to build in the radio.
- Use anything else you like. Oliver Voss (Germany) used an old Fire Extinguisher

Figure 82 Some examples of radio cases





Take a look at the constructor's gallery at https://bobrathbone.com/raspberrypi/pi_internet_radio.html to get some ideas that other constructors have used.



Note: Don't forget to make sure that there is adequate airflow through the radio housing to allow cooling of the Raspberry Pi and other components. If necessary, drill at least five or six holes at the top and bottom of the housing.



If you decide to use a metal case (not advised) you will need a Wi-Fi dongle with an aerial mounted externally to the case. Also, the case must be earthed at the main supply both for safety reasons and to prevent interference with sound and/or the LCD screen

Building in an IR sensor and remote control

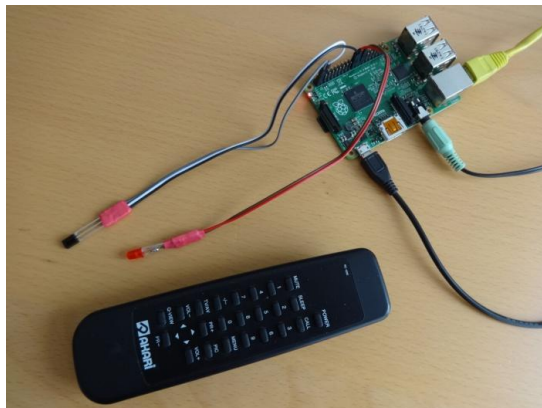


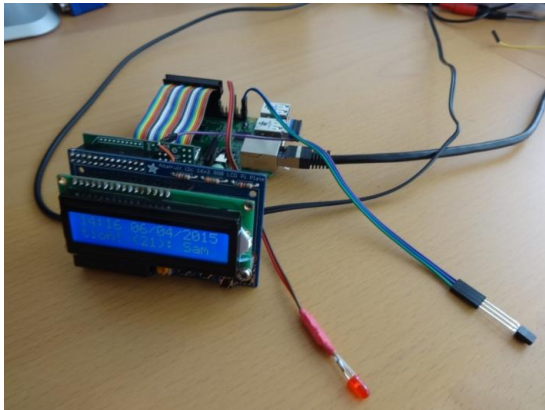
Figure 83 IR Sensor and Remote control

The radio can be built with an IR Sensor and remote control. Also included is an activity LED which flashes when the remote control is used.

A **TSOP382xx** series IR Sensor is used in conjunction with almost any remote control. An activity LED can also be added which flashes every time remote control signal is detected. The remote control provides the same functionality as the buttons or rotary encoders.



Although nearly any remote control that you happen to have lying around can be used, it is recommended to use the **Raspberry Pi CR2025 Mini IR Remote Control** available from suppliers such as **PiHut**. Apart from its size and price advantage you will find a definition for this device called **mini.toml** in the **/usr/share/radio/remotes** directory which will save the effort of setting up your own.



The **AdaFruit** RGB plate can also be fitted with an IR sensor and activity LED but needs a model B+, 2B or 3B (40 GPIO pins) and 26 pin extender.



Note that a 40 pin Raspberry PI is needed as the Adafruit Plate occupies the first 26 pins of the Raspberry Pi GPIO header.

Figure 84 Adafruit and IR sensor and activity LED

Suggested button assignments for the CR2025 mini-IR remote control

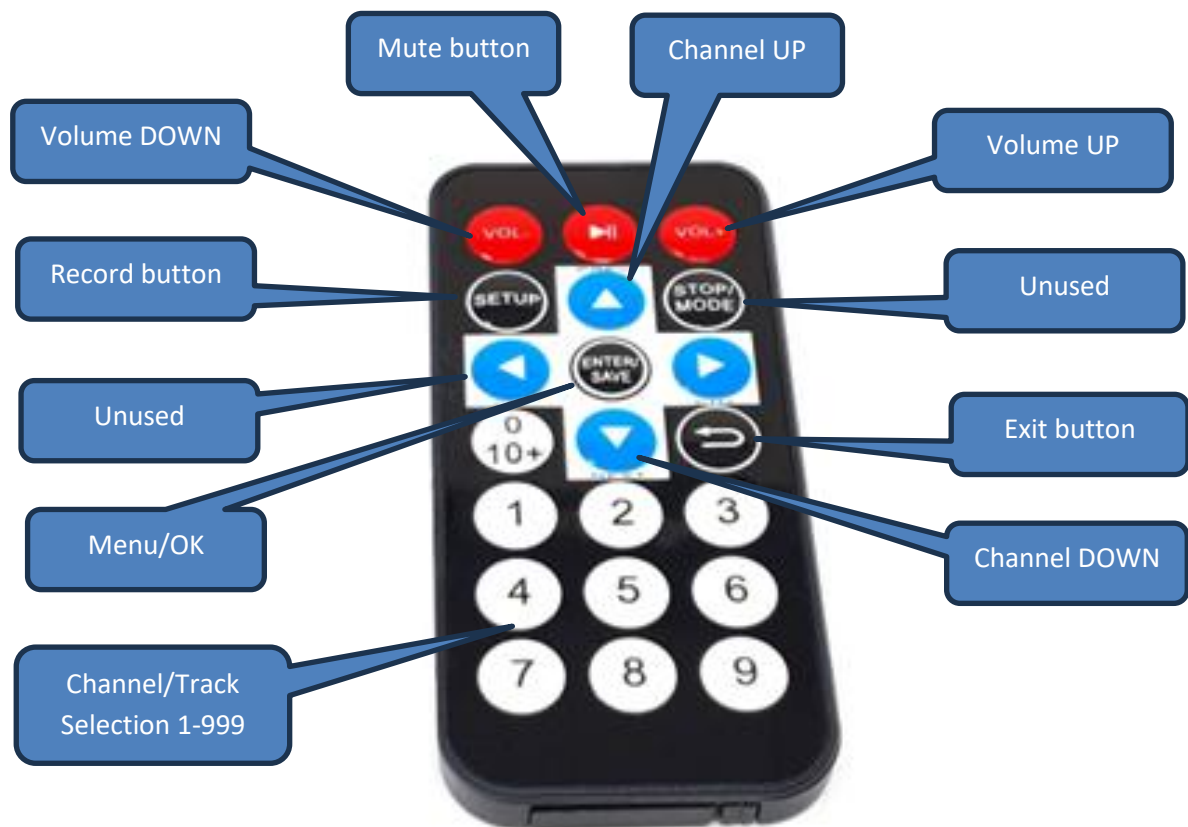


Figure 85 Mini IR-Remote control buttons

Chapter 3 – Wiring and Hardware

Contents chapter 3	Page
All Raspberry Pi models using 40-pin GPIO headers	48
Wiring Push Button radios and the record button	49
Radios using rotary encoder	50
LCD Module Wiring	57
Power supply considerations	60
GPIO Hardware Notes	61

All Raspberry Pi models using 40-pin GPIO headers

Table 4 Radio and DAC devices 40-pin wiring

Pin	Description	Radio Function	Name	Audio DAC Function	LCD Pin	Push Button	Encoder (Tuner)	Encoder (Volume)
1	3V3	+3.3V supply	+3.3V	+3.3V		+3.3V		
2	5V	5V for LCD	+5V	+5V	2,15			
3	GPIO2	I2C Data	I2C Data	I2C Data				
4	5V			+5V				
5	GPIO3	I2C Clock	I2C Clock	I2C Clock				
6	GND	Zero volts	0V	0V	1,3*,5,16		Common	Common
7	GPIO 4	Mute volume				MUTE		Knob Switch
8	GPIO 14	Volume down	UART TX			LEFT		Output A
9	GND	Zero Volts		0V				
10	GPIO 15	Volume up	UART RX			RIGHT		Output B
11	GPIO 17	Menu switch				MENU	Knob Switch	
12	GPIO 18			I2S CLK				
13	GPIO 27	Record button						
14	GND	Zero Volts		0V				
15	GPIO 22			Mute				
16	GPIO 23	Channel down		Rotary enc A		DOWN	Output A	
17	3V3	+3.3V supply		0V				
18	GPIO 24	Channel up		Rotary Enc B		UP	Output B	
19	GPIO 10		SPI-MOSI					
20	GND	Zero Volts						
21	GPIO9		SPI-MISO					
22	GPIO 25	IR Sensor		IR sensor				
23	GPIO 11		SPI-SCLK					
24	GPIO 8	LCD E	SPI-CE0		6			
25	GND	Zero Volts		0V				
26	GPIO 7	LCD RS	SPI-CE1		4			
27	DNC			PiDac+ Eprom				
28	DNC			PiDac+ Eprom				
29	GPIO5	LCD Data 4			11			
30	GND	Zero Volts						
31	GPIO6	LCD Data 5			12			
32	GPIO12	LCD Data 6			13			
33	GPIO 13	LCD Data 7			14			
34	GND	Zero Volts						
35	GPIO 19	IQaudio DAC+	I2S	I2S				
36	GPIO 16	IR LED out						
37	GPIO 26							
38	GPIO 20	IQaudio DAC+	I2S DIN	I2S DIN				
39	GND	Zero Volts						
40	GPIO 21	IQaudio DAC+	I2S DOUT	I2S DOUT				

Wiring Push Button radios and the record button

Wire one side of the push-buttons the GPIO pin as shown in the last column of Table 4 on page 48. Wire the other side of the switches to either +3.3V (Old wiring scheme) or to GND (0V) (Preferred wiring scheme recommended for new projects). Whichever wiring you use; the radio configuration program will ask which wiring scheme is being used. Version 2 onwards boards have internal pull up/down resistors and don't require external resistors. In fact, including these can cause problems.

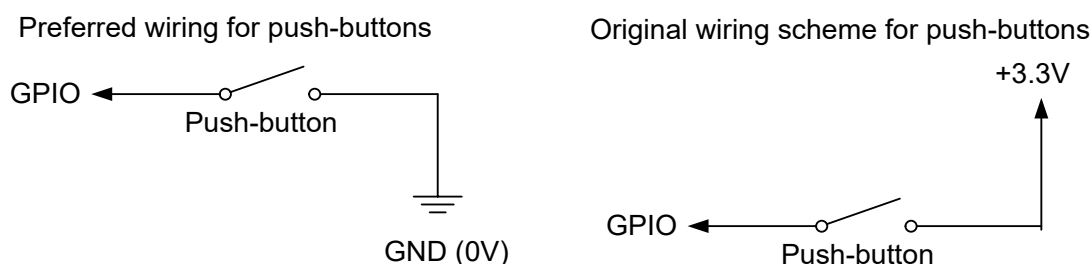


Figure 86 Push-button wiring version 2 onwards boards

The scheme chosen must be configured using the **pull_up_down** parameter in **/etc/radiod.conf** to 'up' or 'down'.

5-way Direction Combination Button Pad

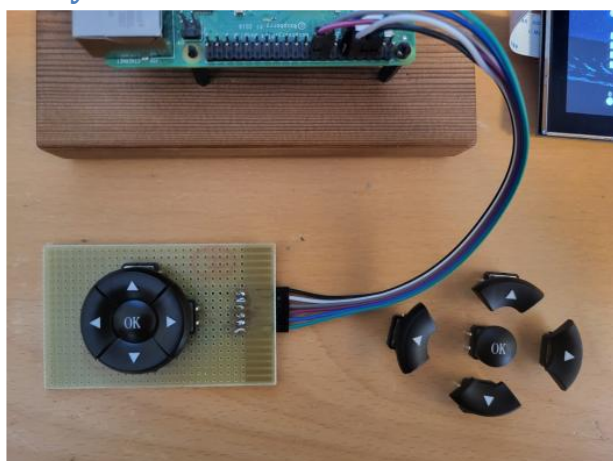


Figure 87 5-way button pad

These types of push button pads have four direction buttons and a single OK button. By convention in the radio they have the following functions:

- UP – Channel Up**
- DOWN – Channel Down**
- LEFT – Volume Down**
- RIGHT – Volume Up**
- OK – Menu Switch**

There isn't a **Mute** button so just use the Volume Down button.

These button pads come as either separate buttons which you have to solder into a standard prototype board or they come ready mounted on a custom-made PCB.

When soldering these buttons into the prototype board you will find that the LEFT and RIGHT buttons won't fit as they want to use the same holes as the OK button. The solution is to snap off one pair of connection pins (there are two pairs) and only use the remaining two pins. Connect one side of each switch to GND 0V and the other to the appropriate GPIO pin for that switch.

When asked to specify the type of user interface select Buttons and set the pull-up to UP.

Radios using rotary encoders

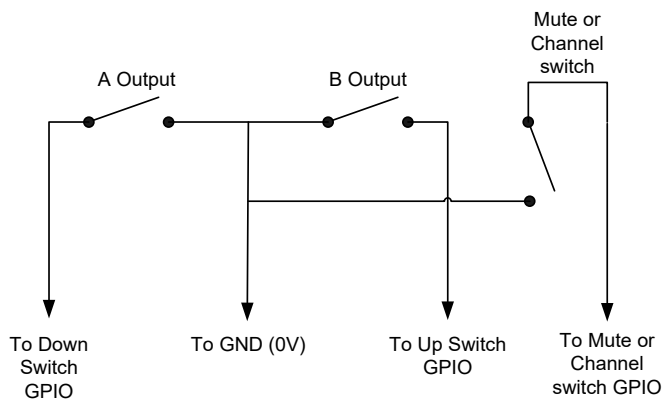


Figure 88 Rotary Encoder Diagram

Rotary encoders have three inputs namely Ground, Pin A and B as shown in the diagram on the left. Wire the encoders according to that shown in *Table 10 LCD module wiring for 40-pin Raspberry Pi's* on page 58. If the encoder also has a push button knob, then wire one side to ground and the other to the GPIO pin. In the case of the mute switch this will be pin 7 (GPIO 4). Version 1 boards are not supported but will probably work.



Warning: The push switches (if fitted) on the rotary encoder are wired differently from the push buttons in the earlier push button versions of the radio. For these encoders one side of the push button is wired to GND (not 3.3V) and the other to the relevant GPIO.

If using a Revision 1 board it is necessary to use 10K pull up resistors connected between the GPIO inputs of the rotary encoder outputs and the 3.3-volt line. Do not add resistors if using revision 2 boards and onwards.



Note: Although version 1 boards with 26-pin GPIO headers are no longer supported, you can still find wiring details for these in the appendices for the *Raspberry Pi Internet Radio – Technical Reference* manual.



Figure 89 Rotary encoder with push switch

This project originally uses a COM-09117 12-step rotary encoder or PEC11R series encoders. It also has a select switch (by pushing in on the knob). These are “Incremental Rotary Encoders”. An incremental rotary encoder provides cyclical outputs (only) when the encoder is rotated. The other type is an absolute rotary encoder which maintains position information even when switched off (See Wikipedia article on rotary encoders).

The rotary encoders used in this project are wired with the COMMON or GND pin in the middle and the A and B outputs either side. However, some rotary encoders are wired with A and B as the first two pins and GND (COM) as the third pin. Note that not all encoders come with a switch, so separate switches for the Menu and Mute button will need to be installed. Check the specification first.

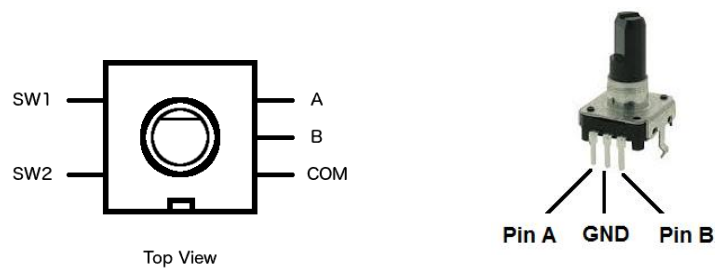


Figure 90 Rotary encoder pin-outs



Note: Not all manufacturers' rotary encoders will work with this project. If they work then fine if not regrettably you will need to purchase the recommended encoders.

Using KY-040 Rotary encoders

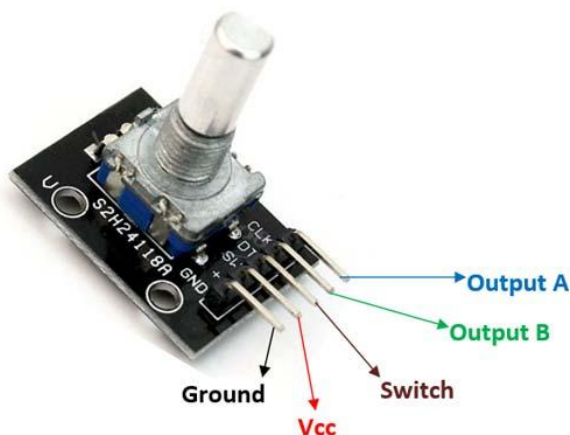


Figure 91 KY-040 Rotary Encoder

These cost-effective Rotary Encoders from Handson Technology originally designed for use with Arduino are now being used more and more by constructors. The KY-040 Rotary Encoder specification shows that these are powered by +5V to the **VCC** pin however VCC must be connected to **+3.3V** and not +5V otherwise you will damage the Raspberry Pi



Figure 92 KY-040 with three 10K resistors

Originally these rotary encoders had three 10K pull-up resistors connected between outputs CLK (R2), DT(R3) and SW(R1) to VCC supply. However, manufacturers discovered that developers did not necessarily want to have the pull-up resistor between the switch connection and VCC as they might want to connect the switch to GND (0V) when operated.



Figure 93 KY-040 with resistor R1 omitted

As a result, manufacturers now tend to supply these rotary encoders with the R1 10K resistor omitted leaving the decision on how to wire up the switch (SW) to the developer. From version 8.0 onwards an internal 10K resistor will be added by the radio configuration software.

During set-up if using KY040 encoders, you will be asked to select the option for either the two or three resistor version.

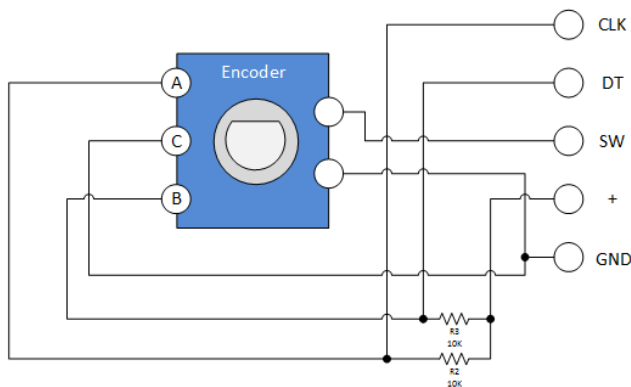


Figure 94 KY-040 Circuit Diagram

The specification shows the rotary encoders are labelled CLK(Clock), DT(Data) and + (VCC) however it is more usual to label these A, B and C.

Connect + to the +3.3V supply. Do **not** connect to +5V.

Note: Some early KY-040 encoders have a 10K resistor (R1) between SW and VCC (+3.3V).

From version 7.2 onwards the internal pull-up resistors can be disabled with the radio configuration program as they are not required for the KY-040 encoder as it has its own pull-up resistors. See Figure 171 *Rotary encoder* on page 112.

Waveshare Rotation Sensor

Just another name for Rotary Encoder. Waveshare make a very similar encoder to the KY-040, however the **GND** and **VCC** connections are reversed.

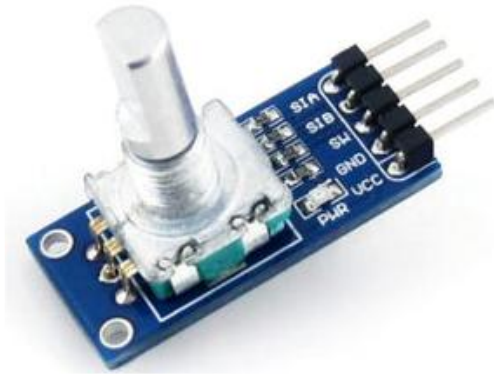


Figure 95 Waveshare Rotation Sensor

The wiring is as follows:

S1A – Serial Input A

S1B - Serial Input B

SW - Knob Switch

GND – GND (0V) – Warning: Opposite to the KY-040

VCC - +3.3V “ “ “ “

Make sure that VCC and GND are connected the correct way around otherwise you will damage the Raspberry Pi.

See *Table 8 Waveshare rotation sensor wiring* on page 56 for wiring information.

Rotary encoders with RGB LEDs



Figure 96 Sparkfun RGB rotary encoder

There are also rotary encoders available with RGB (Red, Green, Blue) LEDs which light up the transparent shaft. These also have a press switch. There are also versions which just have Red and Green LEDs. These encoders are also supported by the radio software from version 7.4 onwards.

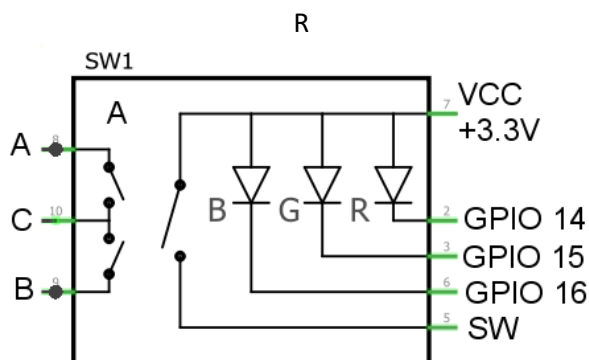


Figure 97 RGB Rotary Encoder circuit

For this type of rotary encoder, the cathodes of the LEDs are connected to VCC (3.3V).

Unlike the standard encoders one side of the switch is also connected to VCC which means that the push-switch works the opposite way around to the on the standard rotary encoder. In this case the **rgb_rotary** device driver has to be selected during configuration.



Figure 98 Transparent knob

There are transparent plastic knobs available which on the standard 6mm shaft. These simply press-fit onto the shaft of the rotary encoder.

They are clear so that they can be used with illuminated rotary encoders such as the Sparkfun RGB encoder.

The LEDs can be driven by configuring the following parameters in `/etc/radiod.conf` as shown below:

```
rgb_red=14
rgb_green=15
rgb_blue=16
```

The LEDs are driven by the `status_led_class.py` driver software. The LEDs connect directly to the above GPIO pins 14, 15 and 16 and are driven by 3.3 volts so do not particularly need resistors but 100 Ohm resistors can be used to reduce the brightness.



Warning: Do not, under any circumstances connect these rotary encoders to +5 Volts, even if the specification says so, otherwise the Raspberry Pi will be irreparably damaged!

RGB I2C rotary encoders

These Rotary Encoders use three Red, Green and Blue (RGB) LEDs in the Rotary Encoder Shaft. This type of rotary encoder is entirely driven by the I2C interface. One drawback of these encoders is that they do not have a push button meaning that these must be installed extra to the encoders.



Figure 99 Pimoroni RGB I2C Rotary Encoders

These encoders contain Nuvoton MS51XB9AE microcontroller which let you directly control the RGB LEDs inside the encoder. The default I2C address is 0x0F but it is possible to configure a second device for the Channel encoder with a separate I2C address (0x1F for the radio project). The encoders also use an interrupt pin (INT). The default interrupts are GPIO22 for the Volume control and GPIO23 for the Channel interrupt. See *Table 9 RGB I2C Rotary Encoders wiring* on page 56.

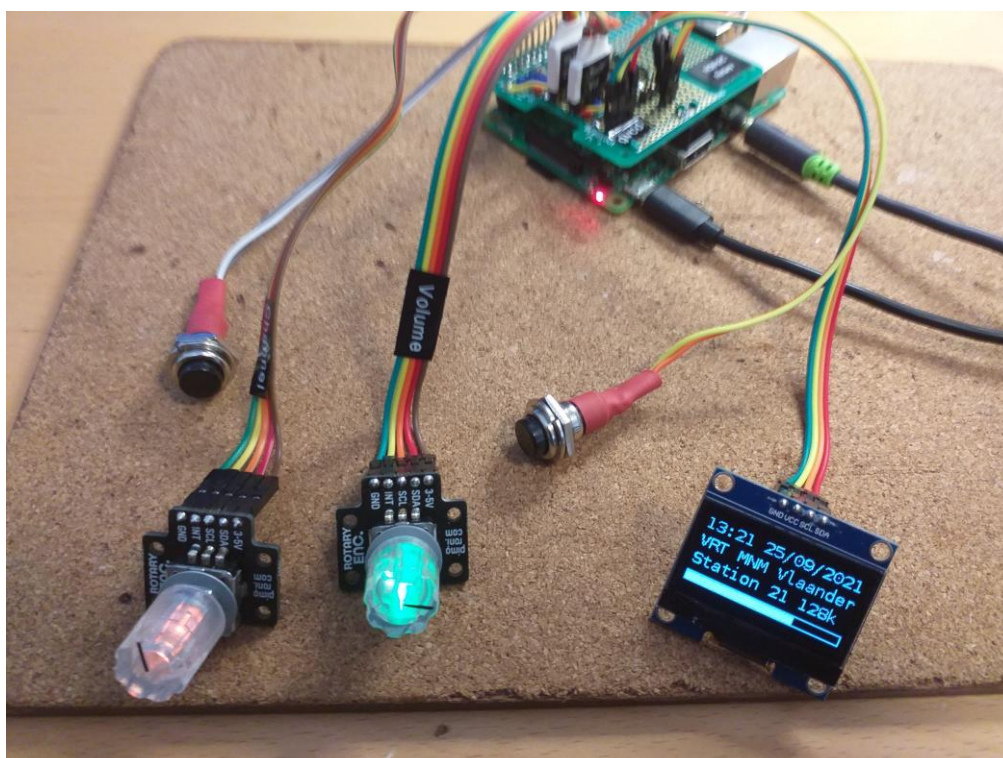


Figure 100 Radio project using RGB I2C rotary encoders

The above photo shows two RGB I2C encoders (Volume 0x0F and Channel 0x1F) running the radio software. The Mute and Menu buttons are wired the same as the standard encoders. Also shown is 128x64 pixel OLED also connected by I2C (In this case with 0x3C as the address).

Rotary encoder wiring

The following tables show how to wire up the four types of rotary encoders used in this project. The first table shows how to wire up conventional rotary encoders (without pull-up resistors). These have five connections A, B, C and two connections to the switch. See *Figure 90* on page 51.

Table 5 Radio A, B, C Rotary Encoder Wiring

GPIO Pin	Description	Radio Function	Volume Rotary Encoder	Channel Rotary Encoder
6	GND	Zero volts	Common C	
8	GPIO 14	Volume up/down	Output A	
10	GPIO 15	Volume up/down	Output B	
7	GPIO 4	Mute volume	Knob Switch 1 *	
GND	GND 0V	Mute volume	Knob Switch 2 *	
9	GND	Zero volts		Common C
16	GPIO 23	Channel up/down		Output A
18	GPIO 24	Channel up/down		Output B
11	GPIO 17	Menu switch		Knob Switch 1 *
GND	GND 0V	Menu switch GND		Knob Switch 2 *

GND is found on physical pins 6, 9, 14, 20, 25, 30, 34 and 39.

* Note: In the case of RGB LED Rotary Encoders one side of the push switch is wired to the VCC (3.3V) supply internally in the switch. The GPIO goes from low to high when the button is pushed.

Table 6 RGB LED Rotary Encoder switch wiring

7	GPIO 4	Mute volume	Knob Switch 1	
+3.3V	VCC	“ “	Internal connection	
11	GPIO 17	Menu switch		Knob Switch 1
+3.3V	VCC	“ “		Internal connection

The second table shows how to wire up KY-040 rotary encoders (fitted with pull-up resistors). These have five connections namely CLK, DT, SW, VCC and GND. See *Figure 94 KY-040 Circuit Diagram* on page 52.

Table 7 KY-040 Rotary Encoder Wiring

GPIO Pin	Description	Radio Function	Volume Rotary Encoder	Channel Rotary Encoder
1	3V3	+3.3V supply	+3.3V (VCC)	
6	GND	Zero volts	Common (GND)	
7	GPIO 4	Mute volume	Knob Switch (SW)	
8	GPIO 14 (TX)	Volume up/down	Output A (CLK)	
10	GPIO 15 (RX)	Volume up/down	Output B (DT)	
17	3V3	+3.3V supply		+3.3V (VCC 3.3V)
9	GND	Zero volts		Common C (GND)
11	GPIO 17	Menu switch		Knob Switch (SW)
16	GPIO 23	Channel up/down		Output A (CLK)
18	GPIO 24	Channel up/down		Output B (DT)

Table 8 Waveshare rotation sensor wiring

GPIO Pin	Description	Radio Function	Volume Rotary Encoder	Channel Rotary Encoder
1	3V3	+3.3V supply	+3.3V (VCC)	
6	GND	Zero volts	Common (GND)	
7	GPIO 4	Mute volume	Knob Switch (SW)	
8	GPIO 14 (TX)	Volume up/down	Input B (SIB)	
10	GPIO 15 (RX)	Volume up/down	Input A (SIA)	
17	3V3	+3.3V supply		+3.3V (VCC 3.3V)
9	GND	Zero volts		Common C (GND)
11	GPIO 17	Menu switch		Knob Switch (SW)
16	GPIO 23	Channel up/down		Input B (SIB)
18	GPIO 24	Channel up/down		Input A (SIA)

Warning: GND and VCC are the opposite way around to a KY-040 Rotary Encoder.

Table 9 RGB I2C Rotary Encoders wiring

GPIO Pin	Description	Radio Function	Volume RGB I2C Rotary Encoder	Channel RGB I2C Rotary Encoder	Optional I2C display
1	3V3	+3.3V supply	VCC +3.3V	VCC +3.3V	VCC +3.3V
3	GPIO2	SDA	SDA	SDA	SDA
5	GPIO3	SCL	SCL	SCL	SCL

22	Volume Interrupt	n/a	INT		
23	Channel Interrupt	n/a		INT	
6 or 9	GND	Zero volts	GND	GND	GND
3,5	I2C convection	I2C address	Hex 0x0F	Hex 0x1F	Hex 0x3C*
7	GPIO4	Mute	Mute Button		
11	GPIO17	Menu		Menu Button	

* Example OLED I2C address which will differ depending upon the attached device.

The above table shows two RGB I2C encoders. The I2C address for the Channel Rotary encoder is configured with the **gb_set_i2c_address.py** utility.

```
$ ./rgb_set_i2c_address.py
RGB I2C Rotary Encoder set new I2C address 0xf

Usage: sudo ./rgb_set_i2c_address.py --help
       --i2c_address=<i2c_address>
       --new_i2c_address=<new_i2c_address>

Recommended values for current and new addresses are 0x0F and 0x1F
Run "i2cdetect -y 1" to check I2C address before and after
```

First only plug in one RGB I2C Rotary encoder with hex address **0x0F**.

```
$ cd /usr/share/radiod
$ sudo ./rgb_set_i2c_address.py --i2c_address=0x0f -new_i2c_address=0x1F
```

Below is the output of the **i2cdetect -y 1** utility with two RGB I2C rotary encoders (**0x0F** and **0x1F**) and an OLED display (0x3C)

```
$ i2cdetect -y 1
   0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
00:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  -- 0f
10:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  -- 1f
20:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
30:  --  --  --  --  --  --  --  --  --  --  --  -- 3c  --  --  --
40:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
50:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
60:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
70:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --
```

Optical Rotary encoders

Optical rotary encoders are much more expensive than the mechanical (conductive) rotary encoders previously shown. They offer the higher resolution compared to mechanical ones. They are usually used for scientific and industrial applications. They are overkill for this project but may be used. This software has been successfully tested with an HRPG-ASCA #16F optical encoder from Avago. See https://media.digikey.com/pdf/Data%20Sheets/Avago%20PDFs/HRPG_Series.pdf

LCD Module Wiring

The following shows the wiring for a directly wired HD44780U LCD controller. It has 16 or 18 pins.

Table 10 LCD module wiring fo 40-pin Raspberry Pi's

LCD Pin	GPIO 40-pin	Pin 40 #	Description
1	n/a	6	Ground (0V) – Wire this directly to LCD pin 5
2	n/a	2	VCC +5V
3	n/a	Note1	Contrast adjustment (0V gives maximum contrast)
4	GPIO7	26	Register Select (RS). RS=0: Command, RS=1: Data
5	n/a	6 or 9	Read/Write (RW). Very important this pin must be grounded! R/W=0 (GND): Write, R/W=1 (+5V): Read. Will damage the PI if not grounded (0V). Wire LCD pin 5 and 1 together
6	GPIO8	24	Enable (EN)
7			Data Bit 0 (Not required in 4-bit operation)
8			Data Bit 1 (Not required in 4-bit operation)
9			Data Bit 2 (Not required in 4-bit operation)
10			Data Bit 3 (Not required in 4-bit operation)
11	GPIO5	29	Data Bit 4 (D4)
12	GPIO6	31	Data Bit 5 (D5) Note if using IQaudio products GPIO22 conflicts !!
13	GPIO12	32	Data Bit 6 (D6)
14	GPIO13	33	Data Bit 7 (D7)
15	n/a	2	LED Backlight Anode (+5V) or Red LED (Adafruit RGB plate) [2] or VEE negative voltage on Midas HD44780
16	n/a	6 or 9	LED Backlight Cathode (GND) or Red LED [2]
17			Optional Green LED (Adafruit RGB plate) [2]
18			Optional Blue LED (Adafruit RGB plate) [2]



Note 1: If using the Midas display with VEE on pin 15 do not connect this pin to the +5V supply or you will damage the display. Connect pin 15 as shown in the section called *Midas LCD display* on page 21.



Note 2: The contrast pin 3 (VE) should be connected to the center pin of a 10K potentiometer. Connect the other two pins of the potentiometer to 5v (VDD) and 0v (VSS) respectively. Adjust the preset potentiometer for the best contrast.



Note 3: The standard LCD comes with 16 pins. Adafruit supply an RGB backlit LCD with two extra pins namely pins 17 and 18. These are non-standard. These must be wired to ground (0 Volts) to work. For more information see the section *Using the Adafruit backlit RGB LCD display* on page 80.

The following diagram (Courtesy protostack.com) shows the electrical connections for the standard 16 pin LCD. Do not use this diagram for Midas displays. See instead *Midas LCD display* on page 21.

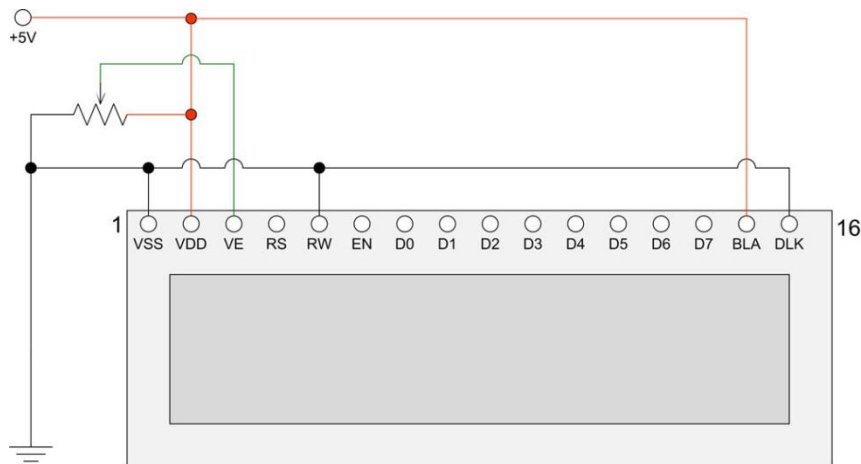


Figure 101 HD44780U LCD electrical circuit



The standard LCD comes with 16 pins. Adafruit supply an RGB backlit LCD with two extra pins namely pins 17 and 18. These are non-standard. These must be wired to ground (GND 0 Volts) to work. For more information see the section *Using the Adafruit backlit RGB LCD display* on page 80.

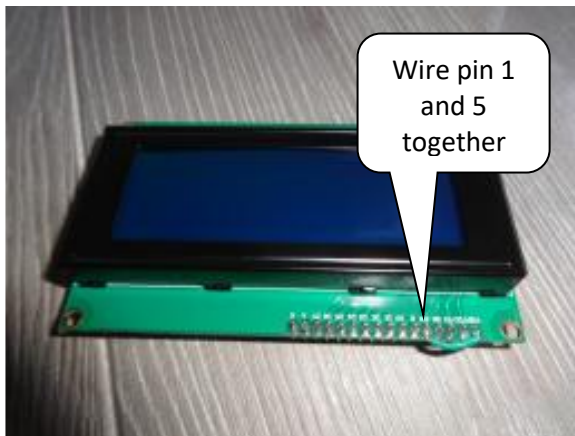


Figure 102 Wire LCD pin 1 (GND) and 5 (RW) together



The Read/Write (RW) pin 5 must be connected to pin 1 (0V). It is very important that this pin is grounded!

If pin 5 is not grounded it will damage the Raspberry Pi. Always wire LCD pin 5 and 1 directly together. Do not rely on grounding pin 5 with a GND wire on the connector. If this wire drops off then the LCD data lines will be put into write mode putting +5V on the GPIO pins which will probably cause irreparable damage to the Raspberry Pi. If using an I2C backpack this step is not necessary as it is already done in the backpack.



Warning – Some LCD displays such as the Midas with VEE have a different voltage arrangement for Pin 15 and Pin 5 (Contrast). Pin 15 is an output which provides a negative voltage (VEE) which connects to one end of the 10K contrast potentiometer and the other end to +5V (VDD). Connecting +5 Volts to pin 15 will destroy the LCD device. See section called *Midas LCD display* on page 21 for further information.



Note: All settings in the `/etc/radiod.conf` file use GPIO numbers and NOT physical pin numbers. So, in the above example `lcd_width` is GPIO 16 (Physical pin 36).

There is a useful program called **wiring.py** which will display physical the wiring required for the settings found in `/etc/radiod.conf`.



Also, there is another program called **test_gpios.py** which will test for button or rotary encoder events. For example, you are not sure how you have wired a button or rotary encoder this program will confirm this for you.

Power supply considerations

The Raspberry Pi except for the model 4B uses a standard Micro USB (type B) power connector, which runs at +5V. The model 4B uses a 5 Volt 3 Ampere power supply with a USB-C adaptor. In general, the Raspberry Pi can draw up to 2.5. Many telephone adapters are not suitable. You also need to consider the LCD screen which can also need up to 20mA but depends on the type of backlight.

Try to find a power adapter that delivers at least 1.5 Ampere. As mentioned above a 2.5A supply will be required to run a Raspberry Pi model 5.

http://elinux.org/RPi_VerifiedPeripherals#Power_adapters

The Raspberry Pi can be powered either the USB port or via the GPIO header (Pin 2 or 4). Some prototyping boards used to provide power in this way.

If using an adaptor or separate 5-volt Power Supply try to use a switched-mode power supply adaptor. This takes less current and generate less heat than a power dissipation device. If a power supply is designed to be earthed then use a 3-core cable with live, neutral and earth wires.

Things not to do:

- Do not try to tap off power from the Power supply or transformer used by the speaker's amplifier. This won't work (earth loops) and can cause damage to the PI and peripherals.
- Do not tap off (cascade) from the amplifier DC supply (12 volts for example) with another 5V voltage regulator. This will most likely cause interference.
- Do not feed power to the PI from two sources (USB hub and Power adapter). Try to use USB hubs that don't feed 5 volts back through the USB ports of the Raspberry PI
- Do not connect an untested power supply to the Raspberry PI without checking the voltage first.

Things to do:

- Use double pole mains switches for isolating the mains supply when switched off. A lot of European plugs can be reversed leaving the live wire un-switched if using a single pole switch.
- If using a metal case always earth it and use a three-pin plug with earth pin.
- Use the official Raspberry Pi power supply. If you want to feed the 5-volt supply directly to the GPIO header you can do so but you must ensure that you are only using a highly regulated +5V supply and typically must be able to deliver 3-Amps and must be connected to physical pin 2 or 4.

You should try to use a single power supply switch for the radio. Connect the AC power supply of the adaptor to the mains switch. This switch can also provide the mains supply to the speaker amplifier. Also see the section on *preventing electrical interference* on page 76



Always consider safety first and make sure that no-one including yourself can receive an electric shock from your project including when the case is open. Also see disclaimer on page 239.

Also see *Appendix B – Using a battery pack* on page 247

GPIO Hardware Notes

The following shows the pin outs for the GPIO pins on revision 1 and 2 boards. For more information see: http://elinux.org/RPi_Low-level_peripherals.

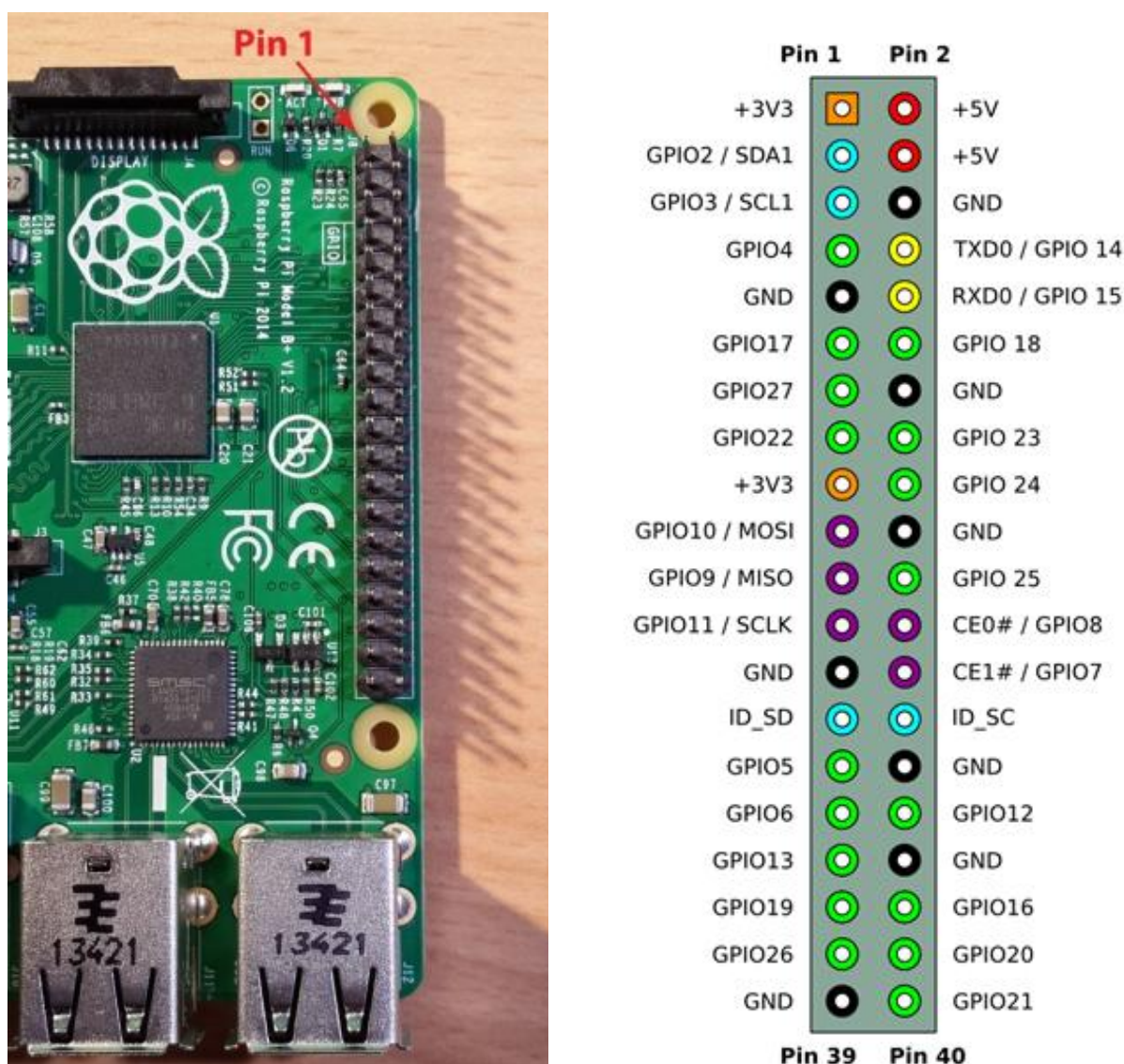


Figure 103 GPIO Pin Numbers



Note: All 40-pin Raspberry Pi's have the same the same pin-outs.

Chapter 4 - Construction details

Contents chapter 4	Page
Construction using an interface board	63
Construction using breakout boards	65
Construction using an Adafruit LCD plate	66
Construction using an I2C LCD backpack	67
Fitting a wake-up button	69
Installing an IR sensor and remote control	70
Construction using an IQaudIO Cosmic Controller	72
Construction using the Pimoroni Pirate radio	73
Construction using the PiFace CAD	73
Construction Tips and Tricks	74
Selecting an audio amplifier	75
Connecting up a USB power adapter	78
Cooling the Raspberry Pi	78
Booting from a USB drive or stick	79
Miscellaneous	80

Construction using an interface board

It isn't necessary to construct the radio connect via an interface board but it does make maintenance easier and is much more reliable and will be needed if you wish to connect to the Raspberry Pi using a ribbon cable. Always bring the ribbon cable into the top of the board as shown in Figure 104 below. If the ribbon cable is connected to the back (underside) of the board the two rows of the 40-pin connector will be swapped over. In the next section how to use breakout boards is covered which may be an easier alternative for many constructors rather than building your own.

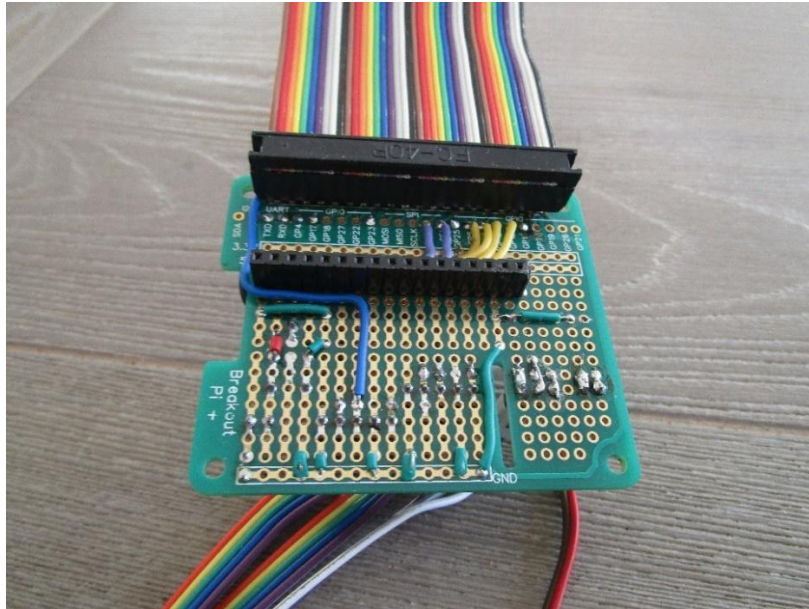


Figure 104 40-pin Interface board with ribbon cable

The figure below shows a 4x20 HD44780U LCD plugged into the interface board.

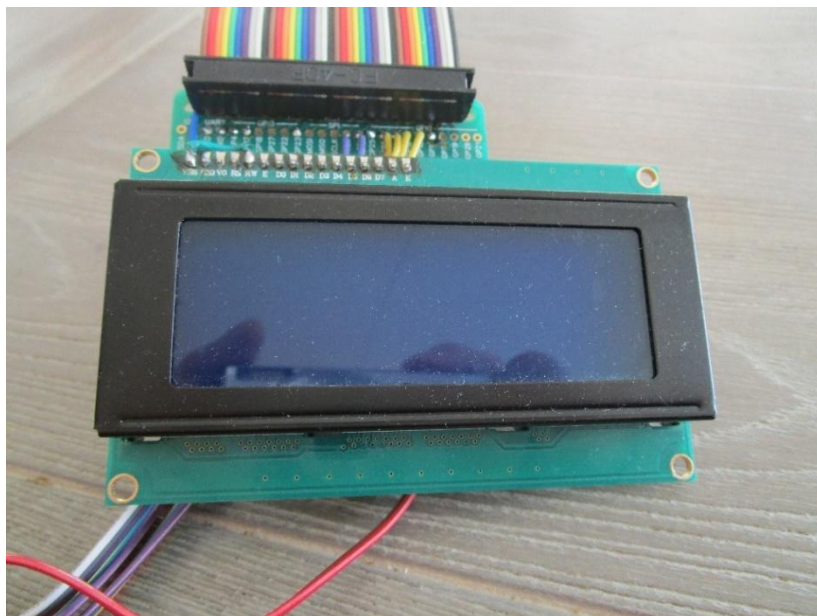


Figure 105 Interface board with LCD screen attached

Below is the other side of the interface board showing the connections to the rotary encoders, an IR sensor the IR activity LED.

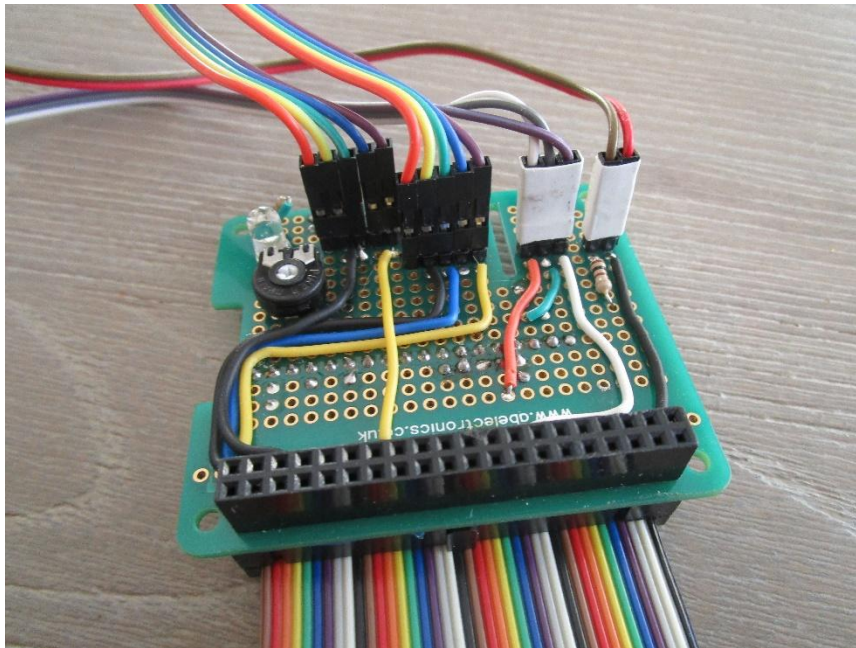


Figure 106 Radio controls connections

Below is the complete overview of the interface card with some test rotary encoders, an IR sensor the IR activity LED.



Figure 107 Interface board overview

There are various interface boards available on the market for both 26-pin and 40-pin Raspberry PIs.

Construction using breakout boards

When this project was begun in the early days of Raspberry Pi there was very little in the way of breakout boards. It was necessary to make connections to buttons, LCDs and Rotary encoders either direct on the GPIO header or via a specially constructed breakout board. Things became more complex when digital sound cards (DAC) were introduced as these occupied the GPIO header and either did not extend the header pins, or if they did so, only extended a few of them.

This has now changed and there is a wide variety of breakout boards available.

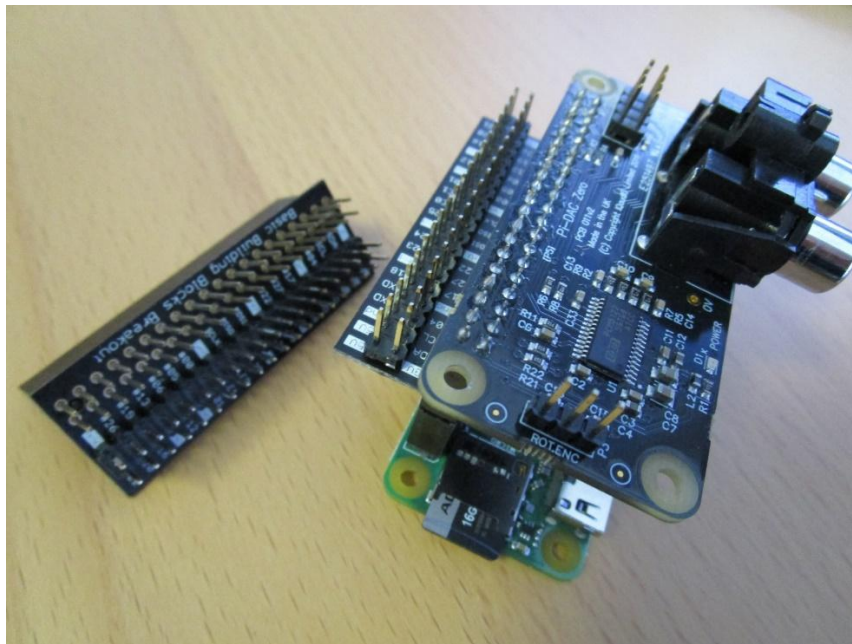


Figure 108 GPIO header breakout board

On the left of the above photo is an example of the 4Tronix GPIO breakout and extender board. On the right of the same photo is the break-out board used with a Raspberry Pi Zero W and an IQaudio DAC plus. All 40-pins are now made available, except for those used by the DAC, to attach buttons and the like to the GPIO pins. You need to install **Bookworm OS** and not **Trixie**. Download Bookworm from: <https://www.raspberrypi.com/software/operating-systems/>

These boards are available from <http://4tronix.co.uk>

See:

<https://shop.4tronix.co.uk/products/gpio-interceptor-gpio-breakout-for-40-pin-raspberry-pi>



Note: Soldering skills are required to solder the 40-pin header to the breakout board. The above breakout board is only shown as an example and many more are available on the Internet.

Construction using an Adafruit LCD plate

This section is for the radio using an Adafruit RGB-backlit LCD plate for Raspberry PI. The complete instructions for both ordering and building this product are available from the following Web site: <http://www.adafruit.com/products/1110> (See tutorials)

Note: Don't confuse this product (which has an I2C interface chip) with the two-line and four-line RGB LCDs which Adafruit also sell.



Figure 109 Adafruit LCD plate

The Adafruit LCD plate is designed to directly into the GPIO header on the Raspberry PI. These fit into a female 26 way header. If you want to connect the Adafruit LCD via a ribbon cable you will need to mount a 26 way male header instead of the female header and you will also need to construct a reversing board (Shown on the left of the picture on the left). Because ribbon cable swaps the two rows of pins over the reversing card is required to swap the two rows of pins back to their correct orientation.

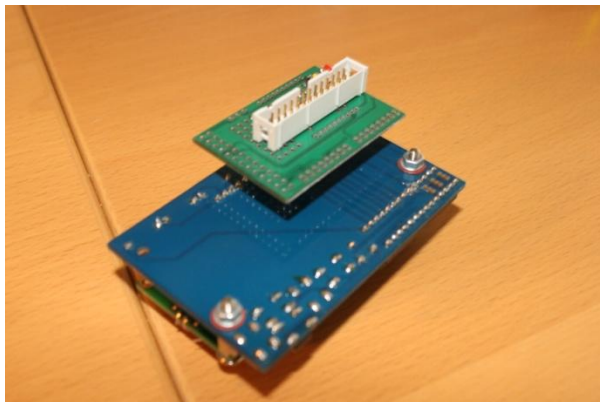


Figure 110 Adafruit LCD plate with ribbon cable adapter

Back view of the reversing board plugged into the Adafruit LCD plate.

The GPIO pins used are:

1. 3.3 Volts
2. 5.0 volts
3. SDA0
4. -
5. SCL0
6. Ground



Note 1: If you are going to plug the Adafruit LCD plate directly into the GPIO header on the Raspberry PI then you don't need the above reversing plate. Just follow the construction instructions on the tutorial on the Adafruit site.



Note 2: The "Select" button on the Adafruit plate is the "Menu" button for the radio.

Using other switches

The Adafruit Plate comes with five 4-pin switches which are mounted on the interface board. You will almost certainly want to use other switches say mounted on a front panel. It doesn't matter which type of switch you use as long as it is a push to make type. The only reason that a four-connector switch is used is for mechanical strength. If you look closely, you will see push button symbol between pins 2 and 4 and 1 or 3 on the component side for four of the switches. Either 2 and 4 and 1 or 3 should be connected to the switches.

It is advisable to solder two posts (male pins) for each switch on the reverse side of the board (The non-component side). Don't solder wires directly into the board. It is better to use push-on jumper wires connected to the switches to connect to the posts on the card.



Note: Rotary encoders cannot be used with the Adafruit Plate as these require three connections and the Adafruit routines to utilise them are not supplied by Adafruit.

Using the Adafruit LCD plate with the model B+, 2B and 3B

The plate is designed for revisions of the Raspberry Pi. It uses the I2C (SDA/SCL) pins. Adafruit supply a special extra tall 26-pin header so the plate sits above the USB and Ethernet jacks. For Pi Model B+, the resistors sit right above the new set of USB ports. To keep them from shorting against the metal, a piece of electrical tape must be placed on the top of the USB ports.

Construction using an I2C LCD backpack

Skip this section if you are not using an I2C backpack. There are two versions of the backpack supported:

1. Adafruit I2C backpack using an MCP23017 port expander – Hex address 0x20
2. Arduino I2C backpack using a PCF8574 port expander – Hex address 0x27 or 0x37

The I2C interface only requires two signals namely the I2C Data and Clock. This saves six GPIO pins when compared with the directly wired LCD interface. See <https://www.adafruit.com/product/292>.

The radio software also supports the more common PCF8574 chip-based backpack popular with the Arduino hobby computer may also be used. See <http://www.play-zone.ch/en/i2c-backpack-pcf8574t-fur-1602-lcds-5v.html> for example.

Adafruit I2C Backpack

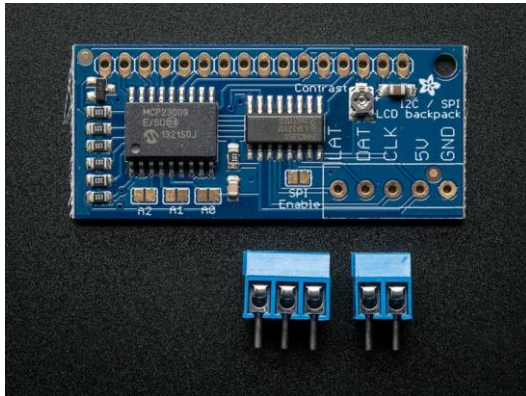


Figure 111 Adafruit I2C Backpack

The Adafruit I2C/SPI backpack interface is shipped as shown in the diagram opposite. There are no connectors shipped to connect to the LCD itself to this interface. These must be ordered separately.

Order a 16 in-line connector.

The Adafruit backpack also supports the SPI interface but is not used in this project

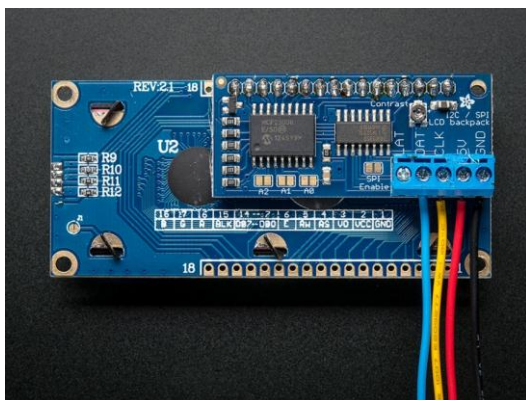


Figure 112 LCD connected to an Adafruit I2C backpack

The diagram shown on the left shows a 2x16 character LCD connected to the I2C backpack.

The wiring right to left is:

1. LAT (Unused)
2. Blue: I2C Data – GPIO 2 (pin 3)
3. Yellow: I2C Clock – GPIO 3 (pin 5)
4. Red: +5 volts – GPIO header pin 2
5. Black: GND (0v) – GND GPIO header pin 6

The I2C Data (DAT) connects to pin 3 on the Raspberry Pi GPIO header and the I2C Clock (CLK) to pin 5 on the GPIO header.

Arduino PCF8574 I2C backpacks

These types of backpack are popular with Arduino users. The device address is usually hex 0x27. Another manufacture may use hex 0x37. This is configurable in the radio configuration program.



Figure 113 Arduino I2C backpack

The wiring From top to bottom is:

7. GND (0 volts) – GPIO header pin 6
8. VCC +5 volts – GPIO header pin 2
9. SDA I2C Data – GPIO 2 (pin 3)
10. SCL I2C Clock – GPIO 3 (pin 5)

The blue potentiometer on the right is the contrast adjustment.

```
i2c_backpack=PCF8574
#i2c_backpack=ADAFRUIT
```

To use this device either amend the **i2c_backpack** parameter in **/etc/radiod.conf** (Comment out the ADAFRUIT line) or run the **configure_radio.sh** program.

Table 11 I2C Backpack connections

Backpack	Label	Description	GPIO	Physical pin
1	GND	Zero volts	-	14
2	VCC	+5 Volts	-	4
3	SDA	I2C Data	2	3
4	SCL	I2C Clock	3	5
5	LAT	Latch – Not used	-	-

The LAT pin is for the Adafruit backpack only and is for the SPI interface but isn't used by I2C.

Creating the interface board for the I2C back pack

An interface board is recommended to connect the I2C backpack and rotary encoders etc. to the GPIO interface. Any number of Raspberry Pi prototyping boards are available for all versions of the Raspberry Pi. The Ciseco Humble Pi prototype board shown in Figure 114 has been discontinued.

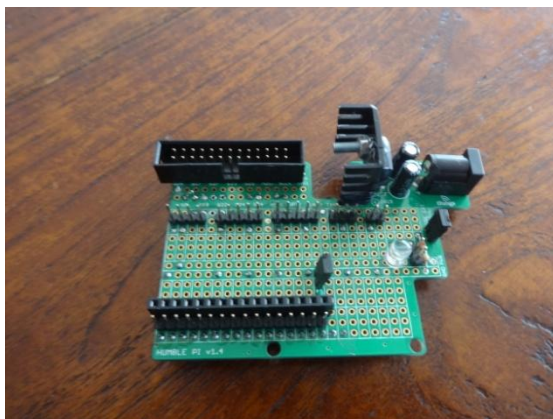


Figure 114 Ciseco Humble Pi I2C interface board

The above figure shows the I2C interface board using the Ciseco Humble Pi (Discontinued). The header pins in the centre from left to right are, I2C interface connector (4 pins), Volume rotary encoder (5 pins), Channel rotary encoder (5 pins), IR sensor (3 pins) and front panel LED (2 pins). In this version there are two rows of 18 pins (male and female) to allow different I2C backpack to be connected. You will normally only need one or the other.



Figure 115 The I2C backpack interface board

The above diagram shows the Adafruit I2C backpack connected to the interface board along with the rotary encoders. The 26-pin male header connects to the GPIO ribbon cable on the Raspberry Pi. On the left is a 6V to 9V power input feeding a 5V regulator.

Fitting a wake-up button

One of the features of this radio's design is that the menu button (LCD versions) or a special key (Touchscreen version) can do an orderly system shutdown. This is more desirable, and certainly safer and more convenient than pulling the power plug out. The system when properly shutdown goes into a so-called halt state. If the power is left connected the Raspberry Pi, it can be woken up by a button connected between pins 5(GPIO3) and 6(GND). GPIO3 can still be used as normal for example for I2C connections.

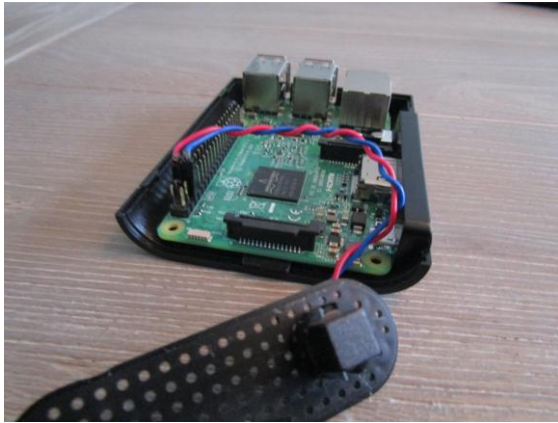


Figure 116 Wake-up button

On the left is the Raspberry Pi unit which is running the radio on a TV with HDMI inputs as shown in Figure 3 on page 7. A small black wake-up button is fitted to the case and connects to physical pins 5 and 6. When pressed with the Raspberry Pi in a halt state but power still applied it will start its boot-up sequence. It should be noted that pin 5 (GPIO3) is also used as the I2C data line. Although the button could still be fitted it is probably not a good idea as it will disrupt the I2C signal if the wake-up button is pressed.

Installing an IR sensor and remote control

IR Sensor

If you wish to use an IR remote control with other variants of the radio then purchase an IR sensor TSOP38238 or similar. The output pin connectivity depends on the exact hardware being used. See Table 15 IR Remote Control Sensor Pin outs on page 149 for the GPIO pin connection.

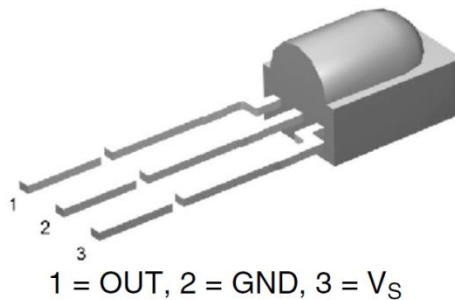


Figure 117 TSOP38238 IR sensor

The TSOP38xxx series works from 2.5 to 5.5 volts and is ideal for use the Raspberry Pi.

IR sensor	Description	RPi
Pin 1	Signal Out	GPIO in *
Pin 2	Ground	Pin 6
Pin 3 **	Vs 3.3 Volts	Pin 1

* See on Table 15 IR Remote Control Sensor Pin outs page 149.

** Caution; Do not accidentally connect to 5 volts

There are equivalent devices on the market such as the TSOP4838 which operate on 3.3 volts only.

See <http://www.vishay.com/docs/82491/tsop382.pdf> for more information on these IR sensors.



Tip: These IR sensors are very prone to damage by heat when soldering them. It is a good idea to use a 3-pin female connector and push the legs of the IR detector into them. If you solder the IR detector directly into a circuit then take precautions by connecting a crocodile clip across each pin in turn whilst soldering it. See figure below:

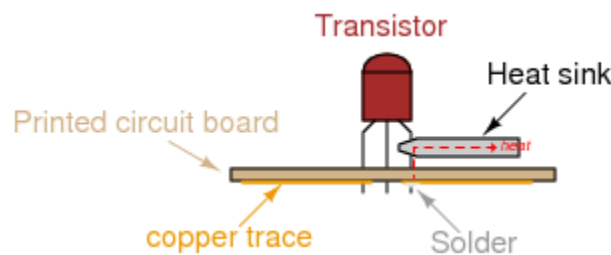


Figure 118 Soldering precautions

Remote control



Almost any surplus IR remote control can be used with this project although it is recommended to use the Raspberry Pi Mini IR remote control. Later on, it is explained how to set up the remote control with the radio software. You will need to install the software for IR sensor. See the section called *Installing the IR remote control software* on page 148.

Remote Control Activity LED

If wanted an activity LED can be connected to GPIO 11 or 13 depending on the type of radio. This flashes every remote control activity is detected. It is a good idea to include this.

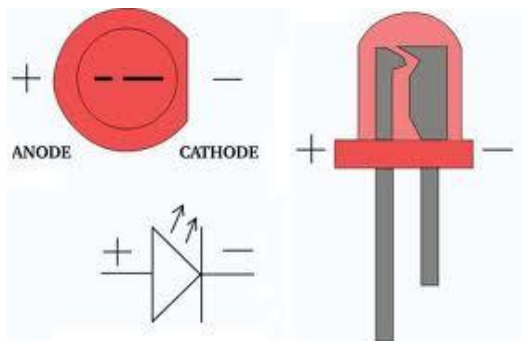


Figure 119 LED polarity

LEDs have polarity and must be wired correctly to work. The diagram shows the polarity of a typical LED. The longer lead is the positive (+) connection and connects to the Anode (The smaller terminal inside the LED)

Also, the LED must be wired in series with a resistor to limit the current, typically 100 Ohms is OK. Failure to do this may cause the LED to burn brightly for a while then burn out. Connect the cathode to GND (RPi Pin 6) and the Anode (+) to the GPIO pin shown in the following table via a 100 Ohm resistor.

The following table shows the GPIO pin used for the LED connections.

Table 12 Remote Control Activity LED

Radio Type	Pin	GPIO	Type of Raspberry PI
Activity LED not fitted	none	n/a	Not applicable
Two- or Four-line LCD with Push Buttons	23	11	26 or 40-pin version
Two- or Four-line LCD with Rotary encoders	23	11	26 or 40-pin version
Two- or Four-line LCD with I2C backpack	23	11	26 or 40-pin version
Adafruit RGB plate with push buttons	33	13	40-pin version only
Vintage radio with no LCD display	16	23	26 or 40-pin version
Designs using IQaudIO etc. sound boards	36	16	40-pin version only
PiFace CAD with IR sensor	16	23	26 or 40-pin version

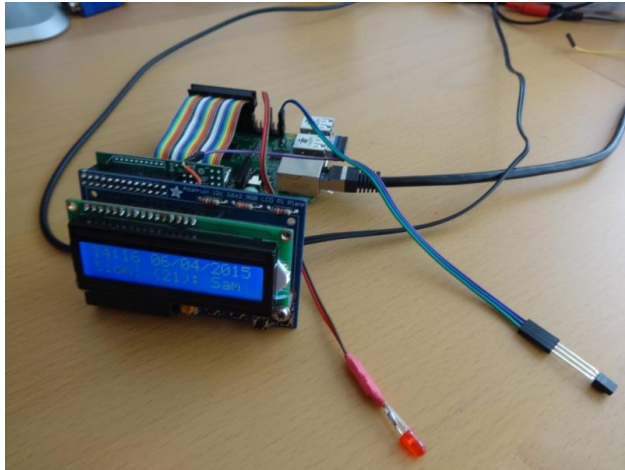


Figure 120 Adafruit plate with IR sensor and activity LED.

The illustration on the left shows an Adafruit RGB plate with IR sensor and activity LED.

The IR sensor picks up 3.3 volts from the reversing plate and connects the signal output to Pin 40 (GPIO 21) and GND (pin 39).

The LED connects to pin 33 (GPIO 13) and ground (pin 34).

Construction using an IQaudio Cosmic Controller

IQaudio manufacture a comprehensive range of sound devices and controller boards. See their Web site at: <http://www.iqaudio.co.uk/>

The radio software provides support for the IQaudio Cosmic controller.

The IQaudio Cosmic controller consists of the following:

- A three push-button interface (Channel UP/DOWN and Menu)
- A rotary encoder (Used as volume control)
- Three status LEDs (Normal, Busy and Error)
- A 128 by 64 pixel OLED display (I2C interface)
- Optional IR detector (Ordered separately)



Figure 121 IQaudio Cosmic controller and OLED display

The main advantage of this hardware, is that it contains everything required by the radio software.



Figure 122 Lego radio with IQaudio Cosmic controller and OLED

Construction using the Pimoroni Pirate radio

A full set of instruction for building the Pimoroni Pirate radio with pHat BEAT can be found here: <https://learn.pimoroni.com/tutorial/sandyj/assembling-pirate-radio>

Soldering skills are required.

Construction using the PiFace CAD

Fortunately, no soldering or construction is required with the PiFace CAD. Just plug it in and install and run the software. The PiFace CAD also has an IR sensor which means that it can be used with a remote control. It is however more sluggish in its operation when compared to other variants of the radio as the SPI interface on the Raspberry Pi is fairly slow. It also has the disadvantage that the push buttons are on the bottom of the unit.



Figure 123 PiFace CAD and Raspberry Pi



Figure 124 PiFace CAD in a case

Various ready-made cases are available from various suppliers. Warning: not all fit properly and might require some modification.



Note: PiFace CAD is an end-of-line product whilst stocks last and is only supported on Bullseye OS.

The **PiFace CAD** uses the SPI interface (from Motorola) See

http://en.wikipedia.org/wiki/Serial_Peripheral_Interface_Bus for further information on SPI.

Construction Tips and Tricks

This section contains some construction tips which may be useful. It goes without saying that having the correct tools such as a good fine tipped soldering iron, wire strippers and the like will greatly help constructing the radio.



Figure 125 Using wire strippers

Cut the plugs off the end that is to be soldered to the rotary encoder or switch. Leave the other end with *female* connectors. Using good wire strippers, strip a few millimetres off the wires. Separate the wires for about 30 millimetres.

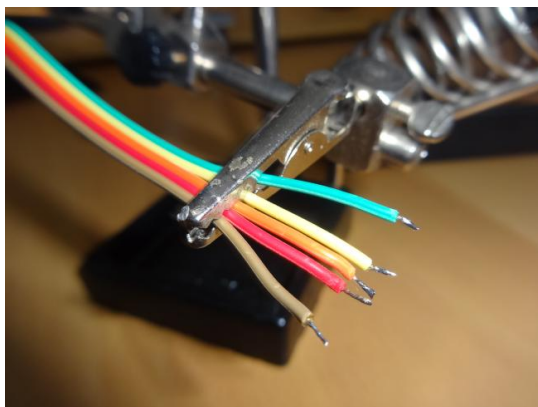


Figure 126 Tinning the wires with solder

Twist the copper strands together as tightly as possible and tin the wires with a little solder. A so called “Extra pair of hands” is very useful for gripping the wires using crocodile clips.

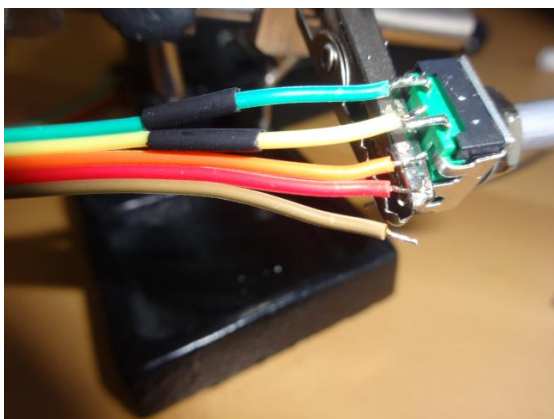


Figure 127 Soldering up the switch

Tin the switch connections with solder. Cut a few millimetres of shrink wrap and slide onto the wires to be soldered. Make sure that the shrink wrap sleeves are well away from the heat of soldering iron as these will shrink easily with the slightest bit of heat. Tack the wire onto the top of the switch connector. Don't attempt to twist the wire around the connector. Just tack it on top with a little bit of heat from the soldering iron.



Figure 128 Shrink shrink-wrap with a hair dryer

After soldering on all wires push the shrink wrap over the newly soldered wires. Using a hair dryer heat the shrink wrap until it has completely shrunk tightly over the wires.

The finished switch can either be connected directly to the Raspberry Pi or to an interface board.

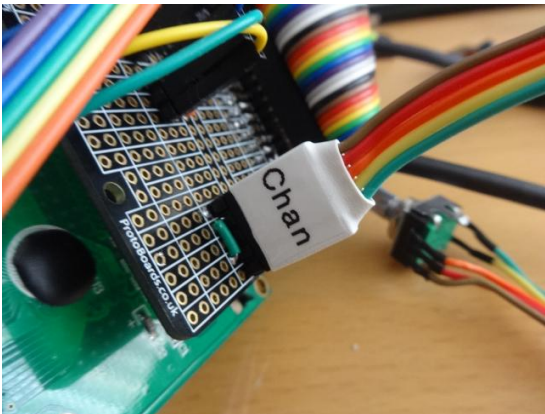


Figure 129 Connecting the rotary encoder an interface board

If using an interface board, the female connectors can be bound together to form a plug using a larger piece of shrink-wrap. Slip the shrink wrap over the female pins and heat with a hair dryer. This can then be labelled up using Dymo tape machine or a CD marker pen. Push the newly created plug onto the male pins on the interface board.

Selecting an audio amplifier

There is a wide range of amplifiers that can be used with the radio which fall into five main categories:

1. A set of PC speakers with amplifier (Logitech or similar) – the simplest option
2. A dedicated AB or Class D stereo amplifier (Velleman or similar)
3. A combined DAC (I2S) and amplifier from manufacturers such as **IQaudio** or **HiFiBerry**
4. The audio stage of an existing (vintage) radio. Use the PA or record player input (Usually mono only)
5. Bluetooth speakers or headset



Figure 130 PC speakers

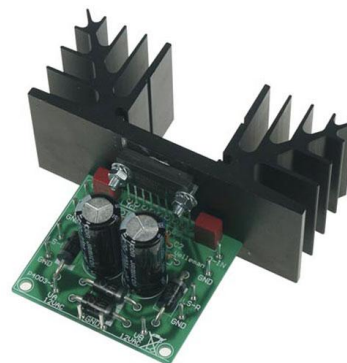


Figure 131 Velleman 30W stereo amplifier



Figure 132 IQaudIO DAC and 20W Amplifier



Figure 133 Vintage radio PA input



Figure 134 JVC Bluetooth Speakers



Figure 135 Audio output jack amplifier



The above manufacturer's products are examples only and do not imply any specific recommendations. There are an enormous number of solutions available. The choice is determined by, amongst other things, price, mono/stereo, power output and quality.

Preventing electrical interference

One of the most irritating faults that one can have is the LCD screen occasionally either going blank or displaying hieroglyphics especially when switching on and off other apparatus or lights on the same circuit. This is due to Electromagnetic Interference (EMI).

See https://en.wikipedia.org/wiki/Electromagnetic_interference for more information.

EMI can be caused by any number of sources such as fluorescent lighting, switching on and off equipment on the same circuit as the radio or even electrical storms. If you are using a standard Raspberry Pi USB power supply then you will probably not experience this problem as nearly all are fitted with a ferrite core (This is the big lump in the cable or may be built into the adapter itself). If you do experience this problem then try the following solutions one at a time in the order shown below. They can all be used together if required.

Using a clip-on ferrite core on the +5 volt cable



Figure 136 Clip on ferrite core

One of the most effective solutions is to put a clip-on ferrite core on the +5V cable going to both the Raspberry Pi and USB hub. Loop the wire through at least once. Even a single loop seems to be enough. Try this first!



Figure 137 Loop +5V supply around the core

Fit a mains filter



Figure 138 Various mains filters

Try using a mains filter. This has the advantage that it can prevent spikes coming in from the mains and protect against electrical storms. The picture on the right shows an integrated filter and panel mount mains socket.



Figure 139 Integrated mains socket and filter

Use an I2C LCD backpack

If all else fails replace the directly wired LCD wiring with an I2C backpack. See the section called *Construction using an I2C LCD backpack* on page 67 for further information.

Preventing ground loops



Figure 140 3.5mm Jack Ground Loop Isolator

Avoid creating ground loops in the first place during construction. Ground loop issues usually cause a humming or electronic noise. Trying to tap off the Raspberry power supply from the power supply for the amplifier (if used) is one sure way to create a ground loop. If you experience such a problem then a 3.5mm Jack Ground Loop Isolator available from suppliers such as Kenable (<http://www.kenable.co.uk>) can prevent unwanted hum on the audio system. Place the isolator between the Audio output of the Raspberry Pi or sound card and the amplifier input.

For further information on ground loops:
[https://en.wikipedia.org/wiki/Ground_loop_\(electricity\)](https://en.wikipedia.org/wiki/Ground_loop_(electricity))

Connecting up a USB power adapter



Figure 141 Connecting up a USB power adapter

It is convenient to connect all power supply components for the Raspberry Pi, amplifier and USB hub etc via a single mains switch. USB 240V AC to +5V power adapters are designed to connect directly to the mains and not via a mains switch. One idea is to purchase a European round pin adapter and use standard electrical connector blocks to connect the incoming AC power cable to the two pins of the power adapter. AC cables to other components such as the amplifier can also be connected to the connector blocks. Use electrical tape or shrink-wrap to isolate the connector blocks.

Cooling the Raspberry Pi

The Raspberry Pi is built from commercial chips which are qualified to different temperature ranges; the LAN9512 is specified by the manufacturers being qualified from 0°C to 70°C, while the Application Processor (AP) is qualified from -40°C to 85°C. Operation outside these temperatures is not guaranteed. The temperature of the CPU should not really go above 80°C. If it does the CPU will throttle back its processor clock speed to reduce the temperature. The official line from the Raspberry Pi is that does not require cooling even though you can get temperature warnings when using the Raspberry Pi 7-inch touch screen.

The **vcgencmd** command can be used to check the CPU temperature.

```
$ vcgencmd measure_temp  
temp=67.1'C
```

For most of the radio designs in this document no specific cooling is required. If you need to cool the Raspberry Pi (Those with touchscreens in particular) then a variety of heat sinks and cooling fans are available as shown in Figure 142 below. Even with a heat sink good ventilation is necessary.



Figure 142 Heat sink kit

A variety of heat sink and cooling fan kits are available for the Raspberry Pi. Fans are a less good idea as they will produce a background hum. Also, as the CPU gets hot and the RPi is mounted vertically the adhesive softens and the fan and its heat sink tend to fall off unless secured in some way. Try to using a heat sink first.



Figure 143 Cooling fans

Booting from a USB drive or stick

The Raspberry Pi is normally booted from an SD card; however, it is also possible to boot from a USB 3.0 disk drive or stick. The drive can be either a USB drive with a spindle or a Solid-State Drive (SSD) or a USB stick. This procedure works with Raspberry Pi's with USB 2.x or 3.x ports such as models 3B, 3B+ or 4B+. The Pi 400/500, 4B and Model 5 may already be USB boot enabled and can usually boot from a USB drive without modifying the firmware.

The advantages of booting from a USB/SSD disk drive are:

- A USB/SSD disk drive is a lot less susceptible to corruption than a SD card
- A USB/SDD disk drive generally has greater capacity than a SD card for media files etc.
- USB disk drives are a lot faster than a SD card. SDD disks are even faster but limited to USB 3.0 speeds.



Netac produce a portable SSD USB 3.2 Gen 2 (10 Gbps, Type-C) in a range of sizes from 128GB up to Terabyte sizes. The illustration on the left shows their external Solid-State Drive PSSD 250GB. It connects to the Raspberry Pi USB 3.0 port via a USB cable supplied with the disk drive. SSD drives are more reliable than USB sticks.

You need to copy the Raspberry Pi OS to the drive in the same way you make a SD card using Raspberry Pi Imager software.

To enable booting from a USB connected Disk Drive first run **raspi-config**.

```
$ sudo raspi-config
```

Select option **6 Advanced Options** **Configure advanced settings**

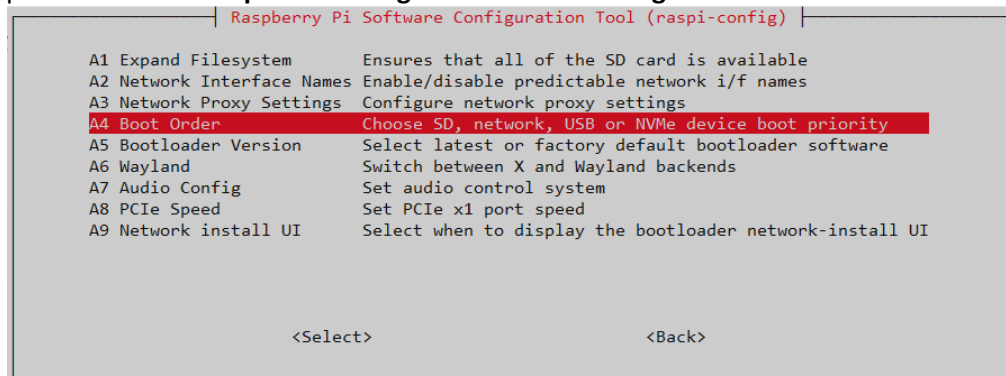
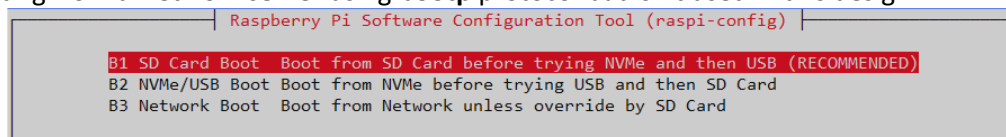


Figure 144 Advanced Options

Next select option **B1** to boot from the SSD disk drive. Option **B2** is also a viable choice. Option **B3** is for booting from a Network server using **bootp** protocol but isn't used in this design.



Confirm your choice. Plug the SSD drive or USB stick into one of the USB 3.0 ports and reboot.

NVMe stands for Non-Volatile Memory Express, the new generation of Solid-State Drives (SSD)

Miscellaneous

Simple tone regulator

It may be that you wish to fit a tone regulator to the radio. Below is one option.

The following diagram and modified text came from Jack Orman at:

<http://www.muzique.com/lab/swtc.htm>

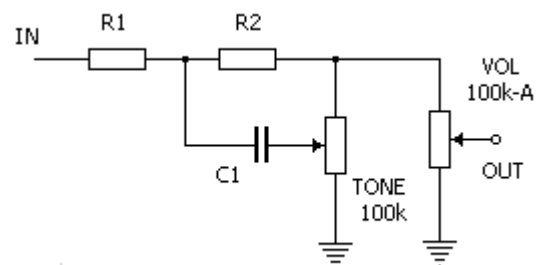


Figure 145 Simple tone control circuit

This tone control circuit that has a response that can be altered from high cut to high boost as the knob is turned. The output resistance is constant so the volume does not vary as the tone control is adjusted.

Suggested values for beginning experimentation with, are $R1=10k$, $R2=47k$, $C1=0.022\mu F$ and $100k$ for the tone and volume pots.



Note that the above circuit has a lot of attenuation of the audio output so using the onboard audio output of the Raspberry Pi might result in a disappointing level of volume. It is recommended to use a sound output DAC or USB sound dongle.

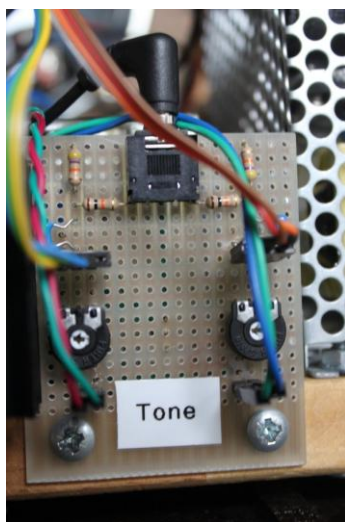


Figure 146 Tone control board

The illustration on the left shows a simple passive tone regulator board using the above circuit.

The audio output from the Raspberry Pi or DAC is fed into the board via a standard Audio socket.

Below the input are the connections to the tone regulator potentiometer mounted on the front panel of the radio. Below the potentiometer connections are the two $100K$ presets for adjusting the output level to the Audio Amplifier.

Below these the Left and Right audio outputs connect to the Amplifier.

Using the Adafruit backlit RGB LCD display

The Adafruit backlit RGB LCD has three LED backlights (Red, Blue and Green) which can either be switched on individually or in various combinations together as shown in the table below:

Table 13 Adafruit backlit RGB display wiring

Switch pin	Red (Pin 16)	Green (Pin 17)	Blue (Pin 18)	Colour	Diodes required
1	0	0	0	Off	0
2	0	0	1	Blue	0
3	0	1	0	Green	0
4	0	1	1	Light Blue	2
5	1	0	0	Red	0
6	1	0	1	Purple	2
7	1	1	0	Yellow	2
8	1	1	1	White	3
Common	GND			Total diodes	9



The diodes used are any low voltage low current diodes such as the IN4148. So to use all of the above combinations would require a single pole 8 way rotary switch and logic and nine diodes. The first switch position is off.

Figure 147 IN4148 diode

- Do not wire anything to position 1.
- Wire pin 16 (Blue) of the LCD backlight to switch position 2.
- Wire pin 17 (Green) of the LCD to switch position 3
- Wire pin 18 (Red) of the LCD to switch position 5
- Wire pin 17 and 18 via two diodes to pin 4 to give the colour light blue
- Do the same for the other two-colour combinations
- Wire pin 16, 17 and 18 to pin 8 via three diodes to give the colour white
- Wire the centre pin of the switch to 0v (GND)

Chapter 5 – System Software Installation

Contents chapter 5	Page
Conventions used in this tutorial	83
What version of the OS works with my Raspberry Pi model	83
Useful installation tools	84
Entering system commands	85
Editing configuration files	86
System Software installation	86
SD card creation	86
Booting the Raspberry Pi for the first time	95
Preparing the Operating System for software installation	98
Configuring the Operating System	99

Conventions used in this tutorial

Installation of the radio program requires you to enter lines at the command line prompt. This requires you to log into the Raspberry Pi as user '**pi**'. The default password is **raspberrypi**.



Note: Don't carry out any of the following commands just yet. They are just examples.

```
Raspberrypi login: pi
Password: raspberrypi
pi@raspberrypi:~$ Last login: Sun Apr  6 10:18:18 2014 from 192.168.2.100
pi@raspberrypi:~$
```

The prompt line is displayed ending with a \$ sign. The **pi@raspberrypi:~** string means user 'pi' on host machine called 'raspberrypi'. The ~ character means the user 'pi' home directory **/home/pi**. In this tutorial if you are required to do something as user **pi** then only the \$ sign will be shown followed by the command as shown in the example below:

```
$ mpc status
```

Some commands produce output which does not need to be shown. In such a case a ':' is used to indicate that some output has been omitted.

```
$ aplay -l
**** List of PLAYBACK Hardware Devices ****
: {Omitted output}
card 0: ALSA [bcm2835 ALSA], device 1: bcm2835 ALSA [bcm2835 IEC958/HDMI]
  Subdevices: 1/1
  Subdevice #0: subdevice #0
card 1: Device [USB PnP Sound Device], device 0: USB Audio [USB Audio]
  Subdevices: 0/1
  Subdevice #0: subdevice #0
```

END OF EXAMPLE COMMANDS.

What version of the OS works with my Raspberry Pi model

The following article contains a table showing which model Raspberry Pi's work with which Raspberry Pi OS:

https://en.wikipedia.org/wiki/Raspberry_Pi_OS



As of the 1st October of October the Raspberry Pi Foundation have introduced a new version of the operating system called **Trixie**. Version **8.1** and earlier of the software does not work on **Trixie**. You must use **Bookworm 64-Bit Desktop**. A new version **8.2** onwards is required to work with **Trixie**. The latest version of the radio is version **8.3**.

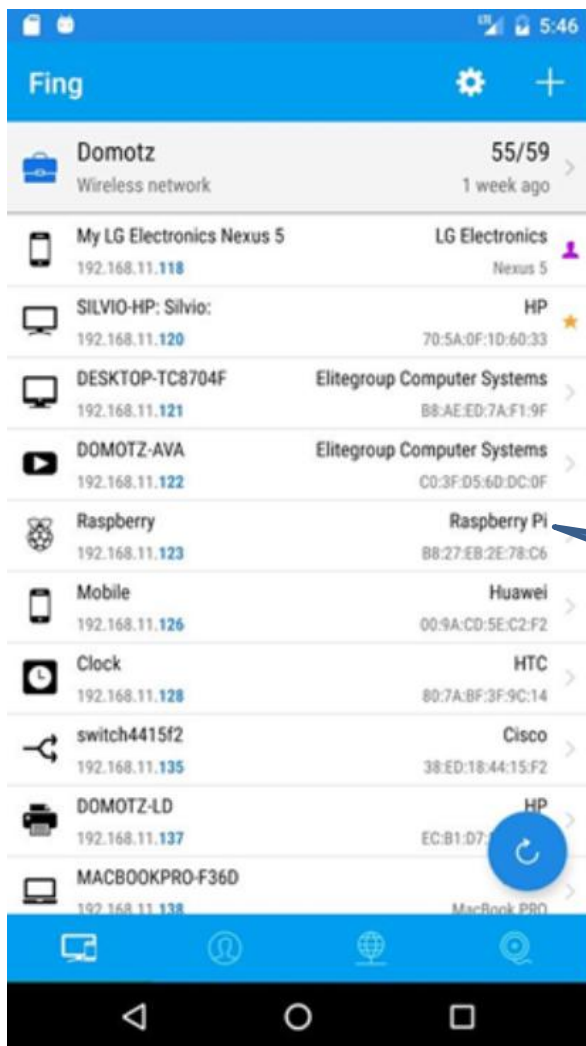
You will find a useful getting-started guide at

<https://www.raspberrypi.com/documentation/computers/getting-started.html>

You should still use the instructions in this guide to install the Operating System (OS) and radio software.

Useful installation tools

Finding the Raspberry Pi on a network using Fing



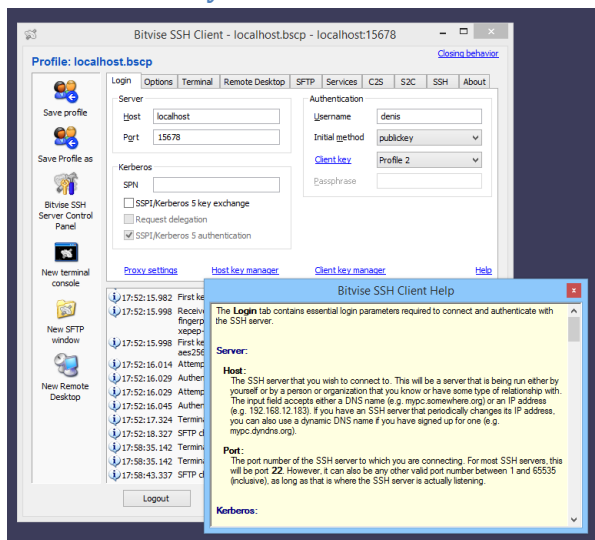
The **Fing** App is a free network toolkit and scanner for iOS (Apple) and Android. It will discover all the devices on your network, identify intruders and can also run Internet speed tests.

In this context it is very useful for finding out the IP address of any Raspberry Pi connected to the network. It will only see it of course once the Raspberry Pi has been set up first on the Wireless Network or hard-wired in. It scans both Wireless and hard-wired Ethernet devices.

Once detected it is possible to SSH to the Raspberry Pi using **Bitvise** or **Putty**.

See <https://www.fing.com/products/fing-app> for further information.

Bitvise and Putty



Bitvise is a free SSH client available for Windows or Mac to connect SSH server enabled Unix or Linux operating systems. It is a graphical based SSH client just like Putty with more features. It supports the File Transfer using SFTP Secure File Transfer Protocol). Once installed you can easily make a terminal connection to the Raspberry Pi using the IP address discovered by Fing above as well as easily transferring files.

See <https://www.bitvise.com/ssh-client> for more information.

Entering system commands

If you are new to Linux there are a couple of things that may cause confusion.

1. Entering program names on the command line
2. File path names
3. File permissions

Take the following examples:

```
$ raspi-config
```

The following command fails

```
$ create_stations.py
$ -bash: create_stations.py: command not found
```

The following two commands both work.

```
$ cd /usr/share/radio
$ ./create_stations.py
```

```
$ /usr/share/radio/create_stations.py
```

The third command has a `./` in front of it, the first one doesn't. Why?

The reason is that the **raspi-config** program is in the **/usr/bin** directory which is in the **PATH** environment directive. This can be seen with the following command.

```
$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/local/games
:/usr/games
```

The second program is located in the **/usr/share/radio** directory which is not in the **PATH** directive. The `./` in front of the command means that the program or script will be found in the current directory.

So, this means for programs not in directories specified in the **PATH** environment directive, you must either specify the full path name to the command or change to its directory and then enter the command with a `./` in front of it.

In the system prompt for user `pi` you will see a `~` character. The `~` character means the home directory for the current user, in this case `pi`. So `~` is the same as **/home/pi**.

```
pi@raspberrypi3:~ $
```

For information on file permissions see the following link:

https://wiki.archlinux.org/index.php/File_permissions_and_attributes

Editing configuration files

At various points during the installation procedures in this manual you will be asked to edit certain configuration files such as `/etc/radiod.conf`. To see how to do this see the section called *Editing configuration files manually* on page 167.

System Software installation

Raspberry Pi OS (previously called **Raspbian**) is the Foundation's official supported operating system. The latest version of **Raspberry Pi OS** is called **Bookworm**. Create a new SD card with **Bookworm** or **Bookworm Lite**. There is also a "Full" version of **Bookworm** however this is unnecessary for this project.

A lot of very useful Raspberry Pi documentation will be found at:

<https://www.raspberrypi.org/documentation>



Note: The touch-screen or HDMI TV version of the software requires a desktop version of the operating system so use **Raspberry Pi Bookworm** and not the **Lite** version. Only use **Lite** for LCD versions of the radio.



Warning: Version **8.3** is designed to work on primarily on **Raspberry Pi Trixie** but can run with restrictions on **Bookworm**. **Trixie** uses Music Player Daemon (MPD) version **0.24.4**. **Bookworm** uses MPD version **0.23.12**.

SD card creation

Use at least a 16 Gigabyte Card (Class 10) for **Trixie Lite** or 32 Gigabyte for **Trixie Desktop/Full**. Create an SD card running the latest version of **Raspberry Pi Trixie** or **Trixie Lite**. A Class 10 SD card is a memory card that can transfer data at a minimum rate of 10 Megabytes per second (MB/s).



Note: This version of the radio only **64-bit** architecture of the Raspberry Pi Operating System is supported and has been tested. Several features of the radio such as the recording of radio stations only work with the **Trixie 64-bit OS**. Also, there is no good reason to use a **32-bit OS**.



WARNING: During installation you will see the following message:
The following packages were automatically installed and are no longer required:
..... List of packages

Use 'apt-get autoremove' to remove them.

DO NOT DO THIS as under the **Trixie OS** it may remove the **rpd-wayland-core** package which prevents you from logging into the Raspberry Pi Desktop. These packages are not doing any harm other than taking some room up on disk.

The following Raspberry Pi models have a 64-bit architecture and run Linux:

- Raspberry Pi 4
- Raspberry Pi 400
- Raspberry Pi 3B
- Raspberry Pi 3B+
- Raspberry Pi 3A+
- Raspberry Pi 5
- Raspberry Pi Zero 2 W

The **Raspberry Pi 1, 2, and Zero** models are not **64-bit** because their processors only have a **32-bit** architecture width. Use **Trixie** or **Bookworm 32-bit** OS on these Raspberry Pi models but be aware that **32-bit** installations are no longer supported as certain radio components don't work with **32-bit** systems.

There are a couple of ways of creating the SD card but in this tutorial, we are using the **Raspberry Pi Imager** software to create the SD card. You will need a Windows PC or Laptop with a SD Card Reader as shown in Figure 148 below.



Figure 148 Windows Laptop with SD card reader

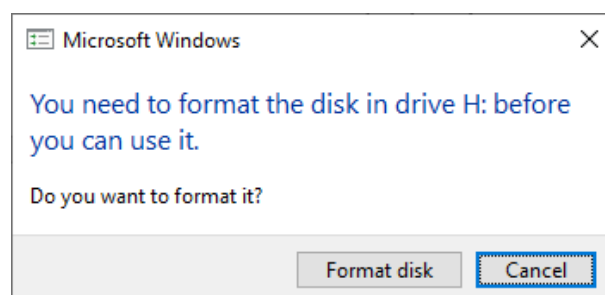


Figure 149 USB SD Card reader

If your PC does not have an SD Card Reader then you can use a USB SD Card reader as shown on the left. The Raspberry Pi Imager software can then be used to write the Raspberry Pi OS to SD Card.

Use a **Class 10** SD card such as **San Disk Ultra**.

An Apple Mac PC can also be used but will require that you download the Mac OS version of the RPi imager software. First insert the SD card into the SD card reader on your PC. When you insert an SD Card that already has an OS other than Windows you will see the following:



Ignore this message and close the above dialogue box. The above may occur a number of times during this process. There are two architectures supported in this version of the radio software:

- **64-Bit Bookworm** Operating Systems
- **64-Bit for Trixie** Operating System



Note 1: If you chose the Raspberry Pi **Model 5** then version **1.9.6** onwards of the **Raspberry Pi** imager will only offer the **Trixie** Operating System and not **Bookworm**. Even if you are installing on a **Model 5** select the **4B**. This will allow you to select **Bookworm** if required. The **Model 5** only runs with **Bookworm** or **Trixie** (default).



Note 2: It is possible to boot from a **USB 3.0** disk drive or stick instead of from an SD card. This is described in section called *Booting from a USB drive or stick* on page 79.

Using a Web browser, go to <https://www.raspberrypi.org/software/> Download and install the **Raspberry Pi Imager** software for your PC Operating System (Normally Microsoft Windows or Mac OS). Once installed you will see the Raspberry Pi Imager Icon on your desk-top. Click on the desktop Imager icon.

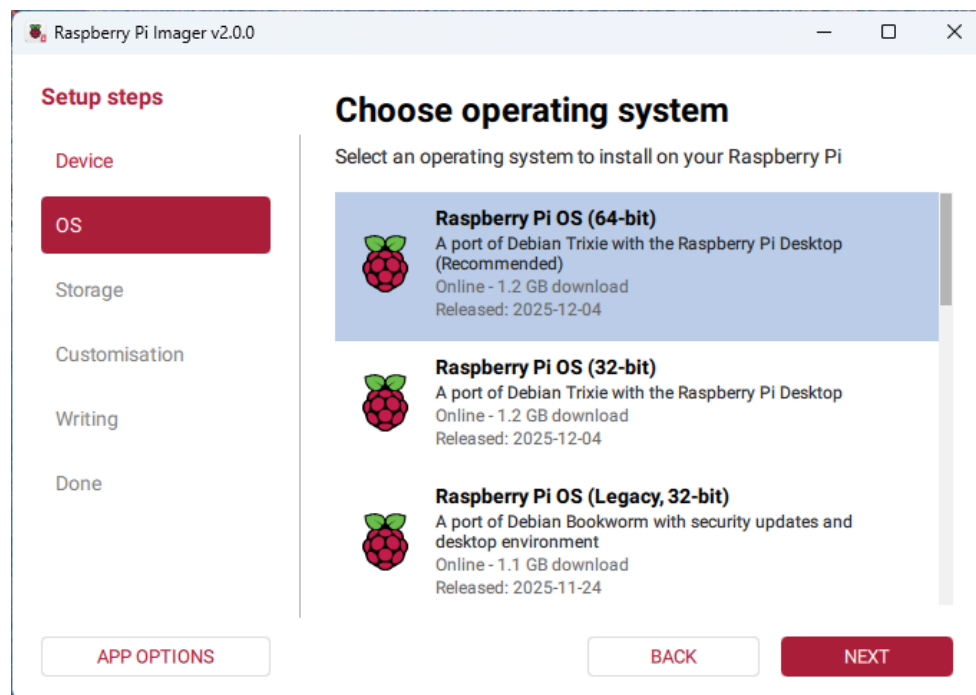
The latest version of the software now re-directs all **RPI.GPIO** calls to **Igpio** in the **/usr/share/radio/RPi** directory when running on **Bookworm** or **Trixie OS**. This mechanism is enabled by the Radio Configuration installation program.

Installing the OS using Raspberry Pi Imager

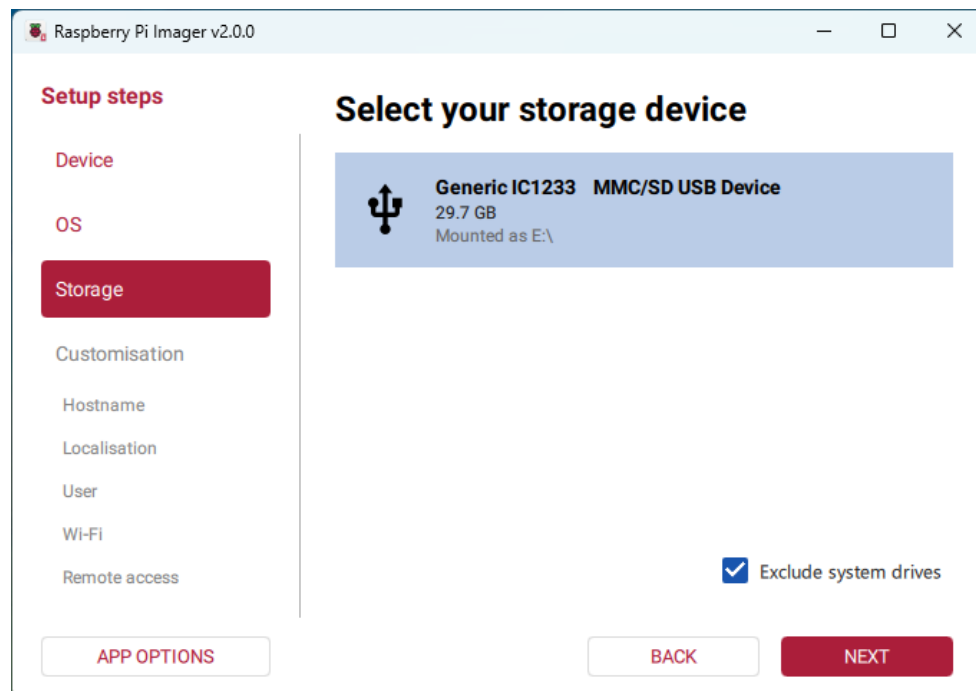
Download the **Raspberry Pi Imager** software from <https://www.google.com/search?q=raspberrypi+imager+download> : Insert the SD card into your card reader. Ignore any Windows messages to format the card as it isn't necessary. Select your Raspberry Pi model from the list. For example, Raspberry Pi 4.



The OS versions for your RPi will be displayed. Select **Raspberry Pi OS (64-bit) Trixie Desktop**.



Now select your SD card. Note that the **Exclude system drives** tick box is selected to prevent you from selecting any other USB drives which may have an OS on them.



Note: If there are multiple choices of USB drives make sure that you select the correct one. Do not proceed until you are absolutely sure you are selecting the correct drive and not for example a USB backup drive.

Once both the OS and USB drive have been selected the following screen is displayed:

The screenshot shows the 'Customisation: Choose hostname' screen in the Raspberry Pi Imager v2.0.0 application. On the left, a sidebar lists various customisation options: Device, OS, Storage, Customisation (highlighted), Hostname, Localisation, User, Wi-Fi, Remote access, and Raspberry Pi Connect. The main area is titled 'Customisation: Choose hostname' and features a text input field labeled 'Enter your hostname' with the value 'piradio' entered. Below the input field, a note states: 'A hostname is a unique name that identifies your Raspberry Pi on the network. It should contain only letters, numbers, and hyphens.' At the bottom, there are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT'.

Enter the host name you want for your Raspberry Pi such as **piradio** and press **NEXT**.

The screenshot shows the 'Customisation: Localisation' screen in the Raspberry Pi Imager v2.0.0 application. The sidebar on the left is similar to the previous screen, but 'Localisation' is now highlighted. The main area is titled 'Customisation: Localisation' and includes the instruction 'Select your location for suggested time zone and keyboard layout'. It contains three dropdown menus: 'Capital city' (set to 'London (United Kingdom)'), 'Time zone' (set to 'Europe/London'), and 'Keyboard layout' (set to 'gb'). At the bottom, the same four buttons are present: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT'.

Select your geographic region from the **Capital city** drop down box. The **Time Zone** and **Keyboard Layout** will be selected automatically but can be changed if required. Press **NEXT**

The following screen will be displayed. Initially the screen will show the **General** tab. Enter your username and password for user **pi**. You can change the username from **pi** but it is recommended to use user **pi**. Make a note of the username and password.

Raspberry Pi Imager v2.0.0

Customisation: Choose username

Create a user account for your Raspberry Pi

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

Username:

Password:

Confirm password:

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

APP OPTIONS SKIP CUSTOMISATION BACK NEXT

Enter the **SSID** and **Password** of your home router. This will be on the router itself otherwise see your router documentation.

Raspberry Pi Imager v2.0.0

Customisation: Choose Wi-Fi

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

SECURE NETWORK OPEN NETWORK

SSID:

Password:

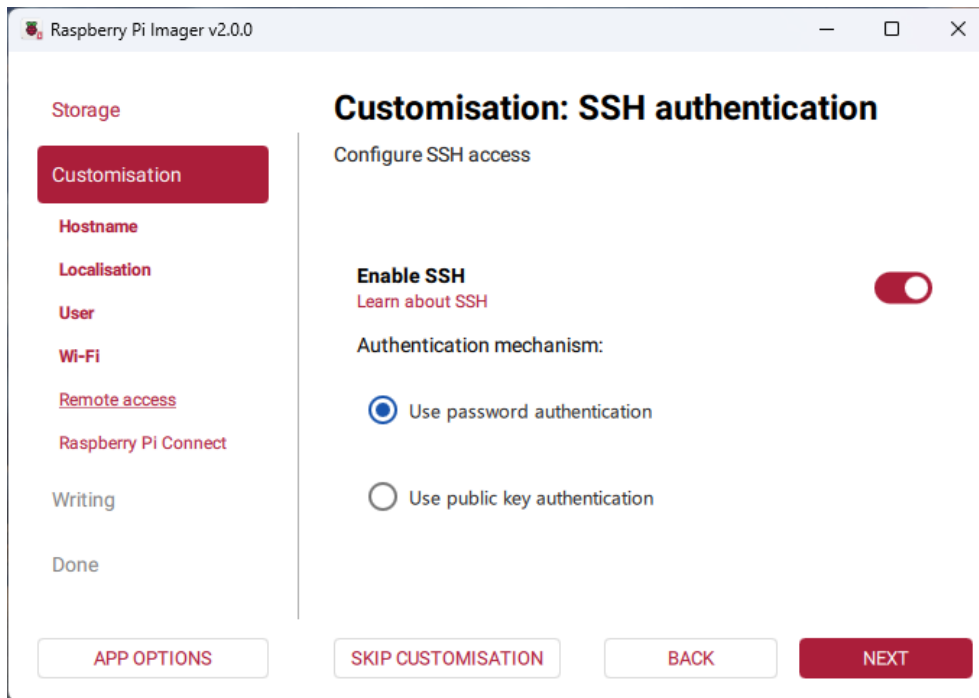
Confirm password:

☐ Hidden SSID

APP OPTIONS SKIP CUSTOMISATION BACK NEXT

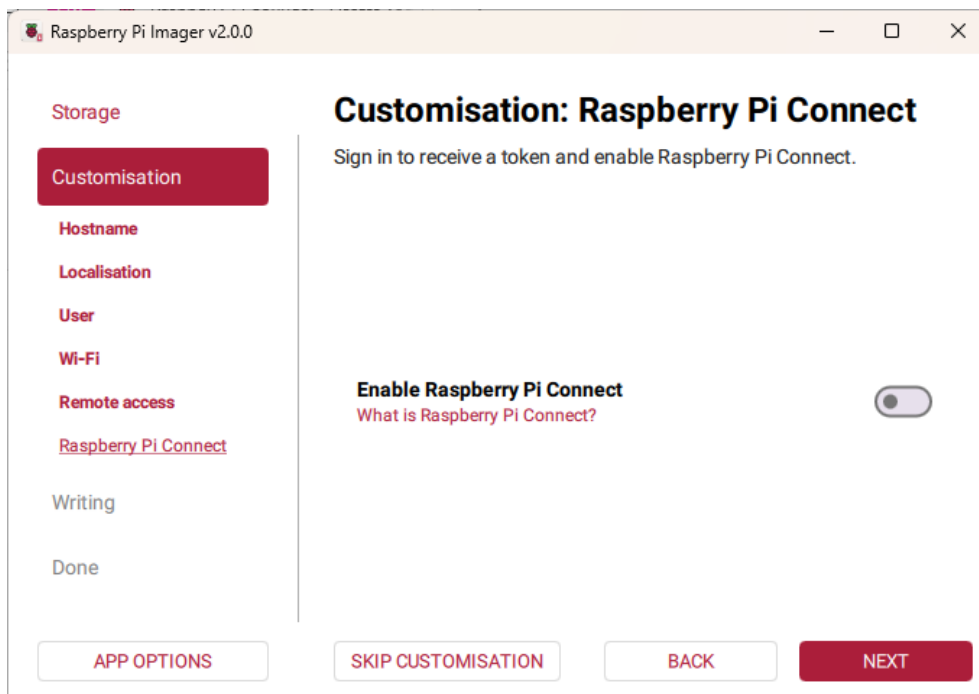
TIP: Unfortunately, there isn't the option to display the password as you are typing it. Key your router password into **Notepad** and then cut and paste it into the Password and password confirmation fields.

Now enable SSH so that you can log into the Raspberry Pi using **Bitvise** the press **ENTER**.

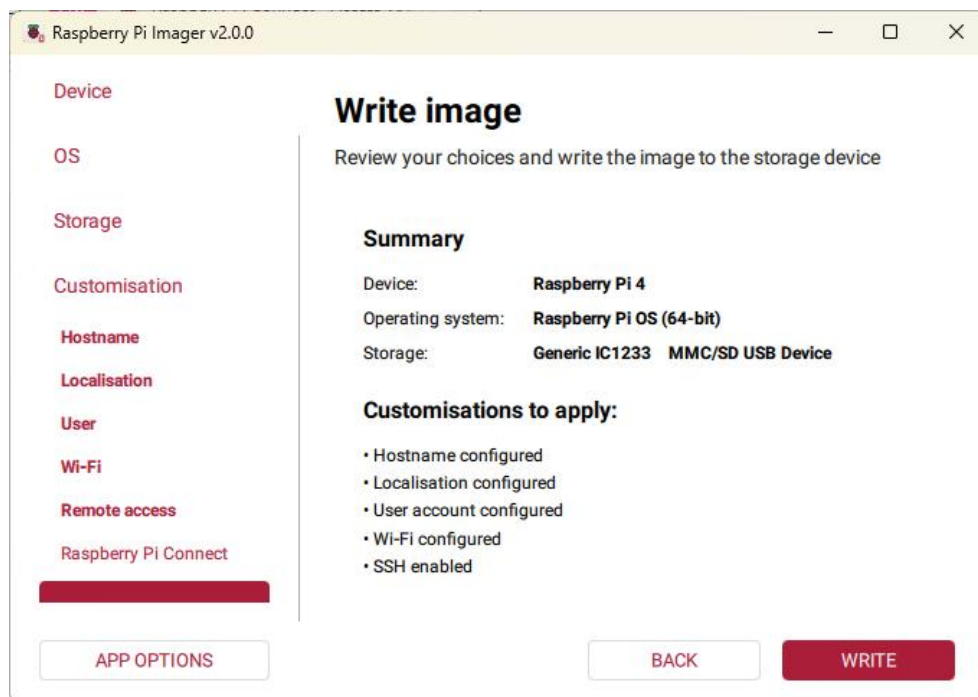


Important: Enable the **SSH** option “Use password authentication”. If you select “Use public key authentication” you will have to set up SSH keys between the PC and the RPi which is more complicated and not covered in this manual.

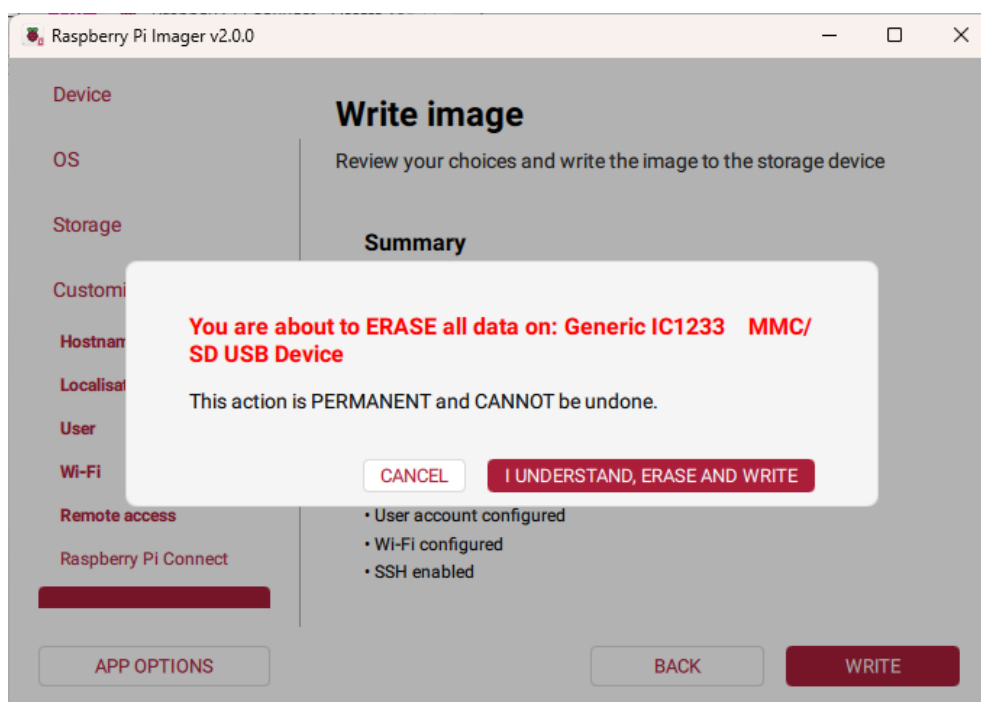
You will now be asked to enable Raspberry Pi Connect which is a remote connection facility over the Internet. It is not required and also it is not free. Press **NEXT** to skip this option.



The following summary screen displays a summary of your choices.

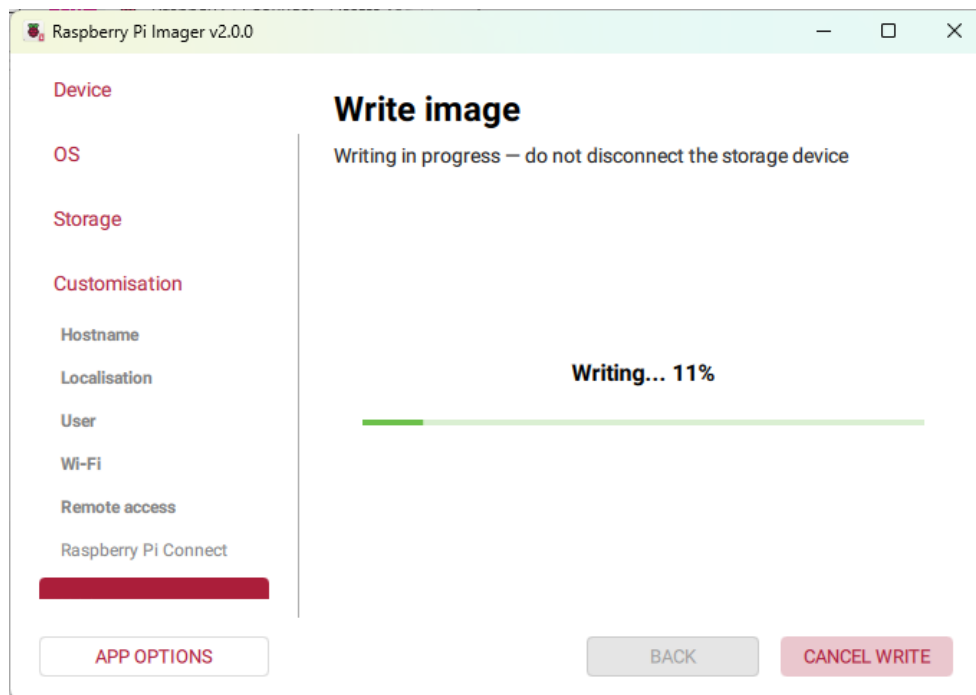


Press **WRITE** to continue.

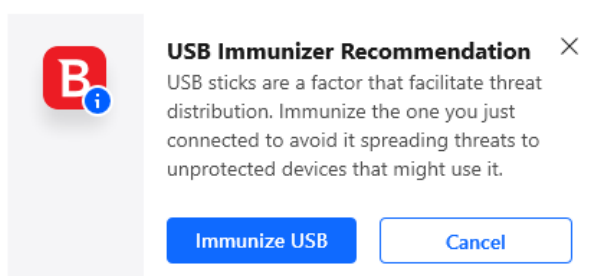
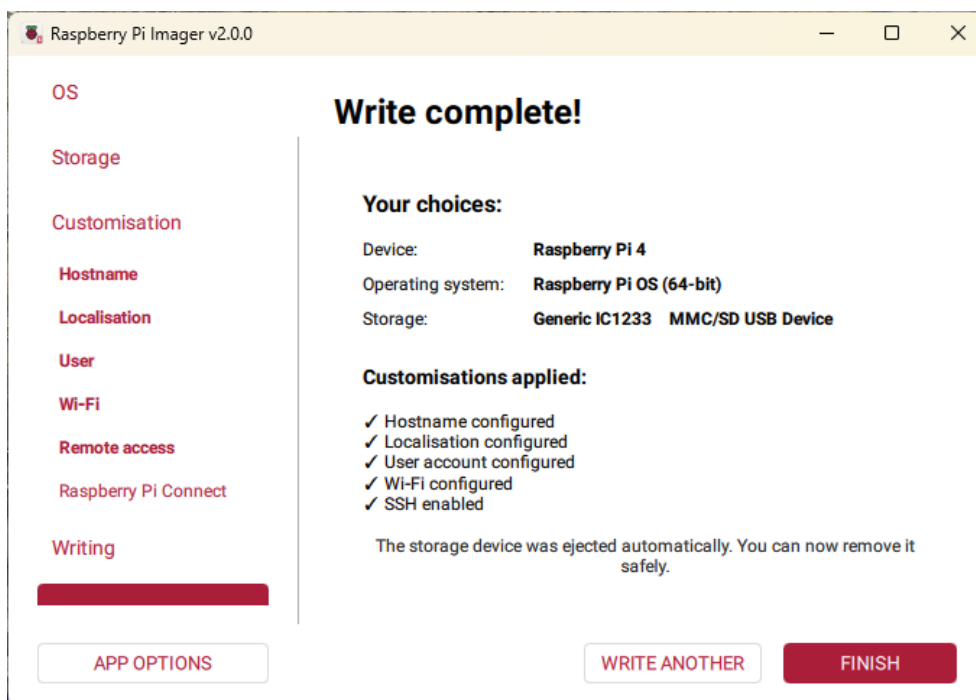


This is the point of no return. Pressing “I UNDERSTAND ERASE AND WRITE” will continue to overwrite all existing data on your USB device/SD card and replace it with the Operating System that you selected.

The program will now copy the Operating System to the SD card. This will take about 5 minutes to complete including the verification stage.



When the write operation is complete you will see the following screen. Press **FINISH** and remove the SD card. Insert it into the Raspberry Pi (Remove power from the Raspberry Pi first).



After this operation you will see the pop-up dialogue box on the left. You can safely ignore this warning.

Click cancel to continue

Booting the Raspberry Pi for the first time

Power up the Raspberry Pi. The first boot will take a little longer as there are a few jobs that the OS must carry out, such as re-sizing the file system on the SD card.



Note: If you are using a USB SSD device with, say 250GB, the Raspberry Pi will take a few minutes to resize the file system to the capacity of the disk. Be patient whilst it is doing this.

Log into the Raspberry Pi for the first time

There are two ways of logging into the Raspberry Pi:

1. Using the graphical Linux Windows desktop
2. Using SSH to log into the RPi over the network

Logging in with the Raspberry Pi desktop

If you installed the OS with the Linux desktop and you want to log in via the Desktop, you will need an HDMI monitor, a USB keyboard and mouse. Boot up the Raspberry Pi with the new SD card with the **Raspberry Pi Trixie** Operating System (OS). If you have used **Trixie** or **Bookworm** desktop then a graphical desktop will be displayed. Start a terminal session by clicking on the black terminal icon  on the top left of the screen to the left of the "Welcome" message. With **Raspberry Pi Lite** only a log in prompt will be displayed. In such a case log into the Raspberry Pi as user **pi** and using the password **raspberry**. Alternatively log in using SSH (See following section).

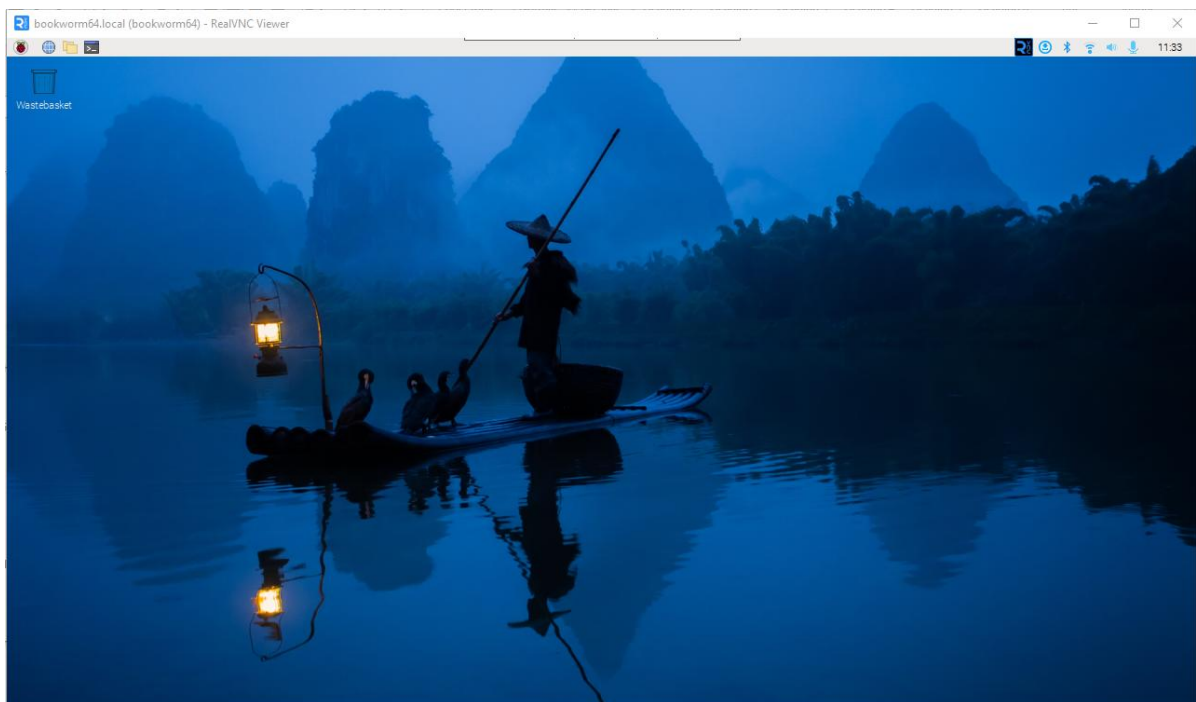
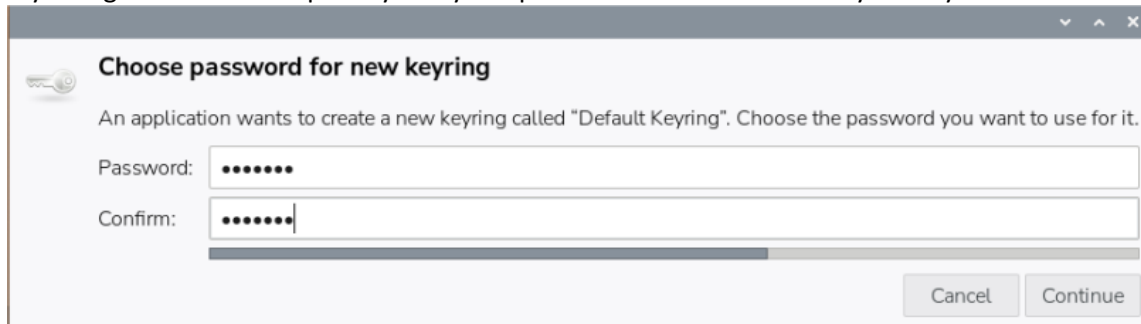


Figure 150 Raspberry Pi Bookworm desktop

If you created the SD card with the Lite version of the OS then you will not see a desktop. Instead, you will see a terminal session which will allow you to carry out the next steps of the installation.

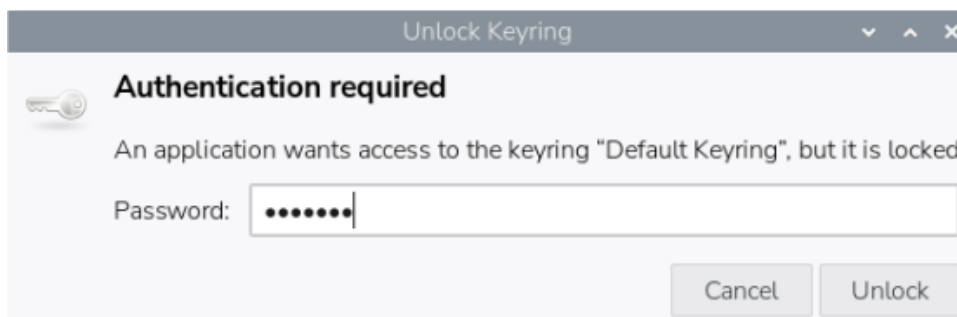
Using Keyrings

If you log into the Desktop and you try to open the Chromium browser you may see the following.



If you entered a password (use the same one as your login) the keyrings facility is enabled and password keys are stored in the encrypted `.local/share/keyrings/Default_Keyring.keyring` file in the user's home directory. You can see this with the `file` command.

```
$ file .local/share/keyrings/Default_Keyring.keyring
.local/share/keyrings/Default_Keyring.keyring: GNOME keyring, major version
0, minor version 0, crypto type 0 (AES), hash type 0 (MD5), name "Default
Keyring", last modified Fri Apr 17 09:53:40 2026, created Thu Jan 1
00:00:00 1970, not locked if idle, hash iterations 1407, salt
2083600801244101775, 1 item(s)
```



You can disable keyrings by removing the `Default_Keyring.keyring` file

Disable sudo password protection

As of the 14th of April 2026, the Raspberry Pi foundation released version 6.2 of the operating system. This has a new security feature which requires you to enter your user (usually `pi`) password the first time you use `sudo`. The `sudo` command gives the user extra privileges required to restricted access files and folders. Whilst any improvements in security are welcome, this new feature will cause problems with both the radio programs and associated scripts. For the moment this security feature needs to be disabled. Run `sudo raspi-config` to disable it. This may change in the future.

```
$ sudo raspi-config
[sudo] password for pi: *****
```

Then select:

1 System Options **Configure system settings**

S10 Admin Password Enable or disable sudo password

Would you like the admin (sudo) password to be enabled? No

The admin password is disabled

Logging in with SSH (Bitvise)

If you don't have a USB keyboard, mouse and screen then you can log into the RPi using the Bitvise or Putty utility running on a Personal Computer. Run the **Bitvise** interface and using **Fing** to find the IP address of the Raspberry Pi as shown in the section *Finding the Raspberry Pi on a network using Fing* on page 84. Enter the IP address from Fing into the Host field (192.168.1.36 in this example). Also enter the user's name **pi** and the password you set up.

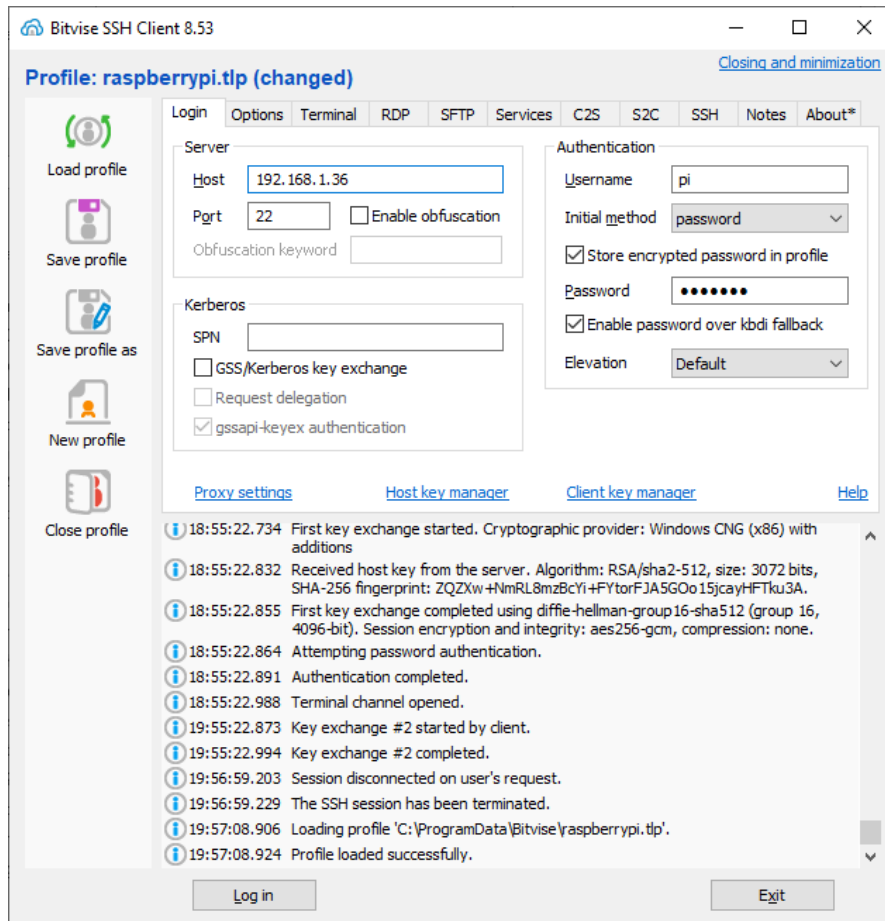


Figure 151 Bitvise connection dialogue



Note: You can also use the host name of the Raspberry Pi instead of its IP address. If you set the SD card up according to the instructions shown earlier. For example, if the hostname was “raspberrypi” then try the name **raspberrypi** first. If that doesn't work then try **raspberrypi.lan**, **raspberrypi.local** or **raspberrypi.home** in the host field. If the hostname was “piradio” then try the names **piradio**, **piradio.lan**, **piradio.local** or **piradio.home**. If none of these work, just use the IP address or use **Fing** to find the name. See *Finding the Raspberry Pi on a network using Fing* on page 84.

Enabling SSH using raspi-config

If SSH was enabled when creating the SD card this step should not be necessary and you can skip it. You will need a USB keyboard, mouse and HDMI display screen. Once logged in run **raspi-config**.

```
$ sudo raspi-config
[sudo] password for pi:
```

The first time that you log into the Raspberry Pi after each and every reboot and attempt to use **sudo** you may be asked to enter the password for your login username (normally **pi** but can be any name you used when you created your SD card).

Select option 3 – Interface options. The following screen will be displayed

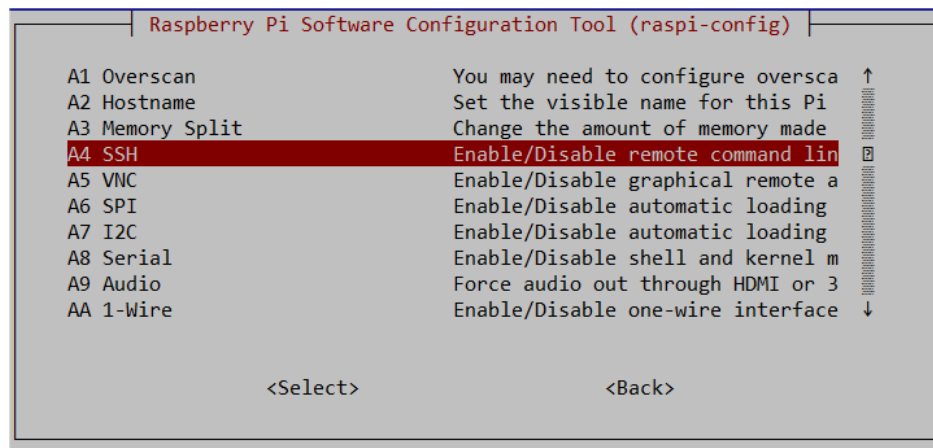


Figure 152 Enabling SSH

Reboot the Raspberry Pi after which it will be possible to log into the Raspberry Pi using SSH.

```
$ sudo reboot
```

Now log into the Raspberry Pi using SSH.

```
$ sudo reboot
```

After logging back in the following message may be displayed if a new password wasn't set.

```
SSH is enabled and the default password for the 'pi' user has not been
changed. This is a security risk - please login as the 'pi' user and type
'passwd' to set a new password.
```



Note: Security is becoming more and more of an issue for devices connected to the internet. If SSH has been enabled then please change the user password at the first opportunity.

Preparing the Operating System for software installation

Update to the latest the system packages

Run the following command to update the library list.

```
$ sudo apt update
```

The above command will take some time! If you see a message similar to the following.

```
E: Repository 'http://raspbian.raspberrypi.org/raspbian Bookworm InRelease'
changed its 'Suite' value from 'testing' to 'stable'
```

Run the following command

```
$ sudo apt update --allow-releaseinfo-change
```

Now re-run the update.

```
$ sudo apt update
```

Run the following command to upgrade to the latest packages for this release.

```
$ sudo apt upgrade
```



IMPORTANT: Do not, under any circumstances, run the apt **full-upgrade** or **rpi-update** command especially if installing on **Trixie** as the kernel in these updates isn't stable yet. Do not use **rpi-update** with **Bookworm** or **Trixie** unless specifically advised to do so by an official source.

Reboot the Raspberry Pi.

```
$ sudo reboot
```



Important: After upgrading the system, the repository locations may no longer be valid. Re-run **apt update** to refresh the package list. Failing to do this may result in packages failing to install.

Re-run the update command to update the library list.

```
$ sudo apt update
```

Once you have finished updating the operating system login to the system and run **raspi-config**.

```
$ sudo raspi-config
```



Warning: If you are intending to run the touch-screen/HDMI version of the radio, do not be tempted to start removing components of the **pygame** software such as games as this may unfortunately remove graphic libraries used by the radio software.

Configuring the Operating System

Disable booting to the desktop environment

If you are planning to use a touch-screen or HDMI display skip this section.

The desktop environment is not required for the LCD, TFT or OLED versions of the Radio and takes a lot of processing power. It is enabled by default in **Trixie** but is not installed with **Trixie Lite**. If you are not planning to use the touch-screen or HDMI version of the radio or you don't otherwise plan to use it then disable it.

Run raspi-config:


```
$ sudo raspi-config
```

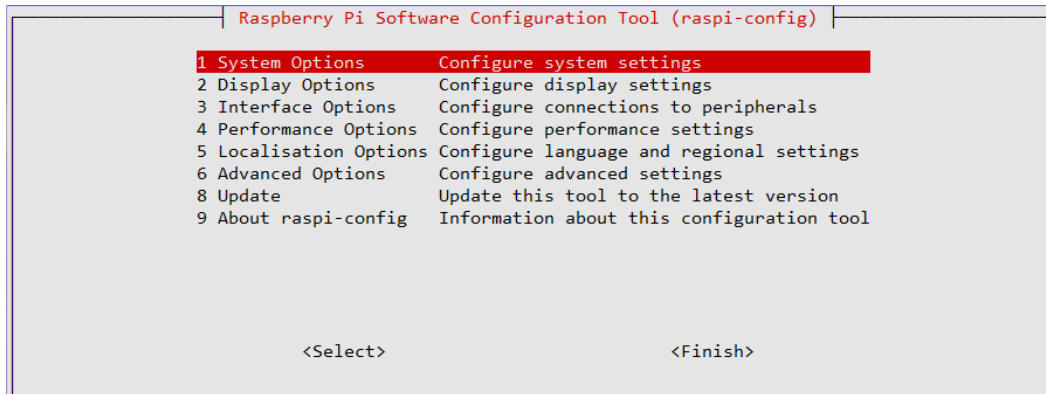


Figure 153 raspi-config main screen

Select option **1 System options**

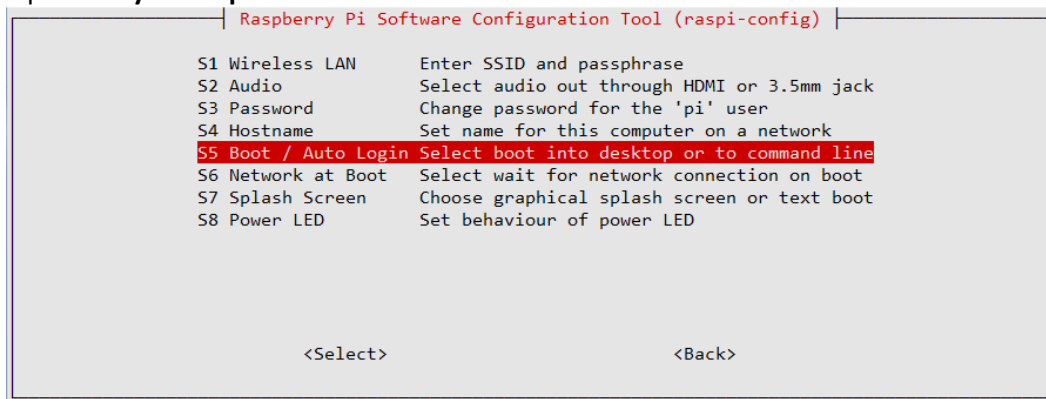


Figure 154 Disabling the graphical desktop

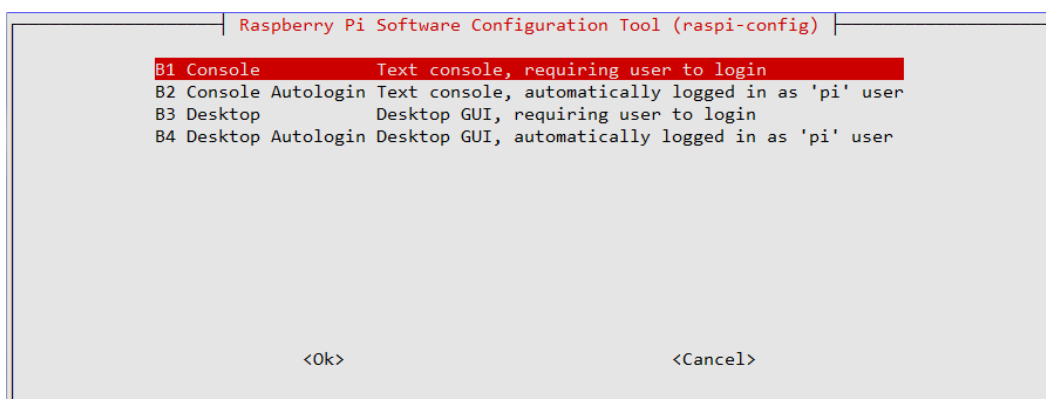


Figure 155 Desktop enable/disable selection

Select option **B1**(secure) or **B2**(insecure) to disable the desktop and select OK

Setting the time zone

The **Raspberry Pi OS** operating system is usually set to UK time or the time zone that you set with the **Raspberry Pi OS imager** settings. If you still need to set the time zone then the easiest way to set the time zone for your country if you are in a different time zone is to use the **raspi-config** program and select option 5 “Localisation Options”:

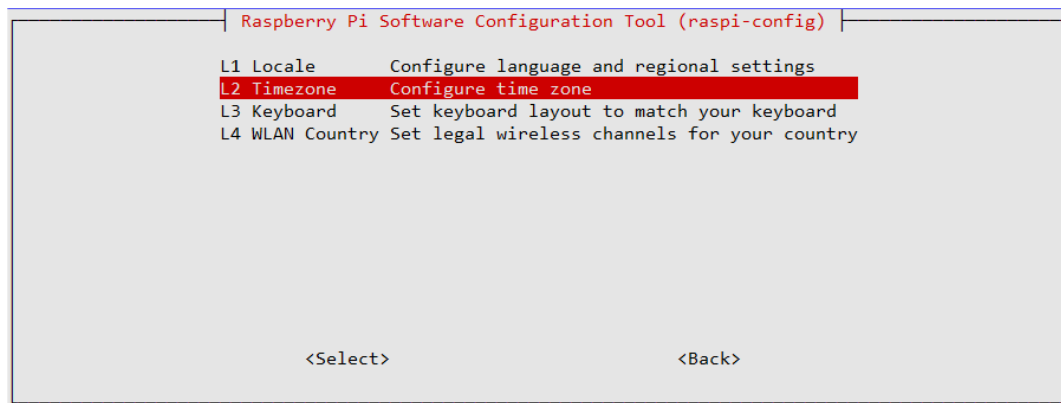


Figure 156 Setting the time zone

Select option L2 “Timezone”:



Figure 157 Saving the time zone

Select the region your country is in, Europe for example. Use the tab key to move to <OK> and press enter. The program will then display a list of time zones for the selected region.



Figure 158 Time zone country selection

Select the correct one and save it by tabbing to <ok> and pressing the enter key. The time zone will be updated. Exit the program once finished.

Configuring the Wi-fi Connection via raspi-config



If the **RaspberryPi** Imager was used to create the SD card it may be that the Wi-Fi interface has already been configured, in which case this option can be skipped.

Select the country you are in for legal Wi-Fi channel selection. From the first **raspi-config** screen select **1 System Options** followed by option S1 Wireless LAN:

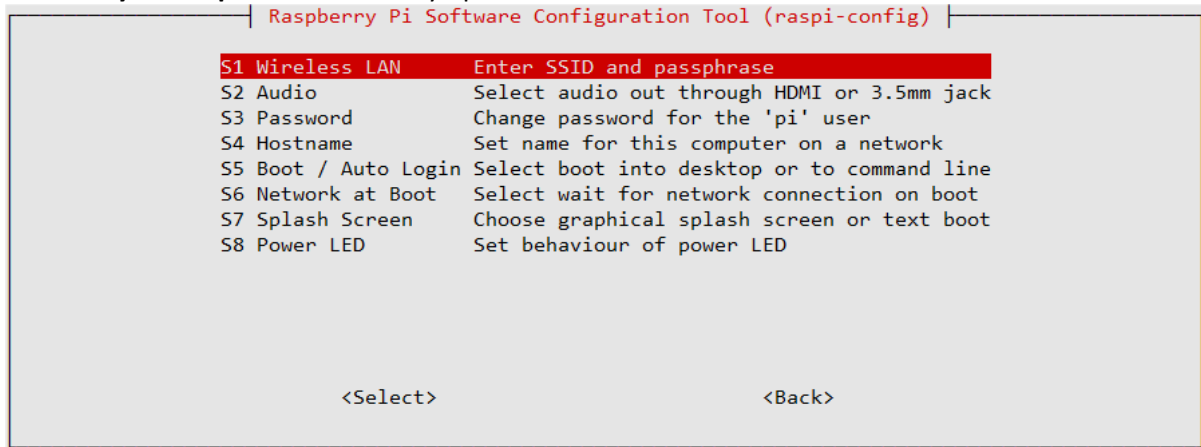


Figure 159 Setting up the Wi-Fi in raspi-config

Now enter the SSID and passphrase for your network. Save the settings and reboot the Raspberry PI. If this option is not available then use the procedure

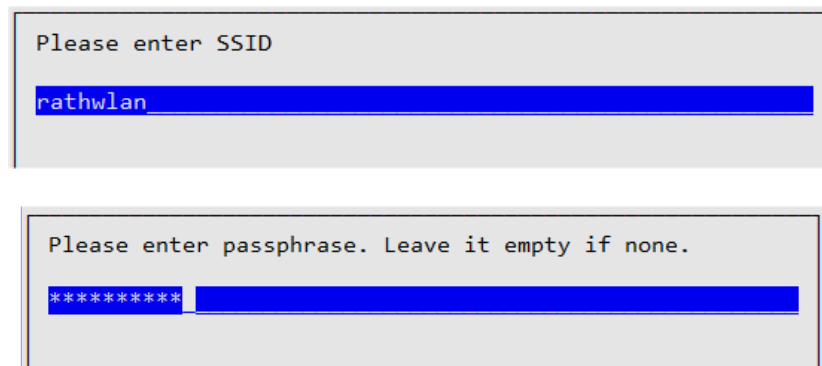


Figure 160 Entering Wi-Fi credentials

Reboot the system. After reboot it is possible that you may see the following message:

```
Wi-Fi is disabled because the country is not set.  
Use raspi-config to set the country before use.
```

In such a case re-run **raspi-config**. Select localisation options and select I4 to set Wi-Fi country:

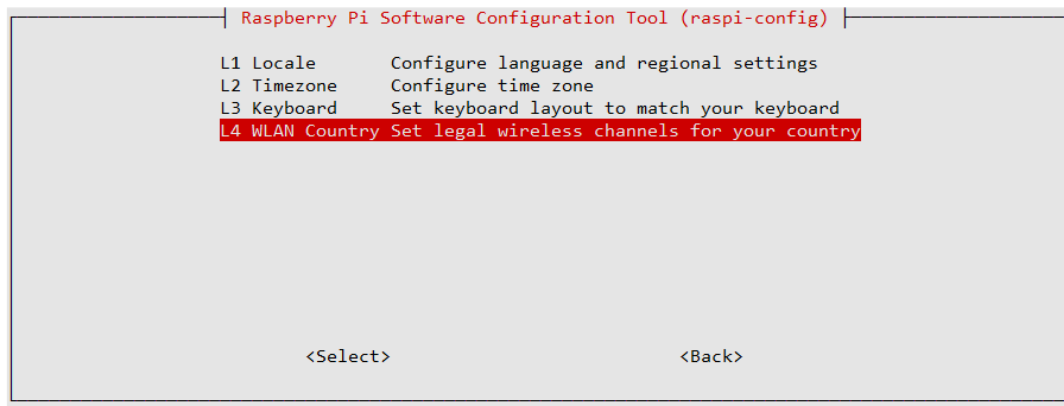


Figure 161 Setting up the Wi-fi country

Select your country from the dropdown menu and exit the **raspi-config** to save the setting.

Changing the system hostname and password



If the **Raspberry Pi Imager** was used to create the SD card it may be that the hostname and password may have already been set, in which case this option can be skipped.

It is a good idea to change the system password for security reasons especially if your raspberry PI is connected to a network with a wireless (Wi-Fi) network. Changing the hostname is also a good idea as it makes identifying your radio on the network much easier. If you wish to do this, change the default hostname from 'raspberrypi' to something like 'piradio' or 'myradio'.

Both the password and hostname can be changed using the **raspi-config** program. Select option **1 System options**

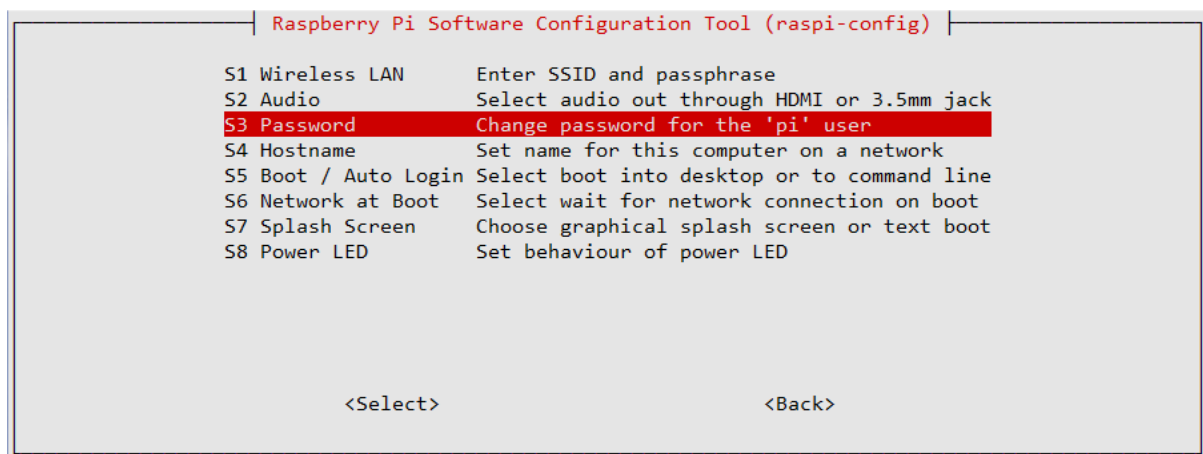


Figure 162 Changing the Raspberry PI password

Option **S3** is used to change the password. Make sure you record your new password somewhere safe (It is easy to forget it).

The hostname is changed in option S4 Hostname:

```
Enter new UNIX password:
Retype new UNIX password:
```

As the password is entered nothing is displayed. This is normal. You will be asked if you wish to reboot the system. After you reboot the system you will see the new hostname at the login prompt.

```
pi@piradio:~$
```

In the above example the new hostname is **piradio**. Once the hostname has been changed the program will ask if you wish to reboot. Answer “yes” to reboot.

Setting up the locale

As default the language used by Raspbian is English. To change this in Advanced options select option I1 Change Locale.

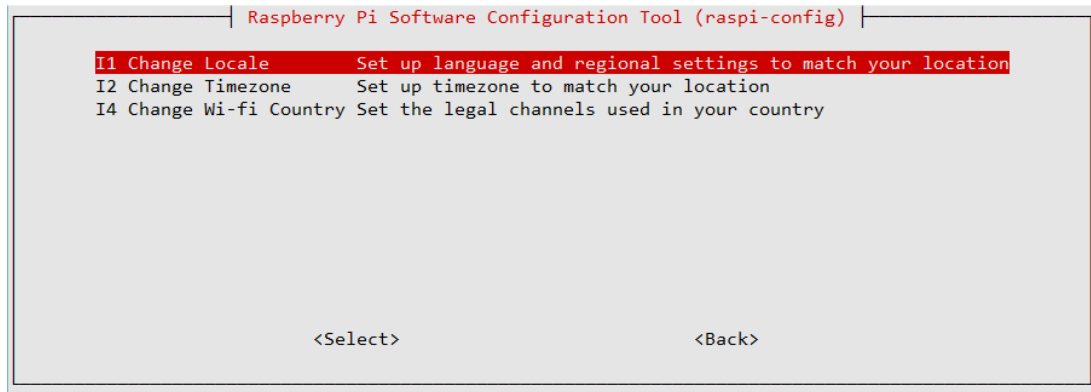


Figure 163 Selecting change locale

Select a locale beginning with the two-letter international code for your country and language.

Usually this will be a locale containing the string ISO or UTF-8.

- fi_FI.ISO-8859-1 (Try this one first)
- fi_FI.ISO-8859-15@euro
- fi_FI.UTF-8

In the following example for the Netherlands the locale is this **nl_AW UTF-8** for the Netherlands (nl).

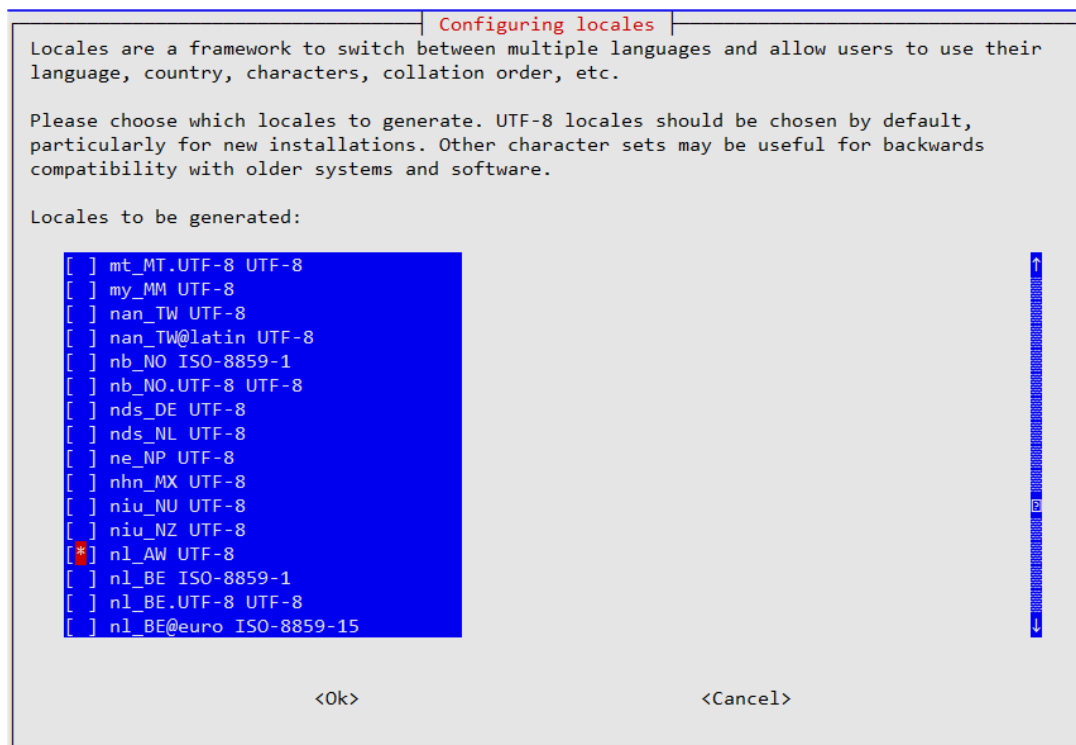


Figure 164 Generating the locale.



Warning: Do not be tempted to generate “All locales” as this is not only unnecessary and futile and will take a very long time.

Press OK to continue. The program will now generate the selected locales.

```
Generating locales (this might take a while)...
  en_GB.UTF-8... done
  nl_AW.UTF-8... done
Generation complete.
```

The following screen is displayed:

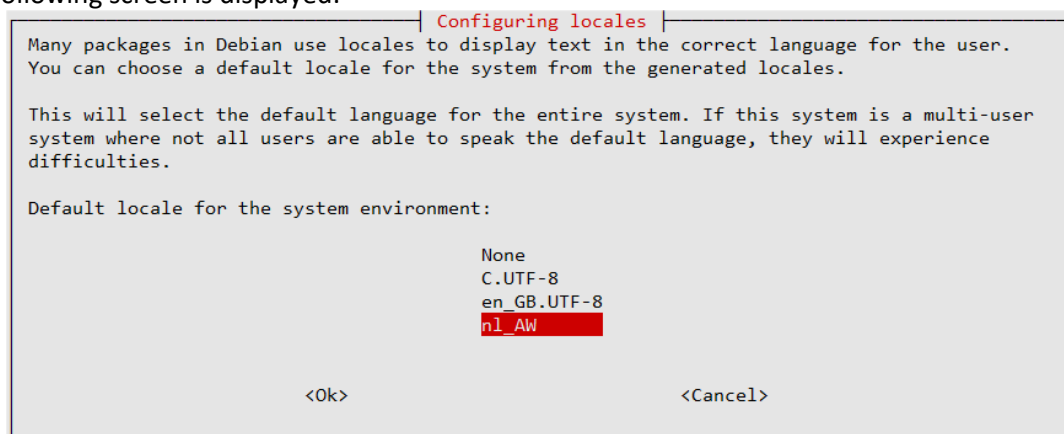


Figure 165 Selecting the locale

Select the required locale **nl AW** in this example and press OK to save. The new locale will become active after the next reboot.

Disabling the soft keyboard

As of early 2026 the Raspberry Pi desktop comes with a soft keyboard which is enabled by default as shown in the following illustration. This is undoubtedly a useful feature if you don't have a physical keyboard attached but at other times it may get in the way. Fortunately, it is easy to disable it using `raspi-config`.

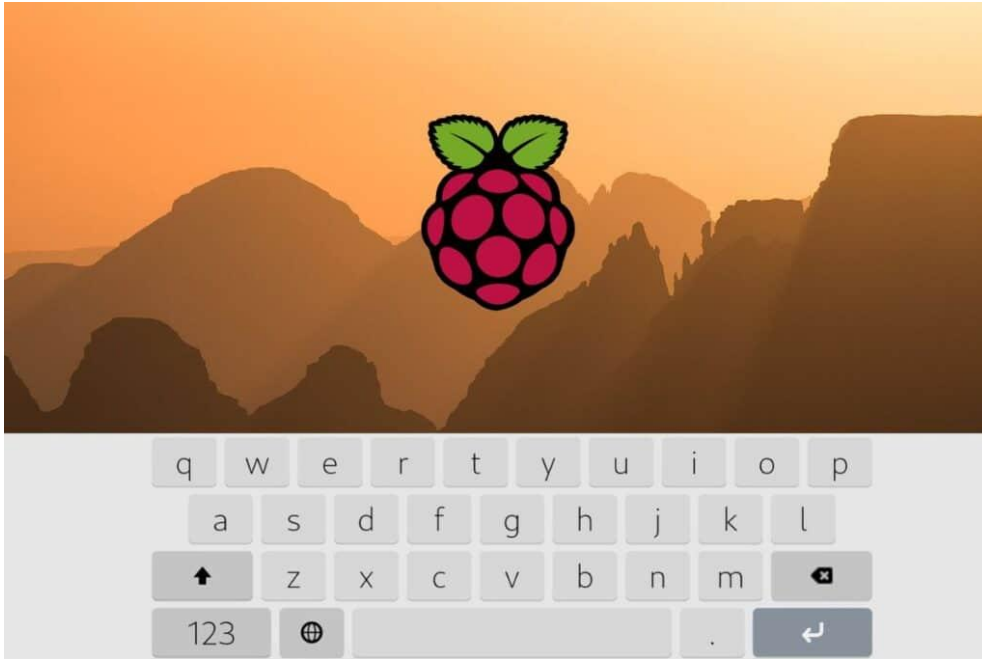


Figure 166 Desktop soft keyboard

```
$ sudo raspi-config
```

Select option **2 Display Options Configure display settings**

```
Raspberry Pi Software Configuration Tool (raspi-config)

D2 Screen Blanking   Enable/disable screen blanking
D4 Composite         Enable/disable composite output
D5 4Kp60 HDMI        Enable 4Kp60 resolution on HDMI0
D6 Onscreen Keyboard Enable on-screen keyboard
D7 Keyboard Output   Select monitor used for on-screen keyboard
```

Select option **D6 Onscreen Keyboard Enable on-screen keyboard**

```
Raspberry Pi Software Configuration Tool (raspi-config)

S1 Always On  On-screen keyboard always enabled
S2 Autodetect On-screen keyboard enabled if touch device found
S3 Always Off On-screen keyboard disabled
```

Finally select option **S3 Always Off On-screen keyboard disabled** to disable it. Press **ESC** to exit.

Installing VNC

This section is only really relevant if you are running the graphical versions of the radio. Normally you will need a screen, keyboard and mouse to see the and use Debian Linux desktop. However, you can use VNC as an alternative. This consists of two parts. The **RealVNC** client running on the PC and the VNC (wayvnc) running on the Raspberry Pi.

Install the RealVNC client software on the PC

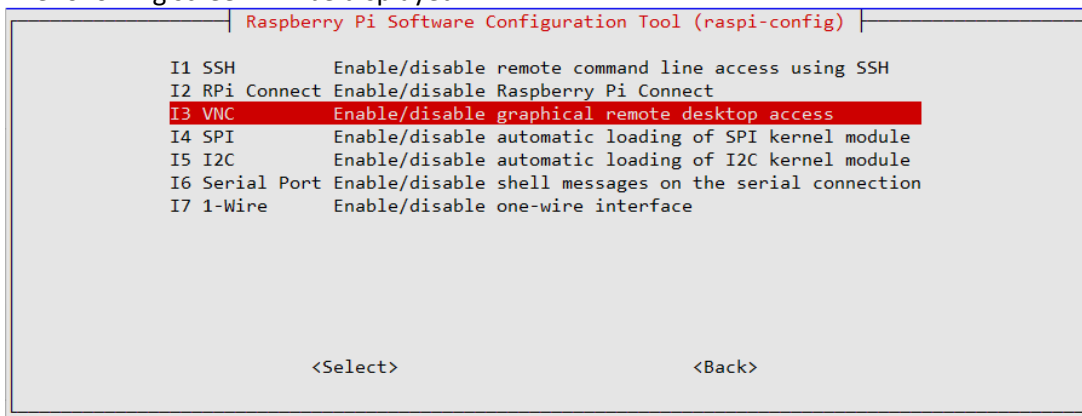
It is necessary to first install **VNC** client (**RealVNC**) on your PC. Download and install **RealVnc** from <https://www.realvnc.com/en/connect/download/>

Enabling the VNC server on the Raspberry Pi

Run radio-config and select the following option:

3 Interface Options Configure connections to peripherals

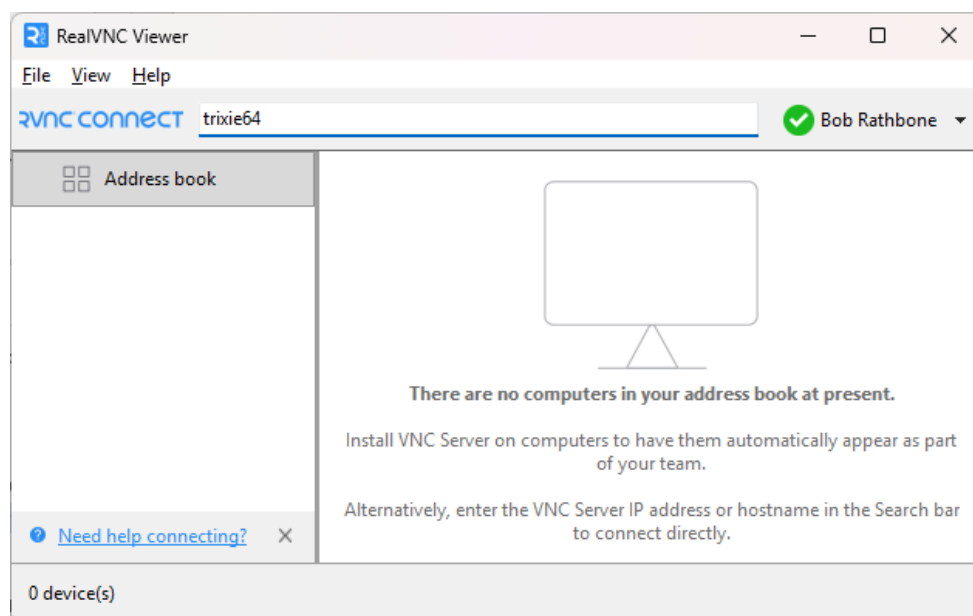
The following screen will be displayed:

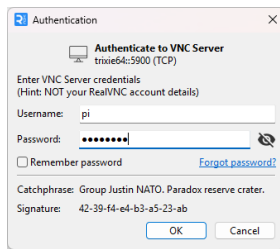


Select option **I3 VNC Enable/disable graphical remote desktop access**

Would you like the VNC Server to be enabled? **Yes**

Connecting to the Raspberry Pi with VNC





Chapter 6 - Installing the radio Software

Contents chapter 6	Page
Installing the Radio Daemon	110
Configuring the radio	110
Configuring HDMI monitor and Touchscreens	121
Configuring sound devices	130
Installing a Bluetooth audio device	137
Manually configuring sound cards	140
Setting the mixer volume	141
Keeping the Radio software up to date	142

Installing the Radio Daemon



WARNING: Do not use the **root** account (**su -**) to install any part of this software. Always use a standard user account such as '**pi**'. All commands that require root privileges should use **sudo**. No installation testing has ever been done with the **root** account and is therefore not supported. You may also create permission problems during operation.

Installing the Music player daemon

After reboot install the [Music Player Daemon](#) (mpd) and its client (mpc) along with the Python3, MPD and **python3-rpi.gpio** libraries. Note that you must install **python3-mpd** and not **python.mpd**.

```
$ sudo apt -y install mpd mpc python3-mpd
```

The Raspberry PI Internet Radio software is distributed as a Debian package. This can be downloaded from http://www.bobrathbone.com/raspberrypi/pi_internet_radio.html

For both 32 and 64-bit systems

```
$ cd
$ curl -L -O -f http://bobrathbone.com/raspberrypi/packages/radiod_8.3_all.deb
```

Run **dpkg** to install the package.

```
$ sudo dpkg -i radiod_8.3_all.deb
```

The **dpkg** program will install the files.

Configuring the radio

Once the basic radio package installation has been completed the installation will automatically run the **configure_radio.sh** script. This installs or updates the program files and will also update settings in the radio configuration file **/etc/radiod.conf** (if selected) and any other relevant configuration files.

When you install the radio package using **dpkg**, the installation script will display the following screen.

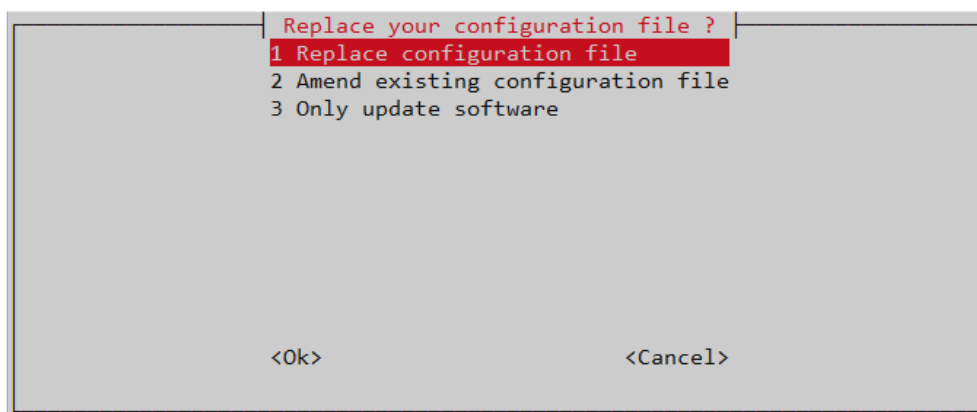


Figure 167 Configure radio

The above options are explained in detail below

Option 1 Replace configuration file will overwrite the existing configuration. Unless you have a heavily modified configuration you may safely overwrite the configuration file. You will normally select this option when installing the radio software the first time.

Option 2 Amend existing configuration file – This is used in the case of re-installs and allows you to keep certain parts of your existing configuration file. This is useful to keep settings that you have previously made such as your audio configuration.

Option 3 Only update software – This installs the software only. The existing configuration is left untouched. Select this option if just upgrading the software.



Note: This configuration program can be re-run at any time. Run the **radio-config** program and select **option 1 Configure radio software**: This will display the screen shown in *Figure 168 Select Raspberry Pi Model* below.

Next the following screen will then be displayed:

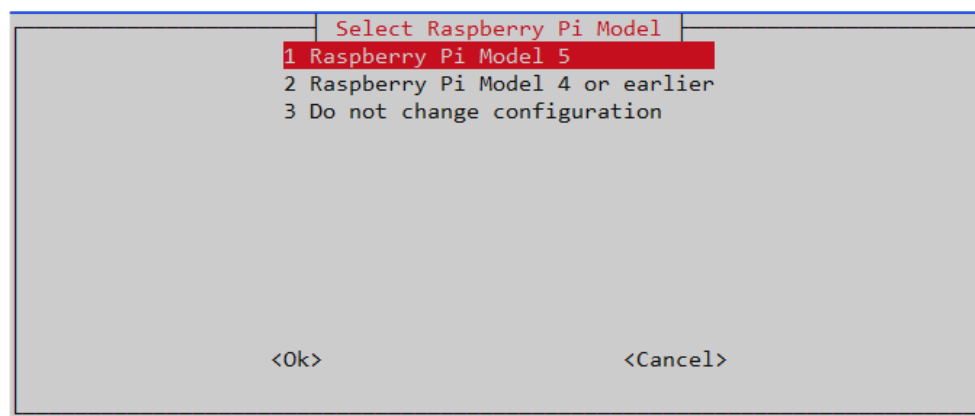
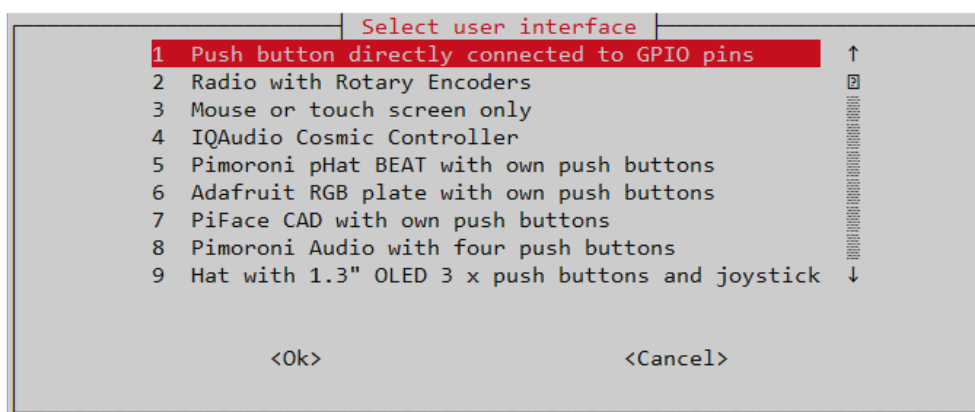


Figure 168 Select Raspberry Pi Model

Select your model of Raspberry Pi, either the Model 5 or the model 4 or earlier.

Confirm selection and continue to the next screen:



Select **option 1** if push buttons are being used or **option 2** if using rotary encoders. Otherwise select the type of user interface being used in **option 3** onwards. If you scroll down, you will be given the option not to change the display configuration.

This screen and all following screens have the option to not change the configuration. The program will always ask you if your choice is correct and give you the opportunity to change the selection.

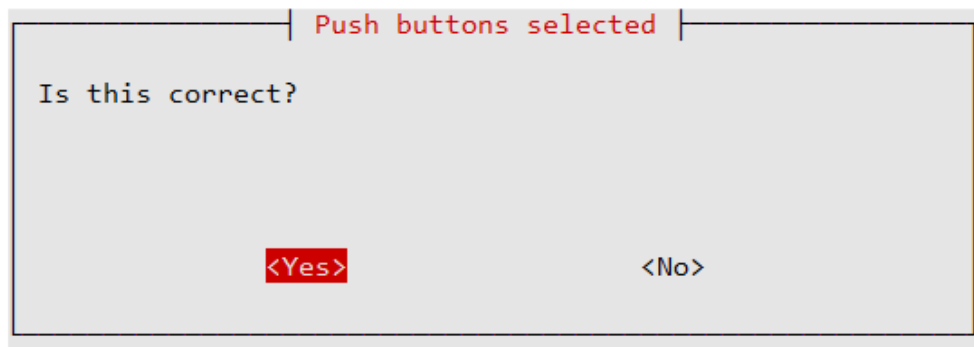


Figure 169 Configure radio - Confirmation screen

If the push-button interface is selected, the program will ask how they have been wired. The push-buttons can be wired to either +3.3V (The original scheme) or GND(0V).

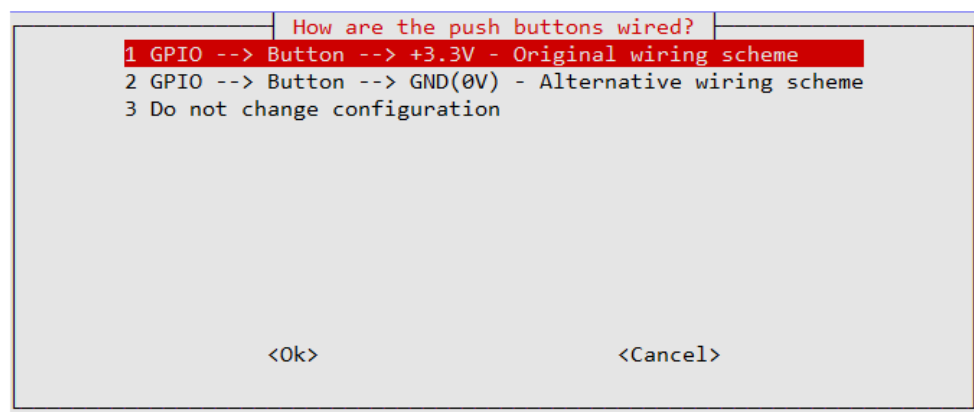


Figure 170 Push-button voltage selection

If option **2 Rotary Encoders** was selected then the following screen will be seen:

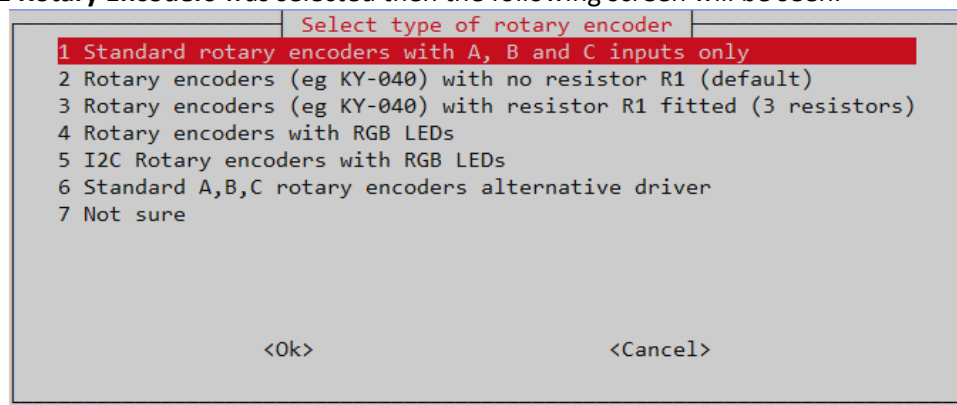


Figure 171 Rotary encoder type selection

Traditional rotary encoders have three connections called A, B and C. They usually also have two connections to a switch (See Radios using rotary encoder on page 50). If these rotary encoders are being used the select option 1 as shown above.

If rotary encoders such as the KY-040 (See *Using KY-040 Rotary encoders* on page 51) are being used, these have their own 10K pull-up resistors and so do not need the internal GPIO pull-up resistors as well. Select option 2 KY-040 encoders to disable the internal GPIO pull-up resistors.



If you are missing the push-button 10K resistor as shown in *Figure 92 KY-040 with three 10K* resistor on page 51 then select option 2 (KY040 with resistor R1 fitted). If resistor R1 is missing select option 2 (KY040 with no resistor R1).

If using rotary encoders with RGB LEDs select option 4.

If there are problems with the standard A, B, C rotary encoders try option 4 - Alternative driver.



Note: Option 2 will not work with traditional A, B and C encoders. If unsure select option 3 and the internal pull-up resistors will be enabled and will work with both types of encoders.

Next select the GPIO header wiring. This is either 26-pin (Older RPi models) or 40-pin wiring. Note that the 26-pin wiring scheme can also be used on 40-pin headers if so wished.

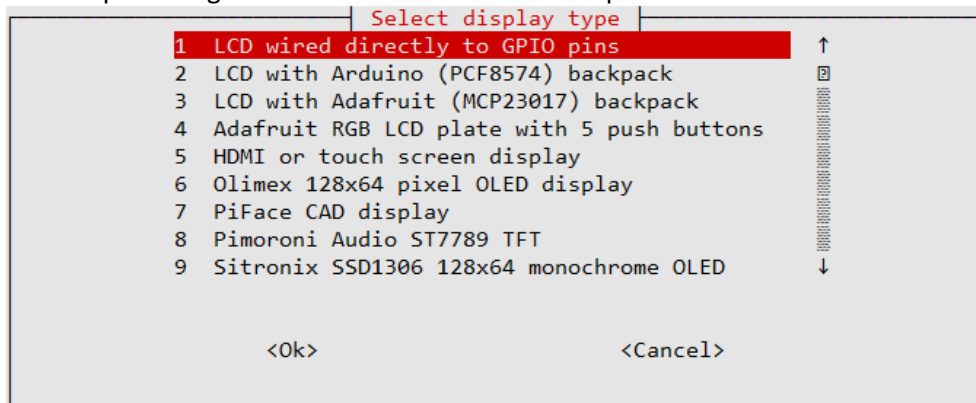


Figure 172 Configure radio - Display interface selection 1

Notice that there is a scroll bar to the right of the options. Use the Up/Down keys to scroll down to the remaining options. Select the option which matches the display you are using.

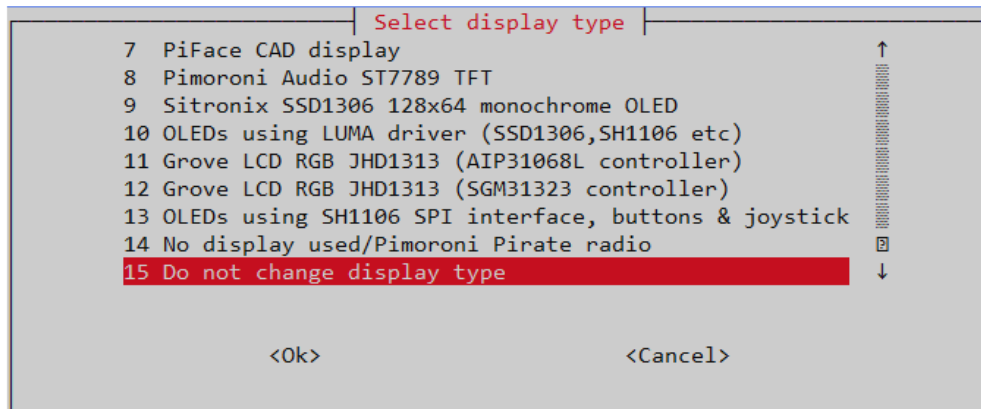


Figure 173 Configure radio - Display selection 2

Select the correct option for the display interface and confirm selection. Again, it is possible leave the configuration unchanged. There is also an option 10 (Scroll down) called “Do not change display type”.

If option 5 – ‘HDMI or touch screen display’ was selected the go to the section called *Configuring HDMI monitor and Touchscreen* on page 121.

If options 6, 8, 9 or 10 were selected go to *Configuring OLEDs* on page 115.

If option 2, 3, 4 or 6 was selected then this will require the hex address to be configured. Otherwise, the program will skip the screens in the next section and go to the section called *Run radio-config if you see the above message*.

Select the type of LCD display on page 114.

If option 6 (Olimex OLED) was selected then the following will be displayed:

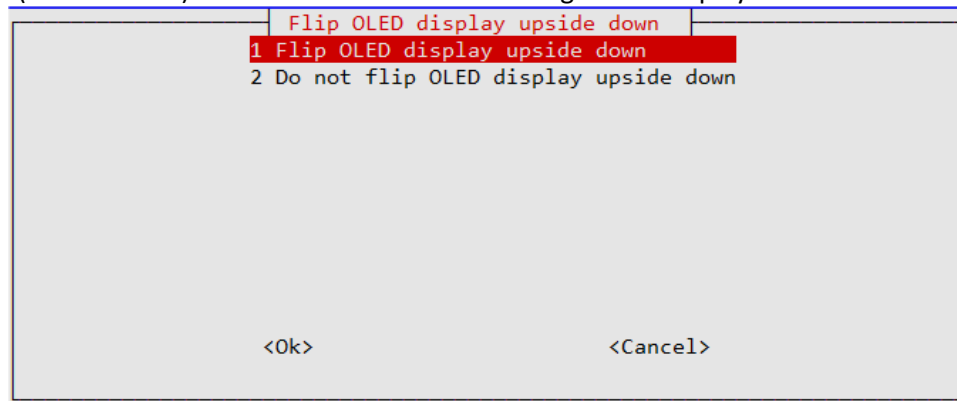


Figure 174 Olimex OLED flip display

This option sets the **flip_display_vertically** parameter in **/etc/radiod.conf** to **yes** or **no** and allows the Olimex OLED display to be flipped vertically. This option doesn't work with the ST7789TFT.

Some display components cannot be configured during the initial installation as they require installation of supporting packages first. In such a case you will see the following example.

```
WARNING!  
You cannot configure Pimoroni Audio with ST7789 controller during the  
initial installation  
Run radio-config from the command line then select 1 configure radio  
software from the menu line and reconfigure the radio software  
  
Press enter to continue:
```

Run **radio-config** if you see the above message.

Select the type of LCD display

Skip this section if you are not using an LCD display directly connected to the GPIO pins. Confirm the selection and continue to the next screen to select the type of LCD display. This section is not relevant for a HDMI or touch screen.

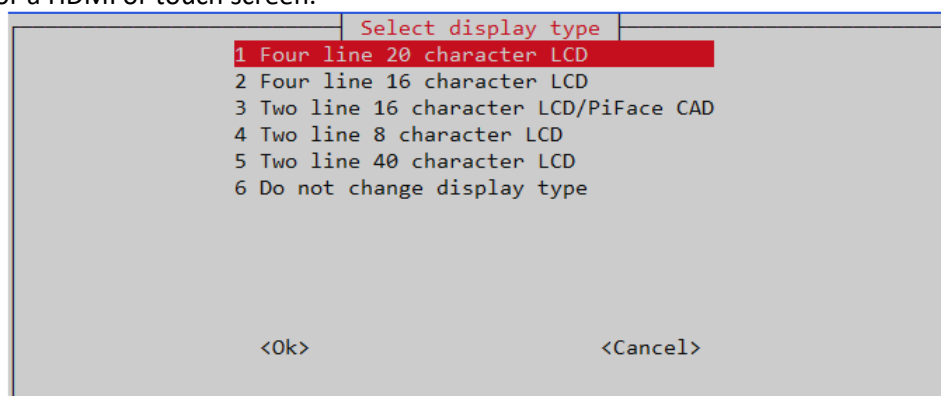


Figure 175 Configure radio - Display type selection

Select the type of display to be used and confirm the selection. The installation script asks if you wish to configure the audio device:

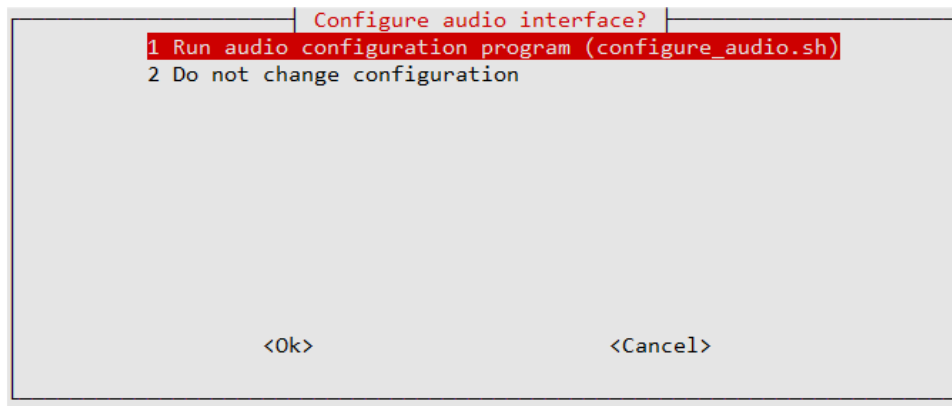


Figure 176 Configure radio audio output option

You should select option 1 to run the audio device configuration program.
If you selected option 1 then go to the section *Configuring the audio output* on page 116.

Configuring OLEDs

If not using an OLED display, then go to the section *Configuring the audio output* on page 116.
There are five drivers available for OLED displays. These can have either the 2-wire **I2C** interface (+5v, GND, Data and Clock) or the 4-wire **SPI** interface. They are selected by the **display_type** parameter in `/etc/radiod.conf`. This is setup by the `configure_radio.sh` configuration program.

Table 14 OLED drivers

display_type	Interface	Lines	Driver	Option	Notes
ST7789TFT	SPI	5	st7789tft_class.py	8	Used by the Pimoroni Pirate Audio hat
OLED_128x64	I2C/SPI	5	oled_class.py	6	The driver automatically detects I2C or SPI
SSD1306	I2C	4	ssd1306_class.py	9	From version 7.2 onwards
LUMA	I2C	4	luma_class.py	10	From version 7.3 onwards
SH1106_SPI	SPI	4	sh1106_class.py	11	From version 7.6 onwards

The Luma device supports multiple devices namely devices using SSD1306, SSD1309, SSD1325, SSD1331, SH1106 or WS0010 hardware driver chips. Only SSD1306 and SH1106 devices have been tested.

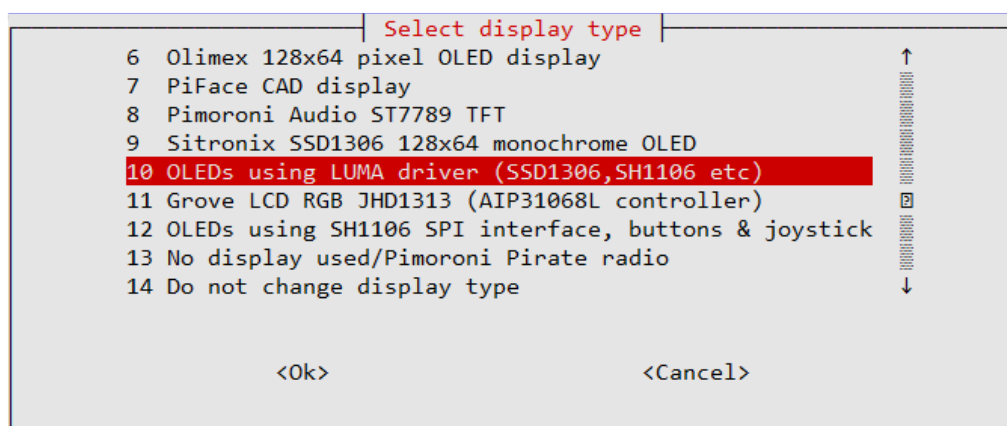


Figure 177 Selecting an OLED display

Select option 6, 8, 9, 10 or 11 depending upon the type of OLED being configured.

If option 10 (Luma device driver) was selected the following screen will be displayed:

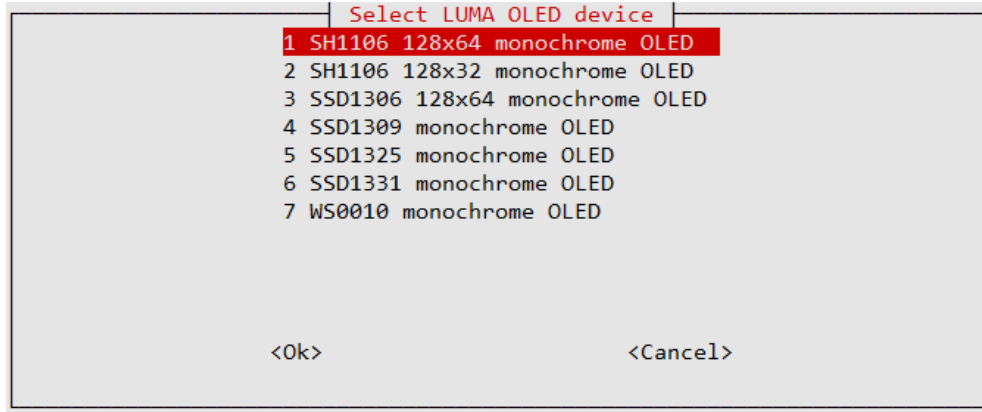


Figure 178 Luma devices

Select the device that your OLED is using.

If using devices with an I2C interface then select the appropriate I2C device address. OLED devices usually use address **0x3C** but check your device specification or run **i2c_detect.py -y 1** to display it.

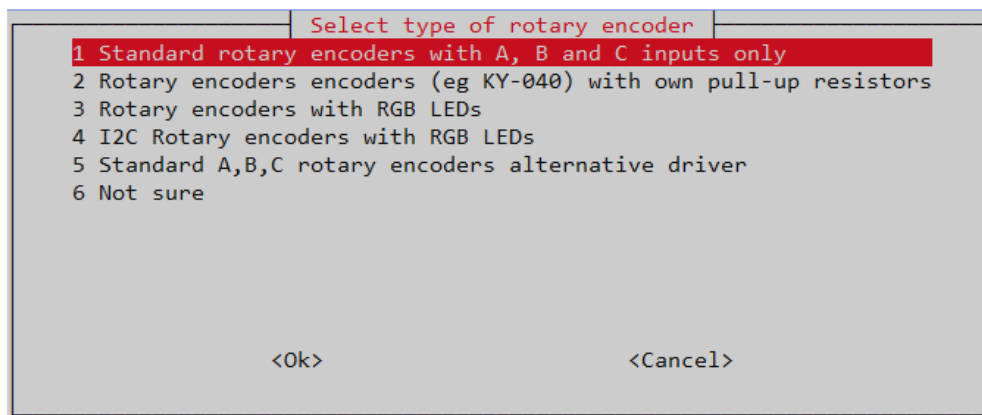


Figure 179 Selecting the OLED I2C address

After configuring I2C address go to the section *Configuring the audio output* on page 116.

Installing the Grove LCD RGB

The **Grove LCD** with RGB backlight is capable of displaying any colour wanted. For instance, the colour AQUA would be RGB (5,195,221). To turn the word "AQUA" into its RGB value the PIL (Python Image Library) package is required. Install it with the following command.

```
$ sudo apt install python3-pil
```

Now carry out the instructions shown in section *Installing the Radio Daemon* on page 110.

Configuring the audio output

At the end of the radio configuration process the radio installation calls the **configure_audio.sh** script which then displays the screen shown below in *Figure 180 Selecting the audio output device*. The following screen will be displayed:

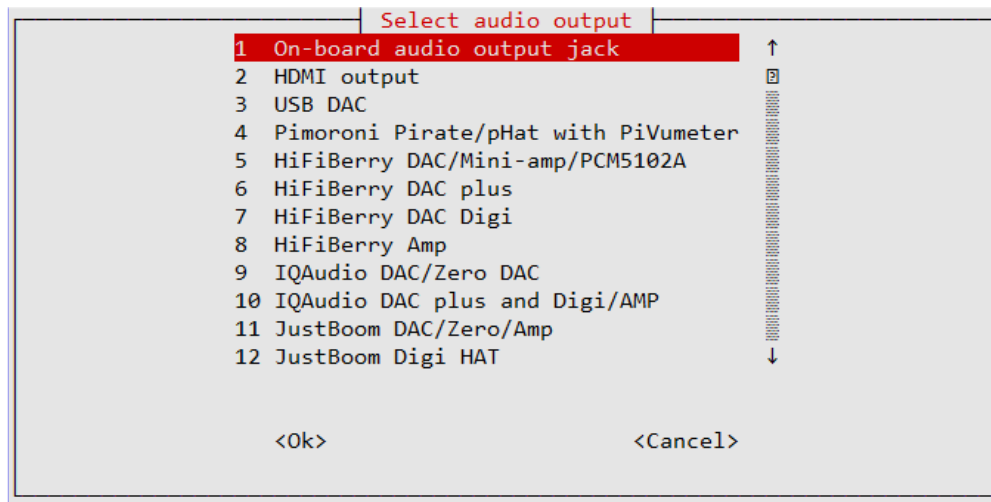
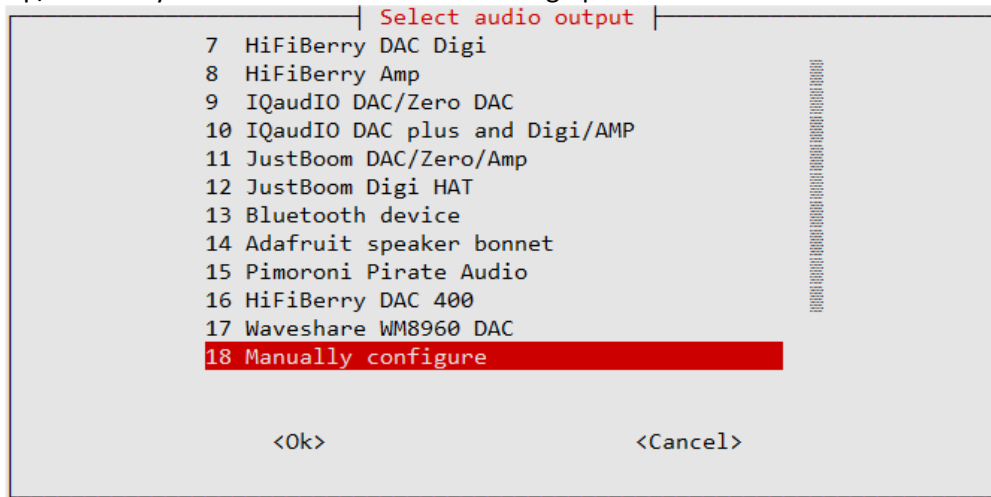


Figure 180 Selecting the audio output device

Use the Up/Down keys to scroll down to the remaining options:



Please note the scroll bar on the right side of the above screen. There are more options after option 12 (Bluetooth device). Use the Up/Down keys to scroll up and down.



Unfortunately, during the initial radio package installation, it is not possible to install many of the sound cards (DACs in particular) which require extra software packages to be installed which cannot be installed whilst the radio software is being installed. This is because the **dpkg** installer locks out any other process from running it whilst it is already busy with an installation.

Options 1, 2 and 3 can be installed at this time. If you want to install a DAC or Bluetooth devices, press **<Cancel>** and allow the rest of the radio package to be installed first.

See the section called *Configuring sound devices* on page 130 for further information on configuring other sound devices.

The installation process creates a Radio Stations playlist which will take a few minutes.

When the installation run is completed, the following text will be displayed.

```
:
PI Radio software successfully installed
See /usr/share/doc/radiod/README for release information
Installation complete
It is necessary to reboot the system to start the radio
Run the radio-config program to further configure the radio software
```



The configuration program can be safely re-run at any time in the future. To do this run the **radio-config** utility from the command line. See the [/usr/share/doc/radiod/README](#) file for release notes.

If you still need to configure a **DAC** sound card, then reboot the Raspberry Pi then run **radio-config** after the Pi has rebooted. Select option **2 Configure audio output devices** to finish installation of the required sound card as previously shown in *Figure 180 Selecting the audio output device* on page 117.

Installation logs

A log of the changes made by the radio configuration program will be written to the **/usr/share/radio/logs/install.log** file. For the audio configuration program changes will be written to the **/usr/share/radio/logs/audio.log** file.

Reboot to enable the software

The software is installed in the **/usr/share/radio** directory. Now reboot the Raspberry Pi.

```
$ sudo reboot
```

Once rebooted the software should run and display the time and station details on the screen. However, sound might not be heard. This is because final audio configuration still has to be done. Restart the radio with the following command:

```
$ sudo systemctl restart radiod
```

Once restarted music should be heard out of the on-board audio jack. If not go to the section called on page 209.

The radio daemon (LCD versions only) can be started and stopped with the **systemctl** command:

```
$ sudo systemctl start radiod
$ sudo systemctl stop radiod
```

This will also stop and start the MPD daemon.

To prevent automatic start-up of the radio at boot time run the following command:

```
$ sudo systemctl disable radiod
```

To re-enable it:

```
$ sudo systemctl enable radiod
```

Testing the I2C interface.



Version 6.13 onwards comes with its own **SMbus** library (smbus2) in the smbus2 sub-directory of the radio package.

The I2C interface should have already been enabled by the installation program. To test the I2C interface, carry out the following:

If you are using a revision 2 Raspberry Pi (Newer boards) carry out the following:

```
$ sudo i2cdetect -y 1
```

If you are using a revision 1 Raspberry Pi (Very old V1 boards) carry out the following:

```
$ sudo i2cdetect -y 0
```

This will search **/dev/i2c-0** (Very old v1 RPIs) or **/dev/i2c-1** (Later RPi versions) for all address, and if correctly connected, it should show up at **0x20** for the Adafruit LCD Plate or normally **0x27** for the Arduino PCF8574 backpack but might be another address such as **0x3F**. The OLED 128x64 pixel display uses address **0x3C**. See Figure 181 *The I2C bus display using the i2cdetect program*.

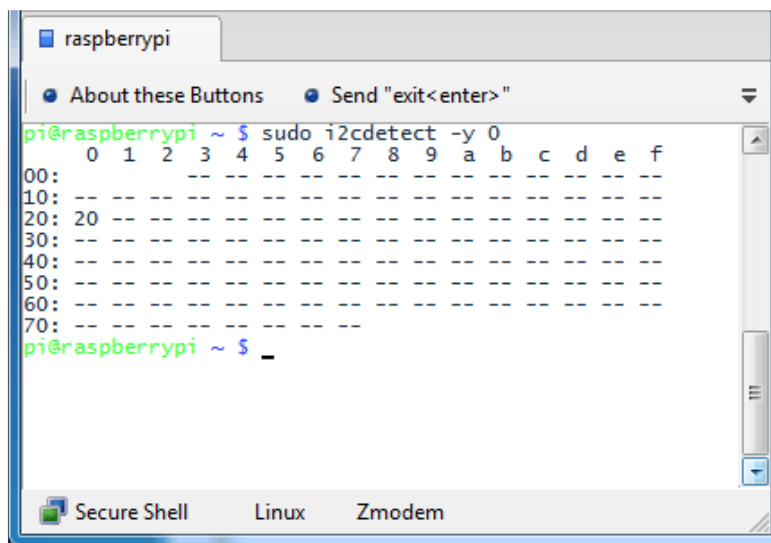


Figure 181 The I2C bus display using the i2cdetect program

If the following is seen instead then it is necessary to run enable the I2C module at boot time using **raspi-config**.

```
Error: Could not open file `/dev/i2c-1' or `/dev/i2c/1': No such file or directory
```

If problems with **i2cdetect** are still encountered, then edit the **/boot/firmware/config.txt** (for **Bookworm** and **Trixie**) file using **sudo nano** and change the following line:

```
#dtparam=i2c_arm=on
```


Change to:

```
dtparam=i2c_arm=on
```

Also, the **i2c-dev** module must be added to the **/etc/modules** file.

```
i2c-dev
```

Reboot and retry the **i2cdetect** program.



Note: If the Arduino **PCF8574** backpack is using another address other than **0x27**, **0x37** or **0x3F** then you must modify the **i2c_address** parameter in **/etc/radiod.conf**. For example, if the backpack is using the address **0x2F** then modify the **i2c_address** parameter to match this as shown in the example below:

```
# The i2c_address parameter overrides the default i2c address. 0x00 = use default
# Some backpacks use other addresses such as 0x3F, then set i2c_address=0x3F
i2c_address=0x2F
```

Once both of these packages have been installed, you have everything you need to get started accessing I2C and SMBus devices in Python. Now reboot the Raspberry PI.

```
$ sudo reboot
```

The Radio should start automatically. If not then go to the section called *Chapter 9 -Troubleshooting* on page 209.

Configuring HDMI monitor and Touchscreens

The configuration program will be run when you first install the radio package, or by running the **radio-config** command.

When you get to this point either during initial installation or by running **radio-config**, you should have selected option **3 Mouse or touch screen only** in the **Select user interface** screen and option **5 HDMI or touch screen display** in the **Select display type** menu as shown in the following two screens.

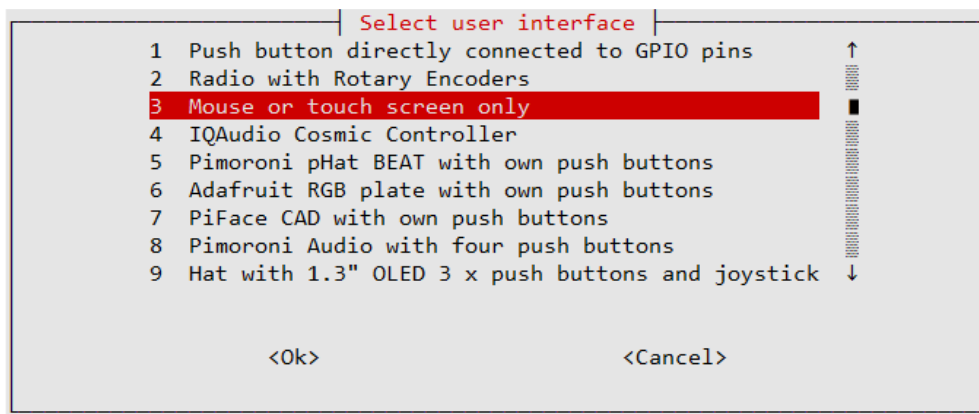


Figure 182 User interface selection menu

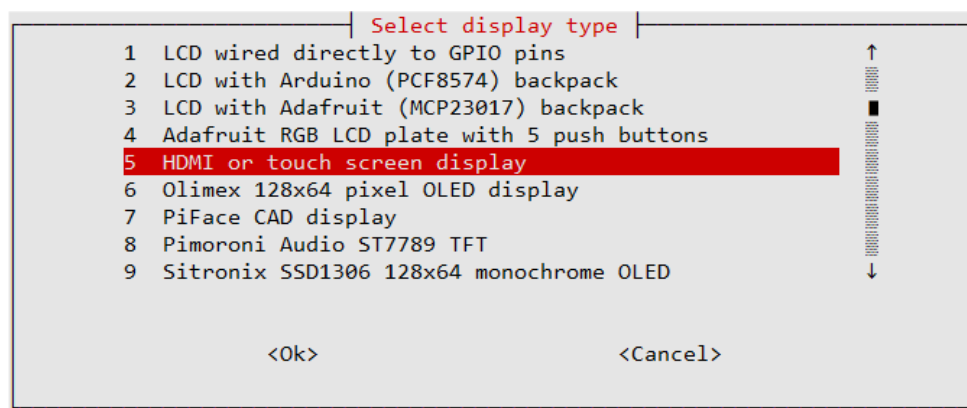
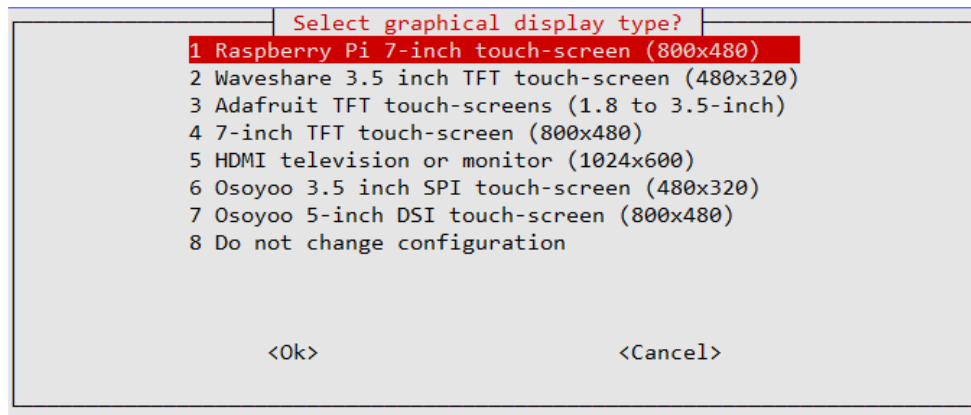


Figure 183 Select display type menu



Note: During the initial package installation, it is necessary to install the HDMI configuration first. This must be done for all other types of graphical screens including touch screens no matter what their size or type. You must run **radio-config** to install all other types of touchscreens or graphical displays.

If you are installing the **radiod** package using the **dpkg** package installer for the first time select **option 5** on the next screen. If updating **radiod** using **radio-config** select the appropriate option on the next screen.



If the screen upside-down then add the following line to the end of `/boot/firmware/config.txt`.

```
# Rotate screen 180
lcd_rotate=2
```

The following screens ask how you want to configure the graphical radio programs at startup.

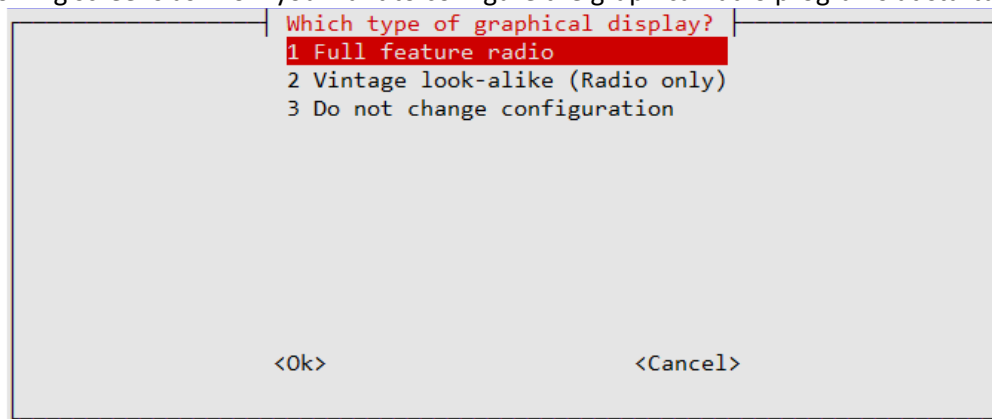


Figure 184 Selecting the type of radio display

Select which radio program to start-up. See *Figure 2 Raspberry pi 7-inch touchscreen radio* and *Figure 4 Vintage tuning touch-screen radio* on page 7.

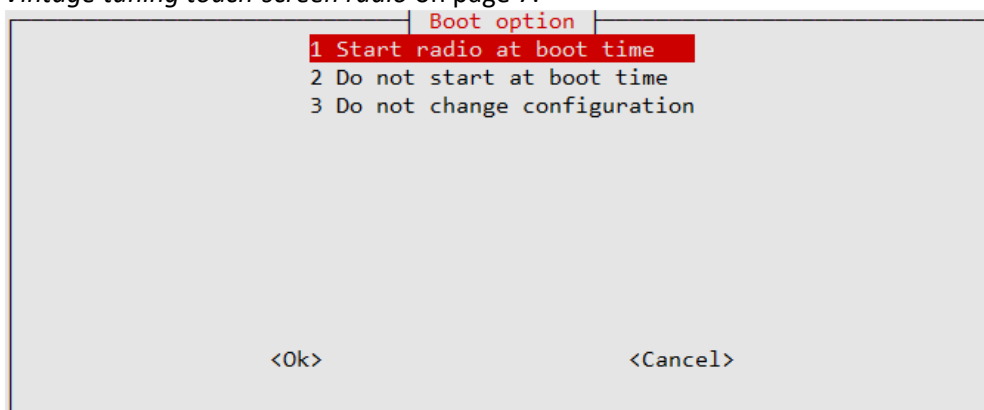


Figure 185 Configuring the HDMI or touch screen display startup

Normally select option **1** to automatically start the **gradio.py** program when the Graphical Desktop is loaded. Depending on the type of graphical radio you selected (**gradio** or **vgradio**) this sets up the appropriate program instruction in `/etc/xdg/autostart/radio.desktop`.

The installation script also copies the graphic screen configuration to **/etc/radiod.conf**. It also disables start-up of the **radiod** service which is only used for the LCD versions of the radio. Now select the option to display the radio full screen (7-inch touchscreen) or in a desktop window (Large HDMI monitor or TV).

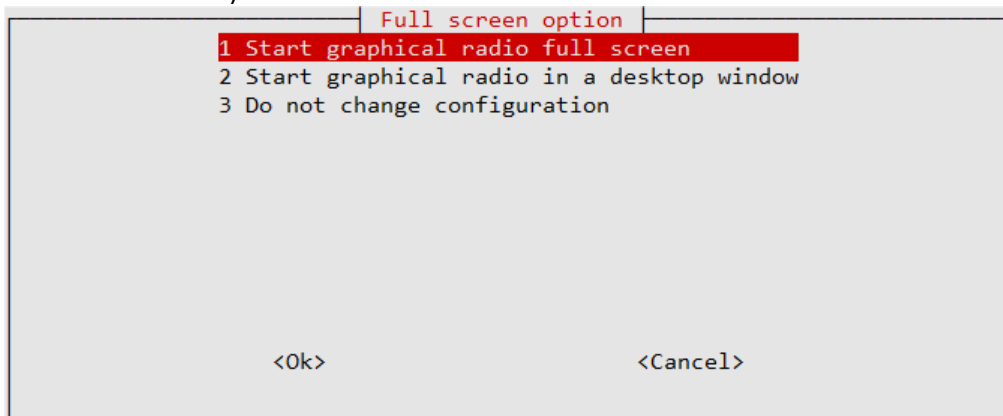


Figure 186 Configuring the graphical radio full screen

Reboot the Raspberry Pi.

```
$ sudo reboot
```

After reboot if you have an HDMI monitor or screen you should see an X-Windows or Wayland (default) desktop. Operation of the HDMI/Touch screen is shown in the section called *Operation of HDMI and touch screen displays* on page 183. Some of options such as for **Adafruit**, **Waveshare** and **Osoyoo** touch screens will bring up extra configuration menus as shown in the following.

Configure Waveshare 3.5" SPI TFT touch screen

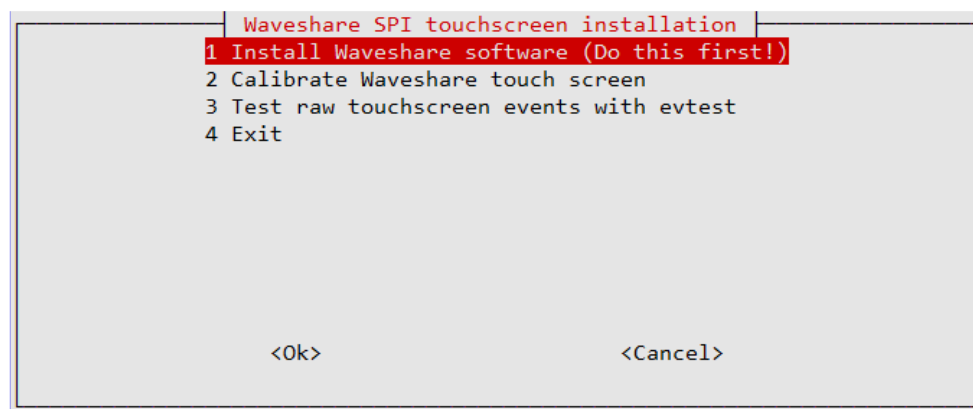


Figure 187 Waveshare 3.5-inch installation

Select option 1 to install and configure the Waveshare software. See the two sections called *Calibrating touchscreens* and *Testing touch screens using evtest* on pages 127 and 129 respectively for further setup and test of the touchscreen.

Configure the Osoyoo 3.5" SPI touch screen

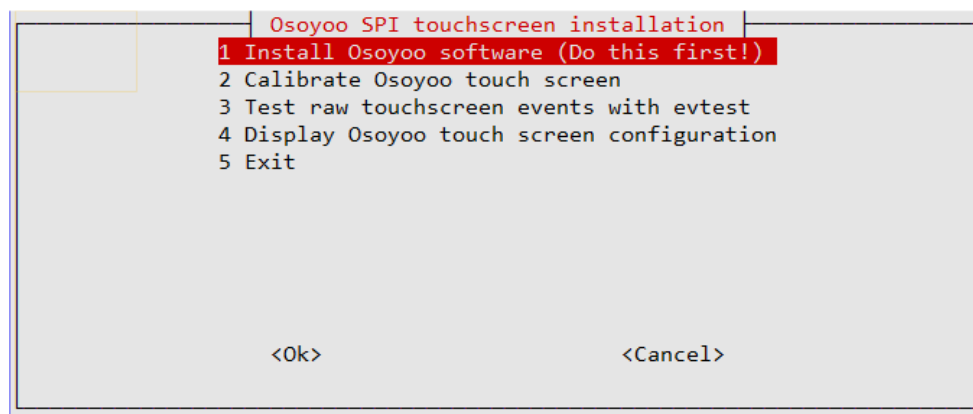


Figure 188 Osoyoo 3.5" SPI touch screen

Select option 1 to install and configure the WaveShare software. See the two sections called *Calibrating touchscreens* and *Testing touch screens using evtest* on pages 127 and 129 respectively for further setup and test of the touchscreen.

Configuring the Osoyoo 5-inch DSI capacitive touchscreen

There isn't currently a script to configure this. For **Bookworm** use the manufacturer's instructions at <https://osoyoo.com/2019/09/20/instruction-for-raspberry-pi-5-dsi-touch-screen/>

For **Trixie** configure the user interface and display type as shown in *Figure 182 User interface selection menu* and *Figure 183 Select display type menu* on page 121.

User interface: Select option **3 Mouse or touch screen only**

Display type: Select **5 HDMI or touch screen display**

Then modify the following entries in the SCREEN section of `/etc/radiod.conf` as shown below:

```
[SCREEN]
:
screen_size=800x480
fullscreen=yes
:
search_window_rows=11
```

Configuring Adafruit 3.5" touch screen

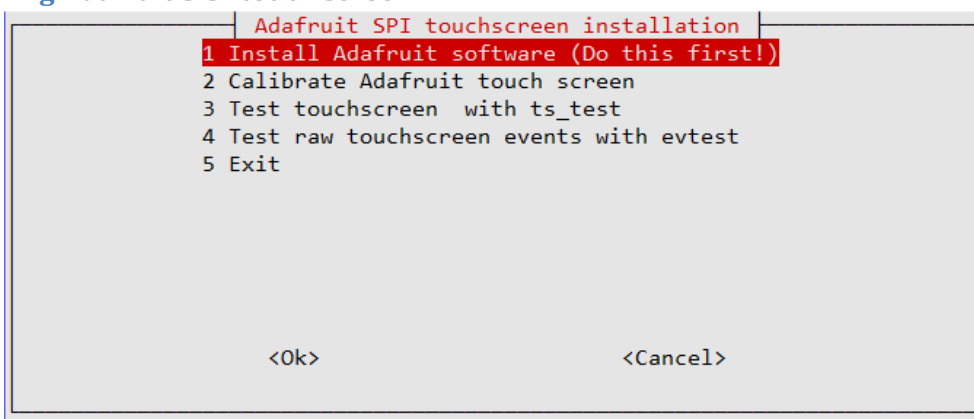


Figure 189 Adafruit touch screen installation

This will install any required software components and then displays the following list.

```
Running on a desktop environment
Desktop manager: labwc

This script downloads and installs
PiTFT Support using userspace touch
controls and a DTO for display drawing.
one of several configuration files.
Run time of up to 5 minutes. Reboot required!

Select configuration:
[1] PiTFT 2.4" V2 resistive (240x320) (320x240)
[2] PiTFT 2.4", 2.8" or 3.2" resistive (240x320) (320x240)
[3] PiTFT 2.2" no touch (320x240)
[4] PiTFT 2.8" capacitive touch (320x240)
[5] PiTFT 3.5" resistive touch (480x320)
[6] PiTFT Mini 1.3" or 1.54" display (240x240)
[7] ST7789V 2.0" no touch (320x240)
[8] MiniPiTFT 1.14" display (240x135)
[9] BrainCraft HAT or 1.3" TFT Bonnet + Joystick (240x240)
[10] Uninstall PiTFT
[11] Quit without installing
```

Note: Only option **[5] PiTFT 3.5" resistive touch (480x320)** is supported by the **radiod** software at the moment, but even this this does not have the touch screen function working. Select option [5]

```
Display Type: PiTFT 3.5" resistive touch
Select rotation:
[1] 90 degrees (landscape)
[2] 180 degrees (portrait)
[3] 270 degrees (landscape)
[4] 0 degrees (portrait)
```

Select option **[1]** or **[3]** (landscape)

```
Select install type:
[1] Setup PiTFT as desktop display (mirror)
[2] Display console on PiTFT (console)
[3] Install Drivers only
1
```

Select option **[1]** (mirror)

```
PiTFT Checking init system...
Found systemd
/boot is mounted
PiTFT System update
Updating apt indexes.....
:
Settings take effect on next boot.
REBOOT NOW? [Y/n]
```

Answer Y. The system will reboot and after a view minutes the Raspberry Pi Desktop should be displayed on the Adafruit 3.5" display. :

See the two sections called *Calibrating touchscreens* and *Testing touch screens using evtest* on pages 127 and 129 respectively for further setup and test of the touchscreen.

MHS 3.5-inch RPi display installation

If not using MHS devices skip this section.

Full instructions for installing and configuring the MHS devices will be found at:

http://www.lcdwiki.com/MHS-3.5inch_RPi_Display

Below are the basic installation instructions extracted from the above site. First clone the Waveshare installation software.

```
$ cd
$ git clone https://github.com/goodtft/LCD-show.git
```

It is now necessary to change to the **LCD-show** directory and run the correct installer for the type of MHS screen being installed. For example:

MHS35-show MHS 3.5-inch RPi display

MHS55-show MHS 5.5-inch RPi display

MHS35B-show MHS 3.5-inch RPi B display

LCD7B-show MHS 7-inch RPi display

There are other installation scripts for different MHS LCD models in the

Below is an example of the installation for the MHS TFT 3.5" RPi display

```
$ cd LCD-show/
$ ./MHS35-show
```

This command to rotate the screen 180 degrees and can be run at any time (reboot required)

```
$ ./MHS35-show 180
```

However, do refer to the MHS documents as the above are only very basic instructions.

Edit **/boot/firmware/config.txt** and force the console to 720 x 480.

```
framebuffer_width=720
framebuffer_height=480
```

Now carry out the instructions shown in section *Installing the Radio Daemon* on page 110.

Using other touch screens

There are various other touch screens on the market but this version of the software does not support screen sizes of less than 480 x 320 pixels. There is a **screen_size** parameter in **/etc/radiod.conf** configuration file.

```
# Size is in pixels. Supported is 800x480 (7" screen) or 720x480 (3.5"
screen)
# or 480x320 (2.8" or 3.5" screen)
screen_size=800x480
```

Another important aspect of screen size are the following parameters in **/boot/firmware/config.txt**.

```
framebuffer_width=1280
framebuffer_height=720
```

Changing the above can force a console size. By default, it will be display's size minus overscan settings in **/boot/firmware/config.txt**.

Now carry out the instructions shown in section *Installing the Radio Daemon* on page 110.

Testing touch screens using evtest

Touch screens use kernel events along with other devices such as mouse and keyboard. This is done with the **evtest** program which is installed by the **radiod** installation scripts as required. You can run **evtest** either from the command line or from one of the touch screen installation scripts called from **radio-config**. In either case you will see a screen similar to the following two examples.

```
$ sudo evtest
```

Example 1 Adafruit devices

```
No device specified, trying to scan all of /dev/input/event*
Not running as root, no devices may be available.
Available devices:
/dev/input/event0:      PixArt USB Optical Mouse
/dev/input/event1:      vc4-hdmi-0
/dev/input/event2:      vc4-hdmi-0 HDMI Jack
/dev/input/event3:      vc4-hdmi-1
/dev/input/event4:      vc4-hdmi-1 HDMI Jack
/dev/input/event5:      stmpe-ts
Select the device event number [0-5]: 5
```

Enter 5 for device **stmpe-ts** device. The following will be displayed

```
Input driver version is 1.0.1
Input device ID: bus 0x18 vendor 0x0 product 0x0 version 0x0
Input device name: "stmpe-ts"
Supported events:
Event type 0 (EV_SYN)
Event type 1 (EV_KEY)
Event code 330 (BTN_TOUCH)
Event type 3 (EV_ABS)
Event code 0 (ABS_X)
Value      0
Min         0
Max        4095
Event code 1 (ABS_Y)
Value      0
Min         0
Max        4095
Event code 24 (ABS_PRESSURE)
Value      0
Min         0
Max        255
Properties:
Testing ... (interrupt to exit)
```

Example 2 Waveshare devices

```
No device specified, trying to scan all of /dev/input/event*
```

```
Available devices:
/dev/input/event0:      RPI Wired Keyboard 1
/dev/input/event1:      RPI Wired Keyboard 1
/dev/input/event2:      Microsoft Compact Optical Mouse 500
/dev/input/event3:      ADS7846 Touchscreen
Select the device event number [0-3]: 3
```

Enter 3 for the ADS7846 touchscreen. The following will be displayed.

```
Input driver version is 1.0.1
Input device ID: bus 0x1c vendor 0x0 product 0x1ea6 version 0x0
Input device name: "ADS7846 Touchscreen"
Supported events:
Event type 0 (EV_SYN)
Event type 1 (EV_KEY)
Event code 330 (BTN_TOUCH)
Event type 3 (EV_ABS)
Event code 0 (ABS_X)
Value      0
Min         0
Max        4095
Event code 1 (ABS_Y)
Value      0
Min         0
Max        4095
Event code 24 (ABS_PRESSURE)
Value      0
Min         0
Max        255
Properties:
Testing ... (interrupt to exit)
```

In both examples when you touch the touchscreen you should see events similar to the following:

```
Event: time 1771244406.111207, type 3 (EV_ABS), code 0 (ABS_X), value 2448
Event: time 1771244406.111207, type 3 (EV_ABS), code 1 (ABS_Y), value 1771
Event: time 1771244406.111207, type 3 (EV_ABS), code 24 (ABS_PRESSURE),
value 170
Event: time 1771244406.111207, type 1 (EV_KEY), code 330 (BTN_TOUCH), value
1
Event: time 1771244406.111207, ----- SYN_REPORT -----
Event: time 1771244406.117647, type 3 (EV_ABS), code 0 (ABS_X), value 2394
Event: time 1771244406.117647, type 3 (EV_ABS), code 1 (ABS_Y), value 1778
Event: time 1771244406.117647, type 3 (EV_ABS), code 24 (ABS_PRESSURE),
value 123
Event: time 1771244406.117647, ----- SYN_REPORT -----
Event: time 1771244406.130099, type 3 (EV_ABS), code 0 (ABS_X), value 2339
Event: time 1771244406.130099, type 3 (EV_ABS), code 1 (ABS_Y), value 1779
Event: time 1771244406.130099, type 3 (EV_ABS), code 24 (ABS_PRESSURE),
value 114
Event: time 1771244406.130099, ----- SYN_REPORT -----
Event: time 1771244406.136289, type 3 (EV_ABS), code 0 (ABS_X), value 2295
Event: time 1771244406.136289, type 3 (EV_ABS), code 1 (ABS_Y), value 1773
Event: time 1771244406.136289, type 3 (EV_ABS), code 24 (ABS_PRESSURE),
value 116
Event: time 1771244406.136289, ----- SYN_REPORT -----
Event: time 1771244406.142459, type 3 (EV_ABS), code 0 (ABS_X), value 2283
Event: time 1771244406.142459, type 3 (EV_ABS), code 1 (ABS_Y), value 1768
Event: time 1771244406.142459, type 3 (EV_ABS), code 24 (ABS_PRESSURE),
value 119
```

The above indicates the Linux kernel is detecting touchscreen events.

Calibrating touchscreens

It isn't always necessary to calibrate a touch screen as default settings are usually provided. Some manufacturer's screens such as Adafruit will need re-calibration during setup. There are two calibration programs used by the touchscreens in this project. These are **xinput_calibrator** and **ts_calibrate**. They must both be run in an X-Terminal although they may be invoked from a normal **SSH** terminal or from the radio installation menu (See individual touchscreen descriptions). They both work in a similar way by touching the cross hairs in turn with a stylus.

The **xinput_calibrator** tool is either called from the **radio-config** (**Ossoyoo** or **Waveshare**) touchscreen installation menu or from the command line:

```
sudo DISPLAY=:0.0 xinput_calibrator --output-filename  
/usr/share/X11/xorg.conf.d/99-calibration.conf
```

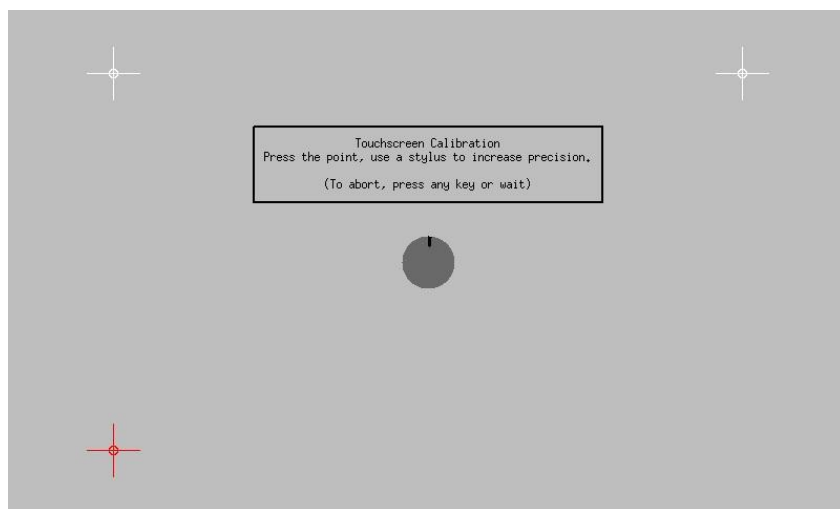


Figure 190 Touchscreen xinput_calibrator tool

Likewise, **ts_calibrate** is called from the Adafruit touchscreen installation menu or from the command line as shown below.

```
$ sudo TSLIB_FBDEVICE=/dev/fb0 TSLIB_TSDEVICE=/dev/input/touchscreen  
ts_calibrate
```



Figure 191 Touchscreen ts_calibrate tool

The **ts_calibrate** tool has an associated test program called **ts_test** as shown in the next illustration.

```
$ sudo TSLIB_FBDEVICE=/dev/fb0 TSLIB_TSDEVICE=/dev/input/touchscreen ts_test
```



Configuring sound devices

There are more sound cards (DACs) on the market than can be listed in this manual. If you find that the sound card you wish to use isn't listed in the configuration program, it should still be possible to use them. See *Manually configuring sound cards* on page 140. Below are the currently supported sound devices:

- CMedia USB speakers or devices (See page 131)
- Sound cards such as **HiFiBerry**, **IQaudio**, **JustBoom**, **RPi audio devices** and **Pimoroni pHat** DAC and DAC+ products (See page 132)
- Bluetooth speakers or headphones (See page 162).
- Waveshare WM8960 DAC
- Adafruit Speaker Bonnet
- HDMI speakers if fitted

To check if the audio device is present run the **aplay** command.

```
$ aplay -l
**** List of PLAYBACK Hardware Devices ****
card 0: ALSA [bcm2835 ALSA], device 0: bcm2835 ALSA [bcm2835 ALSA]
  Subdevices: 8/8
    Subdevice #0: subdevice #0
    Subdevice #1: subdevice #1
    Subdevice #2: subdevice #2
    Subdevice #3: subdevice #3
    Subdevice #4: subdevice #4
    Subdevice #5: subdevice #5
    Subdevice #6: subdevice #6
    Subdevice #7: subdevice #7
card 0: ALSA [bcm2835 ALSA], device 1: bcm2835 ALSA [bcm2835 IEC958/HDMI]
  Subdevices: 1/1
    Subdevice #0: subdevice #0
card 1: Device [USB PnP Sound Device], device 0: USB Audio [USB Audio]
  Subdevices: 0/1
    Subdevice #0: subdevice #0
```

In the above example **Card 0** is the on-board devices namely the audio output jack and HDMI. **Card 1** is a **USB PnP** sound device.

To configure other sound devices run **radio-config** from the command line and select option **2 Configure audio output devices**.

Example: configuring a USB sound device

To configure a USB DAC sound devices such as **CMedia** speakers or sound dongles select option **3 USB DAC**.

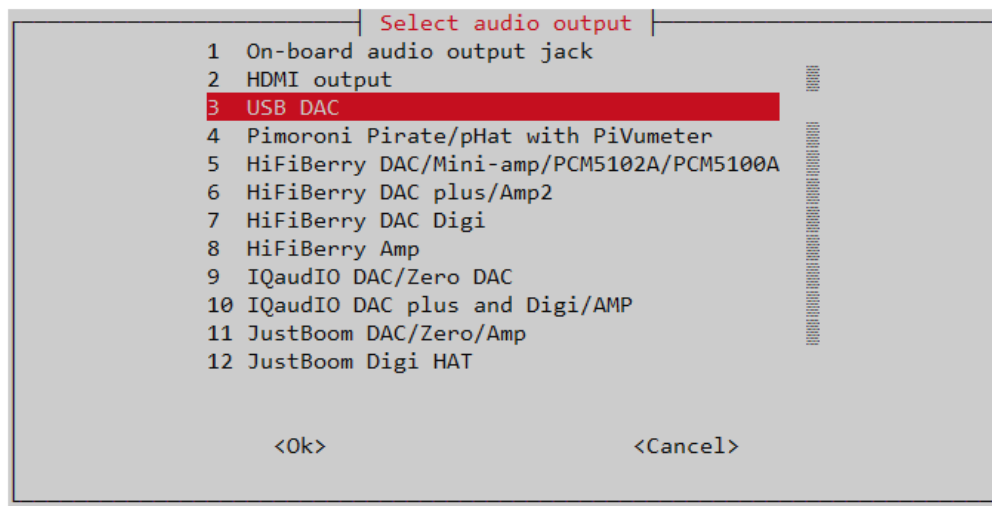


Figure 192 Configure USB DAC

Reboot when prompted. After rebooting the Raspberry Pi run the **alsamixer** program.

```
$ alsamixer
```

The following screen is displayed:

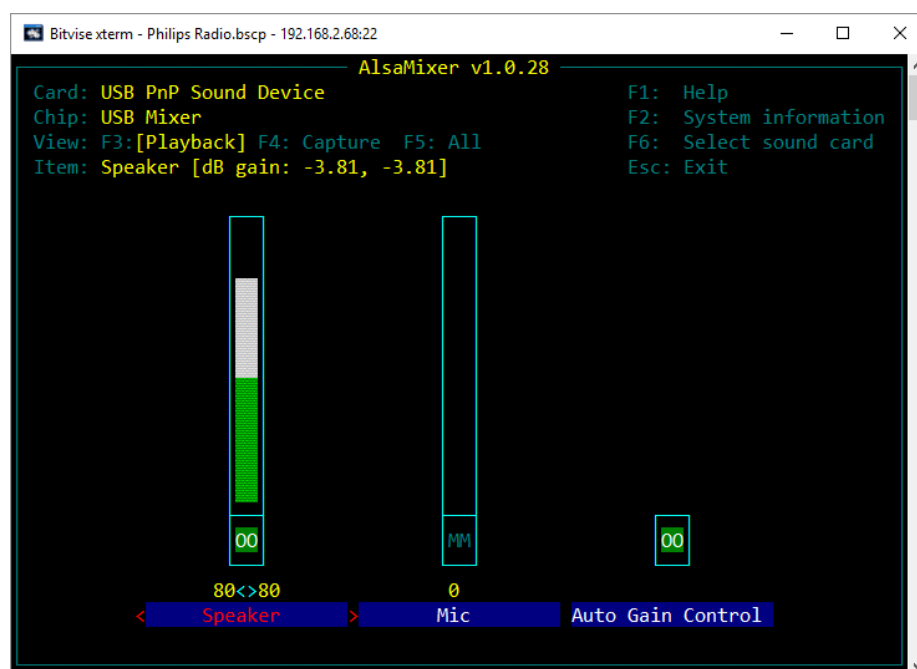


Figure 193 The USB PnP Alsa Mixer

Use the Left and Right keys to position on the 'Speaker field'. Adjust the sound level using the Up and Down keys (80% in the above example). Pres **Esc** key or **Ctrl Z** key to exit.

Configuring a DAC Sound Card

This section covers configuration of add on DAC sound cards such as **HiFiBerry**, **IQaudIO** and **JustBoom DAC**, **DAC+** and Amplifier products. Older versions of the **HiFiBerry DAC** that used the 26-pin GPIO header are not supported.

To configure a DAC sound cards run **radio-config**.

```
$ radio-config
```

Next select option 2 Configure audio output devices. The following screen will be displayed:

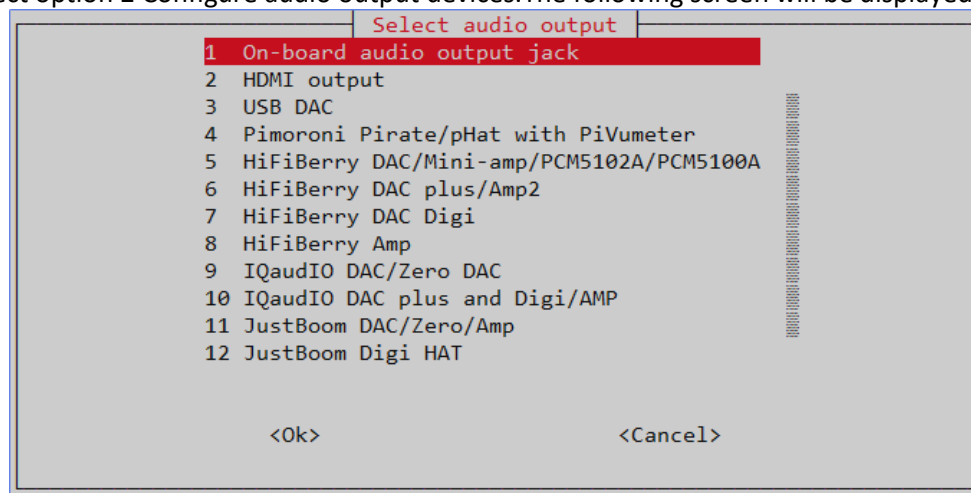


Figure 194 Configuring the onboard audio jack output

More options are available by scrolling down with the down arrow key:

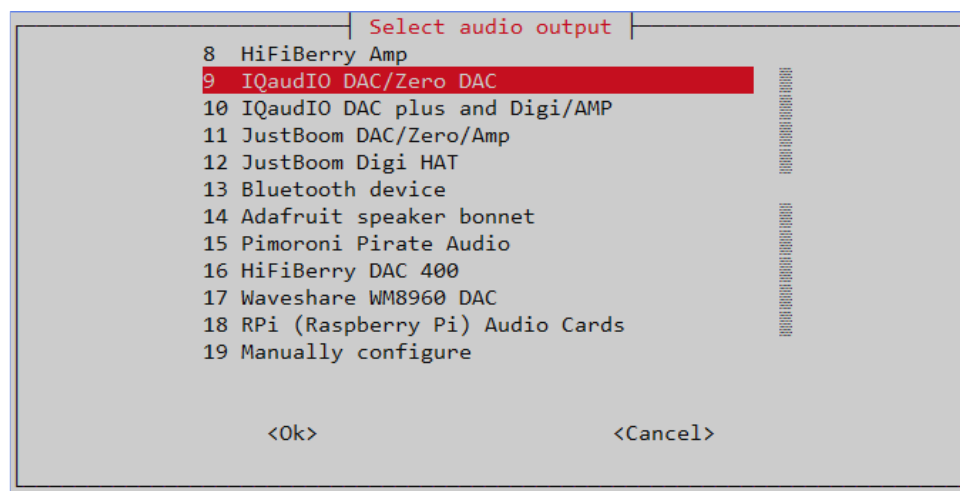


Figure 195 Configuring add-on DAC sound cards

Select option for the DAC being used and press OK. Reboot when prompted by the next screen. If using Bluetooth devices such as speakers or headphones then select option 13 Bluetooth device.

The **Pimoroni pHat** is compatible with HiFiBerry DAC (Not DAC+) and uses the same Device Tree (DT) overlay so select HiFiBerry DAC if using the pHat.

To configure Raspberry Pi Sound Cards select option 18 RPi (Raspberry Pi) Audio Cards.

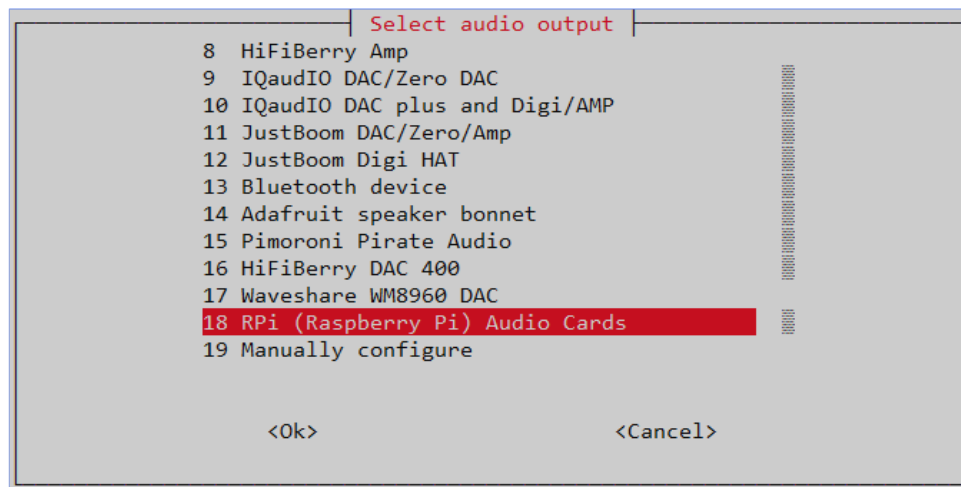


Figure 196 Selecting Raspberry Pi Audio cards

The following screen will be displayed:

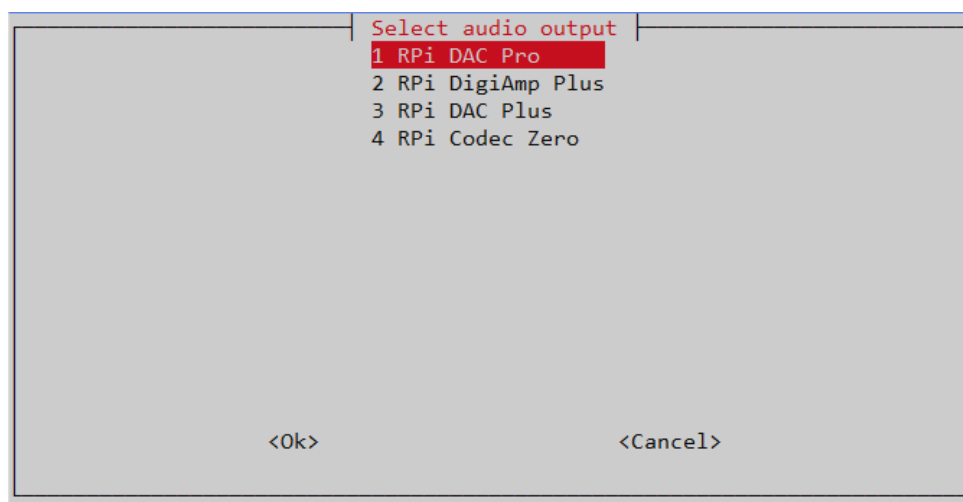


Figure 197 Raspberry Pi DAC options

The program displays the configuration changes it has made. Reboot after it has completed.

```
$ reboot
```

After rebooting run the **alsamixer** program.

```
$ alsamixer
```

Use the left and right keys to select the mixer control (Analogue) and use the up down keys to change the volume to 100%.

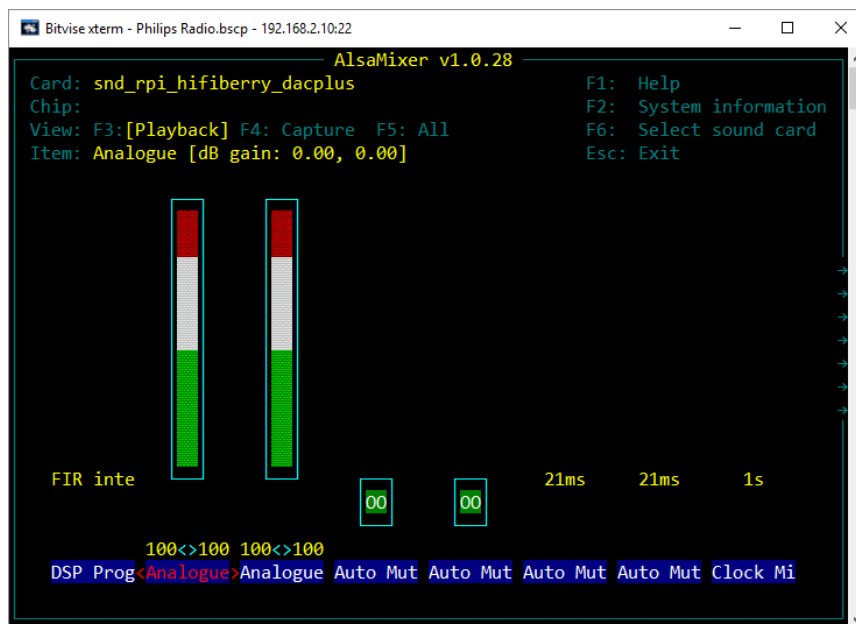


Figure 198 Set mixer analogue volume

Next use the right key to position on the “Digital” mixer control and use the up down keys to change the mixer volume:

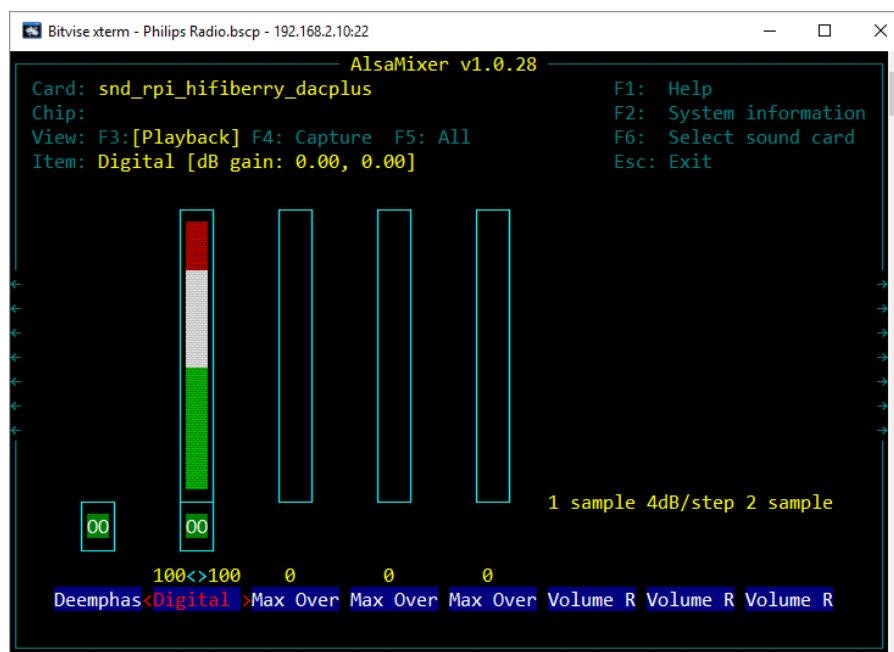


Figure 199 Set mixer digital volume

Configuring Allo Sound Cards

The Allo Piano DAC, has two outputs, one for normal sound ranges and one for a sound woofer. The board comes pre-programmed as 2.1 for the Raspberry Pi. The subwoofer has a right and left output but is mono only.

However, it uses a second **I2S** channel on **GPIO5** and mute signal on GPIO6 both used to control the woofer amplifier. This conflicts with the LCD signals **lcd_data4** and **lcd_data5** pins. Other Allo products are not affected as they only use one I2S channel. There are two ways to correct this.

Method 1

Edit **/etc/radiod.conf** and locate the LCD GPIO connection definitions.

```
# LCD GPIO connections for 40 pin version of the radio
lcd_select=7
lcd_enable=8
lcd_data4=5
lcd_data5=6
lcd_data6=12
lcd_data7=13
```

Change the **lcd_data4/5** assignments in **/etc/radiod.conf** to:

```
lcd_data4=26
lcd_data5=27
```

Wire LCD **data4/5** to GPIO 26 (physical pin 37) and GPIO 27 (physical pin 13).

Method 2

Configure the LCD to use an I2C backpack. This frees up 6 GPIOs including GPIO5 and 6. See *Construction using an I2C LCD backpack* on page 67.

Configuring Allo driver dtoverlay

Allo manufacture several audio products as well as the Piano 2.1 with woofer. Currently these cannot be configured via the radio installation program. It is necessary to load the correct **device tree overlay (dtoverlay)** by adding the following line for the corresponding **Allo** audio card the end of the **/boot/firmware/config.txt**.

Device	Device Tree Overlay in config.txt
Allo Piano HIFI DAC	dtoverlay=allo-piano-dac-pcm512x-audio
Allo Piano 2.1 HIFI DAC with woofer	dtoverlay=allo-piano-dac-plus-pcm512x-audio
Allo Boss HIFI DAC / MINI BOSS HIFI DAC	dtoverlay=allo-boss-dac-pcm512x-audio
Allo DIGIONE	dtoverlay=allo-digione

Also disable the on-board output jack by amending it in the **config.txt** file.

```
dtparam=audio=off
```

Reboot the Raspberry Pi to load the Allo dtoverlay.

```
$ sudo reboot
```

Verification

Check, if the sound card is enabled with “**aplay**”. The output below will vary depending upon which Allo DAC has been selected:

```
$ aplay -l
**** List of PLAYBACK Hardware Devices ****
card 0: PianoDAC [PianoDAC], device 0: PianoDAC multicodec-0 []
Subdevices: 0/1
Subdevice #0: subdevice #0
```

Installing the Waveshare WM8960 DAC

Run the radio-config utility and then select option **2 Configure audio output devices**

```
$ radio-config
```

Scroll down to Select option 17 of audio configuration menu and select it.

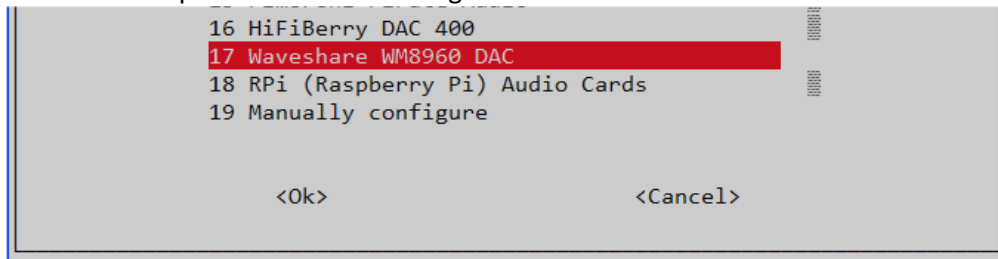


Figure 200 Waveshare WM8960 DAC selection

Setting the Headphone and Speakers Volume

Select option 1 to set up the volume levels for the WM8960 card.

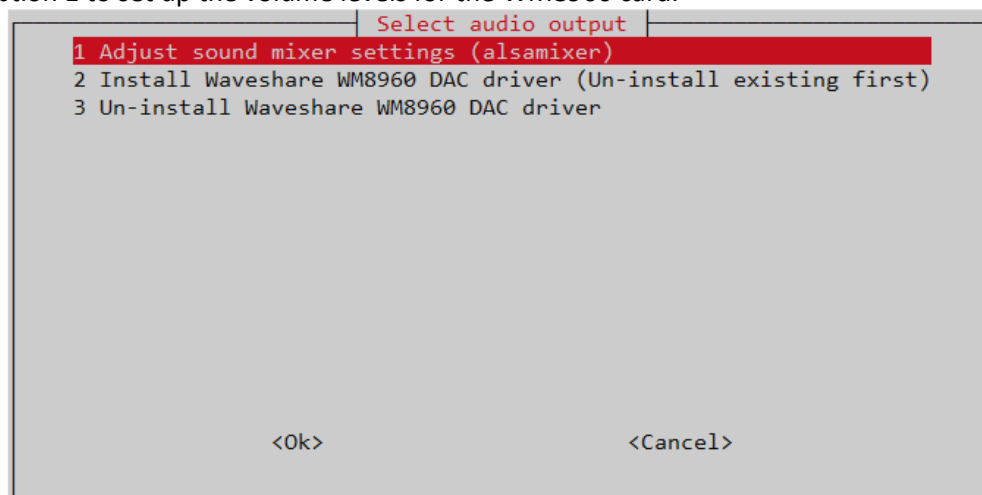


Figure 201 Waveshare 8960 sound card

This will display the **alsamixer**. If you don't see the Card: wm8960-soundcard on the top left of the screen below then press the **F6** key on your keyboard to select the WM9860 sound card.

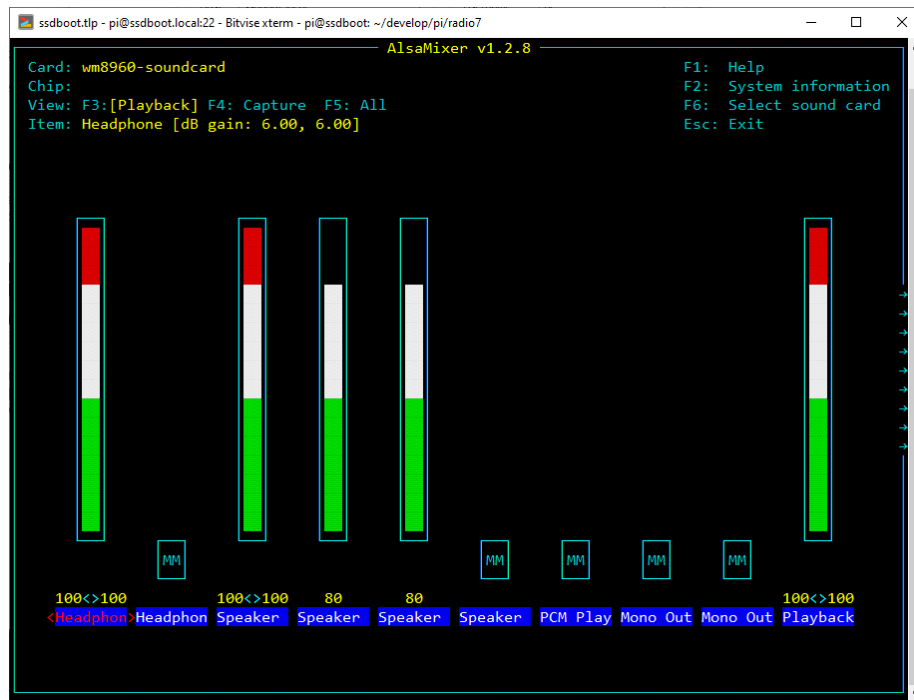


Figure 202 Alsamixer setting the Headphone Volume

Set the headphone volume to maximum, press Esc to exit. Reboot the Raspberry Pi to test.

```
$ sudo reboot
```

Uninstalling the Waveshare WM8960 sound card

Select option **3 Un-install Waveshare WM8960 DAC driver**. Now run the `configure_audio.sh` configuration script as shown in *Configuring the audio output* on page 116 to select the new audio device that you wish to use.

Installing a Bluetooth audio device

Install the Bluetooth software

If you haven't already done so update the operating system first as shown in the section *Update to the latest the system packages* on page 98. To install the Bluetooth software, do the following

- Power on your Bluetooth device
- Run the **radio-config** utility.
- Select option **2 Configure audio output devices** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.
- Select option **13 Bluetooth audio**

The following menu will be displayed:

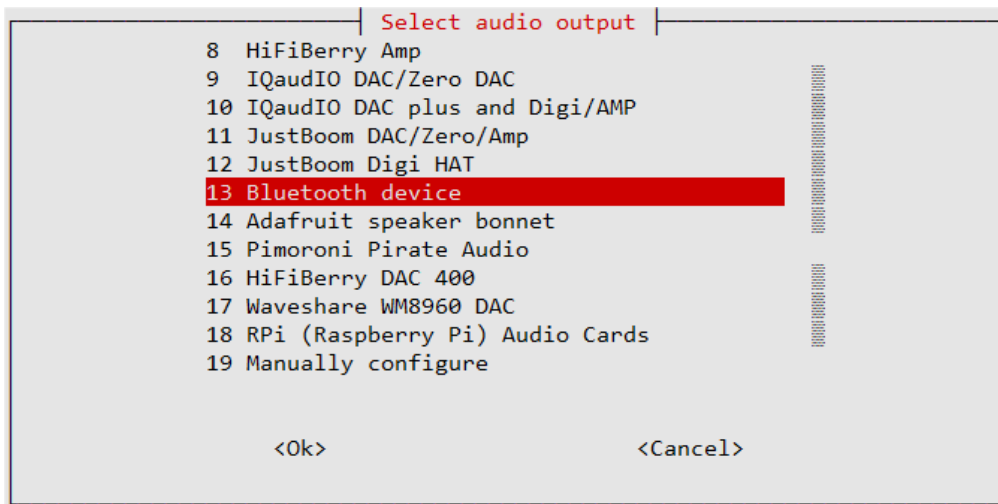


Figure 203 The Bluetooth Configuration Menu

This will display the Bluetooth menu as shown below. The menu is arranged in the order that you need to do things.

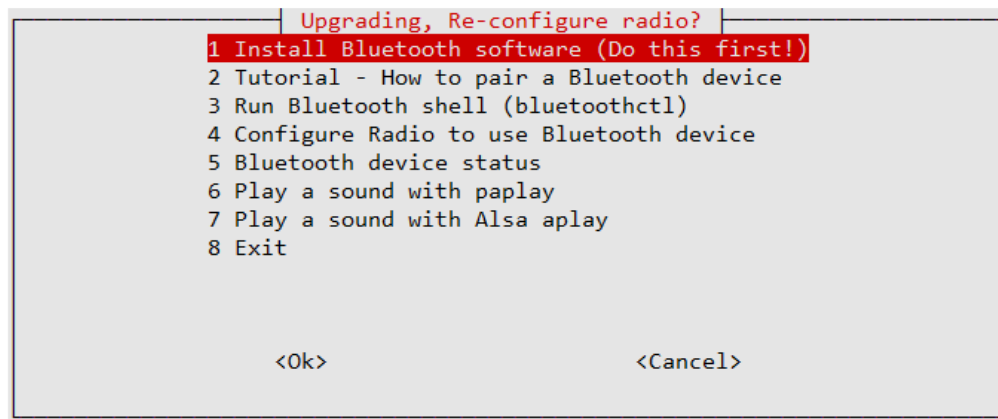


Figure 204 Bluetooth software installation

Start by selecting option **1 Install Bluetooth software**. As it says do this first do this first. The installation software will display the following:

```
/usr/share/radio/scripts/configure_bluetooth.sh Bluetooth configuration log,
Wed Jan 14 08:10:21 GMT 2026
Boot configuration in /boot/firmware/config.txt
Installing Bluetooth packages
sudo apt-get -y install pulseaudio
Reading package lists... Done
:
Bluetooth software installation complete
```

Pairing your Bluetooth device

Read **option 2 Tutorial - How to pair a Bluetooth device**. The operations to pair a device are summarised below:

Select **Option 3 Run Bluetooth shell (bluetoothctl)** from the menu or run **bluetoothctl** from the command line. In the **bluetoothctl** prompt run the following instructions shown in bold:

```

[bluetooth]# power on
[bluetooth]# agent on
Agent registered
[CHG] Controller B8:27:EB:A3:E4:52 Pairable: yes
[bluetooth]# default-agent
Default agent request successful
[bluetooth]# scan on
Discovery started
[CHG] Controller DC:A6:32:05:36:9D Discovering: yes
[NEW] Device C0:48:E6:73:3D:FA [TV] Samsung Q7 Series (65)
[NEW] Device 00:75:58:41:B1:25 SP-AD70-B
[bluetooth]# scan off
:
[CHG] Controller DC:A6:32:05:36:9D Discovering: no
Discovery stopped
[bluetooth]# pair 00:75:58:41:B1:25
Attempting to pair with 00:75:58:41:B1:25
:
Pairing successful
[bluetooth]# connect 00:75:58:41:B1:25
Connection successful
[SP-AD70-B]#trust 00:75:58:41:B1:25

```

Note: The example device in the above output **00:75:58:41:B1:25** will be different for your device.

Identifying your Bluetooth device can be a bit of a challenge. During discovery your device will be displayed in the format **<mac address>:<name>**, for example: **00:75:58:41:B1:25 SP-AD70-B**. The **<name>** will be typically the name or model name of your device. Look at the label on the device or any technical documentation that you might have to identify the name or alias of your device.

Now display the info for your device:

```

[SP-AD70-B]# info 00:75:58:41:B1:25
Device 00:75:58:41:B1:25 (public)
    Name: SP-AD70-B
    Alias: SP-AD70-B
    Class: 0x00240404
    Icon: audio-headset
    Paired: yes
    Bonded: yes
    Trusted: yes
    Blocked: no
    Connected: yes
    LegacyPairing: no
:
[SP-AD70-B]# exit

```

You should see the highlighted options set as above.

Configure radio software for Bluetooth

Select option **4 Configure Radio to use Bluetooth device** to configure the Radio (radiod) and Music Player Daemon (MPD) configuration files to use your Bluetooth device.

```

/usr/share/radio/configure_bluetooth.sh Bluetooth configuration log, Wed 13
Nov 10:52:28 GMT 2024
Device 00:75:58:41:B1:25 SP-AD70-B
Paired 00:75:58:41:B1:25 found
:
Copying /usr/share/radio/asound/asound.conf.dist.blue to /etc/asound.conf
Configuring audio_out parameter in /etc/radiod.conf
audio_out="bluetooth"

```



```
A log of this run will be found in /usr/share/radio/logs/bluetooth.log
```

This completes the installation.

Testing the Bluetooth device

Reboot the Raspberry Pi. If everything is OK you should hear radio on your Bluetooth device. If not try rebooting again. If that doesn't help, stop the radio.

```
$ sudo systemctl stop radiod
```

Re-run radio-config and select the Bluetooth configuration menu as previously shown. There are two tests available from the menu.

- **6 Play a sound with paplay**
- **7 Play a sound with Alsa aplay**

The **paplay** option is the most basic sound test. You should hear a piano playing. If no sound heard, first check that your device is connected and normal by display the status with option **5 Bluetooth device status**.

If it is not connected, power on Bluetooth adapter and connect it with option **3 Run Bluetooth shell**

```
[SP-AD70-B]# power on
[SP-AD70-B]# connect 00:75:58:41:B1:25
```

The **aplay** command relies on the Alsa and MPD configuration being correct. In particular **/etc/asound.conf** and **/etc/mpd.conf** configuration files.

The **/etc/asound.conf** should be a copy of **/usr/share/radio/asound/asound.conf.dist.blue** file configured with your Bluetooth device.

The MPD configuration in **/etc/mpd.conf** should look like the following.

```
audio_output {
    type      "alsa"
    name      "SP-AD70-B"
    device    "bluealsa"
    mixer_type "software"
}
```

Manually configuring sound cards

Unless you have a need to manually configure some other sound card or need to troubleshoot a non-working card you can skip this section. Configuring a **HiFiBerry DAC+ Pro** or **DAC2 Pro** is shown in this example

Edit the **/boot/firmware/config.txt (Bookworm)** and add the following line to the end of the file depending upon the version you are using.

```
dtoverlay=hifiberry-dacplus
```

See <https://www.hifiberry.com/guides/configuring-linux-3-18-x/> for other devices.

Modify the **audio_output** section in **/etc/mpd.conf** to support the HiFiBerry DAC and software mixer.

```
audio_output {  
    type      "alsa"  
    name      "HiFiBerry DAC"  
    device     "hw:0,0"  
#    mixer_type  "hardware"  
    mixer_type "software"  
}
```

Reboot the Raspberry Pi.

```
$ sudo reboot
```

If no music is heard run the **alsamixer** program and set the volume to at least 80% as shown in the previous section on **HiFiBerry** devices.

A list of **HiFiBerry** product software drivers (DT overlays) will be found in *Appendix C – HiFiBerry Device Tree Overlays* on page 248.

Other sound cards may not only require specific DT overlay to be configured in config.txt as shown above but also require additional software and or libraries to be loaded. See the manufacturer's documentation for details.

Setting the mixer volume

All sound output goes through a mixer. After rebooting the Raspberry Pi, for the on-board output jack, run the **alsamixer** program:

```
$ alsamixer
```

The following screen is displayed:

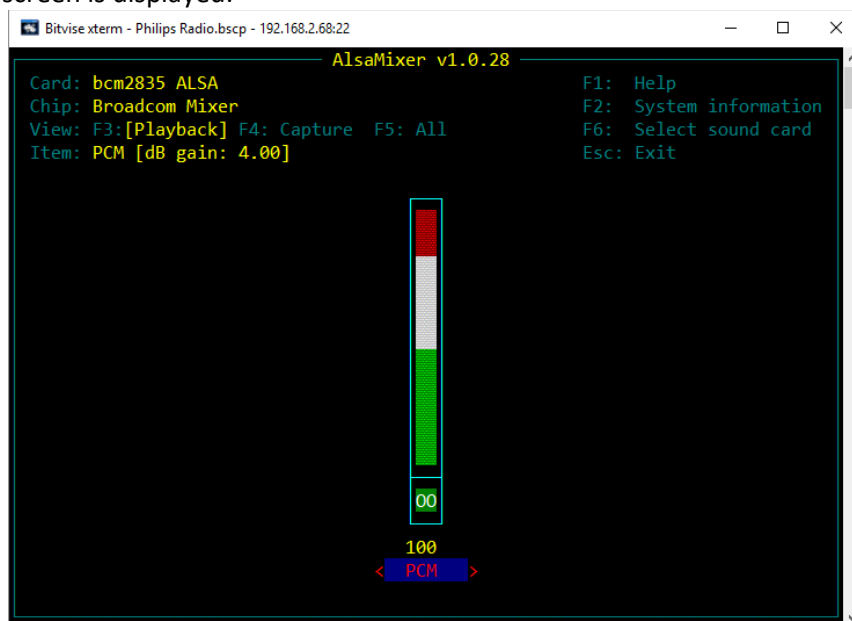


Figure 205 Basic Alsa sound mixer

The above illustration shows the **bcm2835** AlsA Mixer. There is only one mixer control called PCM (Pulse Code Modulated). Adjust the volume to 100% if not already set by using the Up and Down keys on the keyboard. Press the **Esc** key or **Ctl-Z** to exit the program.

It is also possible to set the volume for the on-board mixer volume with the **amixer** program.

```
$ amixer cset numid=1 100%
numid=1,iface=MIXER,name='PCM Playback Volume'
; type=INTEGER,access=rw---R--,values=1,min=-10239,max=400,step=0
: values=400
| dBscale-min=-102.39dB,step=0.01dB,mute=1
.toml
```

Keeping the Radio software up to date

Things are constantly changing with the software. These changes include updates to the Raspberry Pi OS, third-party products and libraries. Also new hardware solutions as well as old hardware being dropped are constantly changing the Raspberry Pi landscape. Things that used to work may be broken by one of these updates or need to be done differently.

Important news, changes and new releases will be announced on **X** (Formerly Twitter) at: https://twitter.com/bob_rathbone or https://x.com/bob_rathbone. For technical reasons the BR Web site will still use the original blue Twitter logo for the time being.

Follow this **X** (Formerly Twitter) feed for announcements about new versions. One of the main things to keep abreast of are the Release Notes which are the best place to view software updates. See http://www.bobrathbone.com/raspberrypi/pi_internet_radio.html

Chapter 7 Configuring additional radio components

Contents chapter 7	Page
The configure radio menu	144
Documents and Tutorials	145
Displaying Documents and Tutorials in a Web Browser	146
Installing the IR remote control software	149
Installing the Web Interface	152
Airplay (shairport-sync) Installation	162
Install Icecast streaming	163
Installing the Speech facility	165
Install Luma OLED/TFT driver	166
Install recording utility	166
Editing configuration files manually	167
Configuring the volume display	169
Creating a new language file	170
Configuring an RSS feed	170
Installation of the FLIRC USB dongle	171
Install PiFace CAD (Bullseye only)	175
Configuring Russian/Cyrillic text	175
Configuring European languages	176
Configuring wallpaper backgrounds	176
Spotify installation	177

The configure radio menu - radio-config

Log into the Raspberry Pi then run **radio-config** from the command line. The following menu will be displayed:

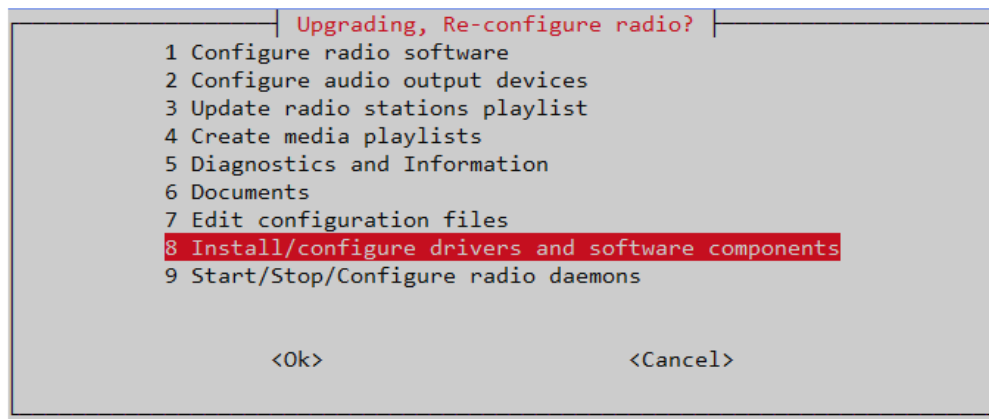


Figure 206 The radio-config menu

Select **8 Install/configure drivers and software components** from the radio-config menu

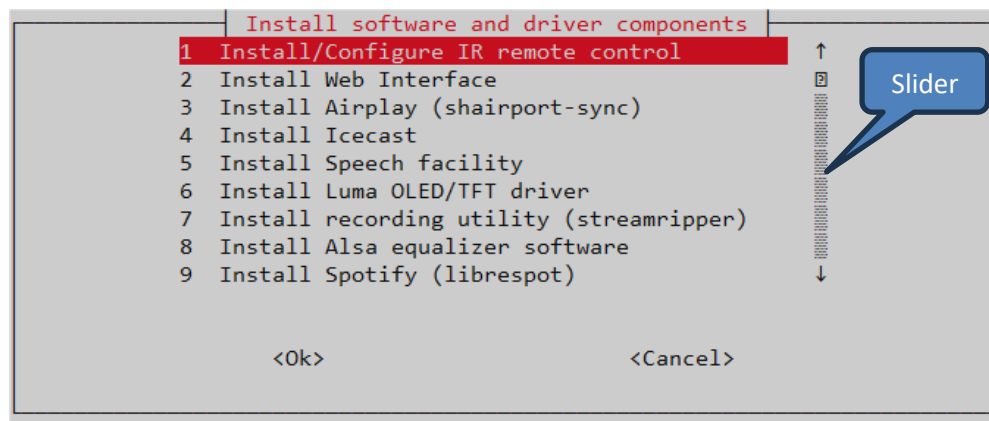


Figure 207 The additional radio software components menu

This displays additional components such as the Web interface and hardware driver software. Note the slider to the right of the menu selections. Use up and down keys on your keyboard to access additional installation selections.

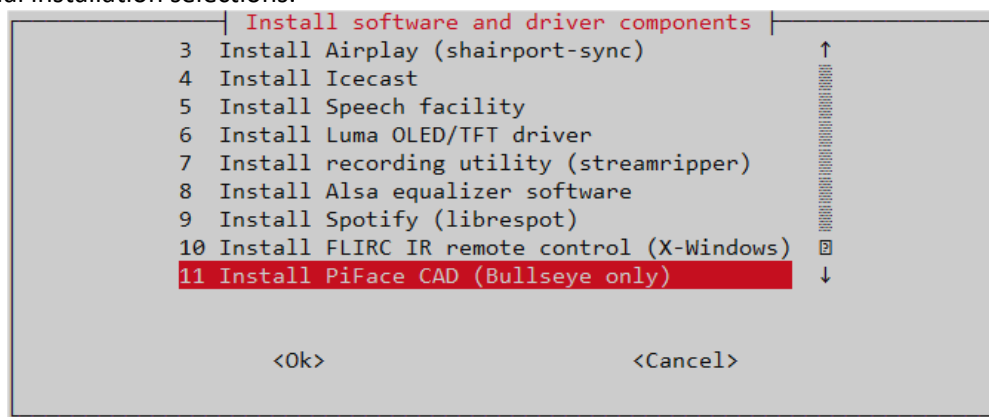


Figure 208 Additional menu selections

Documents and Tutorials

Much of the installation and configuration instructions have been moved to online tutorials. To access these tutorials, do the following:

- Run the **radio-config** utility as shown in the previous section.
- Select option **6 Documents and Tutorials** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.

The following menu will be displayed:

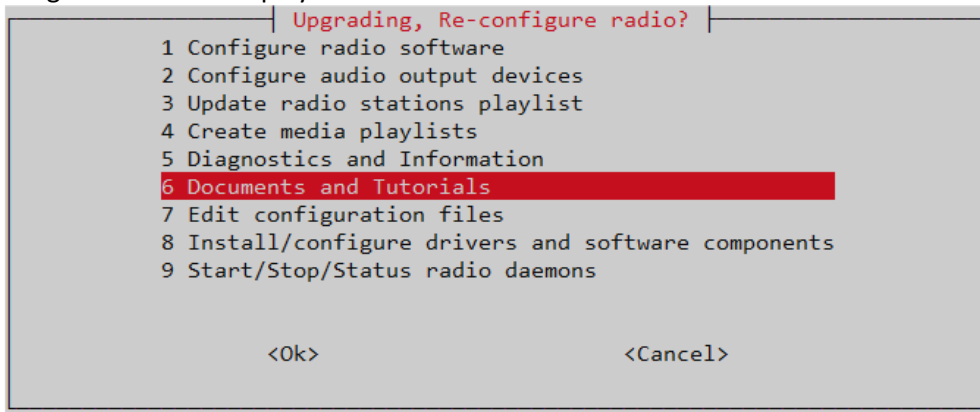


Figure 209 Online Documents and Tutorials

All available Documents and Tutorials will be displayed in the following menu:

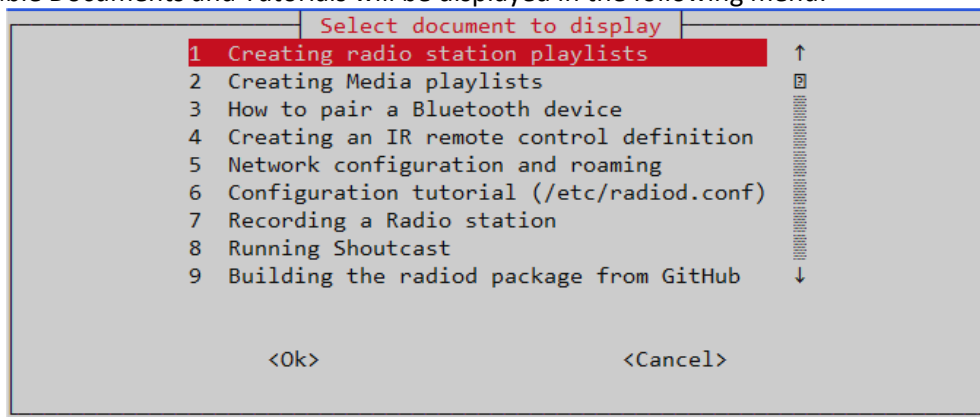
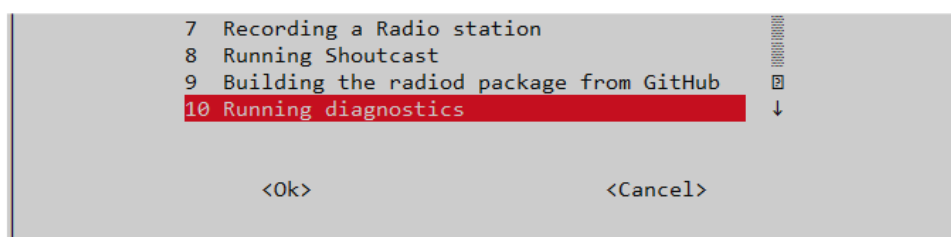


Figure 210 Documents and Tutorials

For example, selecting option 2 will display the Tutorial for creating media playlists. On the right side of the above menu, you will see a vertical scroll bar. If you scroll down using the keyboard up and down keys you will see further options displayed.



Below is an example tutorial **4 Creating an IR remote control definition**.


```
Creating Media playlists

To create a Media Playlist select option 4 Create media playlists from the radio-config
menu. This will give you five options.
Create playlist
1 From USB stick
2 From network share
3 From SD card
4 From USB Disk Drive
5 Recordings

Files

/var/lib/mpd/playlists Location of MPD playlists

/var/lib/mpd/playlists/ Where of media playlist such as USB_Stck.m3u

Running the create_playlist program from the command line

Usually you will call the create_stations program from radio-config

radio-config --> Update radio stations playlist

However there may be occasions that you want to run the create_playlist.sh program from the
command line or in a script. First populate the media with music files and then run
create_playlist.sh
cd /usr/share/radio
./create_stations.py

Commands: Use arrow keys to move, '?' for help, 'q' to quit, '<-' to go back.
Arrow keys: Up and Down to move. Right to follow a link; Left to go back.
H)elp O)ptions P)rint G)o M)ain screen Q)uit /=search [delete]=history list
```

Figure 211 Example on-line tutorial

Displaying Documents and Tutorials in a Web Browser

This feature is only available once you have installed the Web Interface as shown in section called *Installing the Web Interface* on page 152.

Open any Web Browser and type in: **http://<name of your Raspberry Pi>.<network name>**
The **<network name>** will be something like **net**, **home** or **local**. See your router documentation or use **Fing** as shown in section called *Finding the Raspberry Pi on a network using Fing* on page 84.
Alternatively enter **http://<IP address>**. Note that you must use the **http**: protocol and not **https**:

Examples:
http://piradio.net
http://192.168.1.67



Internet Radio Web Interface

Select the required radio station or track from the above window. You can also select a track from the above window.

Source Selection

Select source ▼ [Select source](#)

Select source from above box and then click the Select source button

[Tutorials and documents](#) - Guides to installation and maintenance

For my main site go to www.bobrathbone.com

Technology Overview

The Raspberry Pi is accessed through these pages running [Apache Web Server](#). The low level interface of the Raspberry Pi is accessed using CGI scripts written in Python.

Click on Tutorials
and Documents

The documents index will then be displayed:

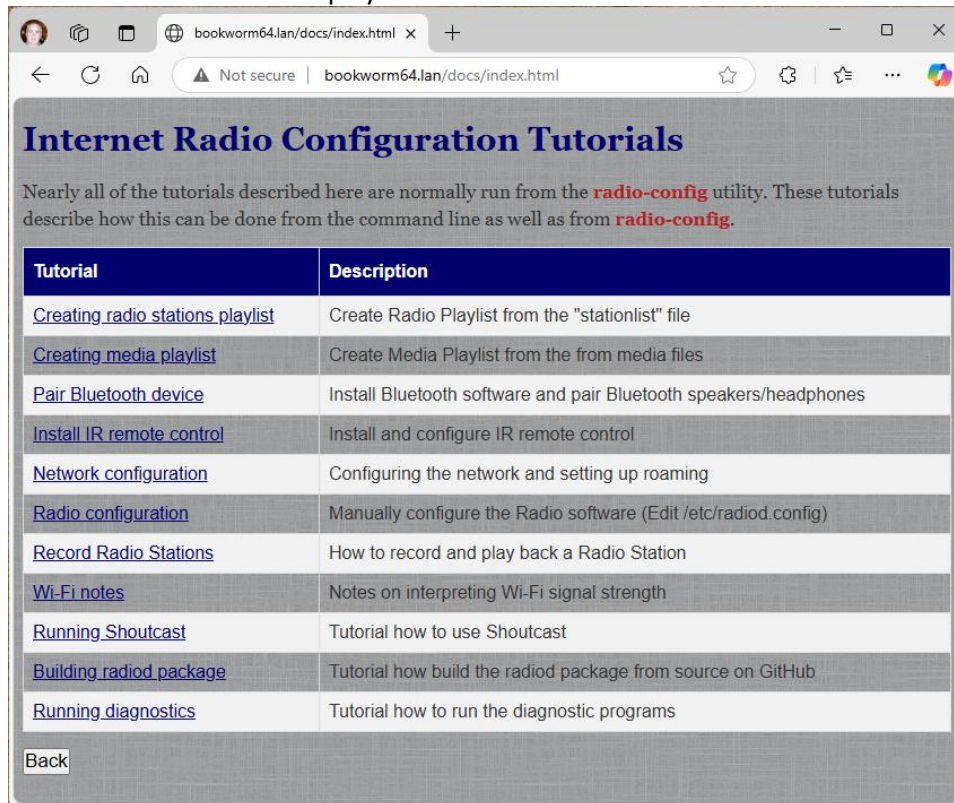


Figure 212 Tutorials and Documents index

In the following example the **media_playlist.html** file has been selected.

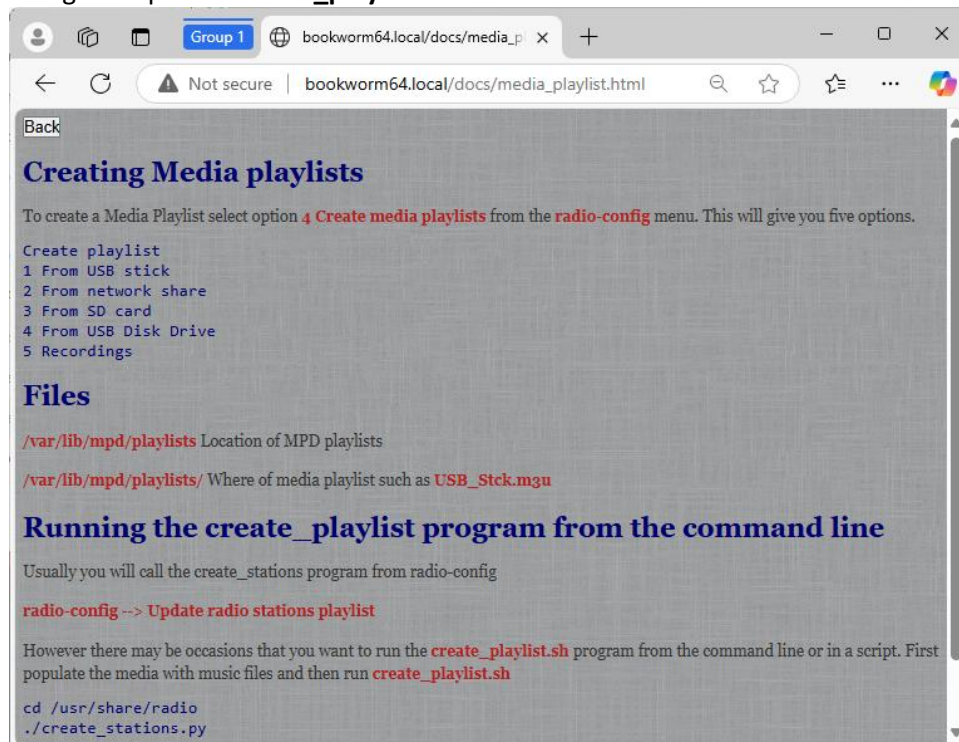


Figure 213 Example Web tutorial

The “Back” buttons at the top and bottom of the page can be used to return to the document index.

Installing the IR remote control software

Before starting, the **TSOP382xx** series IR Sensor needs to be wired to the correct GPIO pin. The following table shows the correct GPIO pin assignment for the IR receiver depending upon the hardware being used. Configuration commands shown later use the GPIO number shown in bold in the table below.

Table 15 IR Remote Control Sensor Pin outs

Radio Type	Pin	GPIO	Type of Raspberry PI
Two- or Four-line LCD with Push Buttons	21	9	Any (No DAC)
Two- or Four-line LCD with Rotary encoders	21	9	Any (No DAC)
Two- or Four-line LCD with I2C backpack	21	9	Any (No DAC)
Adafruit RGB plate with push buttons	36	16	40-pin version only
All versions using DAC sound cards	22	25	40-pin version only
IQaudio Cosmic Controller and OLED display	22	25	40-pin version only



Note: If you have wired your IR sensor to a different GPIO number other than shown in the above table, simply use any of the above GPIO numbers and then later amend the **dtoverlay=gpio-ir,gpio_pin=x** in **/boot/firmware/config.txt** where **x** is the GPIO number you have used.

The service that handles the IR sensor and its associated IR activity LED is called **ireventd**. It is installed at the same time as the radio software but needs to be configured as shown in the following pages.

If you haven't already done so update the operating system first as shown in the section Update to the latest the system packages on page 98. To configure the IR remote control software:

- Run the **radio-config** utility.
- Select option **8 Install/configure drivers and software components** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.
- Select option **1 Install IR remote control** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.

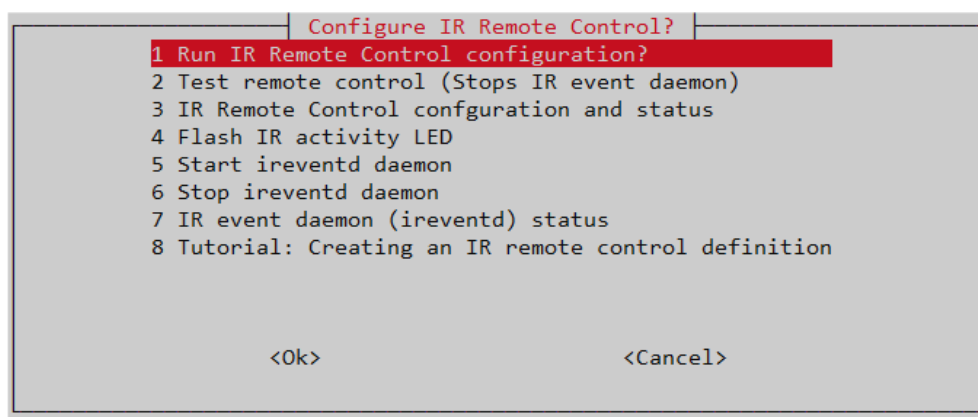


Figure 214 IR Remote Control Installation program

It is now necessary to select which GPIO is to be used for the IR sensor. This is either 9, 16 or 25 as shown in *Table 15 IR Remote Control Sensor Pin outs* above.

Select option **1 Run IR Remote Control configuration** from the menu. Select the correct GPIO pin configuration. This must match how you have physically wired up the IR sensor.

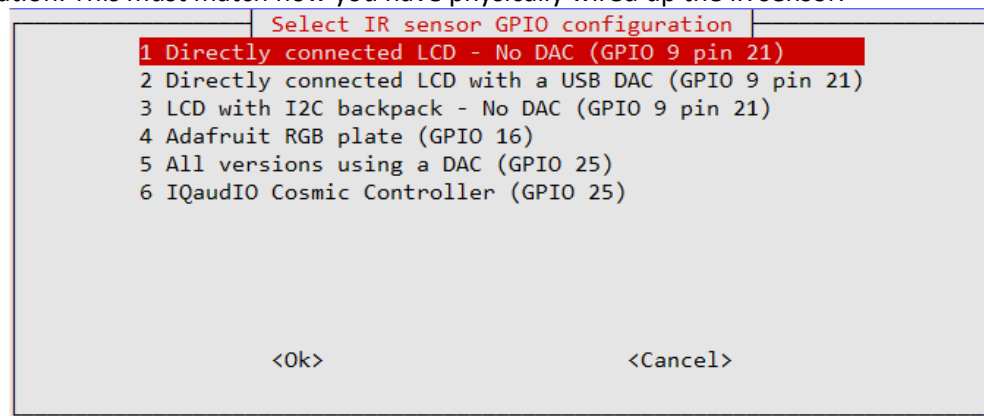


Figure 215 IR configuration IR sensor GPIO selection

Now select the Remote Activity LED GPIO. This is either GPIO 11, 13, 14 or 16. Again this must match up how you have physically wired the IR activity LED.

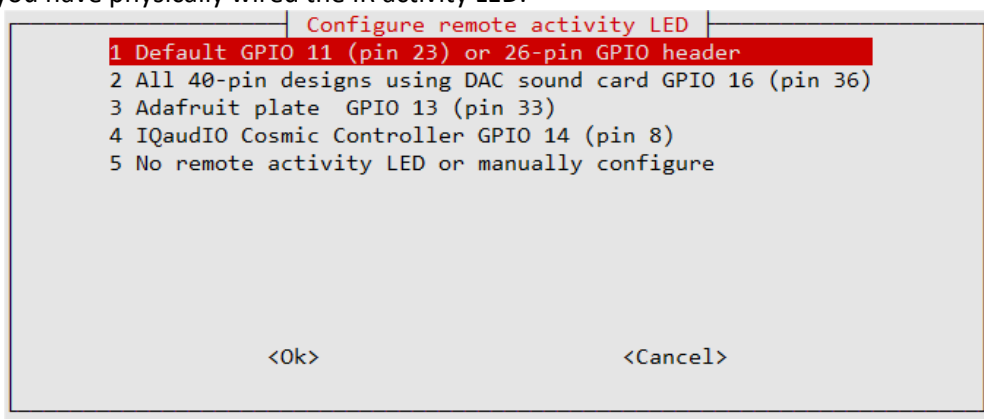


Figure 216 IR configuration Activity LED GPIO selection

Once configuration selection has been confirmed the program displays the IR remote control definition file selection. It lists the file that it finds in the **/usr/share/radio/remotes** directory.

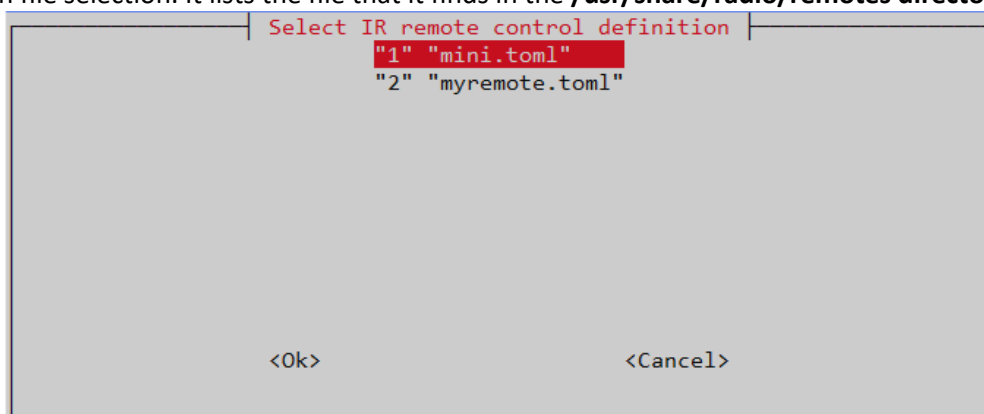


Figure 217 IR remote control definition file selection

If you are using the Raspberry Pi mini IR remote control then select "mini.toml".

Once the file selection has been confirmed the program will install the Kernel Event components and configure the **/boot/firmware/config.txt** file. It installs **ir-keytable** and **python3-evdev** packages.

The program will display the following instructions to complete the set-up process. In the following it gives the example for the Kernel events service.

```
:  
A log of this run will be found in /usr/share/radio/logs/install_ir.log  
Reboot the Raspberry Pi or run the following command:  
sudo systemctl restart ireventd radiod
```

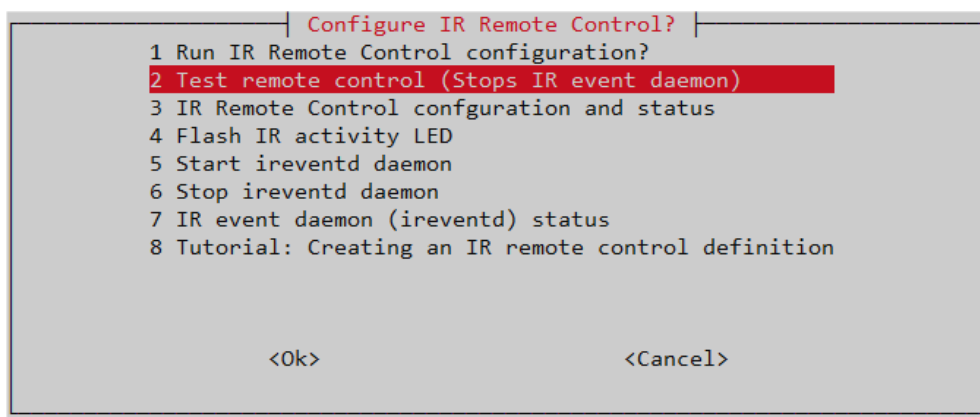
The program adds the **gpio-ir** dtoverlay to the **/boot/firmware/config.txt** file. The **gpio_pin** varies with the selection you made. For example:

```
dtoverlay=gpio-ir,gpio_pin=25
```

Reboot the radio:

```
$ sudo reboot
```

After reboot check that the correct IR daemon is running. Re-run radio-config and select the IR Remote Control configuration menu as previously shown. Now select option **2 Test remote control**. Note that this will stop the IR event daemon (ireventd)



This runs the test_events.py program which will display the following

```
Press Ctl-C to end test  
/dev/input/event0 vc4-hdmi-0 vc4-hdmi-0/input0  
/dev/input/event1 vc4-hdmi-0 HDMI Jack ALSA  
/dev/input/event2 vc4-hdmi-1 vc4-hdmi-1/input0  
/dev/input/event3 vc4-hdmi-1 HDMI Jack ALSA  
/dev/input/event4 gpio_ir_recv gpio_ir_recv/input0  
IR input device found at /dev/input/event4  
Boot configuration in /boot/firmware/config.txt  
dtoverlay=gpio-ir,gpio_pin=25  
Press Ctl-C to end events test  
Waiting for IR events
```

Now press the buttons on the Remote Control that you configured. Output similar to the following should be displayed.


```
KEY_CHANNELUP 0x192 1
KEY_CHANNELDOWN 0x193 1
KEY_RIGHT 0x6a 1
KEY_LEFT 0x69 1
KEY_VOLUMEUP 0x73 1
KEY_VOLUMEDOWN 0x72 1
KEY_MUTE 0x71 1
```

If you are not using the Raspberry Pi mini remote control the you will have to create an IR remote control definition using **ir-keytable** program to handle IR remote control events. See the tutorial in the IR Remote Control configuration menu.

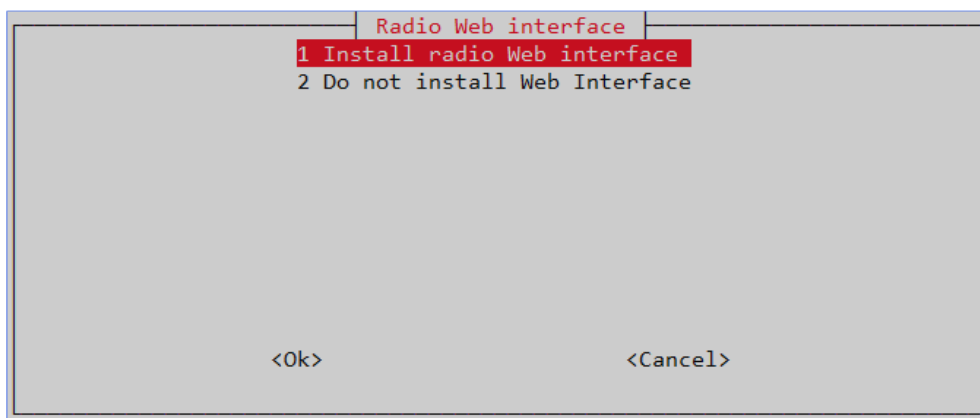
The configuration files for **ir-keytable** have the **toml** extension. These can be listed with the following command:

```
$ ls /lib/udev/rc_keymaps/
adstech_dvb_t_pci.toml  encore_enltv.toml  pixelview_002t.toml
af9005.toml             evga_indtube.toml  pixelview_mk12.toml
alink_dtu_m.toml       eztv.toml          pixelview_new.toml
:
```

This will list over 430 different remote controls and if you are lucky, it may find one that matches the remote control you are using. If not, create one using the following method.

Installing the Web Interface

- Run the **radio-config** utility.
- Select option **8 Install/configure drivers and software components** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.
- Select option **2 Install Web Interface** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.



Select option **1 Install radio Web interface** or option 2 to exit. The following output will be displayed as Apache Web Server and components are installed.

```

/usr/share/radio/scripts/install_web_interface.sh configuration log, Sun 18
Apr 08:03:25 GMT 2025
Installing radio Web interface for Trixie OS
Installing Apache Web server
Reading package lists...
Building dependency tree...
:
: {Output omitted}
:
Reading package lists...
Building dependency tree...
Reading state information...
0 upgraded, 0 newly installed, 0 to remove and 107 not upgraded.
Setting up MariaDB database
PASSWORD('raspberrypi')
*1844F2B11CCA3F3B31F573A1384F608BB6DE3DF9
A log of these changes has been written to
/usr/share/radio/logs/install_web.log

```

Starting the radio Web interface

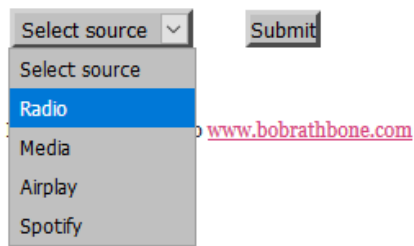
Version 3.x is used in this example. Point your web browser at the IP address of the Raspberry Pi. For example: <http://192.168.1.251> you should see the following display :



Figure 218 Version 3.x Radio Web interface

Use the drop down box under “Select source” as shown in the following section:

Source Selection



The example on the left shows four music sources that can be selected namely Radio, Media, Airplay and Spotify. If installed it will also display Shoutcast.

The desired source can be selected from the source drop-down selection box. Click on the required source then click on 'Submit' button to load the selected source in the radio. If you have more than one Media or Radio playlist, then repeatedly clicking on the appropriate source and Submit button will cycle through the playlists for that source. The name of the new playlist, however, is not displayed.

The Shoutcast tab is explained in *Using the Shoutcast Web Interface* on page 201.

Now select the **Radio** tab. This will display **O!MPD** login screen Logging in as user **admin**, password **admin**.

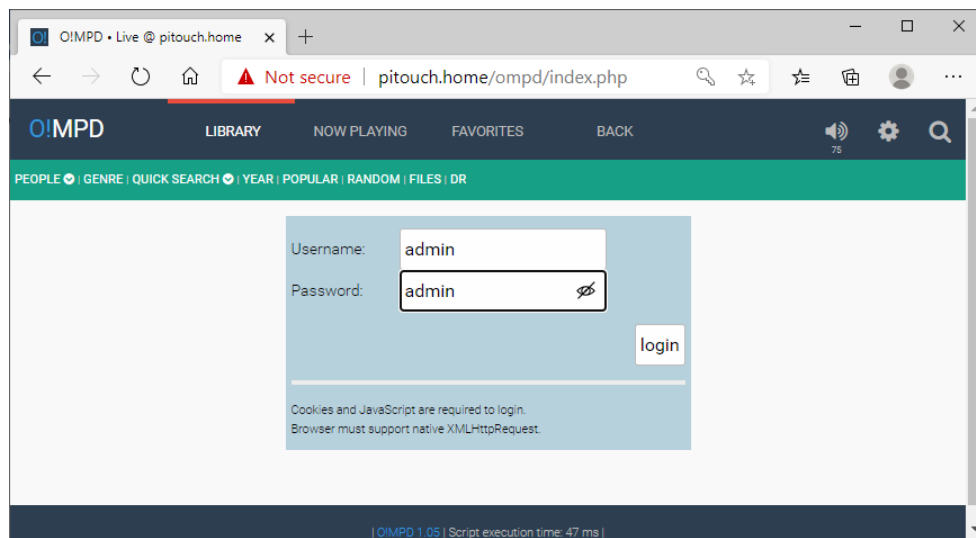
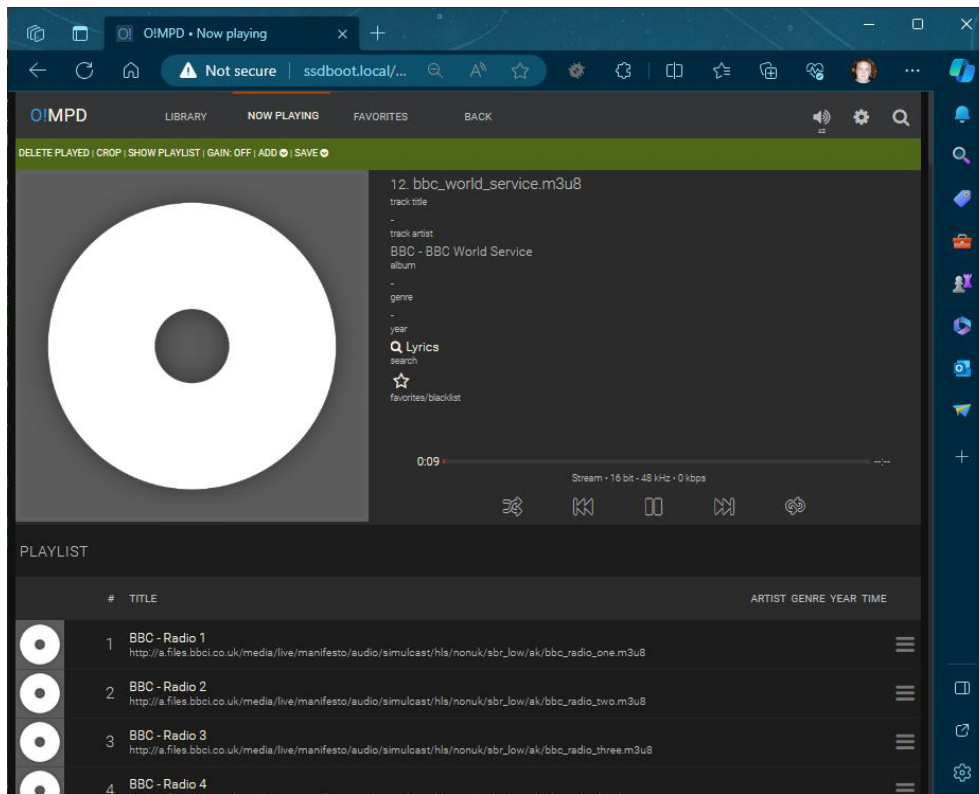
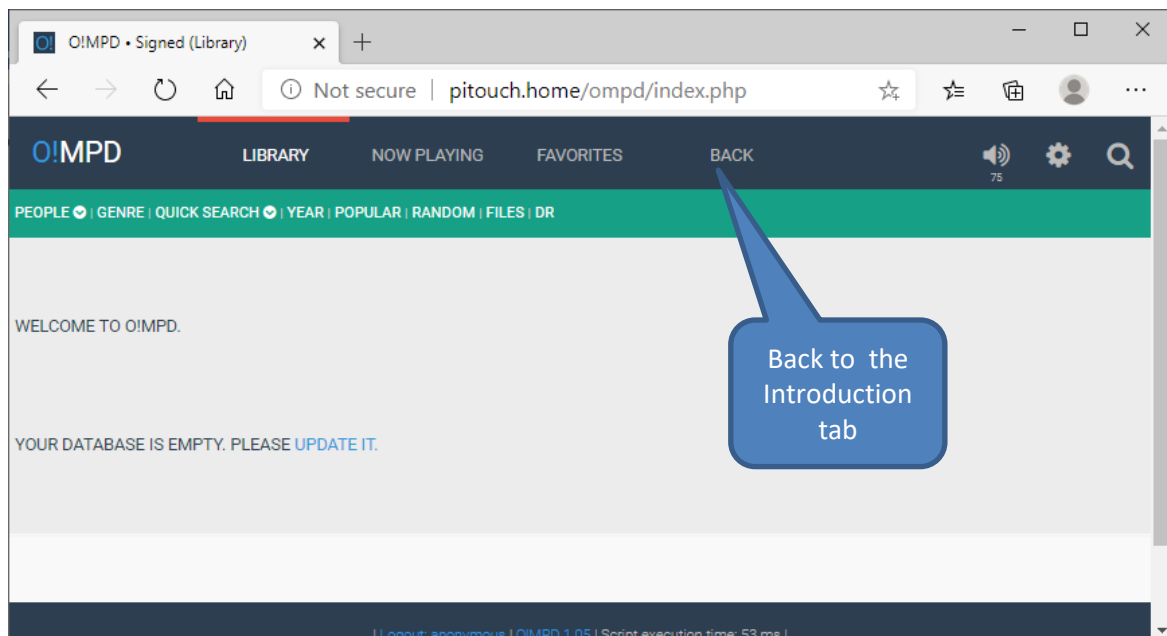


Figure 219 O!MPD Login screen

Now select the **NOW PLAYING** tab.



Note: The radio tab only displays radio stations or media tracks from MPD. It is not currently capable of displaying **Spotify** or **Airplay** details which can only be seen on the radio itself.



Note that there is an extra tab called **O!MPD** when compared to the **Snoopy** interface. If you have already created a music library by running the **create_playlist.sh** program then it is necessary to update the **O!MPD MySQL** database. Otherwise skip these instructions.



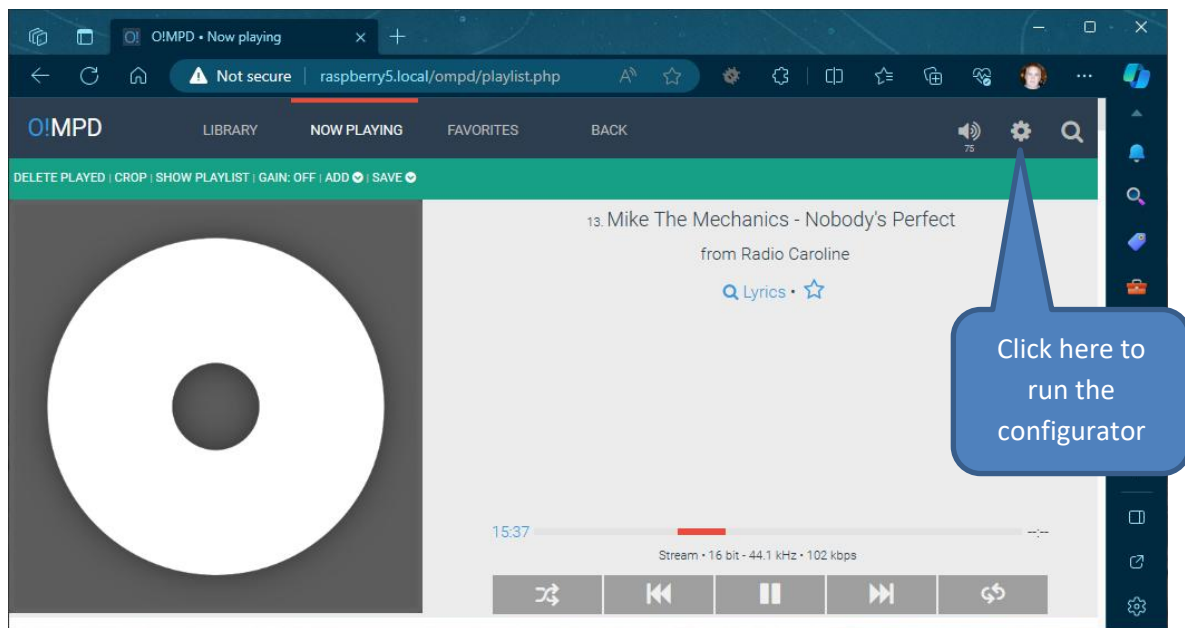
Note: The “(i) Not secure” message in the above screens can be ignored. It is because the web pages do not have a security certificate to verify that the pages come from a legitimate source and also are not using the HTTPS protocol. Since these pages are not coming from a source on the Internet and are locally installed, they do not pose a security risk and come from a reliable source namely the **Bob Rathbone radioweb** package.

Setting up the Music directory in OM!PD

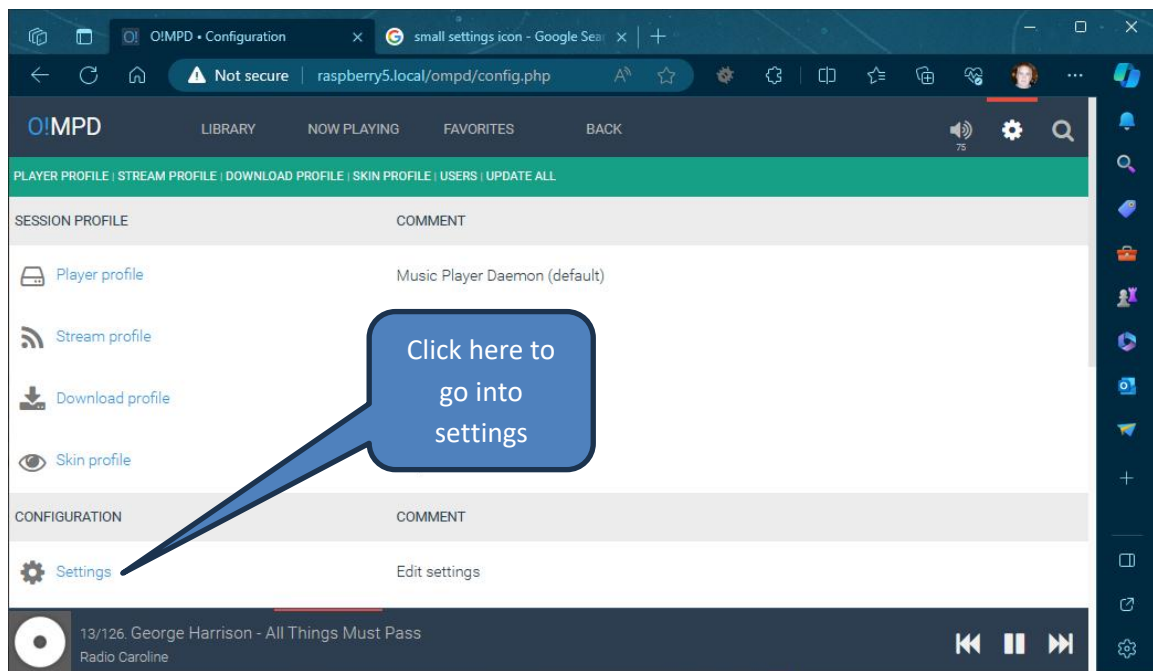
Before you can access music in the media libraries (US Stick, SDcard and remote shares) you need to tell **OM!MPD** where the music is stored (or at least check it). MPD by default stores its music in **/var/lib/mpd/music**. The radio software adds soft links to the actual locations of the media files as shown in the example below.

```
$ ls -la /var/lib/mpd/music
total 8
drwxr-xr-x 2 root root 4096 Mar 15 07:32 .
drwxr-xr-x 5 mpd audio 4096 Mar 15 07:42 ..
lrwxrwxrwx 1 root root 9 Mar 15 07:32 media -> /media/pi
lrwxrwxrwx 1 root root 16 Mar 15 07:20 sdcard -> /usr/share/Music
lrwxrwxrwx 1 root root 6 Mar 15 07:32 share -> /share
```

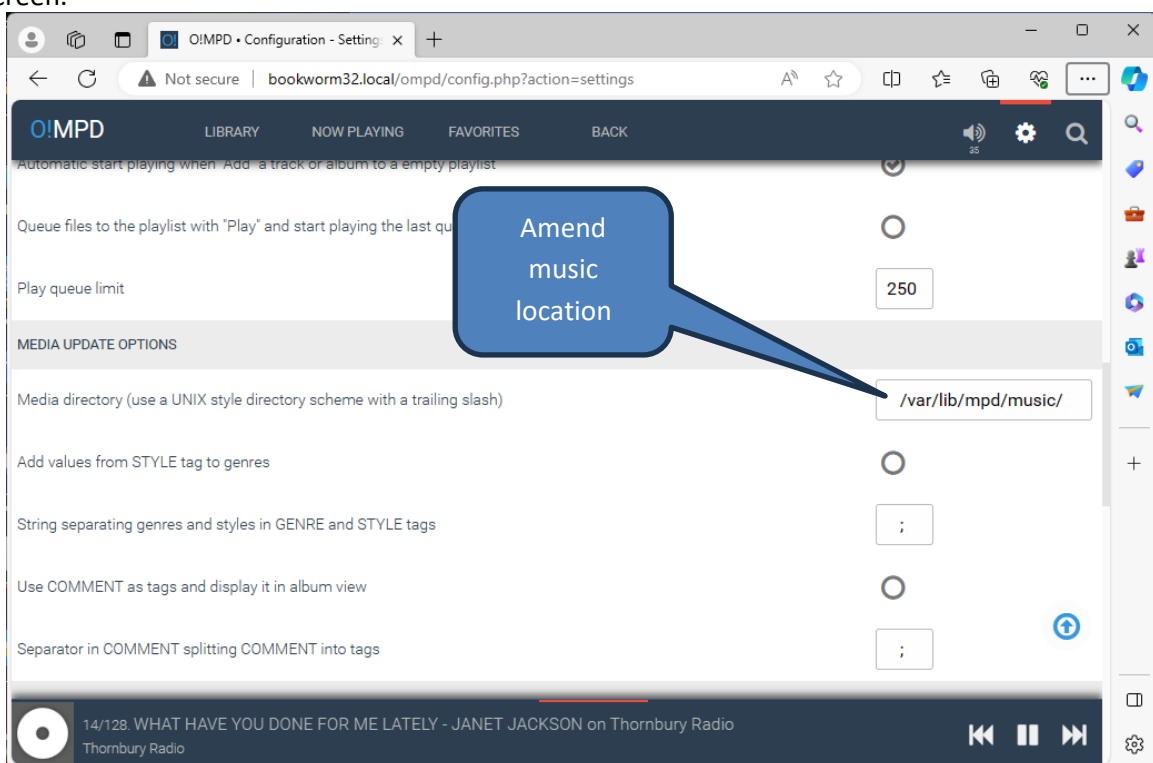
OM!MPD needs to be told where the music media is stored. This is now pre-configured but you should check it. Click on the settings icon as shown in the next screen.



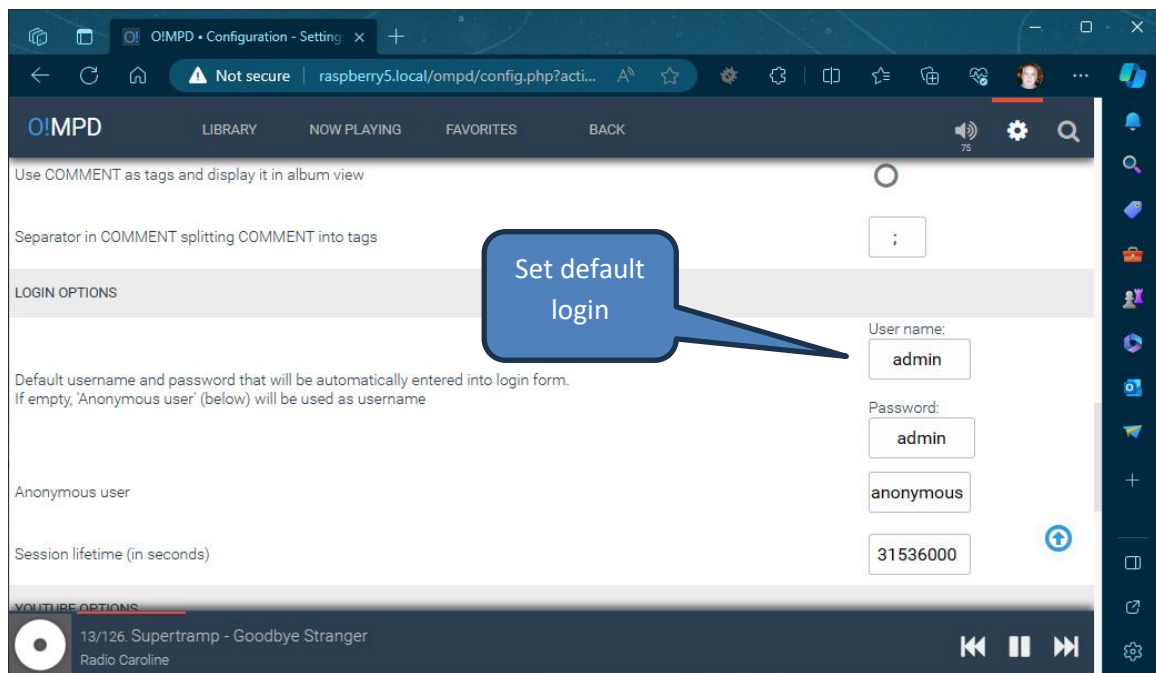
Now click on the CONFIGURATION → Settings Icon.



Once the configuration screen opens, scroll down to MEDIA UPDATE OPTIONS as shown in the next screen.



If missing, enter `/var/lib/mpd/music/` into the location box as shown above. Do not forget to add the trailing `/` character to the end of the name.

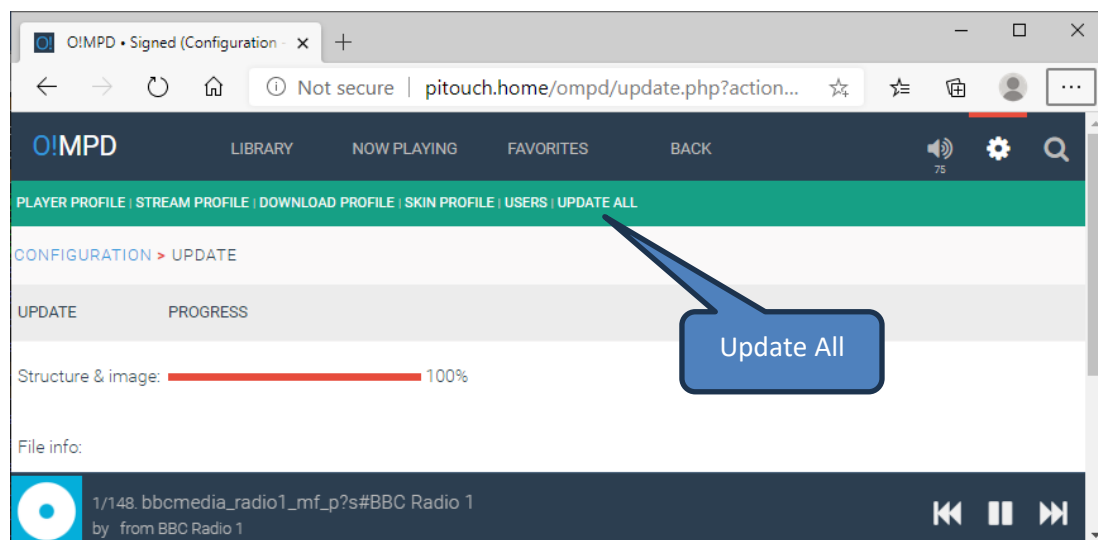


You can also set the default LOGIN OPTIONS on the same screen. Once done you will automatically be logged in as **admin** password **admin**.

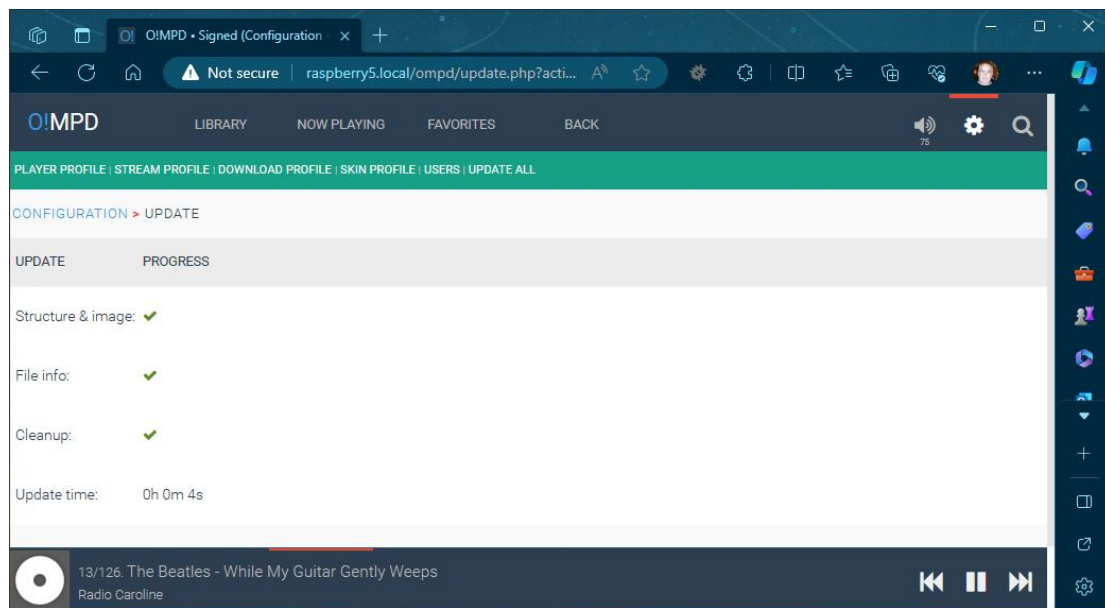


Now scroll down to the to the SAVE SETTINGS button at the bottom of the page and click on it.

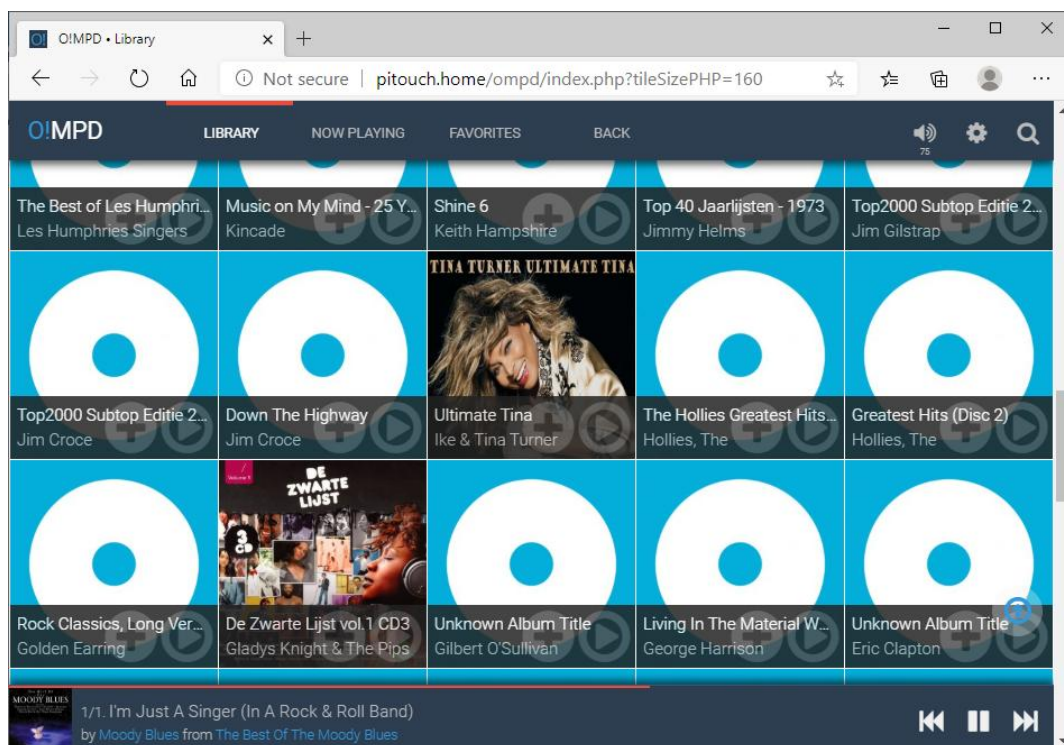
Finally scroll back up to the top of the screen and click on the UPDATE ALL menu item. The media library will now be updated:



When the update is finished the following screen will be displayed.



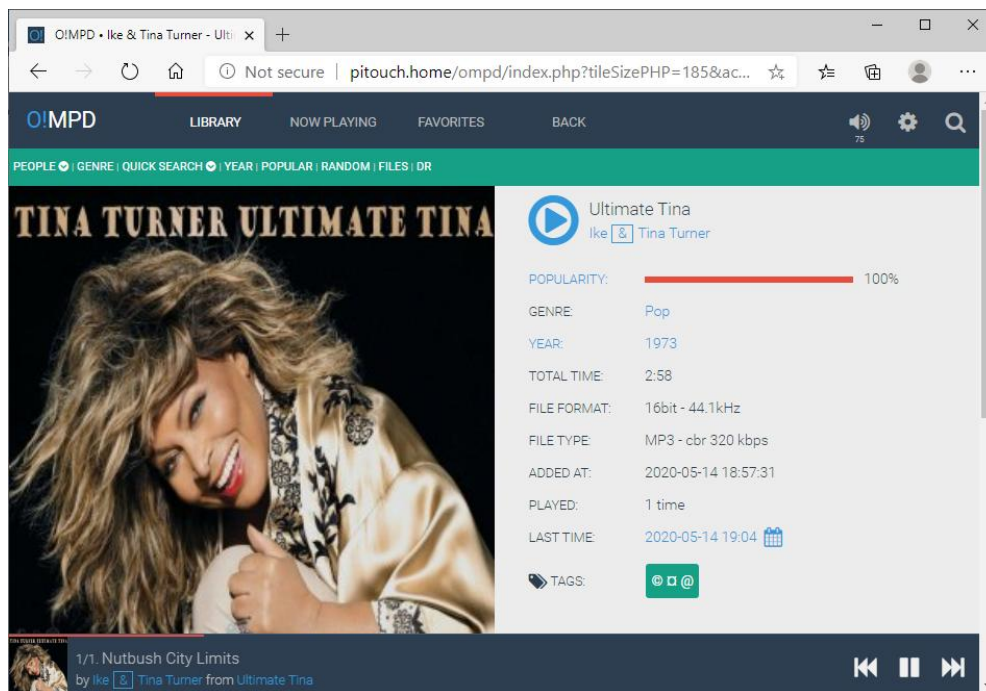
When the update is finished click on the **[NOW PLAYING]** tab to display the contents of the media library.



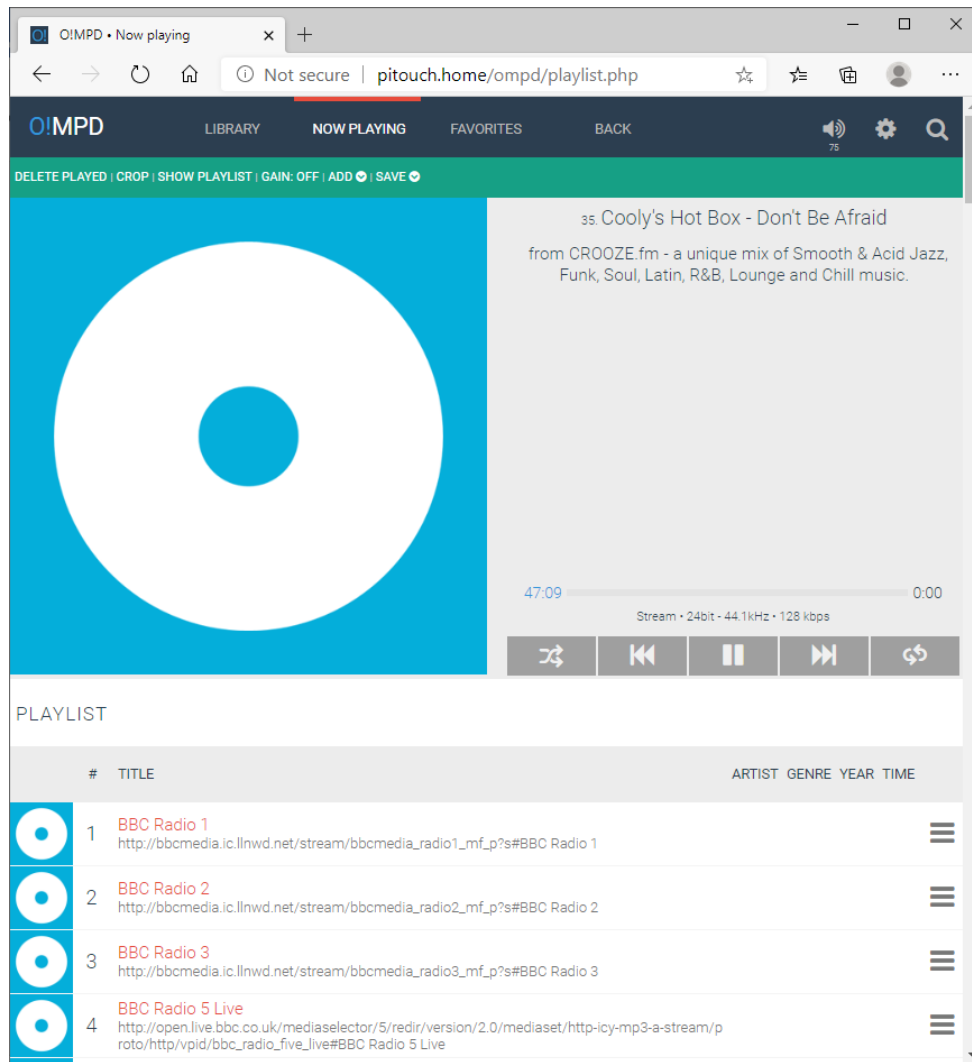
Clicking on one of the tracks will display its details along with the album artwork.



Note: The **O!MPD** Web Interface was written by **Artur Sierżant**, from Poland. **O!MPD** extracts its artwork from the media file that it is currently playing. However, the full-feature graphics version of the Bob Rathbone software (**gradio.py**) gets its artwork from the **Discogs** database. See section *Radio Station Artwork display* on page 187. This is because, as you can see from the illustration above, all too often no artwork is included in the stream.



Clicking on the blue play icon will play the track. To play a radio station click on the “NOW PLAYING” top menu item. Select the required radio station to start playing the stream.



Changing the Web Interface Radio photo

If you want to change the photo displayed by the Web interface on the first page, then replace the **jpeg** photo file at **/var/www/html/images/radio.jpg**. Try to adjust the size on disk to about 50K using a suitable photo editor such as Photo Shop. Copy the new jpeg photo to the pi home directory with any ftp program.

Now copy it to the Web pages image directory using **sudo**.

```
$ sudo cp radio.jpg /var/www/html/images/.
```

If the new image looks stretched then it may also be necessary to change image proportions in the `<img.>` statement in **/var/www/html/index.html** (v2.3) or **/var/www/html/index.html** (v3.x). Find the following line in the index file and adjust the width/height values to display the photo with the correct proportions.

```
</td>
```

Airplay (shairport-sync) Installation

If you have not already done so, carry out a system and firmware upgrade, as shown in *Preparing the Operating System* on page 98. To install Airplay which uses the **shairport-sync** package carry out the following.

- Run the **radio-config** utility.
- option **8 Install/configure drivers and software components** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.
- Select option **8 Install Airplay (shairport-sync)** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144. The following menu will be displayed.

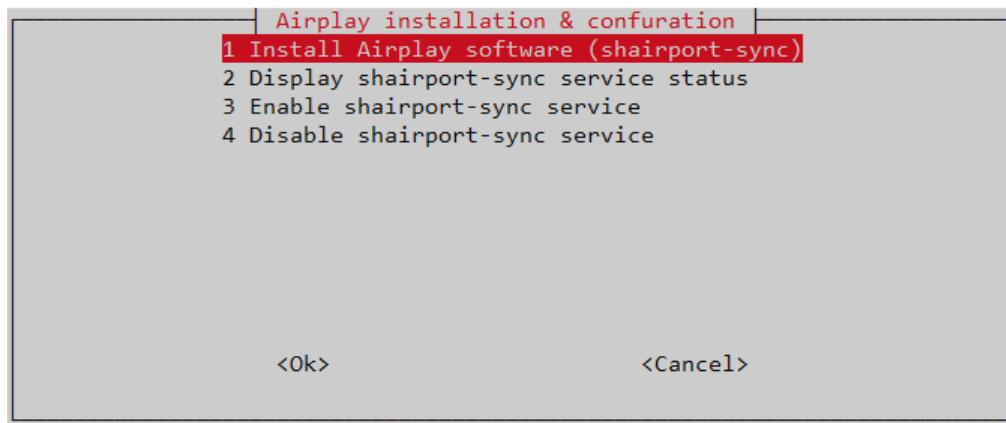


Figure 220 Airplay (shairport-sync) installation menu

Select option 1 Install Airplay software (shairport-sync) to install the Airplay software.

Using Airplay on the radio

Using Airplay on a HDMI/Touchscreen is described in the section called *Running Airplay on the HDMI touchscreen* on page 190. LCD versions of the radio are described here.



Figure 221 Airplay source selection

Press the menu button until **Input Source:** is displayed.

Turn the channel button (or Up/Down switches on a push-button radio) until **Airplay receiver** appears.

Press the menu button one more time. The word Airplay will be displayed on the bottom line along with 'Unknown artist' and 'Unknown title'.

Now use the Airplay device to connect to the raspberry PI (varies according to device software). Start playing the music tracks and this should start being heard on the radio which also displays the Artist, Track and Album on the LCD display. The volume is adjustable if correctly set-up. The mute also works in the normal way but does not pause or stop the Airplay stream as this can only be done from the device running Airplay.



Figure 222 Running an Airplay device on the radio with Cloudbreak

The above example is using an evaluation copy of CloudBreak running on an Android mobile telephone.

Unfortunately, Cloudbreak is no longer available however there are a number of Airplay Apps available for Android telephones such as “Double twist”.

Install Icecast streaming

You may wish to play the output of the Radio through your PC speakers or a mobile device such as a tablet or telephone. This is possible to do this using **Icecast**. For more information on **Icecast** see the following Wikipedia article: <http://en.wikipedia.org/wiki/Icecast> .

Please also refer to *Intellectual Property, Copyright, and Streaming Media* on page 237.

To install Icecast streaming:

- Run the **radio-config** utility.
- Select option **8 Install/configure drivers and software components** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.
- Select option **4 Install Icecast** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.

```
Starting Icecast2 integration with the Music Player Daemon
The Icecast2 installation progra will ask if you wish to configure Icecast2.
Answer 'yes' to this. Configure Icecast as follows:
Icecast2 hostname: localhost
Icecast2 source password: mympd
Icecast2 relay password: mympd
Icecast2 administration password: mympd
Continue y/n: y
```


Enter 'y' to continue. The Icecast2 installation program will ask if you wish to configure Icecast2:

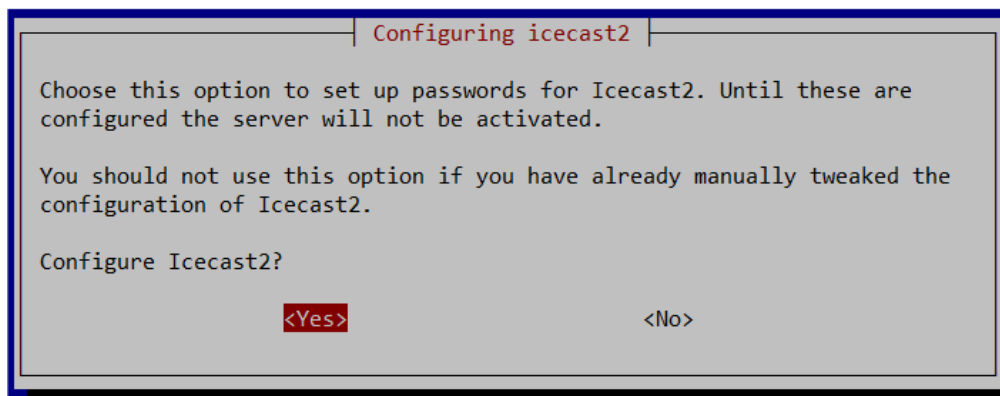


Figure 223 Configuring Icecast2

Answer '**yes**' to this. Configure Icecast as follows:

Icecast2 hostname: **localhost** (or the hostname of the Raspberry Pi)

Icecast2 source password: **mympd**

Icecast2 relay password: **mympd**

Icecast2 administration password: **mympd**

It is important that you replace the default password 'hackme' with 'mympd'. The installation program continues configuration. The icecast2 server will be started:

```
Done Configuring icecast2..
Processing triggers for libc-bin (2.19-18+deb8u6) ...
Processing triggers for systemd (215-17+deb8u5) ...
Configuring Icecast2
Copying /etc/icecast2/icecast.xml to /etc/icecast2/icecast.xml.orig
```

The Icecast2 `install_streaming.sh` script sets the **streaming_on** parameter in `/etc/radiod.conf` to yes.

```
streaming_on=yes
```

It also adds the following configuration to `/etc/mpd.conf`.

```
# MPD Radio Stream
audio_output {
    type          "shout"
    name          "PI Radio MPD Stream"
    description    "MPD stream on Raspberry Pi Radio"
    host          "localhost"
    port          "8000"
    mount         "/mpd"
    password      "mympd"
    bitrate       "128"
    format        "44100:16:2"
    encoding      "mp3"
}
```

This completes the installation of Icecast2

Installing the Speech facility(espeak)

It is possible to configure speech for visually impaired and blind persons who cannot read the display. As channels are changed or stepping through the menu the radio will “speak” to you.

- To install and configure **espeak** run **radio-config** from the command line
- Select option **8 Install/configure drivers and software components**
- Then select option **5 Install Speech facility**

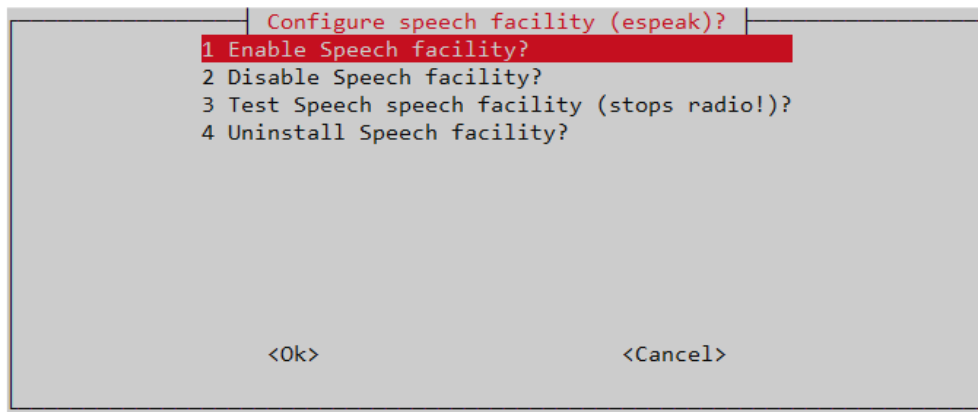


Figure 224: Enabling the speech facility

Select option **1 Enable Speech facility?** to install and enable the **espeak** software. The following screen will be displayed.

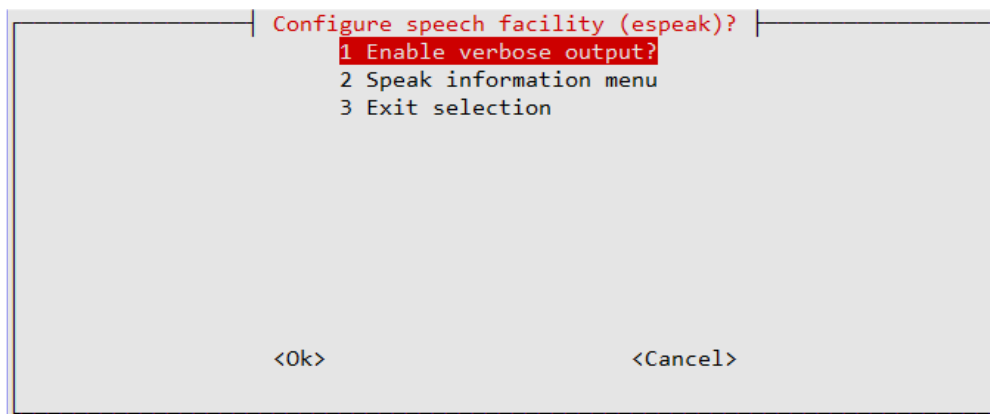


Figure 225 Speech verbose levels

Select option **1 Enable verbose output?** To enable basic level of speech (menu's and station/track names), or option **2 Speak information menu** to speak contents of information menu when selected. Option **3** exits without changing anything.

Once the **espeak** software has been installed it can be tested using option **3 Test Speech speech facility**. When option **3** is selected you should hear a voice saying 'Speech facility enabled'.

You can disable the speech facility without uninstalling the software by selecting option **2 Disable Speech facility**. For further information on **espeak** see <https://espeak.sourceforge.net/>

Install Luma OLED/TFT driver:

To install the Luma TFT drivers:

- Run the **radio-config** utility.
- Select option **8 Install/configure drivers and software components** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.
- Select option **7 Install Luma OLED/TFT driver** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.

On confirmation the following will be displayed

```
/usr/share/radio/scripts/install_luma.sh configuration log, Thu Nov 28
12:06:35 GMT 2024
Using /usr/share/radio
Reading package lists...
Building dependency tree...
Reading state information...
:
A log of these changes has been written to
/usr/share/radio/logs/install_luma.log
```

This completes the installation of the Luma drivers.

Install recording utility (liquidsoap)

From version 8.0 onwards it is possible to record Radio stations which are transmitted using Shout cast or Icecast format. To do this it is necessary to install the **liquidsoap** package and to select the **GPIO** setting for the **Record** button.

Note: **liquidsoap** only works on **Bookworm 64-bit OS systems** and not on **32-bit** systems. Please also see the section called *Legality of recording radio streams* on page 238.

To install the Radio station recording utility (liquidsoap):

- Run the **radio-config** utility.
- Select option **8 Install/configure drivers and software components** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.
- Select option **7 Install recording utility (liquidsoap)** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.

After confirmation the following screen will be displayed:

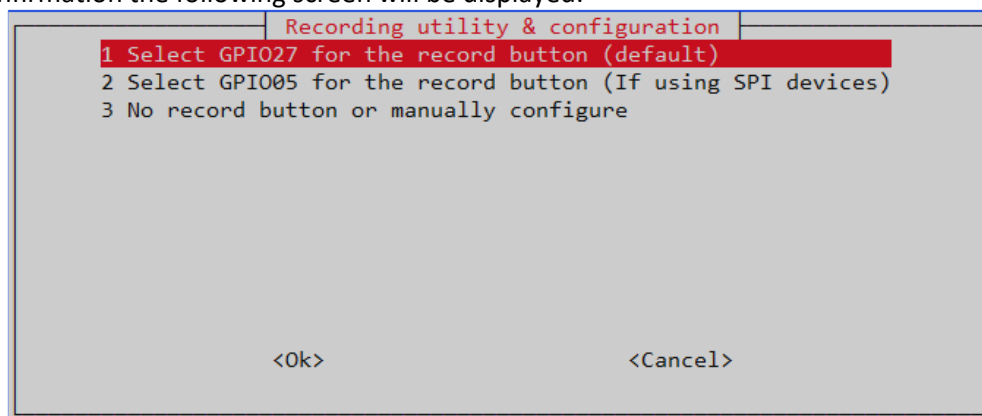


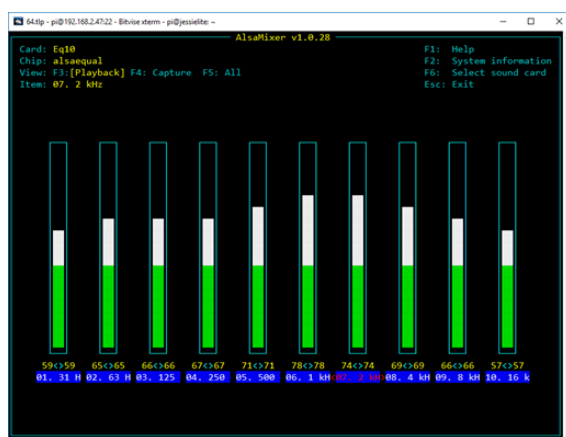
Figure 226 Installing the station recording utility (liquidsoap)

The radio design now includes an additional 6th button for recording. Normally this uses the **GPIO27** pin (physical pin 13). Select this when using LEDs, OLEDs or I2C displays. If using any displays that use the SPI interface such as the **Waveshare 2.4"** or **1.5" TFTs** using the **SPI** interface select option 2 **GPIO5** (physical pin 29). If you do not want to configure a Record button or wish to configure this manually later, select option 3. Confirm your selection. The installation program will then install **liquidsoap** and configure the **record_switch** parameter to use the selected GPIO setting.

```
Installing liquidsoap recording software
sudo apt-get install liquidsoap
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
:
0 upgraded, 1 newly installed, 0 to remove and 56 not upgraded.
Configured record_switch in /etc/radiod.conf
record_switch=27
End of installation. Press enter to continue:
```

Reboot the radio. Now when either the Record button on the radio or IR remote control are pressed, these will start the recording process. At the end of recording process creates the **Recordings** playlist which can then be accessed via the Radio menu the same as any other playlist.

Install the Alsa Mixer software



The Alsa mixer (**alsamixer**) only runs in an **X-Windows** or **Wayland** desktop. It is normally launched from the **gradio** or **vgradio** radio programs. It allows individual adjustment of the audio frequencies.

It cannot currently be used with Bluetooth audio devices nor with the Waveshare WM8960 audio DAC as this requires its own special **/etc/asound.conf** file.

To install the Alsa mixer software (**alsamixer**):

- Run the **radio-config** utility.
- Select option **8 Install/configure drivers and software components** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.
- Select option **8 Install Alsa equalizer software** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.

Editing configuration files manually

Whilst most configuration of the **radiod** software is done **radio-config** utility it will occasionally be necessary to configure the configuration files manually. To do this do the following:

- Run the **radio-config** utility.
- Select option **7 Edit configuration files** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.

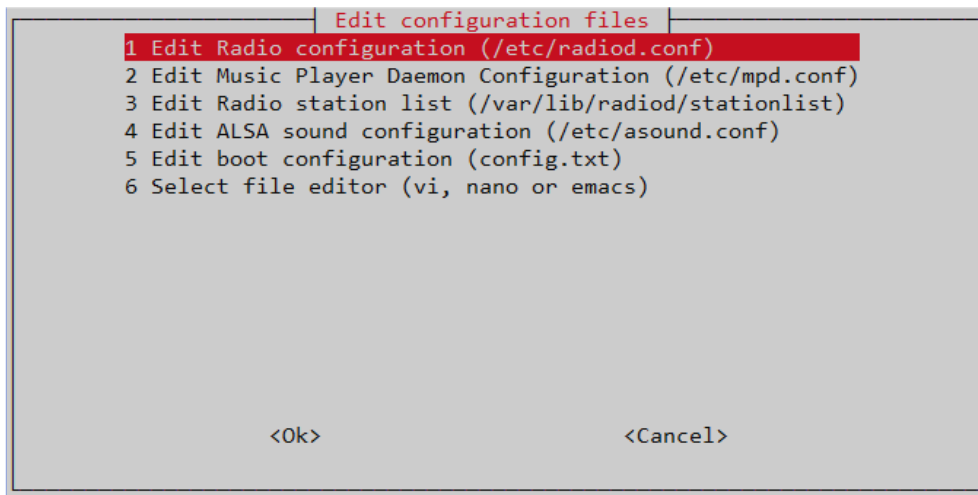
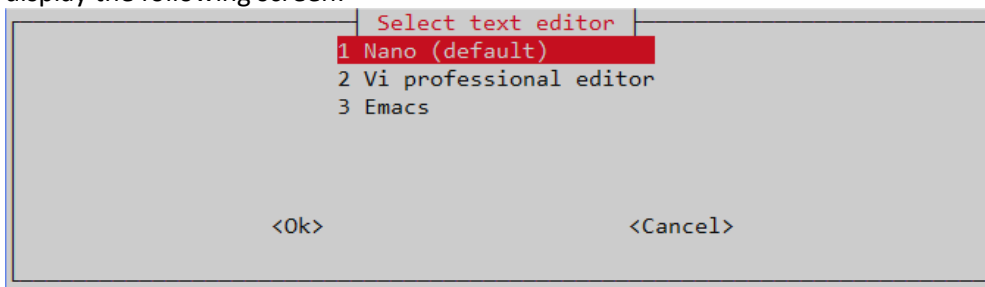


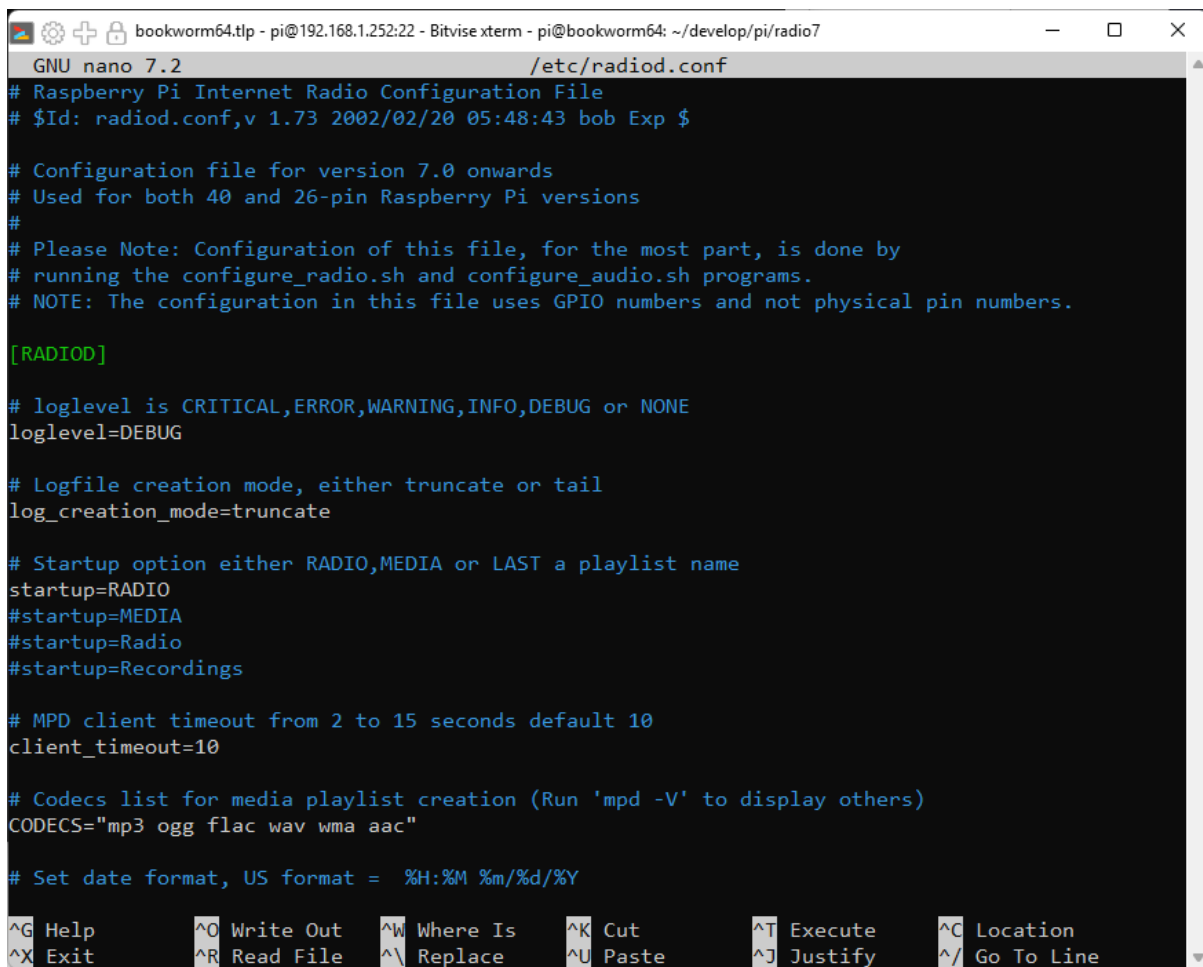
Figure 227 Editing configuration files

Before you can edit any of the files you must select the text editor that you wish to use. There are three choices; **Nano** (for beginners), **Vi** (Advanced users) and **Emacs** (Very advanced editor)

Select option **6 Select file editor (vi, nano or emacs)** to select which text editor you wish to use. This will display the following screen:



If you are a beginner then you should select option **1 Nano (default)** and then select **<Ok>**. Once you have selected the text editor that you wish to use then return to the previous screen. Select the configuration file you wish to edit for example: **1 Edit Radio configuration (/etc/radiod.conf)** The **/etc/radiod.conf configuration filer** will be displayed in the editor with some basic instructions at the foot of the page.



```
GNU nano 7.2 /etc/radiod.conf
# Raspberry Pi Internet Radio Configuration File
# $Id: radiod.conf,v 1.73 2002/02/20 05:48:43 bob Exp $

# Configuration file for version 7.0 onwards
# Used for both 40 and 26-pin Raspberry Pi versions
#
# Please Note: Configuration of this file, for the most part, is done by
# running the configure_radio.sh and configure_audio.sh programs.
# NOTE: The configuration in this file uses GPIO numbers and not physical pin numbers.

[RADIOD]

# loglevel is CRITICAL,ERROR,WARNING,INFO,DEBUG or NONE
loglevel=DEBUG

# Logfile creation mode, either truncate or tail
log_creation_mode=truncate

# Startup option either RADIO,MEDIA or LAST a playlist name
startup=RADIO
#startup=MEDIA
#startup=Radio
#startup=Recordings

# MPD client timeout from 2 to 15 seconds default 10
client_timeout=10

# Codecs list for media playlist creation (Run 'mpd -V' to display others)
CODECS="mp3 ogg flac wav wma aac"

# Set date format, US format = %H:%M %m/%d/%Y

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

Figure 228 Editing /etc/radiod.conf configuration file with Nano

Use the four arrow keys on the keypad to navigate around the file. Use Ctrl-X to exit. It is also possible to call Nano directly from the command line, for example:

```
$ sudo nano /etc/radiod.conf
```

Configuring the volume display

The volume can be displayed as either text or as a series of blocks. This is configured in /etc/radiod.conf using the **volume_display** parameter. The default is text.

```
# Volume display text or blocks
volume_display=text
```

```
12:01 30/08/2017
WPJR Country
Lonestar - Mr. Mom
Volume 75
```

To display the volume as a series of blocks change this to 'blocks':

```
volume_display=blocks
```



```
12:01 30/08/2017
WPJR Country
Lonestar - Mr. Mom
■■■■■■■■■■
```



If the timer or alarm functions are being used then the volume display reverts back to text display so as to allow display of the alarm or timer values.

Creating a new language file

To create a new language file by running the `language_class.py` program and redirecting the output to a file called `language.<new>` where `<new>` is the country code. For example, to create a language file in Dutch, the country code is `nl` (**Note:** There is already a `language.nl` file so don't overwrite it).

```
$ cd /usr/share/radio/language
$ sudo ./language_class.py > language/language.nl
```

Now edit the text (Not the labels) in the `language/language.nl` file. It isn't necessary to change every message. Lines beginning with `#` are for any comments.

```
# Nederlands text for uitspraak
main_display: Hoofd menu
search_menu: Zoek menu
select_source: Media selecteren
options_menu: Opties menu
rss_display: RSS beeld
information: Informatie beeld
the_time: De tijd is
loading_radio: Radio zenders laden
loading_media: Media laden
search: Zoek
source_radio: Internet Radio
source_media: Muziek selectie
sleeping: Slaapen
```

Finally copy the new language file to `/var/lib/radiod/language` (Omit the country code) and restart the radio.

```
$ sudo cp language/language.nl /var/lib/radiod/language
$ sudo systemctl restart radiod
```

Configuring an RSS feed

To display an RSS feed it is necessary to create the `/var/lib/radiod/rss` file with a valid RSS URL. For example:

```
http://feeds.bbc.co.uk/news/uk/rss.xml?edition=int
```

The above is the RSS for the BBC news feed however any valid RSS feed may be used. If the `/var/lib/radiod/rss` is missing or contains an invalid RSS URL then this mode is skipped when stepping through the menu. The software comes with a valid BBC RSS feed file in the

`/var/lib/radio/rss` file. You can test the feed first by pasting it into your PC's Web browser URL and pressing enter. If configured, the RSS feed will be automatically displayed by stepping through the menus. A useful site with thousands of RSS feeds is <https://rss.feedspace.com>

Installation of the FLIRC USB dongle



Note: This installation procedure is only for the HDMI or Touchscreen display of the radio. If using an LCD display see *Installing an IR sensor and remote control* on page 70.



The FLIRC USB dongle allows the use of any remote control with your Raspberry Pi. In this design it is intended for use with the graphical version of the radio (Touch-screen or HDMI displays). It allows button presses on a remote control to be mapped to the keyboard input of the Raspberry Pi. For example, pressing the volume up button on the remote control will act just like pressing the + key on keyboard (if so mapped). The graphical version of the radio accepts key presses. FLIRC will not work with the LCD versions of the program.

More can be found at the FLIRC Web site: <https://flirc.tv>

Installation documentation can be found at: <https://flirc.tv/ubuntu-software-installation-guide>

To install the FLIRC IR remote control software:

- Run the **radio-config** utility.
- Select option **8 Install/configure drivers and software components** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.
- Select option **10 Install FLIRC IR remote control (X-Windows)** from the **radio-config** menu shown in *Figure 229 Install FLIRC IR remote control* on page 171. Note that you will need to use the down arrow key on your keyboard to see option 10.

The following screen will be displayed:

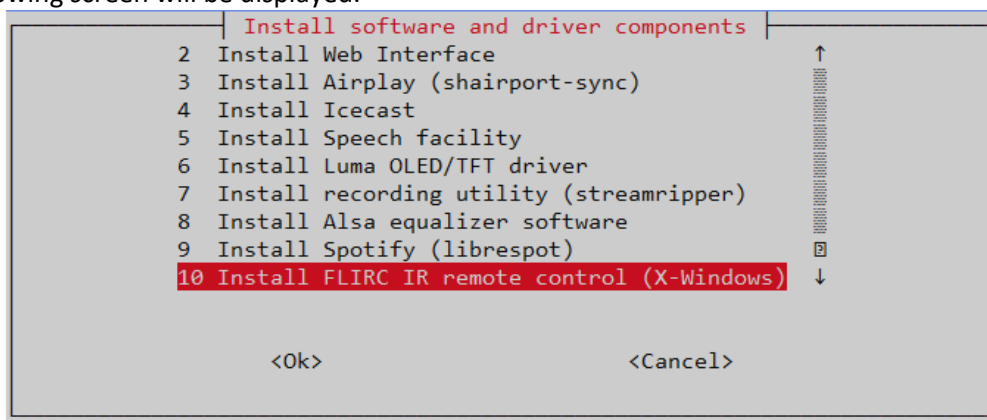


Figure 229 Install FLIRC IR remote control

Select **<Ok>** to continue. The program will display the progress of the installation procedure. When finished reboot the Raspberry pi.

On the desktop open **Programs** (Top left Raspberry Pi icon) → **Accessories** → **Flirc**



Note: At this stage you may see a pop-up window offering a Firmware Upgrade.

Answer Yes to upgrade the firmware in the FLIRC dongle. Failure to do so may result in the Dongle not connecting.

Click on the left-hand program icon (A Raspberry) and select Accessories. In Accessories select **Flirc**. The following screen will be displayed. However, on a 7-inch touchscreen you may not be able to see the whole FLIRC window. In this case use the procedure called Configuring FLIRC from the command line on page 173. The first time you run this program it may ask you if you want to upgrade the firmware. Always upgrade the firmware:

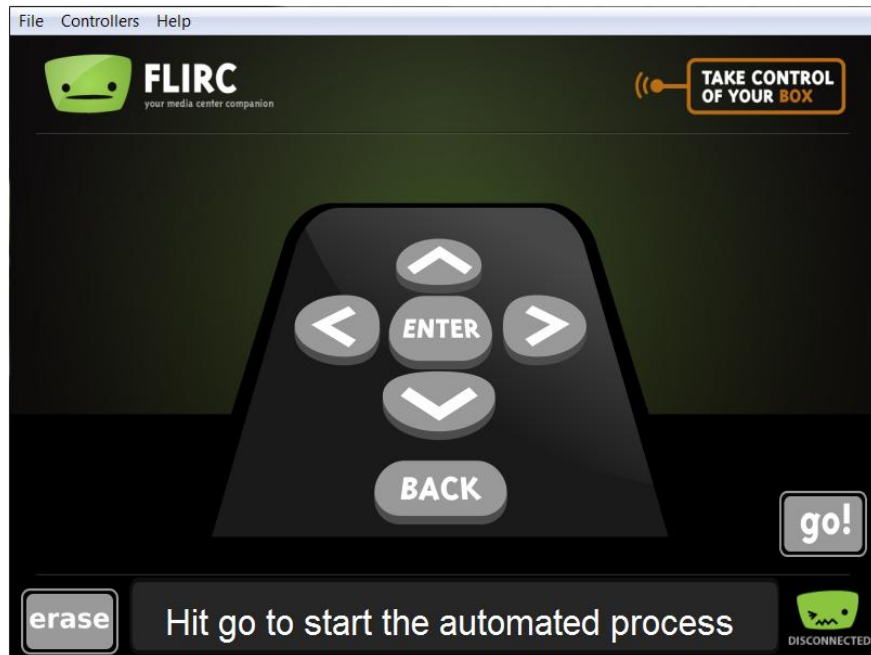


Figure 230 FLIRC setup program

On the **Controllers** drop down menu select the **Full Keyboard** controller.



Figure 231 FLIRC keyboard controller

Now map the buttons on the remote control to the keys shown in Table 17 Graphic screen keyboard command on page 192. For example, press the letter **m** on the above keyboard and then press the Mute button on the Remote Control. For volume control up press Shift key followed by the + key on

the keyboard, then press the volume up button on the remote control. Do the same with the – key for volume down. Full instructions for configuring FLIRC are to be found at:

<https://flirc.gitbooks.io/flirc-instructions/>

Configuring FLIRC from the command line

If using a small touchscreen there may not be enough room to see the Flirc screen. If so, do the following:

- 1) Amend the **fullscreen=yes** parameter to **fullscreen=no** in **/etc/radiod.conf**
- 2) Reboot the Raspberry PI
- 3) When rebooted open a terminal session on the desktop (Don't use remote SSH).
- 4) In the terminal window on the command line run the following:

```
$ flirc_util format
```

Now record the buttons:

```
$ flirc_util record up
Press any button on the remote to link it with '+'
```

'up' is the name of the key. Now press the Channel Up key. The following will be displayed:

```
Successfully recorded button.
```

Repeat the command for each key name.

They are:

```
pageup, pagedown,
+, -,
left, right,
up, down,
return,
l (small letter L), p, a,
r, t, c, s, m and d.
```

In the case of the + and – keys press shift first, followed by the + or – key.

Test and if necessary, repeat key-mapping. If configuring on an HDMI Television do not configure volume (+-) or mute (m) keys as the TV will provide these functions.

The configured keys can be displayed with the **flirc_util keys** command, however this command may be missing from the latest version of **flirc_util**.

```
$ flirc_util keys

Recorded Keys:
Index  hash          key
-----
0      7D14E297      down
1      ED385097      up
2      58C86297      right
3      41787497      left
4      BF8F6297      return
```

```

5 AB616762 r
6 2676D097 t
7 B6536297 c
8 9206E297 s
9 590C3E97 l
10 E8E8D097 p
11 C49C5097 a
12 F1EFD097 e
13 B9F03963 escape
14 D1F15097 pageup
15 53DA6297 pagedown
16 9F5BE297 escape
17 A8DDF497 left_ctrl Q
18 66FFBE97 d

```

Saving the configuration:

```

$ flirc_util saveconfig my_flirc_config

Saving Configuration File 'my_flirc_config.fcfg' to Disk
[=====>] 100%

Configuration File saved

```

There is also a **loadconfig** command.

What if a key does not work after configuring it.

First delete the key by its index. In this example the key d (Display Window) command isn't working.

```

$ flirc_util delete_index 18

```

Re-record the key

```

$ flirc_util record d
Press any button on the remote to link it with '+'
Successfully recorded button.

```

Re-test and repeat until a reliable 'hash' is received from the remote control.

If a key is multiply defined delete the first one you see by its index.

```

13 B9F03963 escape
14 D1F15097 pageup
15 53DA6297 pagedown
16 9F5BE297 escape

```

```

$ flirc_util delete_index 13

```

Re-test and if necessary, re-record.

There is a help facility for the flirc_utility.

```
$ flirc_util help
```

Install PiFace CAD (Bullseye only)

Unfortunately, the **PiFace CAD** display is obsolete hardware and only available to purchase whilst stocks last. For the same reason the software to install it is also out-of-date. In particular the IR remote control software which uses **lirc** is badly out of date. In the case of the radio software, it is only supported on **Bullseye**. To install the FLIRC IR remote control software:

- Run the **radio-config** utility.
- Select option **8 Install/configure drivers and software components** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144.
- Select option **11 Install PiFace CAD (Bullseye only)** from the **radio-config** menu shown in *Figure 229 Install FLIRC IR remote control* on page 171. Note that you will need to use the down arrow key on your keyboard to see option 10.

The following screen will be displayed:

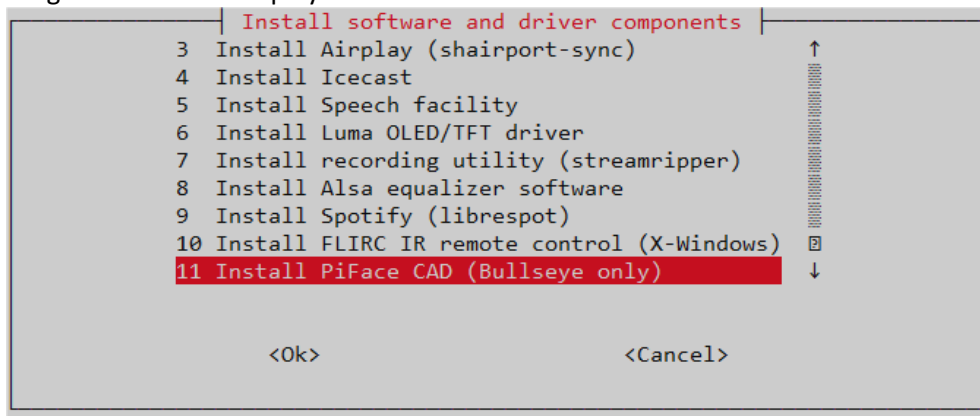


Figure 232 PiFace CAD installation

Because the IR remote control software uses the out-of-date **lirc** and drivers it will not work with the latest **irkeytable** software so do not install the IR remote control software shown earlier. For this reason, no support for the inbuilt IR detector is offered.

Configuring Russian/Cyrillic text

The radio program can display the Russian language either in **Cyrillic** or **Romanized** (convert to Latin) characters. For example, **Радио Пятница** when Romanized becomes **Radio Pyatnica**.

First purchase a character LCD/OLED with a Russian/Cyrillic character ROM. These devices also will display English characters.

To display Russian/Cyrillic text Romanized it is not necessary to change the configuration as this is the default. To display Russian/Cyrillic change the following parameters in **/etc/radiod.conf**.

Change the language to Russian

```
language=Russian
```

Switch off Romanization


```
romanize=off
```

Unless using a HD44780U compatible controller leave the controller setting as it is

```
controller=HD44780U
```

If using an older LCD with an HD44780 (No U at the end) controller set it to HD44780

```
controller=HD44780
```

Leave the codepage setting as 0. This will pick up the correct code page from the language translation file in the **/usr/share/radio/codes** directory.

```
codepage=0
```



The **translate_lcd** parameter must also be set to **on** for Romanization or Cyrillic translation routines to work. Translation is disabled if using an OLED as OLEDs use system fonts.

Configuring European languages for LEDs

First purchase a character LCD/OLED with a Western European character ROM. These devices also will display English characters. To display Western European text Romanized it is not necessary to change the configuration as this is the default. Any LCD/character OLED can be used for this.

Change the language to European and carry out the same instructions, except for language, as shown in *Configuring Russian/Cyrillic text* on page 175.

```
language=European
```

There is a detailed explanation of LCD code pages and program settings in the Technical Reference manual.

Configuring wallpaper backgrounds

This section is only applicable if the HDMI/Touchscreen version of the radio is to be used. There is an option to provide extra backgrounds. If using an LCD or OLED versions of the radio then skip this section. Scratch and rpd-wallpaper used to be installed by default, however, in the latest versions of **Raspberry Pi OS Desktop** they may not be. Extra backgrounds can be used by installing **Scratch**. Scratch is a visual programming tool for children.

The background wallpaper is set by the following parameter in **/etc/radiod.conf** and can be amended to use a different background. This can only be done after installing the radio software in the next section.

The default wallpapers are available in the **/usr/share/rpd-wallpaper** directory. Amend the wallpaper command once you have installed the radio software. This should already be installed, otherwise install **rpd-wallpaper** by running the following:

```
$ sudo apt install rpd-wallpaper
```

The default wallpaper is aurora.jpg

```
wallpaper=/usr/share/rpd-wallpaper/aurora.jpg
```

Spotify installation

- Run the **radio-config** utility.
- Select option **8 Install/configure drivers and software components** from the **radio-config**
- Select option **9 Install Spotify (librespot)** and enter to move to <Ok> and enter

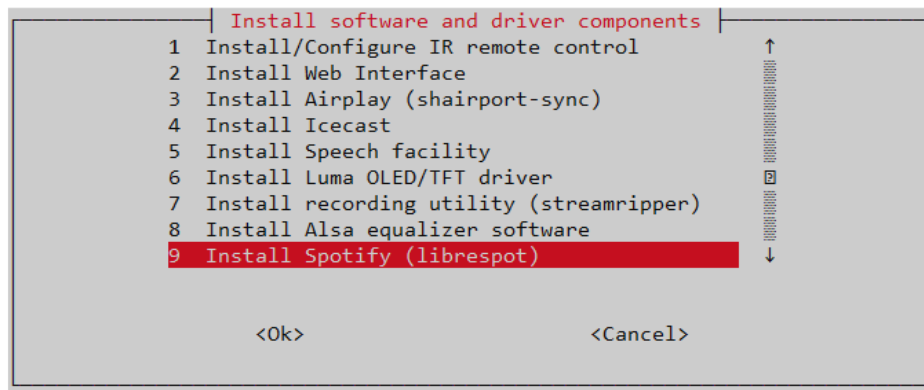


Figure 233 Spotify installation

You will be asked to confirm your choice after which the installation of Spotify will be started. Once completed a log of the installation will be found in **/usr/share/radio/logs/install_spotify.log**.

For operation see *Spotify operation* on page 208

For problems with Spotify see *Troubleshooting Raspotify* on page 230

Chapter 8 – Operation

Contents chapter 8	Page
Operation of LCD and OLED versions	179
Push buttons or Rotary encoders operation	180
The Radio Menu	181
Using the IR Remote Control	182
Operation of HDMI and touch screen displays	183
Playing Media Files	195
The Vintage Graphic Radio	193
Playing Media	195
Adding radio stations via the Web interface	198
Using the Timer and Alarm functions	199
Accessing Shoutcast	200
Overview of media stream URLs	203
Mounting a network drive	204
Controlling the Music Player daemon from Mobile devices	207
Spotify operation	208
Recording radio stations	209
Maintaining the Wi-Fi network and data roaming	211

Operation of LCD and OLED versions

This section assumes that the LCD screen is working correctly, the MPD daemon is installed and tested and that there is an Internet connection available. This section is for **LCD** and **OLED** versions only. For graphical radios see *Operation of HDMI and touch screen displays* on page 183.



Note: Both the **radiod** and **ireventd** daemons (services) are started automatically at boot time by a service called **systemd**.

Starting and stopping the radiod and ireventd services

On the command line run radio-config.

```
$ radio-config
```

Select option **9 Start/Stop/Status radio daemons** from the **radio-config** menu. The following menu will be displayed.

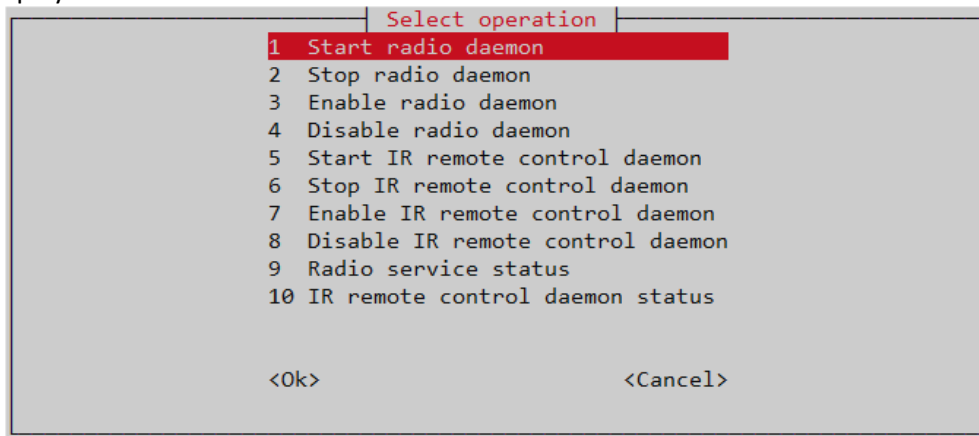


Figure 234 Starting, stopping radio daemons

The choices allow you to *start*, *stop*, *enable*, *disable* and get the *status* of both the **radiod** and **ireventd** (IR remote control) services. For example, selecting option **9 Radio service status** will display the status of the **radiod** service.

```
sudo systemctl status radiod.service
• radiod.service - Radio daemon
  Loaded: loaded (/lib/systemd/system/radiod.service; enabled; preset:
enabled)
  Active: active (running) since Thu 2025-04-17 19:51:26 BST; 16h ago
    Main PID: 774 (python3)
      Tasks: 6 (limit: 1574)
         CPU: 17min 26.507s
    CGroup: /system.slice/radiod.service
            └─774 python3 /usr/share/radio/radiod.py nodaemon

Apr 17 19:51:40 bookworm64 radiod.py[2223]: amixer: Cannot find the given
element from control sysd>
Apr 17 19:51:40 bookworm64 sudo[2222]: pam_unix(sudo:session): session
closed for user root
```

Use the other options as required. By the way, the first line shows the actual system command to run the same operation from the command line.

```
$ sudo systemctl status radiod.service
```

Push buttons or Rotary encoders operation

Radios with push buttons operation

Radios designed with Push Buttons can from five to seven buttons as shown in the following table.

Button	Basic Function	Menu function
Menu Button	Menu selection	Step through Menu
Station/Track UP Button	Select next Radio Station/Track	Cycle through options
Station/Track DOWN Button	Select previous Radio Station/Track	Cycle through options
Mute Button	Mute and un-mute sound, Speak	n/a
Volume UP Button	Increase Volume	Find: Next Artist
Volume DOWN Button	Decrease Volume	Find: Previous Artist
Record Option (Optional)	Stop/Start Recording current station	n/a

Radios with rotary encoders operation

Rotary encoder	Basic Function	Menu function
Menu Button	Menu selection	Step through Menu
Channel turn clockwise	Select next Radio Station/Track	Cycle through options
Channel turn Anti-clockwise	Select previous Radio Station/Track	Cycle through options
Mute Button	Mute and un-mute sound	n/a
Volume turn clockwise	Increase Volume	Find: Next Artist
Volume turn Anti-clockwise	Decrease Volume	Find: Previous Artist
Record Option (Optional)	Stop/Start Recording current station	n/a

Rest of page deliberately left blank

The Radio Menus

Table 16 Radio Menus

Volume control			Tuner selection	
LCD Display Mode	Volume UP	Volume DOWN	Channel UP	Channel DOWN
Mode = TIME Line 1: Time Line 2: Station or Track	Volume Up	Volume Down	Station/Track up	Station/Track down
Mode = SEARCH If source = RADIO Line 1: Search: Line2: Radio Station	Volume Up	Volume Down	Scroll up radio station	Scroll down radio station
Mode = SEARCH If source = MUSIC LIBRARY Line 1: Search Line2: Track/Artist	Scroll up through artists	Scroll down through artists	Scroll up through track	Scroll down through track
Mode = SOURCE Line 1: Input Source: Line2: Radio, Media playlist, Spotify or Airplay	Volume Up Mute	Volume Down Mute	Cycle up through Airplay, Radio Recordings and Media playlists	Cycle down through Airplay, Radio, Recordings and Media playlists
Mode = OPTIONS Line 1: Menu Selection Line 2: <option> Options are Random, Consume, Repeat, Reload Music, Timer, Alarm and Alarm Time (Hours), Alarm Time (Minutes) set and Change colour (1), Streaming on/off. Record time	Toggle selected mode on or off. Set timer and Alarm.	Toggle selected mode on or off. Set timer and Alarm	Cycle through Random, Consume, Repeat, Reload Music, Timer, Alarm Time Set, Streaming, Background colour (1) and Record duration	Cycle through Random, Consume, Repeat, Reload Music, Timer, Alarm , Alarm Time Set, Streaming and Background colour (1) and Record duration
Mode = RSS (2) Line 1: Time Line 2: RSS feed	Volume Up	Volume Down	Station/Track up	Station/Track down
MODE = IP address Line 1: IP address Line 2: Station or Track	Volume Up	Volume Down	Scroll up through track or radio station	Scroll down through track or radio station



Note 1: The colour change option is only available for the AdaFruit RGB plate (ada_radio.py). Note 2: If not available the **RSS Feed Menu** is skipped.

Mute function

If you don't have a separate Mute button, pressing both of the volume UP/DOWN buttons together or in the case of a rotary encoder with a push button (Volume) will mute the radio. If voice is enabled then then operation is slightly different (See section on espeak). Press either the volume up or down switch to un-mute the radio. If you change channel or use the menu switch the radio will also be un-muted. If the alarm is set then the radio will go into sleep mode.

Using the IR Remote Control

If you have fitted the IR remote control depending on the exact IR Remote Control that you have installed you will have the following buttons available:

1. Channel UP and DOWN buttons
2. Volume UP and DOWN buttons
3. Mute and Menu buttons
4. A Record button
5. Buttons 0 - 9

(See *Installing an IR sensor and remote control* on page 70).

The **Volume** and **Channel UP** and **DOWN** buttons normally change the Volume and Station/Track respectively. They have different functions in the **Search** and **Source Selection** menus (See Table 16 Radio Menus on page 181). The **Mute Button** silences the Radio. The **Menu button** steps through the menus.

The 0 - 9 buttons select the Radio station or music track to be played. For numbers higher than 9 press the numbers within 1.5 seconds of each other. For example, 123 pressed in quick succession will play Station/Track 123. The selection will be displayed on the volume line of the display for a couple of seconds as shown below.

```
Play 123
```

If you attempt to select a station/track number greater than the available stations/tracks available then an error will be displayed. For example, if your playlist has 125 radio stations and you select station 145 you will see the following error on the volume line of the display for a couple of seconds.

```
Error > 125
```

Operation of HDMI and touch screen displays

The graphical screen

The HDMI and Touch Screen versions of the program can be started in three separate ways.

1. Automatically when starting the desktop
2. By clicking on the radio icon on the desktop
3. By manually starting the program from the command line (In an X-terminal)

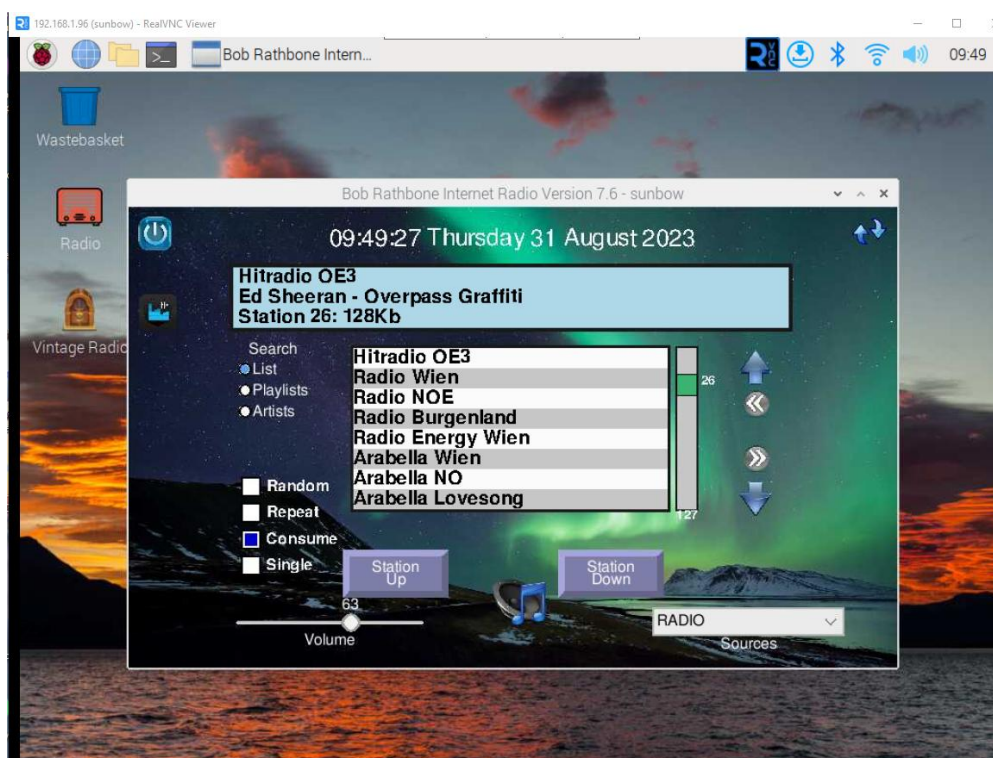
To start the radio from command line run the gradio.py program:

```
$ cd /usr/share/radio  
$ sudo ./gradio.py &
```

Starting the radio from the desktop. Click the icon shown here on the desktop.

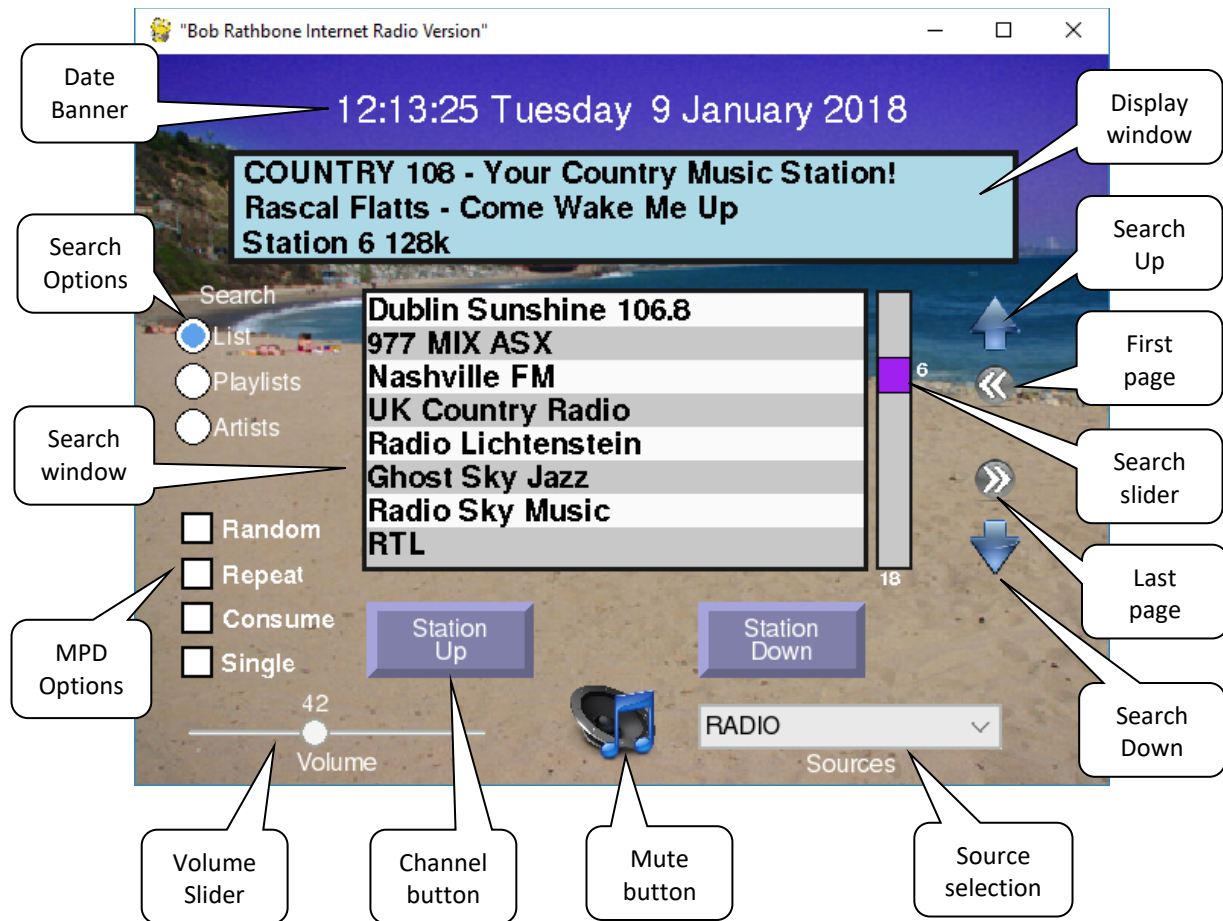


Normally you will need a HDMI monitor or TV with HDMI input and a mouse to operate the graphical radio programs as these programs run on the Raspberry Pi Desktop. However, it is possible to run the RPi desktop on a PC or MAC computer using **VNC** as shown below:



In all cases a screen similar to the following will be displayed. In this example **fullscreen=no**.

Figure 235 HDMI and Touch Screen Display



Clicking the mouse on a control such as station Up/Down or touching it do the same thing. In the following description we will only refer to "clicking". By this, also touching a control is also meant. Note that the "Search Up/Down" Icons mean Up and Down the station/track list respectively.

The display window

The display window normally displays the Radio station or Media rack that is currently playing. Clicking in the display window changes the third line to display the RSS feed if configured. A second click in the same window displays version details, IP address and hostname.

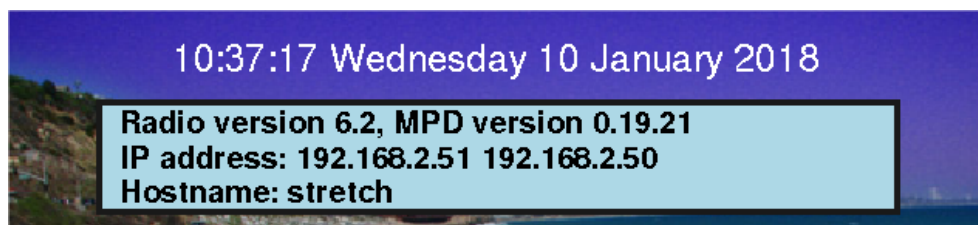


Figure 236 Graphical scree information display

In this example the hostname is 'stretch'. The version number will be different for later releases. Two IP addresses are displayed (Wireless and Ethernet).

The search window

The search window normally displays the contents of the currently selected playlist.



Figure 237 Graphical radio search window

Click on a station in the list selects it. Clicking in the slider window or dragging the slider re-positions the list. The current position, 16 in this example, is displayed next to the slider. The length of the current playlist, 28 for this playlist, is displayed at the bottom of the slider window.

Clicking on the Up and Down arrows travels up and down the list. Clicking on the left double arrow goes to the first page in the list. Clicking on the right double arrow goes to the last page in the list.



Figure 238 Graphical radio search functions

Clicking on the Playlists radio button selects the available playlists. This shows the playlists for radio or media such as the USB stick or Network share. It also shows 'airplay' which is not really a playlist but is a source, but can be selected here. Click on the desired playlist in the search window.



Figure 239 Display playlists

In the following example the USB stick playlist was selected. Once a playlist selected the list is displayed.



Figure 240 Display of media tracks

Clicking the Artists radio button displays the list of artists in the search window. Once clicked the search window positions on the first song of that artist's tracks.



Figure 241 Displaying artists

Note that if you click on the 'Artists' radio button when displaying Radio stations, it will always be forced back to the 'List' display as Artist selection is not relevant for Radio stations.

Smaller TFT screens

Screens with a resolution equal to or less than 420 x 320 pixels will display slightly different than previously shown. Only one line will be displayed in the search window. There are no options for Random, Repeat or Consume due to lack of space.



The search list type (Playlist, Station/Track list or Artist) is cycled through by clicking on the Search list type button. All other controls work the same as shown in Figure 235 on page 184.

Radio Station Artwork display

New in version **8.2** onwards is the Radio station Artwork. If a radio station is supporting Artists Artwork this will be automatically displayed over the top of the search window as shown below. Please note that Artwork is provided by the **Discogs** database. See <https://www.discogs.com> The **Discogs** database isn't perfect and sometimes the incorrect or no artwork will be displayed.

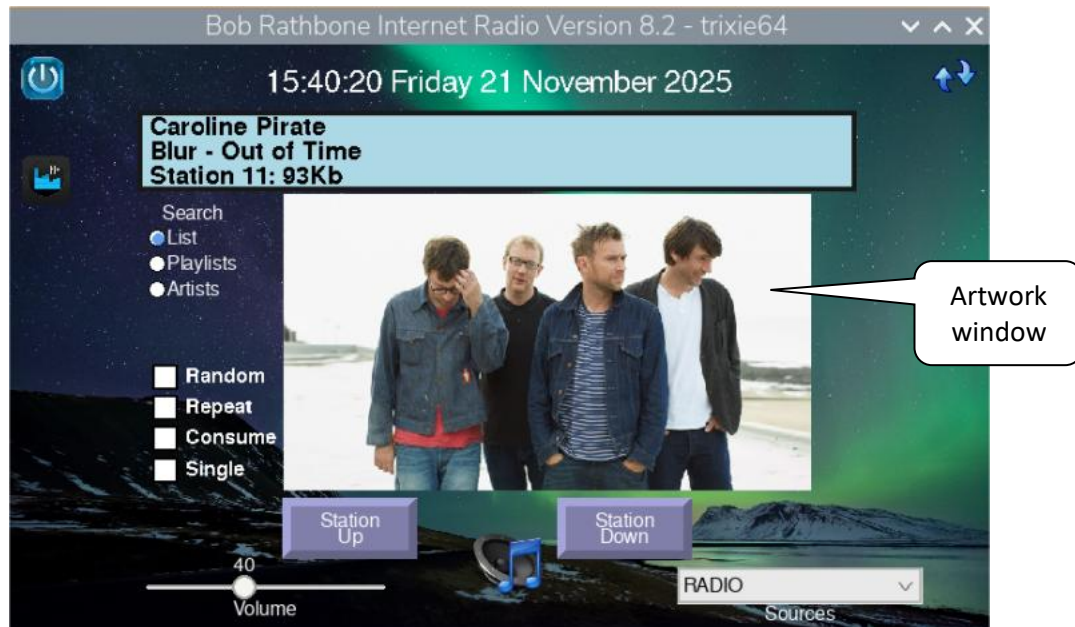


Figure 242 Graphic Radio Artwork display

By clicking anywhere in the Artwork Window, the program will change to displaying the **Search** window as shown below.



Figure 243 Graphic Radio Search Window display

The next time that the transmitting radio station changes the Artwork it will be displayed automatically. However, some radio stations such as BBC don't change their Artwork when a new song is displayed. In such a case click on the currently playing station (Caroline Pirate in this example) and it will redisplay the relevant Artwork, otherwise just wait until the next song change.

Media Artwork display

It used to be the case that if a media music track had artwork this would be displayed in the Graphic version of the program. However, because all too often this artwork would be missing, the radio program now uses the **Discogs** database. Clicking on any of the radio search buttons will re-display the search window. The artwork cannot be displayed until the track is re-selected.

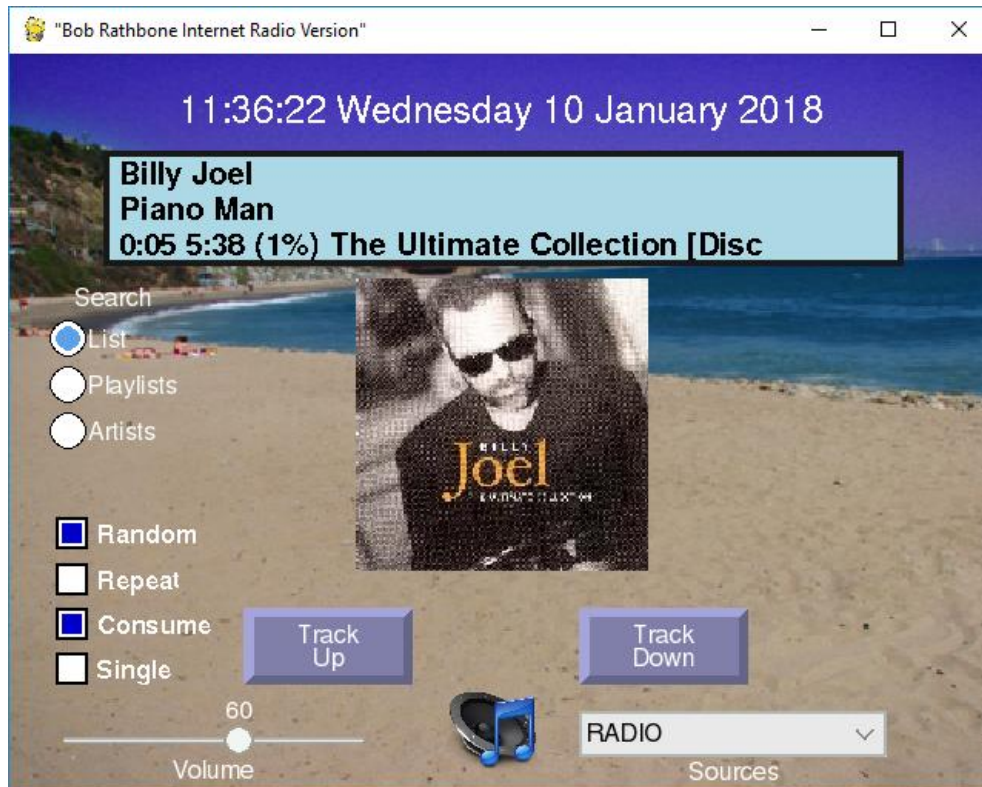


Figure 244 Track artwork display

Note that the two grey push buttons now display 'Track Up/Down' instead of 'Station Up/Down'.

Configuring radio station artwork selection

In both the case of Radio Station and Media streams, the associated artwork that is pulled from the **Discogs** database depends entirely on the Station name and program details or the Artist, Album and Track name that are submitted as the query to the **Discogs** database. If the submitted query is wrong then the artwork retrieved from the **Discogs** database will be wrong.



Note: Unfortunately, the information provided by radio stations about what is currently playing in their stream is chaotic. Normally they should be transmitting the name of radio station and the title of what is playing at the time. Sometimes they only transmit the name and no title or the other way around. In one case the station was sending out the title in the station name field.

Even if a station name is correctly sent out it may not be correct for the **Discogs** Database lookup. For example: A certain Dutch radio sends out the name "Radio 10 NL" in its stream and is correct as far as it is concerned. However, using this name **Discogs** returned the artwork for a radio station in the USA and not Holland. The correct name to get the right artwork is, in this case is "Radio 100% NL". What can be done about this. Fortunately, the search names in the **stationlist** file can be used instead of the name in the stream. Amend the station list entry for the [Radio 10 NL] radio station.

```
$ sudo nano /var/lib/radiod/stationlist
```

Change the name in [...]

```
[Radio 10 NL] https://22733.live.streamtheworld.com/RADIO10.mp3
```

To

```
[Radio 100% NL*] https://22733.live.streamtheworld.com/RADIO10.mp3
```

And add an '*' to the end of the name. The '*' tells the gradio.py program to use the search name to lookup the artwork instead of using the station name transmitted in the radio stream. This will often deliver the in most cases but not always. You can experiment looking up artwork using the lookup facility at: <https://www.discogs.com/>

Configuring media artwork selection

Likewise, a media playlist for example for the USB stick may contain the following entry.

```
media/Hits from Alison Moyet/Singles [US_UK]/11 Ordinary Girl.mp3
```

should be something like the following to have any chance of getting the correct artwork.

```
media/Alison Moyet/Singles [US_UK]/Ordinary Girl.mp3
```

The structure for **Artist**, **Album** name and **Track** is as follows.

Mount **Artist** **Album name** **Track name**
media/Alison Moyet/Singles [US_UK]/Ordinary Girl.mp3

With your media playlist selected by the radio go to the media directory

```
$ cd /var/lib/mpd/music/media/
```

Find the affected file

'Hits from Alison Moyet' and move it to 'Alison Moyet'

```
$ mv 'Hits from Alison Moyet' 'Alison Moyet'
```

Now change directory to 'Alison Moyet' then to '

```
$ cd 'Alison Moyet'/'Singles [US_UK]'
```

Now rename the actual mp3 file.

```
$ mv `11 Ordinary Girl.mp3` `Ordinary Girl.mp3`
```

Finally using **radio-config** select option **4 Create media playlists** and recreate the playlist for the mounted media.



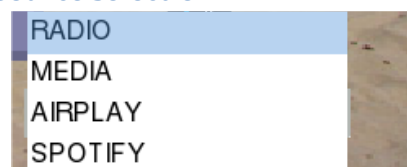
Please note that the **Bob Rathbone Computer Consultancy** does not provide any service to correct any artwork that is displayed incorrectly as this is beyond its control.

Volume and Mute controls



The volume is controlled by a slider at the bottom left of the window. Clicking on the loud-speaker at the bottom of the screen mutes the sound and displays the mute icon as shown on the left. Any volume control change un-mutes the radio.

Source selection



Click on the down arrow on the right of the Source selection to select the Source namely Radio, Media, Airplay or Spotify. The radio will select the first playlist in that source. Re-selecting the same source will select the next playlist for that Source.

Other graphic window controls

Music Player Daemon (MPD) options Random, Repeat, Consume and Single are selected using the square push buttons on the bottom left of the window. Only the Random option is stored for the next time.

Running Airplay on the HDMI touchscreen

Airplay must first of all be installed on the Raspberry Pi. See *Airplay* (shairport-sync) Installation on page 162 for instructions how to do this. To select Airplay either select it from the Sources drop-down box or from the playlists in the search window.

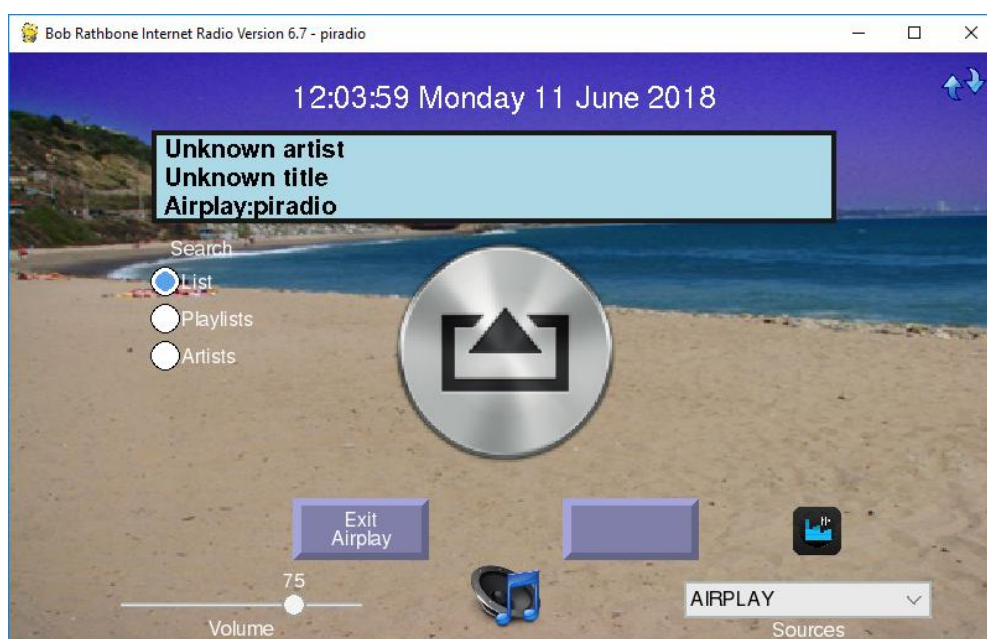


Figure 245 Airplay running on a Graphical screen

Connect to the Raspberry Pi from an Airplay compatible mobile device or run an App such as **CloudBreak**. The hostname to connect to is displayed when Airplay is first opened in the Display Window as shown below:



In this case the hostname is 'piradio'. To exit Airplay, press the left button at the bottom of the screen. The other button on the right has no label and doesn't do anything in Airplay mode.

Changing the graphical radio theme

The colour scheme and background are largely configurable in the [SCREEN] section of the `/etc/radiod.conf` configuration file. Button colours cannot be configured.

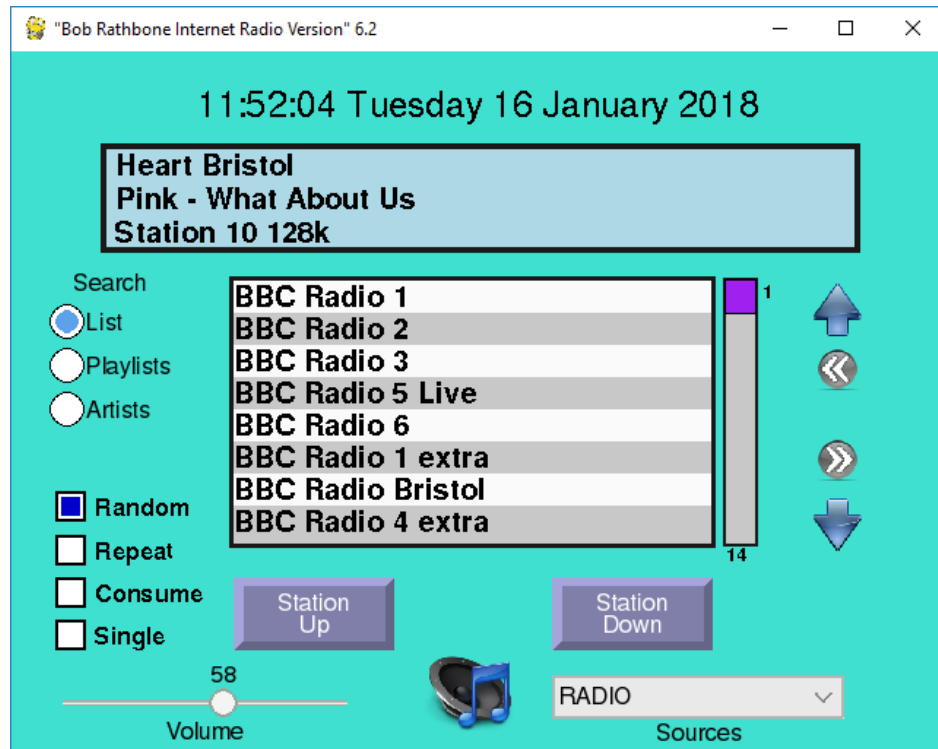


Figure 246 Changing the graphical screen theme

One good personalisation is to use your own favourite holiday picture as the background.

```
wallpaper=<path to your photograph>
```

Window and label colours can be changed to your own preferences. In the above screen the wallpaper option has been disabled, so the `window_color` option is used.

```
# Graphics (touch screen) screen settings
[SCREEN]
fullscreen=yes
window_title="Bob Rathbone Internet Radio Version"
window_color=turquoise
banner_color=black
labels_color=black
display_window_color=lightblue
display_window_labels_color=black
slider_color=purple
display_mouse=yes
switch_programs=yes
screen_saver=0

# Wallpaper backgrounds. See /usr/share/scratch/Media/Backgrounds
#wallpaper=/usr/share/scratch/Media/Backgrounds/Nature/beach-malibu.jpg
# Set date format for graphic screen
dateformat=%H:%M:%S %A %e %B %Y

# The following is specific to the vintage graphical radio
```

```
scale_labels_color=white
stations_per_page=40
display_date=yes
display_title=yes
```

Python pygame colour constants

See https://www.pygame.org/docs/ref/color_list.html

Graphic screen keyboard controls

The HDMI/Touchscreen version accepts input from the keyboard. It is limited and is only included as a keyboard may be connected to the Raspberry Pi when using an HDMI screen. The normal interface is either touch screen or mouse and not the keyboard.

Table 17 Graphic screen keyboard command

Key	Description	Key	Description
Page Up (PgUp)	Channel/Track Up	Up Arrow	Search Up
Page Down (PgDn)	Channel/Track Down	Down Arrow	Search Down
+ Key	Volume increase	Left arrow	Go to first search page
- Key	Volume decrease	Right arrow	Go to last search page
R	Toggle <u>R</u> andom	L	Select Search <u>L</u> ist
T	Toggle Repeat	P	Select Search <u>P</u> laylists
C	Toggle <u>C</u> onsume	A	Select Search <u>A</u> rtists
S	Toggle <u>S</u> ingle	M	Toggle <u>M</u> ute on/off
D	Cycle <u>d</u> isplay window	ESC	Exit program
X	Switch between vgradio and gradio		

Operating the Alsa Equalizer

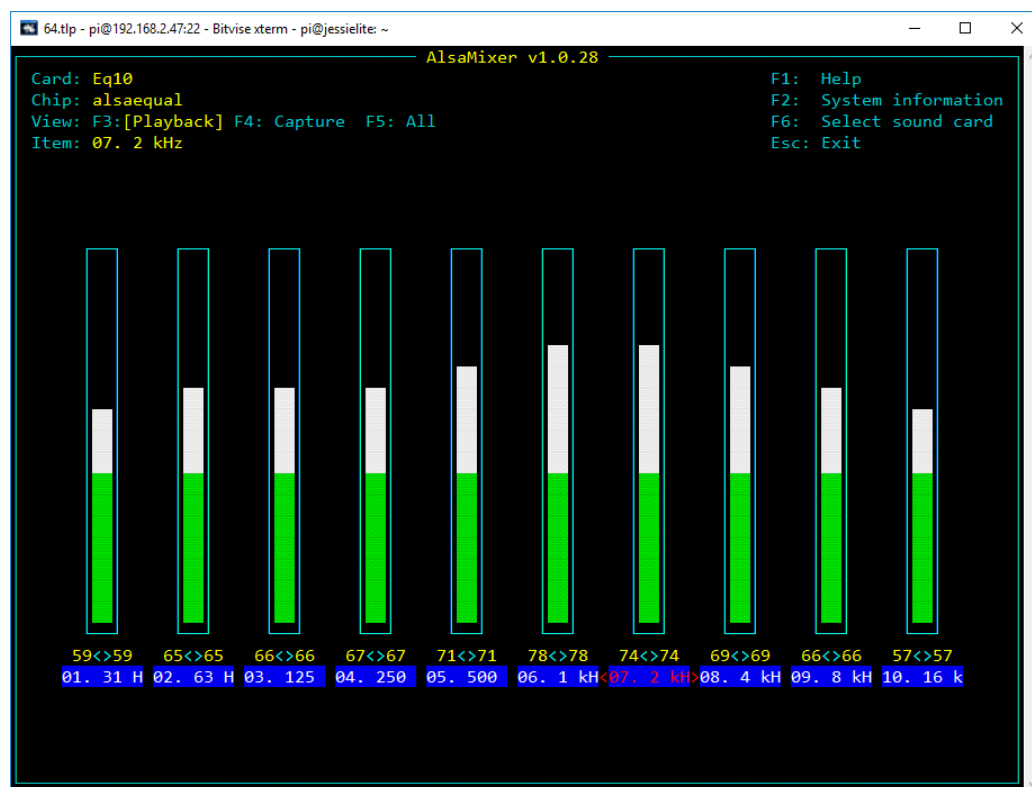


Figure 247 The Alsa

Use the Tab key to move along to the desired frequency to be changed. In this example, it is the <2KHz> block. Use the up and down arrows to adjust the level. The settings are saved in the `/var/lib/mpd/.alsaequal.bin` file. Changes to the sound should be heard.



Note: If you set a particular frequency value too high you will cause unpleasant distortion to the sound output.

The Vintage Graphic Radio

As an alternative to the **gradio.py** program there is a touch-screen version of the radio called **vgradio.py**. This radio program only can play radio stations and not other Media (USB stick for example) or Airplay.



Note: This radio program can only play radio stations and not other Media such as a USB stick or Airplay, nor are there currently any plans to change this. If you want to play media you will need the full feature **gradio.py** program previously described.

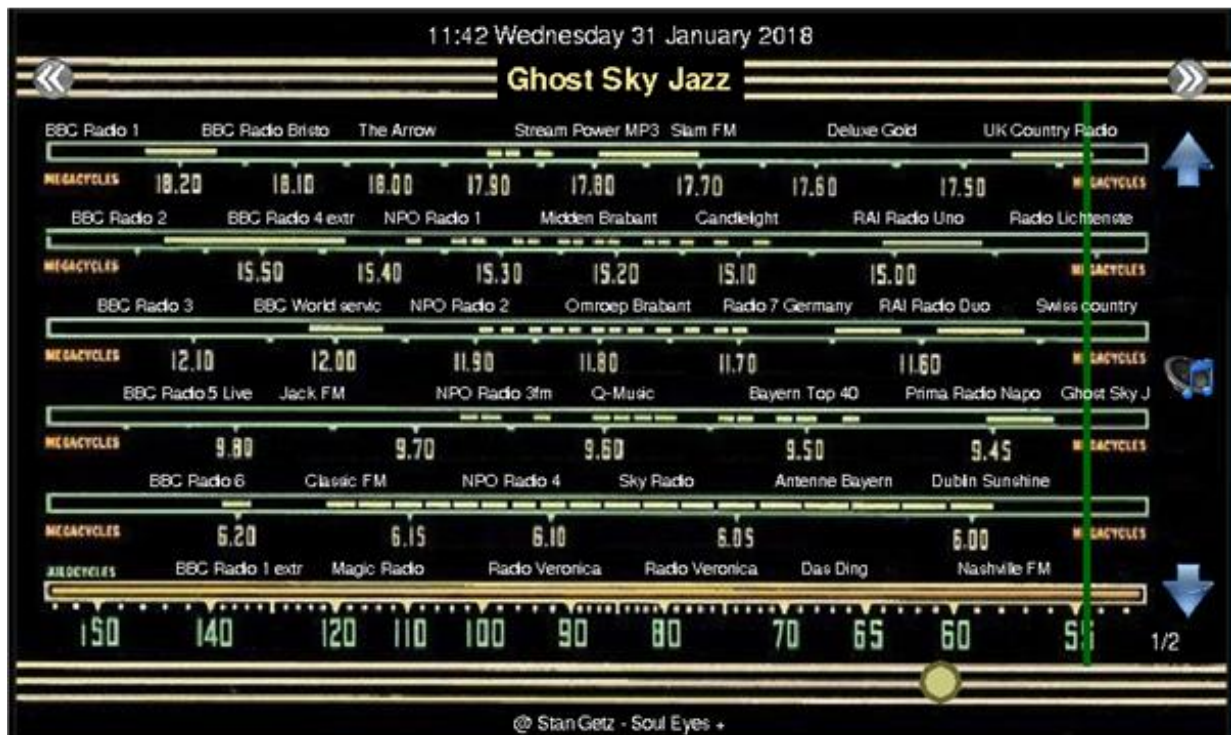


Figure 248 The vintage graphic radio on a touch-screen

This allows a radio to be constructed to look like a vintage radio with a sliding tuning dial. The pages scroll through the stations so hundreds of stations can be added. When you touch the name of a station on the tuner dial the green slider jumps to that location and plays the selected station.

The double arrows at the top of the screen allow you to page through the stations. At the bottom is the round volume slider. Under that is the title of the currently playing song. The blue arrows are used to step through the stations one at a time. The mute button is on the right-hand side of the screen. This design can also be combined with rotary encoders or switches.

To run the graphics versions of the radio **gradio.py** or **vgradio.py** it is necessary to configure this using the Configure Radio utility **configure_radio.sh** and select the Graphics display and user interface options. The Raspberry Pi OS supports either the legacy **X11** or the newer **Wayland** windowing systems. **Wayland** supports either the **wayfire** or **labwc** windows compositors.

This radio is designed to work with a single radio playlist. This is normally the **Radio** playlist. You should configure the radio to start with this playlist by amending the **startup** parameter in **/etc/radiod.conf**.

```
startup=Radio
```

However, this does not mean that you cannot have multiple radio playlists. If you have more than one radio playlist then by using the page up button (Double right arrow) it is possible to scroll through to the current playlist to the end and then onto the next playlist. In this case the new playlist name will be displayed in the very top-left of the screen.

You cannot currently scroll back to the previous playlist but must continue scrolling through the pages until you reached the desired playlist.

If using the FLIRC remote control dongle then it is only necessary to program the following keys: **pageup**, **pagedown**, **left**, **right**, **up**, **down**.

Switching between graphics programs

It is possible to switch between the full feature graphical radio (**gradio.py**) and the vintage graphical radio (**vgradio.py**). First configure the **switch_programs** parameter in the [SCREEN] section of **/etc/radiod.conf**.

```
switch_programs=yes
```



Restart the program. The switch icon on the left will appear towards the top of the right-hand side of the screen. By clicking on it the program will switch between the two versions of the desktop radio programs. There will be a very short pause in the music stream whilst it is doing the switch-over.

Configuring a screen saver



Note: The **xscreensaver** program described here does not appear to work if the radio program is in full screen mode.

Modern LCD displays are not as susceptible to screen burnout as the old cathode ray tubes of old. However continuous static screen displays will eventually cause shadowing. It is therefore a good idea to install a screen saver. The standard one for **Raspbian** is called **xscreensaver**. To install it run the following:

```
$ sudo apt install xscreensaver
```

After installation of the screen saver it can be configured in the desktop preferences menu. This allows configuration of time, screen saver or a blank screen. Choose a not too busy screen saver or the blank screen option.

There is also a program called **xscreensaver-command** for command line manipulation of the screen saver. However, the advice is not to use it as, at the time of writing, it causes severe problems with both the console and desktop display.

Playing Media Files

The radio software also allows you to play music from the following sources:

1. From a USB stick
2. From a Network share
3. From the user Music directory on the SD card
4. From a USB SDD drive
5. From the user Recordings directory

To create media playlists

- Run the **radio-config** utility.
- Select option **4 Create media playlists** from the **radio-config** menu shown in *Figure 206 The radio-config menu* on page 144. The following screen will be displayed.

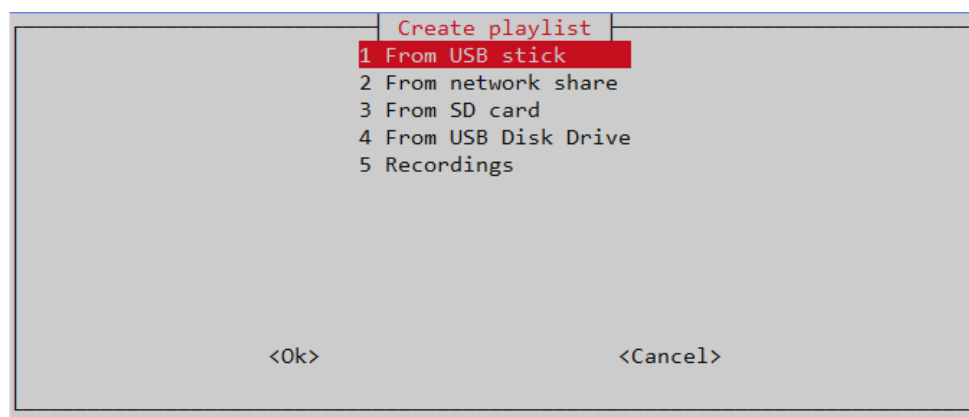


Figure 249 Creating Media playlists

Playing music from a USB stick

Put your music tracks on a USB stick (MP3 and WMA files only) and insert it into the USB port of the Raspberry PI. Select **Option 1 From USB stick**.

Playing music from a Network Attached Storage (NAS)

This is a bit more involved to set up. See the section called *Mounting a network drive* on page 204. Then go to the section called *Playing music from a Network Attached Storage (NAS)* on page 195 and select **Option 2 From network share**.

Playing music from the SD card

With large (32/64GB) SD cards now available music can be stored on the SD card. There is already a directory called **/home/<user>/Music** where music can normally be stored. <user> is normally 'pi'.

Using FTP or any other file transfer program, copy the music from a PC to the **/usr/share/Music** directory and reload the library via the options menu and then select **Option 3 From SD card**.

From USB Disk Drive

This is the same as the option for the SD except that it provides an optional separate link to an external disk drive such as an SSD disk drive so that there can be separate playlists for the internal SD card and an external drive. Select option **4 From USB Disk Drive**.

Playing music from the Recordings directory

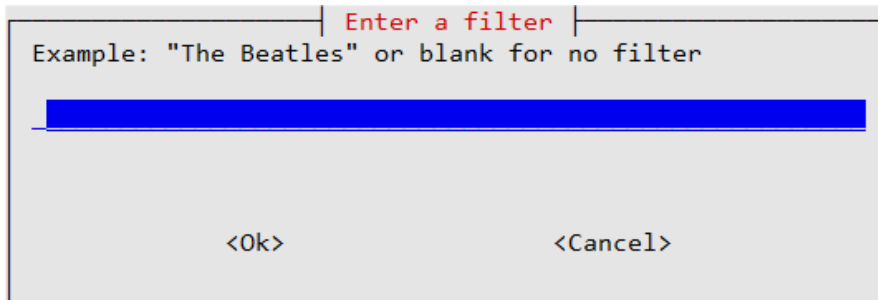
Option 5 Recordings creates a playlist called **Recordings** which can be then selected from the Radio Source Menu. It isn't normally necessary to use this option as the Recordings playlist is updated when the recording session finishes. It is only necessary to run this option if you have re-organised or renamed any of the files in the **/home/<user>/Recordings** directory.

Organising the music files

The search (find menu) routines get Artist and Track name directly from MPD which in turn get them from the music media file itself. The files should be placed in the top-level directory of the USB stick or in the **/usr/share/Music** directory if using the SD card. Any directory structure can be used. For example:

Elvis Presley/The 50 Greatest Hits Disc 1/That's All Right.mp3

In the Radio Search Menu you will only see the first and last parts of the directory structure. In this case it will be **Artist: Elvis Presley** and **Track: That's All Right**.

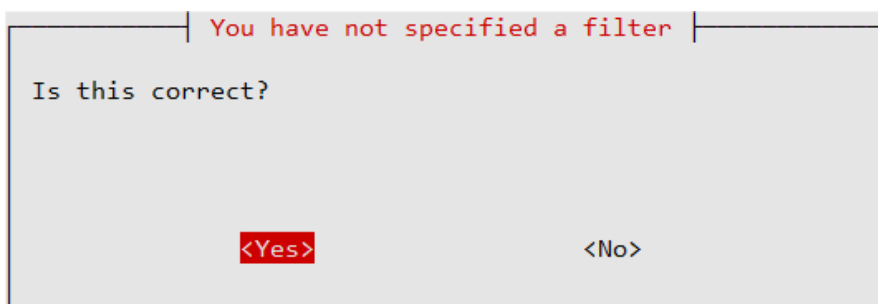


Enter a filter

Example: "The Beatles" or blank for no filter

<Ok> <Cancel>

Enter a filter name or enter for no filter.

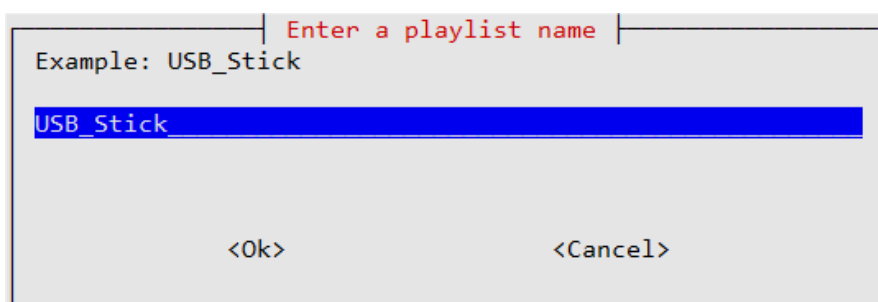


You have not specified a filter

Is this correct?

<Yes> <No>

Select Yes to continue.



Enter a playlist name

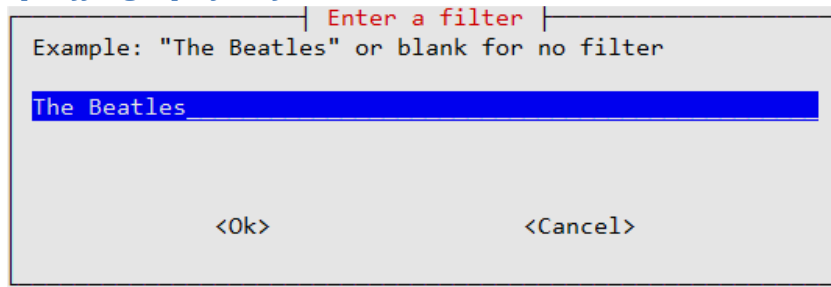
Example: USB_Stick

USB_Stick

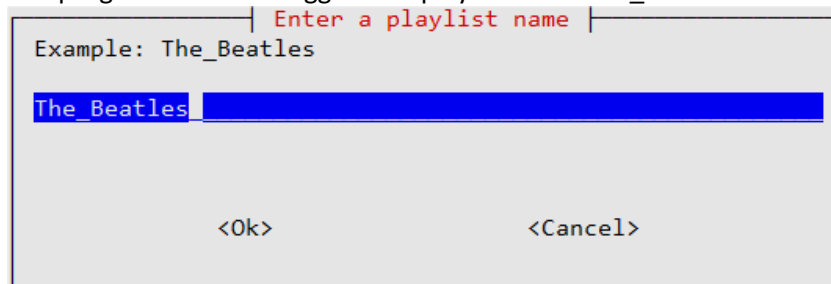
<Ok> <Cancel>

The program will suggest a name for the playlist but you may choose any name (But do not make it too long).

Specifying a playlist filter



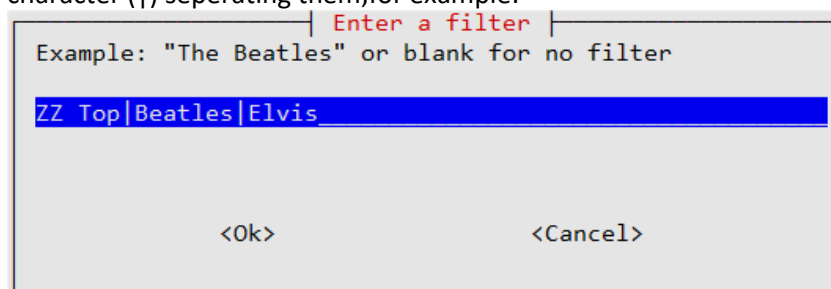
The program will then suggest the playlist name The_Beatles.



You will note that spaces in the playlist name have been replaced with underscores(_). This is just for the file name. When the playlist is displayed in the radio program the underscores will be converted back to spaces. The program will now create a playlist with the name **The_Beatles.m3u** (or whatever name was given).

Specifying multiple filters

More than one string may be specified in a filter. To do this specify the filter strings with a pipe character (|) separating them, for example:

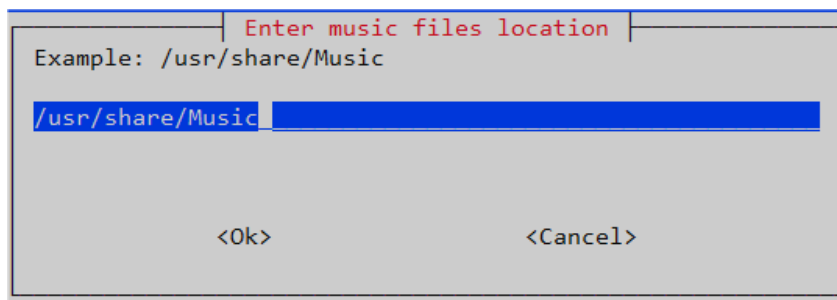


This will filter all songs from ZZ Top, The Beatles and Elvis or any other titles that contain these names. However this may not be what is wanted. Maybe songs by Elvis are wanted and not songs with 'Elvis' in the title. For example *Dire Straits – Calling Elvis*. In such a case use the / character to only look for directory names beginning with 'Elvis'. The above filter becomes:

ZZ Top|Beatles|/Elvis

Restart the radio to reload all new playlists. Using the / character gives a more accurate playlist. Please note that filters are not case sensitive. Filter 'Elvis' and 'elvis' will return the same result.

If you selected option 3 (SD card) you will be prompted for the location where you have installed your music files. This location must pre-exist and have music files. The default is **/usr/share/Music**.



Adding radio stations via the Web interface

It is possible to add a Radio station to the Radio playlist via the **O!MPD** Web client. See <https://www.musicpd.org/clients> for a list of MPD clients from the Music Player Daemon Web site.

In O!MPD select the NOW PLAYING tab and at the top of the page click on **ADD**. Enter the URL of the new Radio station. For example: **<http://uk3.internet-radio.com:8076/live>**

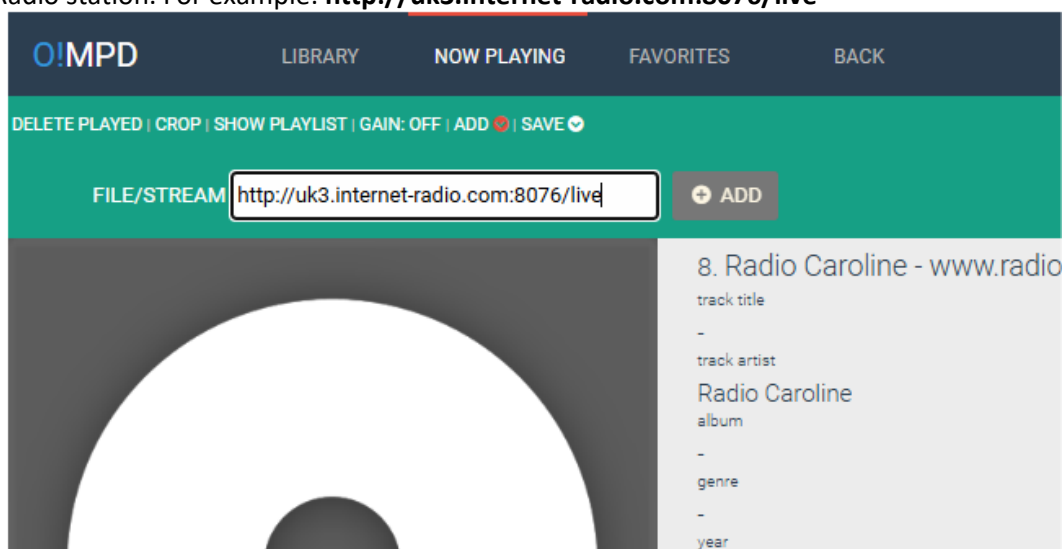
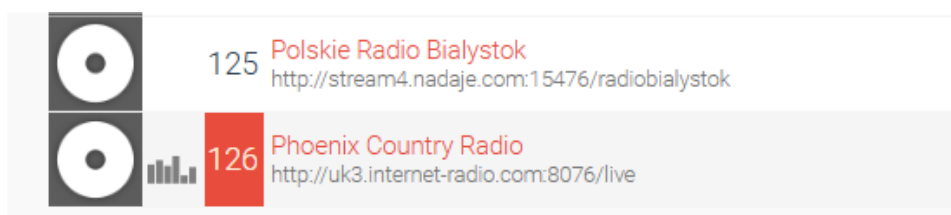


Figure 250 Adding a new station to MPD

Press the ADD button on the right of the URL entry box. The new station will be added to the end of the playlist and start playing. For example: Phoenix Country Radio.



By clicking on the right-hand menu for the new station it is possible to move or delete a URL however this is not supported by the Radio software and changes will be lost on the next restart.

Deleting and moving radio stations can currently only be done by amending the **/var/lib/radiod/stationlist** file.

See **radio-config → 6 Documents and Tutorials → 1 Creating radio station playlists**

What if adding a new station does not work? This is because the URL entered could not be found or is invalid. Find the correct URL for this station or find a URL for another similar station.

Using the Timer and Alarm functions



Note: The Raspbian operating system synchronizes time over the Internet. It does this using the **timesync** service. This service is a light-weight, client only, time synchronisation service, using the Network Time Protocol (NTP). In Bookworm this is implemented the **systemd-timesyncd** service.

There is a timer (Snooze) and alarm function (LCD and OLED versions only). The timer and alarm can operate individually or together. The timer when set will put the radio into Sleep Mode when the timer expires. The Alarm can be set to either On, Repeat or “Weekdays only”.

Setting the Timer (Snooze)

Press the Menu button until the “Menu Selection” is displayed. Press either the channel UP or DOWN control until “Timer off” is displayed on line 2 of the LCD screen. Now push the volume UP button to set the timer. Use volume UP and DOWN to adjust the timer which will be displayed as “Timer hh:mm:ss” where hh=hours, mm=minutes and ss=seconds. The Timer can be set up to 24 hours in increments of one minute. Once the timer is set, press the Menu button; the display will return to TIME mode.

On a four-line LCD display the timer will be seen counting down after the Volume display on line 4. On a two-line LCD display the timer count down will be displayed on line 1 after the time display.

When the timer expires (reaches zero) the radio will enter SLEEP mode. Sleep mode can only be exited by pressing the menu button.

To switch the timer off go back to the timer menu as described above and reduce the timer to 0 using the volume DOWN control. This will switch off the timer.

The timer function uses the `/var/lib/radiod/timer` file which will contain the value of the timer in minutes when it was successfully fired. You do not need to change the contents of this file.

Setting the Alarm

The Alarm menu has three settings:

- The alarm type (On, off, repeat etc)
- The Alarm Hours time (Pressing menu in this mode puts the radio into Sleep mode)
- The Alarm Minutes time (Pressing menu in this mode puts the radio into Sleep mode)

Press the Menu button until the “Menu Selection” is displayed. Press either the channel UP or DOWN (Or rotate rotary encoder) until “Alarm off” is displayed on line 2 of the LCD screen. Using the volume UP control cycle through the options which are

- Alarm off - The Alarm is switched off
- Alarm on – The Alarm is on for one time only. Once the alarm is fired it will return to off.
- Alarm repeat – The Alarm will be repeated every day and not switched off.
- Alarm weekdays only – The Alarm will only fire Monday through Friday. It is not reset.

Now move to “Set alarm time:” using the channel UP control. The current alarm time will be displayed on line 2 of the display. Using the volume UP and DOWN control adjust the alarm time (Hours or Minutes) to the required setting. If you do not wish to put the radio into sleep mode at this stage then use the channel UP/DOWN control to move away from the “Set alarm time:” option and

press the Menu button. If you press the Menu button whilst in the “Set alarm time:” option and the Alarm is set to anything except off then the radio will enter Sleep mode and display the alarm on line 2 for a two-line LCD or on line 4 for a four-line LCD.



Note: Sleep mode can only be exited by pressing the Menu button.

The alarm function uses the `/var/lib/radiod/alarm` file which will contain the current alarm type and time. The format is `t:hh:mm` where t is type (0=off, 1=on, 2=repeat, 3=weekdays only) and hh:mm is hours and minutes (24 hour clock). You do not need to change the contents of this file.



PLEASE NOTE THAT THE ALARM RELIES UPON THE SELECTED RADIO STREAM TO BE AVAILABLE WHEN THE ALARM WAKES UP. THIS CANNOT BE GUARANTEED AS THE STATION FEED MAY BE OFF AIR OR THERE IS A PROBLEM WITH THE INTERNET CONNECTION. YOU SHOULD NOT THEREFORE RELY SOLELY ON THIS ALARM FUNCTION IF YOU HAVE AN IMPORTANT APPOINTMENT OR A PLANE OR TRAIN TO CATCH FOR EXAMPLE. ALSO SEE DISCLAIMER ON PAGE 239.

Using the Alarm and Timer functions together

The Alarm and Timer functions can be used together. For example, you want to set your radio to a 30-minute snooze time before going to sleep and to sound the alarm in the morning. Simply set the Timer to the required elapse time and then set the alarm as described in the previous section. Press the Menu button and the timer will be seen counting down followed by the alarm time on line 4 or line 1 for the four-line and two-line LCD respectively.

Adafruit RGB Plate changing colours

This section is only relevant for the Adafruit RGB plate. When running the radio with an Adafruit RGB plate, it is an option to change the colour of the display. Push the menu button until “Menu selection”. Push the channel button until “Select color” is displayed. Now push the volume button to cycle through the colours. The available colours are red, green, blue, yellow, teal, violet, white or Off (No backlight). Note that the program uses the American spelling ‘color’

Shutting down the radio

You can simply switch the power off. This doesn’t normally harm the PI at all. However, if you want a more orderly shutdown then press the menu button for at least three seconds. This will stop the MPD daemon and issue a shutdown request to the Raspberry PI. Wait at least another ten seconds and then power off the Radio.

Accessing Shoutcast

It is possible to create playlists from the Shoutcast database. See <http://www.shoutcast.com>. Shoutcast provide what can best be described as “fringe” radio stations. They do have a few stations by country but not many. This version of software provides two methods of creating playlists from the Shoutcast database:

1. Using the shoutcast tab in the radio Web interface.
2. Using the `get_shoutcast.py` program

Using the Shoutcast Web Interface

The radio Web interface now has a Shoutcast tab. Click on Shoutcast tab to open the interface. Fill in the search form and press the Submit button once and wait until the summary page is displayed.

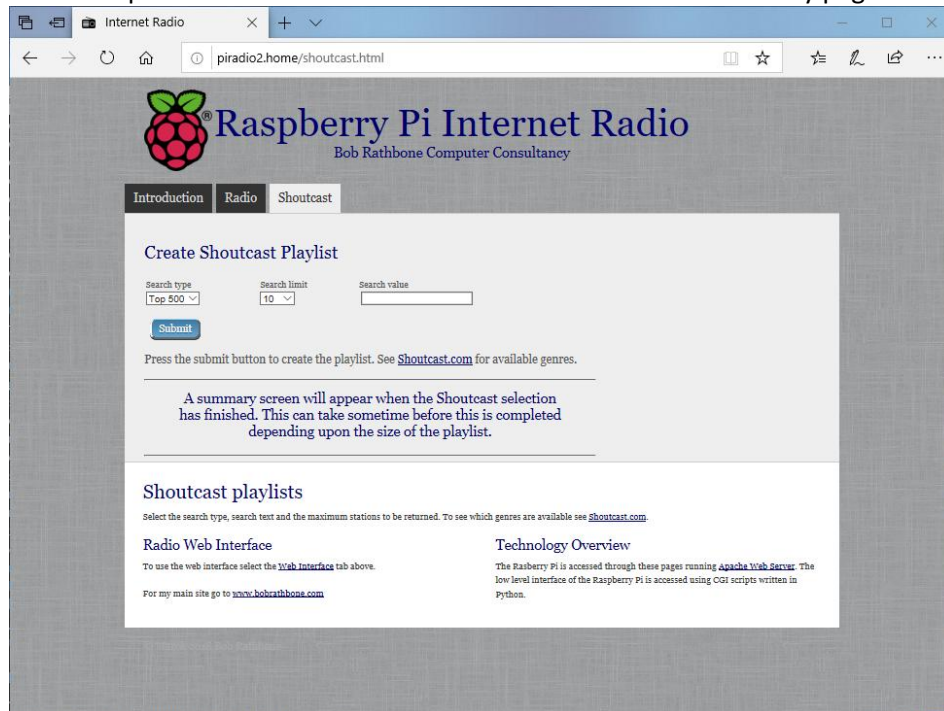


Figure 251 Shoutcast playlist Web page

Select the search type Top 500, Genre or Search from the “search type” drop down box. Select a “limit” for the search then click the “Submit” button. Normally the message “Please wait for selection” is displayed along with a rotating circle wait gif graphic. However, this does not work with **Microsoft Edge** and it is recommended you use the **Firefox** browser.



Figure 252 Shoutcast search selection

The summary page will be displayed. You should see the **Reload playlists: OK** message which means that the new playlist is available in the radio.



Figure 253 Shoutcast playlist summary

Using the get_shoutcast.py program

Log into the Raspberry Pi. Running the program with no parameters will produce the following usage message:

```
$ cd /usr/share/radio
$ ./get_shoutcast.py
Shoutcast UDP connect host localhost port 5100
This program must be run with sudo or root permissions!

Usage: sudo ./get_shoutcast.py id=<id> limit=<limit>
search="<string>"|genre="<genre>" install
    Where:  <id> is a valid shoutcast ID.
           <limit> is the maximum stations that will be returned
           (default 100).
           <string> is the string to search the shoutcast database.
           <genre> is the genre search string.
           install - Install playlist to MPD without prompting.

See http://www.shoutcast.com for availble genres.
```

Examples

```
sudo ./get_shoutcast.py id=anCLSEDQODrElkx1 limit=50 search="Beatles" install
sudo ./get_shoutcast.py limit=100 genre="Country"
```

If id= isn't specified it will be picked up from /etc/radiod.conf. See the shoutcast_key in /etc/radiod.conf for the id parameter. shoutcast_key=anCLSEDQODrElkxI

Radio stream resources on the Internet

There are a lot of resources on the Internet how to find PLS and M3U files so simply search for “PLS or M3U files” through the search machine of your choice. Below are some good sources of radio streams around the world.

<http://radiomap.eu/>

<http://www.publicradiofan.com>

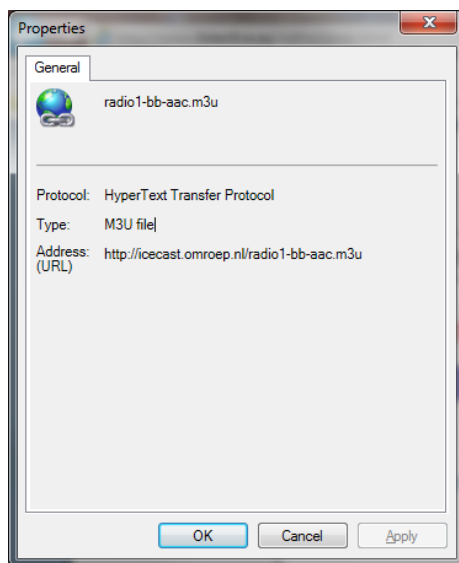
<http://www.radio-locator.com>

<https://www.internet-radio.com/>

<http://bbcstreams.com/>

<http://www.radiofeeds.co.uk/>

Getting a radio stream from a Web browser



To copy a URL open the Web page in any browser on a PC and right click on the URL. Select properties from the drop-down list. For internet explorer will show a window similar to the illustration on the left will be displayed:

Copy and paste the URL into the **/var/lib/radiod/stationlist** file. Add the title in square brackets as shown in the previous section. Other browsers may provide options such as ‘copy link’ or ‘save link as’. This is browser dependant.

Overview of media stream URLs

A deep understanding of this section is not necessary but can be useful when creating playlists. This section is provided for background information only. At first the whole business of how music streams are provided can be quite confusing. The URLs on a radio station Web page can be of different types, for example:

1. A URL pointing to a M3U playlist file (MPEG3 URL). This format is used by MPD.
2. A URL pointing to an HLS (HTTP Live Streaming HLS - M3U8) playlist file
3. A URL pointing to a PLS playlist file (Shoutcast Play List)
4. A URL pointing to an ASX playlist file (Advanced Stream Redirector)
5. A URL which is an actual stream such as MP3 (MPEG 3) or AAC (Advanced Audio Coding)

1, 2 and 3 are so called redirector URLs and point to a playlist file containing one or more URLs to the radio stream(s) itself. The *create_stations.py* program tries to figure out what type of URL that it is and create a playlist from it. This is the facility you should use rather than trying to create your own playlists which can be quite time consuming.

Mounting a network drive

It is very likely that you may have your music on a shared network drive and want to play the music through the radio. There are two main types of network drive protocols used by Linux on the Raspberry Pi namely:

- CIFS – Common Internet File System
- NFS – Network File System

The protocol used for CIFS is SMB (Server Message Block – Microsoft). Previously connections to SMB were via a product called SAMBA but has been largely replaced by the mount using the CIFS option in the Linux. The steps to mount the network drive are as follows:

1. Find out the IP address or network name of your network drive.
2. Create and test the mount command using either NFS or CIFS.
3. Copy the mount command to **/var/lib/radiod/share** file.
4. In the Radio menu select “Music Library” as the source and press “Menu” again to load
5. Update the playlists to include the files on the new share (Network drive).

This procedure assumes that you already have your Network Drive configured and working with your PC and can play music via the PC. In the examples below a Synology Network Drive was used with a volume called Volume1 with a directory called “music”. The IP address for the Synology Network drive used was 192.168.2.6.

First stop the Radio software when creating and testing the mount command.

Don’t configure **/etc/fstab** to do the mount of the network drive. Although this is the usual way of mounting shares however the radio program needs total control of the mount and un-mount process.

The general syntax for the mount command is as follows:

```
mount -t <type> -o option1,option2,... <remote IP address and directory>
<mount point>
```

Where: <type> is either **nfs** or **cifs**.

-o option1,option2 are the mount options.

<remote IP address and directory> Is the IP address and music directory path

<mount point> This will always be /share for this program

Finding the IP address of the network drive

Only general guidance can be given here. Nearly all network drives have a Web interface. The IP address was almost certainly provided from DHCP in your home router. The IP address will be the IP address of the Web Interface. Look at your network drive documentation for further information.

The CIFS mount command

The following example mount command assumes that you have a guest user configured with password ‘guest’. Adapt the command as required.

```
mount -t cifs -o username=guest,password=guest,uid=pi,gid=pi,vers=1.0
//192.168.1.6/music /share
```

The above command is all on one line. The **uid** and **gid** parameters set the ownership of the music files to user **pi**. The **vers** statement is the CIFS version and can be 1.0, 2.0 or 3.0 depending upon the NAS storage. The share directory is created when you first run the Radio program so there is no need to create it. If the command was successful you should be able to display the music from the network drive. Go to section called *Display the share directory* on page 205.

Older NAS drives sec security option

Older NAS drives may also require the **sec=ntlm** option to the **-o** line. The **sec** option is the authentication protocol and determines how passwords are encrypted between the server and client. Security mode **ntlm** used to be the default authentication method but that is now become **ntlmssp**. If you are accessing a network drive which doesn't support **ntlmssp** you have to add **sec=ntlm** to the options as shown below:

```
-o username=guest,password=guest,uid=pi,gid=pi,sec=ntlm
```

Many NAS devices use older technology so they often only use **ntlm** authentication. There are other authentication methods such as **ntlmv2** but most are not currently supported with the Raspberry Pi OS.

The NFS mount command

The following NFS mount example assumes the NFS protocol has been configured for the music directory.

```
mount -t nfs -o ro,nolock 192.168.1.6:/volume1/music /share
```

A few things to note here; the NFS mount command uses the volume name (volume1 – can vary), The CIFS mount command doesn't. The second thing is that the IP address and remote directory are separated by a colon (:). If the command was successful, you should be able to display the music from the network drive.

Display the share directory

If the mount was successful using either CIFS or NFS you should be able to display the **/share** directory with the **ls** command.

```
# ls -la /share
total 4
drwxrwxrwx 85 pi pi 0 May 10 14:18 .
drwxr-xr-x 23 root root 4096 Jul 15 17:57 ..
drwxrwxrwx 4 pi pi 0 May 10 14:16 Albert Hammond
drwxrwxrwx 3 pi pi 0 May 10 14:16 Alexander Curly
drwxrwxrwx 3 pi pi 0 May 10 14:16 Allen Price & Georgie Fame
drwxrwxrwx 3 pi pi 0 May 10 14:16 Al Martino
drwxrwxrwx 3 pi pi 0 May 10 14:16 Animals
drwxrwxrwx 4 pi pi 0 May 10 14:16 Aretha Franklin
drwxrwxrwx 3 pi pi 0 May 10 14:16 Armand
```

The important thing apart from seeing the files is that you should see that the files are owned by pi and group pi or whatever login name that you are using.

Un-mounting the /share directory

To un-mount the share directory use the **umount** command (not unmount).

```
# umount /share
```


Copy the mount command to the configuration

Once the mount command is working copy it to the `/var/lib/radiod/share` file.
For example, for the CIFS mount command:

```
# echo "mount -t cifs -o username=guest,password=guest,uid=pi,gid=pi
//192.168.1.6/music /share" > /var/lib/radiod/share
```

The above command is all on one line.



Note: If you decide to directly edit the `/var/lib/radiod/share` file instead of using the above command then do not include quotations marks around the command.

Loading the media playlists

Now run the radio program. The radio stations will normally be loaded. Cycle through the menu by pressing the menu button until **Input Source:** is displayed. Press the channel up or down buttons or rotate the Channel knob to select one of the media playlists previously created in the Section *Playing Media Files* on page 195.

Once the desired playlist has been selected, press the **Menu** button. The radio program loads the selected playlist and starts playing a track from it. The program does not know the last track number that was played previously for the selected playlist and will select the track number that it finds in the `/var/lib/radiod/current_track` file. If this track number is bigger than the newly selected playlist length it will reset the track number to 1.

Update the playlists for the new share

Select Music Library Now cycle through the menu until **Menu Selection:** is displayed. Press the channel up or down buttons until the **Update list:**No is displayed. Use the Volume buttons to toggle the display to **Update list:**Yes.

Now press the Menu button. This will cause the MPD database to be cleared and updated from all the files loaded in the `/var/lib/mpd/music` directory including the new share. This can take some time (Several minutes) if the Network Drive contains a large amount of music files. During this process the Radio program will ignore any button depressions and you will see the first **Initialising** (Library) and then **Updating** (Library).

Disabling the share

To disable the share simply put a hash character (#) at the beginning of the line in the `/var/lib/radiod/share` file as shown in the example below. Alternatively remove the share file altogether.

```
# mount -t cifs -o username=guest,password=guest,vers=1.0
//192.168.1.6/music /share
```

Further information

For your information if you display the `/var/lib/mpd/music` directory you will see up to five links to the locations where the music files are stored. This will be different for your system.

```
$ ls -la /var/lib/mpd/music/
lrwxrwxrwx 1 root root    9 Jan 25 07:51 media -> /media/pi
lrwxrwxrwx 1 root root   19 Jan  8 07:12 recordings -> /home/pi/Recordings
lrwxrwxrwx 1 root root   14 Nov 26 07:40 sdcard -> /home/pi/Music
lrwxrwxrwx 1 root root    6 Jan 25 06:54 share -> /share
```

```
lrwxrwxrwx 1 root root      14 Jan 25 09:24 usbdrive -> /home/pi/MyDrive
```

Controlling the Music Player daemon from Mobile devices

Android devices



Figure 254 M.A.L.P. play screen

There are a number of Android Apps capable controlling the Music Player Daemon from an Android device such as a smart-phone or tablet. One of the most popular seems to be **M.A.L.P** See the following link: <https://m-a-l-p-mpd-client.en.softonic.com/android> for further information.

Download from the **Android Play Store**. **M.A.L.P** allows you to control a MPD server (Music Player Daemon) and stream from it. The radio daemon is completely integrated with MPD clients such as **mpc** and **M.A.L.P**

Settings

On the top left tap on the menu 
Go to **Settings** → **Profiles**.

In the Edit profile page press the **+** icon in the top right. Enter the name you wish for the profile. Enter the hostname or IP address of Raspberry Pi running the radio. Leave the port number at 6600. Leave the password blank (Don't enter your pi user password). Switch on the remaining three options (*MPD covers*, *Enable streaming* and *Prefer HTTP cover files*).



Finally press the save icon  on the top right to save the profile.



Use the trash icon to delete a profile.



Note: **M.A.L.P** is third party software and no support can be provided by bobrathbone.com.

Apple devices

There are at least three MPD client Apps for **iOS** mentioned on <https://www.musicpd.org/clients/> These are:

MaximumMPD - A MPD client for iOS

Shinobu - Modern and native client for iOS (iPhone / iPad)

Stylophone - A modern, native client for iOS/iPadOS; Also available on Windows!

Unfortunately, the author has no experience of either installing or using these Apps so cannot offer any support on them. You are referred to the above Web site and the Apple Store for further information.

Spotify operation

To start the Radio as a Spotify receiver either select Spotify from the Playlists (LCD or OLED versions) or from the Sources window in the touchscreen version:

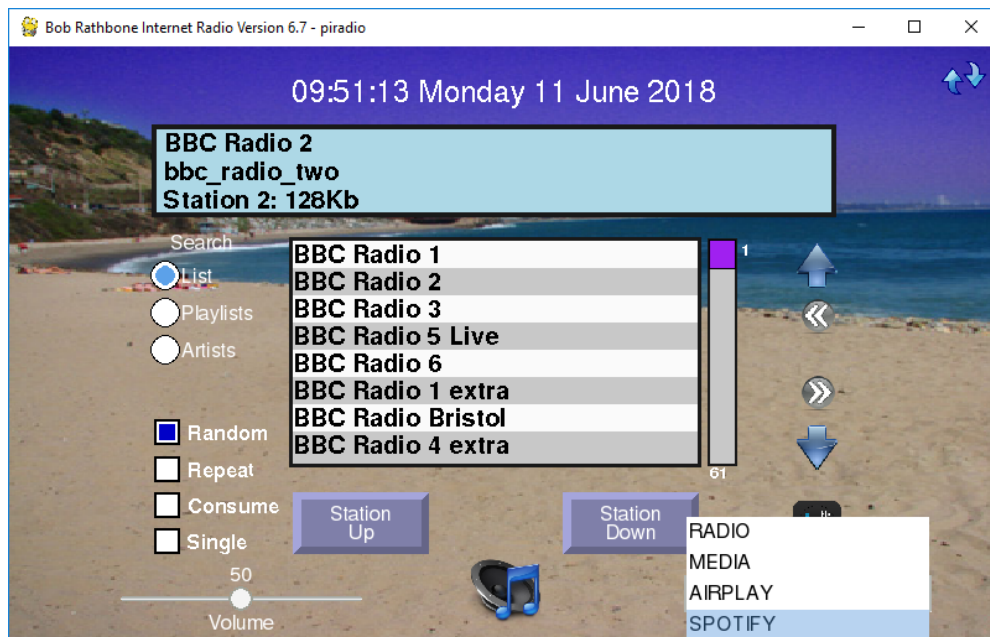


Figure 255 Starting the Spotify Receiver

The following window will appear however a different message may appear on the second line of the display window:



The Spotify application will connect the Raspotify application on the Radio.



Note: If you don't hear any sound then turn the volume up to full.

In the case of the touchscreen version of the program the following screen will be displayed:



Figure 256 Spotify playing a music track

In the case of the LCD or OLED version of the radio the title line above will be displayed on the second line of LCD or OLED screen.



Note: Raspotify unfortunately does not supply the Artist information. Only the track name is supplied by **librespot** which is used by Raspotify. There is currently no solution for this.

Exiting Spotify

For the LCD and OLED versions press the Menu button until the Select source: window is displayed. Select any other source (playlist) to exit.

In the case of the touchscreen version of the radio, press the **Exit Spotify** button at the bottom of the screen.

Recording radio stations

Since version 8.0 it has been possible to record nearly all radio stations. It is possible to record a radio station using either:

- A record button on the Radio if fitted.
- Or a record button on the IR remote control
See example *Figure 85 Mini IR-Remote control buttons* on page 46

First select the radio station you want to record then use the IR remote control or the Record button to start the recording process. If fitted the IR LED should be lit for the duration the recording session. Also, an Asterix '*' will be shown at the end of the Time and Date display on line 1. The recording session can be stopped by pressing the IR remote control or Record button again. Otherwise, the recording session will stop when the record timer has finished. See *Setting the recording time* on page 210.

See the tutorial **Record Radio Stations** for a full explanation. Run **radio-config** program then select option **6 Documents and Tutorials** followed by **7 Recording a Radio station**. Alternatively click **Tutorials and documents** on the first page of the Web interface and select **Record Radio Stations** from the list. See *Documents and Tutorials* on page 145.

Setting the recording time

Setting the recording time is done via the **Menu** button on the Radio. Press the Menu button three times until “Menu option:” is displayed on the first line.

```
Menu option:
Record for:0:45
Caroline Pirate
Volume 75
```

Now turn the **Volume** control to adjust the recording time up and down to the desired recording time. The maximum time allowed is 12 hours. Adjustment steps will vary with the length of time selected. The example on the left shows 45 minutes recording time.

Normally increments/decrements will be 5 minutes when the Volume is changed except when the time is less than 10 minutes when the increments/decrements will be reduced to one minute.

```
12:32 09/06/2025*
Caroline Pirate
Elvis Costello - Wa
Volume 75
```

Press the **Menu** button again to save the new timer setting. When the Record button is pressed you will see the Recording indicator ‘*’ after the date as shown on the left.

Playing back Recordings

Playing back your recordings is like playing back any other media. The recordings are saved in the **/home/<owner>/Recordings** directory. In most cases **<owner>** will be user pi in which case the recordings will be found in **/home/pi/Recordings**. When the recording session is finished the record module of the Radio program creates the playlist **Recordings.m3u** in the MPD playlists directory **/var/lib/mpd/playlists/** along with all the other media playlists that you may have created.

```
Input Source:
Radio
Caroline Pirate
Volume 75
```

To load a Recordings playlist, press the Menu button twice to display the **Input Source:** menu. This will normally be displaying the Radio playlist.

```
Input Source:
Recordings
Caroline Pirate
Volume 75
```

Now turn the **Channel** selection rotary encoder or buttons until **Recordings** playlist appears. Press the Menu button again to play the Recordings.

```
12:45 09/06/2025
The Christians
The Bottle
1:44 3:45 (50%)
```

The **Recordings** playlist will play the same as any other media playlist. Usually, the artist’s name and track will be displayed. The elapsed time and total time (+%) will be displayed on the bottom line.

The parameters that control the recording process will be found in **/etc/radiod.conf**.

```
record_switch=27
record_log=1
record_format=mp3
record_incomplete=no
record_cleanup=yes
load_recordings=no
```

See *Table 18 Liquidsoap recording formats and codecs* on page 211

Table 18 Liquidsoap recording formats and codecs

Parameter	Description
record_switch=<GPIO>	The GPIO number for the Record button. Usually GPIO 27 (physical pin 13) or GPIO5 (pin 29)
record_log=<loglevel>	0 none, 1 critical, 2 severe (default),3 important,4 info,5 debug This is liquidsoap's loggingin and not the radio.
record_format=<format>	Select format: mp4, flac, opus, mp3 Sets codec aac, flac, libopus, libmp3lame
record_incomplete=<yes/no>	Include incomplete tracks when creating the playlist
record_cleanup=<yes/no>	Remove incomplete tracks from the /home/<user>/Recordings directory
load_recordings=<yes/no>	Load new Recordings playlist

Maintaining the Wi-Fi network and data roaming

Normally the network is configured by the **Raspberry Pi Imager** software when creating the SD-card. However, you may wish to add a second or third Wi-Fi access point to enable Wi-Fi roaming for example between your home and office. These extra WiFi network points will be typically another router at a different location for example "office" or a repeater in the same building. To see what Wi-Fi access points are available run the following **iwlist** command:

```
sudo iwlist scan | grep ESSID
lo      Interface doesn't support scanning.

        ESSID:"EE-K50J9R"
        ESSID:"Office"
        ESSID:"NEIGHBOUR"
        ESSID:"TP-Link_D7D4"
:
```

This will display all available Wi-Fi access points available in your immediate vicinity. You will of course only be able to connect to those networks that you have a password for. Note: Your iwlist display will be different from that above.

To configure multiple routers/repeaters

This varies depending upon the version of the OS.

Table 19 Network configuration methods

Operating System	Configuration method	Configuration
Bullseye and earlier	wpa_supplicant.conf	/etc/wpa_supplicant/wpa_supplicant.conf
Bookworm < Dec. 2025	Network Manager	/etc/NetworkManager/system-connections/
Trixie & latest Bookworm	Network Manager	/etc/NetworkManager/system-connections/

For troubleshooting Wi-Fi networks see the section called *Network problems* on page 231.

Bullseye OS network configuration using wpa_supplicant.conf

This is included for historical reasons. **Bullseye** has one or more extra entries in **wpa_supplicant.conf** to implement data roaming. Edit the **/etc/wpa_supplicant/wpa_supplicant.conf** configuration file and add a second network definition. More network definitions can be added as required.

The example **/etc/wpa_supplicant/wpa_supplicant.conf** shows the configuration for two Wi-Fi access points.

```
ctrl_interface=DIR=/var/run/wpa_supplicant GROUP=netdev
update_config=1
country=GB

network={
    ssid="<Your-SSID>"
    psk="<Your-Router-Password>"
    key_mgmt=WPA-PSK
}

network={
    ssid="<Your-second-SSID>"
    psk="<Your-second-router-password>"
    key_mgmt=WPA-PSK
}
```

Network configuration using Network Manager nmcli

Bookworm and **Trixie** do not use **wpa_supplicant.conf** but now use Network Manager using the following **nmcli** commands. There are two methods of configuring roaming. The first is using **nmcli** commands.

Switch on Wi-Fi

```
$ sudo nmcli radio wifi on
```

Add the new Wi-Fi interface

```
$ sudo nmcli connection add con-name "<Your second SSID>" type wifi ifname wlan0 ssid "<Your second SSID>"
```

Set up the password and authentication method

```
sudo nmcli connection modify <Your second SSID> 802-11-wireless-security.key-mgmt wpa-psk wifi-sec.psk "<Your 2nd router password>"
```

Each new Wi-Fi network added has a separate config file in **/etc/NetworkManager/system-connections/** in a well configured system these will have a 256bit WPA PSK rather than a plain text passphrase.

For example, for an SSID **EE-GH6J42** the previous instructions will produce a file called **EE-GH6J42.nmconnection** in the **/etc/NetworkManager/system-connections** directory.

```
[connection]
id=EE-GH6J42
uuid=8e2ed8c9-faa2-4ad2-a8f5-a956d8afc7c1
type=wifi
interface-name=wlan0

[wifi]
mode=infrastructure
ssid= EE-GH6J42

[wifi-security]
key-mgmt=wpa-psk
psk=<Your 2nd router password>

[ipv4]
method=auto

[ipv6]
addr-gen-mode=default
method=auto

[proxy]
```

Connect to new router

To connect and enable data roaming to the second router enter the following command:

```
$ sudo nmcli connection up <Your 2nd router SSID>
```

NB. Do not use quotes around your router SSID for example

sudo nmcli connection up " EE-GH6J42" WRONG!

sudo nmcli connection up EE-GH6J42 CORRECT

You may wonder where the original connection for router is to be found for example **EE-B944TH**. If you look in the **/run/NetworkManager/system-connections/** you will see the following files.

```
$ ls /run/NetworkManager/system-connections
lo.nmconnection netplan-wlan0-EE-B944TH.nmconnection 'Wired connection
1.nmconnection'
```

This is because Debian Linux is now using a product called **netplan** to configure the Raspberry Pi which takes the parameters specified using the **Raspberry Pi Imager software** and configures the initial network configuration in the **/run** directory. It is rather confusing to have network configurations in two separate directories and this may change in the future.

Using the Network Manager nmtui utility

The second method is to use Network manager provides the **nmtui** utility to add, delete and connect to new Wi-Fi access points. To access it run:

```
$ sudo nmtui
```

This will display the **nmtui** utility. Using the direction arrows select **<Add>** and then select Wi-Fi as the type of connection you want to add.

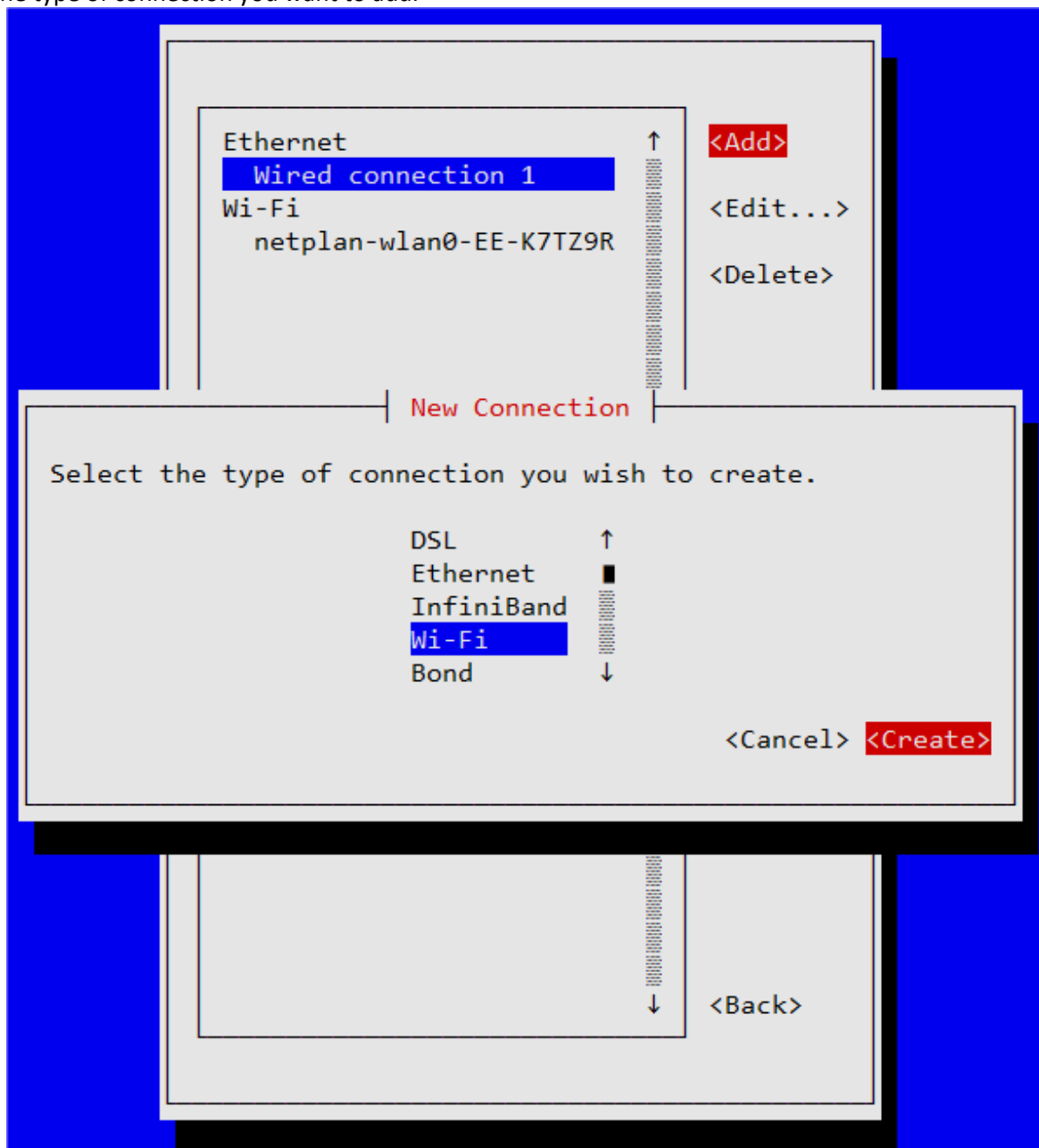


Figure 257 Network Manager nmtui utility

Now move to <Create> and press enter.

Edit Connection

Profile name

Device

WI-FI

SSID <Hide>

Mode

Security

Password [X] Show password

BSSID

Cloned MAC address

MTU

= IPv4 CONFIGURATION <Show>

= IPv6 CONFIGURATION <Show>

[X] Automatically connect

[X] Available to all users

<Cancel> **<OK>**

Figure 258 Adding a new router connection

Use the arrow keys on your keyboard to move around the screen. The space bar selects [X]. The profile name can be anything you like for example 'home' or 'office', but it is recommended to enter your router **SSID** for both the Profile name and **SSID** fields so it is easier to see which configuration file is which.

The Device will always be **wlan0** unless you have an extra Wi-Fi adapter

Select **WPA & WPA2 Personal** in the Security drop-down box (refer to your router documentation)

Enter your router **password** into the Password field. Leave all other fields as they are then press **<OK>**

Unfortunately connecting to the 2nd router doesn't work using **nmcli** as it doesn't show the new connection. Use the following command: NB. Do not use quotes around your router.

```
$ sudo nmcli connection up <Your SSID>
```

Reboot your Raspberry Pi

```
$ sudo reboot
```

Network prioritisation

There are two network management systems in use with modern Linux systems namely:

- **NetworkManager** (Bookworm/Trixie)
- **wpa_supplicant** (Bullseye and earlier)

NetworkManager chooses networks by prioritising wired connections (Ethernet) first and then any known Wi-Fi networks, selecting the most recently used among those, and falling back to scanning for new ones when necessary.

You can change the priority using the following command:

```
sudo nmcli c modify <ID> connection.autoconnect-priority <n>
```

For example:

```
$ sudo nmcli c modify preconfigured connection.autoconnect-priority 100
```

Apply the changes by restarting the connection:

```
$ sudo nmcli connection up <your SSID>
```

wpa_supplicant chooses networks primarily by priority defined in its configuration file, connecting to the highest-priority available network first, and then by the order they appear if priorities are equal, using settings like `priority=X` (where X is the priority number) and BSSID to guide its choice. The higher the number X, the higher the priority.

/etc/wpa_supplicant/wpa_supplicant.conf

```
network={
    ssid="<your SSID>"
    psk=psk
    key_mgmt=WPA-PSK
    priority=80
}

network={
    ssid="<your 2nd SSID>"
    psk=psk
    key_mgmt=WPA-PSK
    priority=100
}
```

In the above example the second defined network takes priority

Chapter 9 -Troubleshooting

Contents chapter 9	Page
General	218
Running the radio software in diagnostic mode	218
HDMI display problems	219
Unable to log into the Raspberry Pi Desktop	220
Debugging the graphical programs gradio.py and vgradio.py	221
LCD display problems	221
Testing Rotary encoders	221
Bluetooth pairing problems	222
Test radios which are using buttons	223
Test events layer	225
Test configured display	225
Testing GPIO inputs	226
Displaying the Radio and Operating System configuration	227
Displaying log files	228
Troubleshooting Raspotify	230
Network problems	231
Radio keeps skipping stations	234
Web interface OM!PD error	235
Graphical radio shows wrong artwork	236

General troubleshooting

Most of the diagnostics can be run from the **radio-config** menu.



Note: When selecting diagnostics from the following menu by necessity need to stop the Radio and IR remote control daemons. Make sure that you are not doing anything important such as recording a radio station before you select any diagnostic programs.

Log into the Raspberry Pi and enter **radio-config** on the command line:

```
$ radio-config
```

Most diagnostics will be found in option **5 Diagnostics and Information** of the **radio-config** menu as shown below.

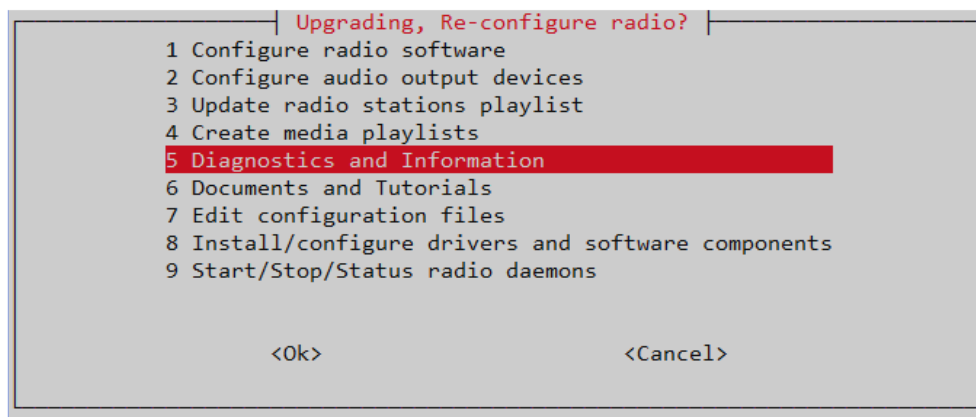


Figure 259 Diagnostics and Information menu

Selecting option 5 will show the following menu:

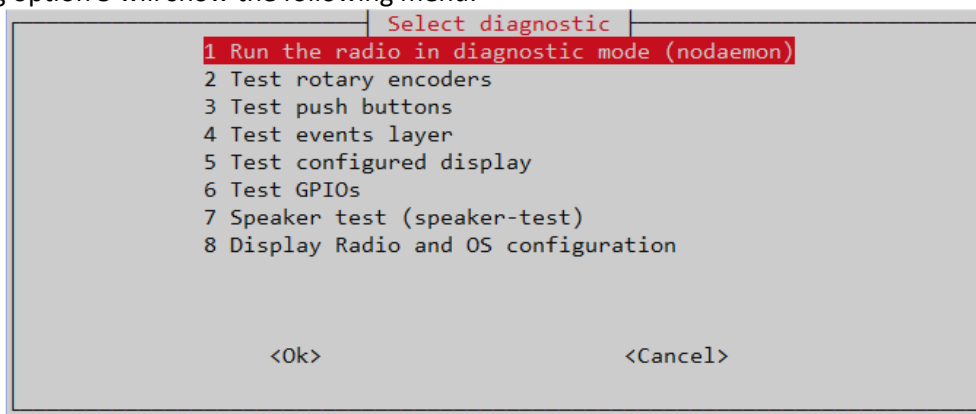


Figure 260 Diagnostics selection menu

Running the radio software in diagnostic mode

The **radiod** program normally runs as a background service (a so-called **daemon**) and is controlled with the **systemctl stop/start** commands. As previously mentioned, the diagnostics, when required, will stop the **radiod daemon** and start these in foreground mode so that you can see what is happening.

To run the radio software in diagnostic mode.

- Select option **1 Run the radio in diagnostic mode (nodaemon)** shown in *Figure 259 Diagnostics and Information menu* on page 218.

The radio program will be started in diagnostic mode.

```
Press Ctrl-C to exit diagnostic mode
radiod.py nodaemon diagnostic log, Fri 13 Dec 12:39:12 GMT 2024
```

If there are any errors they will be displayed immediately under the above message.

To exit and return to the menu press **Ctrl-C**

```
^C
A log of this run has been recorded in
/usr/share/radio/logs/radiod_nodaemon.log
A compressed tar file has been saved in
/usr/share/radio/logs/radiod_nodaemon.log.tar.gz
Send /usr/share/radio/logs/radiod_nodaemon.log.tar.gz to bob@bobrathbone.com
if required
Press enter to continue:
```

Either copy and paste any errors into an email or send the log the above log and send this send to support.

HDMI display problems

The following problem was encountered on a **WaveShare 7-inch HDMI display**. Other displays may also be affected. Instead of displaying an **X-Windows Desktop** the following may be seen.

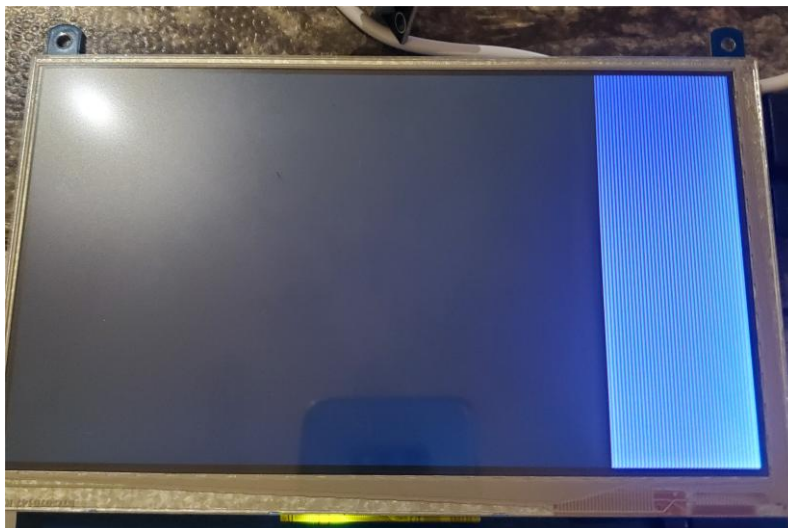


Figure 261 HDMI display problem

To correct this edit the **/boot/firmware/config.txt** file and enable the **vc4-kms-driver** by removing the **#** character in the following line. Reboot the Raspberry Pi. The **X-Windows Desktop** should now be displayed.

```
#dtoverlay=vc4-kms-v3d
```

Unable to log into the Raspberry Pi Desktop

This problem more recently with **Debian Trixie Desktop** version released for the **Raspberry Pi** on the 2nd of October 2025. It can happen if you run **autoremove** to remove packages such as **pipewire** or **pulseaudio-module-bluetooth** that have a dependency on **rpd-wayland-core**. Unfortunately, **autoremove** will also remove **rpd-wayland-core** which is a critical component of the Debian desktop. This problem has now been corrected in versions of Trixie released after December 2025.

The next time the Raspberry Pi is booted into the desktop it goes to the windows login screen. However, despite using the correct username and password you cannot login and get stuck in an endless loop of login requests. Also, the login screen does not look “normal”.

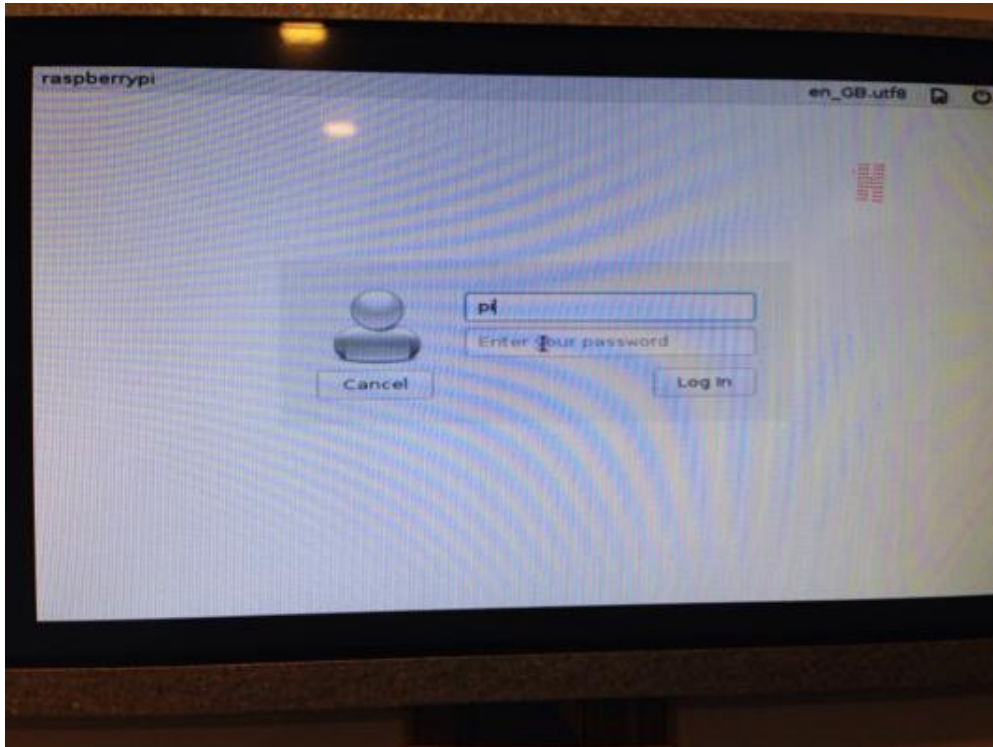


Figure 262 Desktop stuck in an endless login loop

To correct this problem, log into the Raspberry Pi using SSH then run:

```
$ sudo apt install rpd-wayland-core
```

Repeat warning.



WARNING: During installation you will see the following message:
*The following packages were automatically installed and are no longer required:
..... List of packages
Use 'apt-get autoremove' to remove them.*

DO NOT DO THIS as under the **Trixie OS** it may remove the **rpd-wayland-core** package which prevents you from logging into the Raspberry Pi Desktop. These packages are not doing any harm other than taking up some room on disk.

Debugging the graphical programs `gradio.py` and `vgradio.py`

If the graphics versions of the radio are not starting up you can start them from an **SSH** terminal using **Bitwise**. Log into the Raspberry Pi using a **Bitwise** SSH terminal session.

Enter the following command to start the full feature radio.

```
$ DISPLAY=:0.0 sudo /usr/share/radio/gradio.py
```

Or for vintage lookalike graphical radio run

```
$ DISPLAY=:0.0 sudo /usr/share/radio/vgradio.py
```

Either command will start the graphical radio program on your HDMI monitor or touch screen. You will now be able to see where it is crashing on the **SSH** terminal.

If you have two displays, for example, an HDMI screen and a small touch-screen and have **hdmi_force_hotplug=1** configured in **/boot/firmware/config.txt** then you will need to change the **DISPLAY** number from **0.0** to **1.0** as shown below.

```
$ DISPLAY=:1.0 sudo /usr/share/radio/gradio.py
```

LCD display problems

There are a number of reasons why your LCD may not be displaying correctly. The most common causes are:

1. For directly connected LCDs, incorrect wiring is most common – check it and check again
2. Incorrect selection of the display type. Reconfigure using the radio-config utility.
3. If using an I2C backpack check that
 - a. you have configured the correct I2C address. Use **i2cdetect -y 1** to display it
 - b. you have correctly set the brightness potentiometer on the I2C backpack

There are a number of LCD diagnostics you can use to check your display depending upon the type:

lcd_adafruit_class.py, **lcd_class.py**, **lcd_i2c_jhd1313.py**, **lcd_piface_class.py**
lcd_class_gpiozero.py, **lcd_i2c_adafruit.py**, **lcd_i2c_pcf8574.py** **lcd_vfd.py**

For example, for an LCD directly connected to the GPIO pins (no backpack) run:

```
$ cd /usr/share/radio  
$ ./lcd_class.py
```

Testing Rotary encoders

To test the rotary encoders

- Select option **2 Test rotary encoders** from the diagnostics menu. shown in *Figure 259 Diagnostics and Information menu* on page 218.

```
Press Ctrl-C to exit diagnostic mode  
Test standard rotary encoder Class  
Left switch GPIO 14  
Right switch GPIO 15
```

```
Up switch GPIO 24
Down switch GPIO 23
Mute switch GPIO 4
Menu switch GPIO 17
Rotary encoder step size = full
KY040 encoder R1 resistor fitted = no
Waiting for events
```

Now operate each rotary encoder in turn including the push-button on each one. Volume and Tuner events should be seen. If not check your wiring against the configured GPIO settings shown above. Press **Ctrl-C** to end test.

```
Tuner event 2 ANTICLOCKWISE
Tuner event 2 ANTICLOCKWISE
Tuner event 2 ANTICLOCKWISE
Tuner event 1 CLOCKWISE
Tuner event 1 CLOCKWISE
Tuner event 1 CLOCKWISE
Tuner event 1 CLOCKWISE
Tuner event 3 BUTTON DOWN
Volume event 1 CLOCKWISE
Volume event 1 CLOCKWISE
Volume event 1 CLOCKWISE
Volume event 2 ANTICLOCKWISE
Volume event 2 ANTICLOCKWISE
Volume event 3 BUTTON DOWN
```

Bluetooth pairing problems

General

Most problems can be avoided by using **radio-config** to install Bluetooth. See *Installing a Bluetooth audio device* on page 137.

Power on fails

The following message is seen when trying to power on Bluetooth.

```
$ bluetoothctl
Agent registered
[bluetoothctl]> power on
Failed to set power on: org.bluez.Error.Failed
```

Check if Bluetooth is blocked.

```
$ rfkill list all
0: hci0: Bluetooth
   Soft blocked: yes
   Hard blocked: no
1: phy0: Wireless LAN
   Soft blocked: no
   Hard blocked: no
```

Unblock it

```
$ rfkill unblock bluetooth
```

Run **bluetoothctl power on** again. It should now succeed.

```
$ bluetoothctl
Agent registered
[bluetoothctl]> power on
Changing power on succeeded
```

Cannot identify my Bluetooth device

Identifying your Bluetooth device can be a bit of a challenge. During discovery your device will be displayed in the format **<mac address>:<name>**, for example: **00:75:58:41:B1:25 SP-AD70-B**. The **<name>** will be typically the name or model name of your device. Look at the label on the device or any documentation to identify your device name.

It could be that your device is already connected. Check connected devices with **bluetoothctl**.

```
$ bluetoothctl devices Connected
Device 00:75:58:41:B1:25 SP-AD70-B
```

During the scan you might also see your device displaying its RSSI (Received Signal Strength Indicator) and its signal strength (-51 in this case)

```
[CHG] Device 00:75:58:41:B1:25 RSSI: 0xfffffffcd (-51)
```

Signal strength is indicated by a negative number in dBm (decibels relative to one milliwatt); a value closer to zero (e.g., -30 dBm) means a stronger, better signal, while more negative numbers (e.g., -90 dBm) signify a weaker signal

It either case stop the scan remove the device as shown in the following example.

```
[SP-AD70-B]> scan off
Discovery stopped
[SP-AD70-B]> remove 00:75:58:41:B1:25
```

Now retry scanning and pairing your Bluetooth device.

Test radios which are using buttons

To test the rotary encoders

- Select option **2 Test rotary encoders** from the diagnostics menu shown in *Figure 259 Diagnostics and Information menu* on page 218.

```
Test Button Class or Ctl-C to exit:
Left switch GPIO 14
Right switch GPIO 15
Mute switch GPIO 4
Up switch GPIO 24
Down switch GPIO 23
Menu switch GPIO 17
Pull Up/Down resistors UP
Record switch GPIO 27 Pull Up
```

Press each button in turn including the Record button.


```
Button pressed on GPIO 4
Button pressed on GPIO 4
Button pressed on GPIO 4
Button pressed on GPIO 17
Button pressed on GPIO 17
Button pressed on GPIO 27
Button pressed on GPIO 27
Button pressed on GPIO 14
Button pressed on GPIO 14
Button pressed on GPIO 15
Button pressed on GPIO 15
Button pressed on GPIO 24
Button pressed on GPIO 23
Button pressed on GPIO 23
```

Test events layer

All events from lower-level drivers such as the Rotary encoder and Button classes are sent up to the event class. This formats all events into a common set of events which are then passed up to various top level radio programs (radiod, gradio and vgradio).

- Select option **4 Test events layer** from the diagnostics menu shown in *Figure 259 Diagnostics and Information menu* on page 218. This displays the following output

```
Press Ctrl-C to exit diagnostic mode
Waiting for events:
```



Note: If the lower-level Rotary encoder or button tests are not working then there are no events to be seen in the Event layer. Also note that the numbers displayed are Event numbers and not GPIO numbers.

The following output is typical from running this test whilst operating Rotary encoders and the Record switch.

```
Event 6 DOWN_SWITCH
Event 5 UP_SWITCH
Event 2 LEFT_SWITCH
Event 1 RIGHT_SWITCH
Event 3 MUTE_BUTTON_DOWN
Event 7 MENU_BUTTON_DOWN
Event 22 RECORD_BUTTON
```



Note: Also note that events from the IR Remote Control do not use the **event_class** so will not be seen by the event layer. IR events are sent over the network directly to the Radio Class.

Test configured display

To test the configured display:

- Select option **5 Test configured display** from the diagnostics menu shown in *Figure 259 Diagnostics and Information menu* on page 218.



Note: That the success of this test relies upon the display being correctly configured. It is only used for testing **LCDs**, **TFTs** and **OLED** display screens. This test is not relevant for Graphical displays which run in a Windows Desktop.

```
Test display_class.py ESC to exit:
Screen LCD Lines=4 Character width=20
Display type 1 LCD
bg_color 7 White

Enter to continue or ESC to exit:
```

Testing GPIO inputs

The Rotary encoder in particular may not be working but it may not be obvious what the problem is. The GPIOs test program may be useful to locate the problem.

- Select option **6 Test GPIOs** from the diagnostics menu shown in *Figure 259 Diagnostics and Information menu* on page 218.

```
Press Ctrl-C to exit diagnostic mode
Test Rotary encoders and buttons
GPIO: 2 State:High
GPIO: 3 State:High
GPIO: 4 State:High
GPIO: 5 State:High
GPIO 5 rising
GPIO: 6 State:High
GPIO 6 rising
GPIO: 7 State:High
GPIO: 8 State:High
GPIO 8 rising
GPIO: 9 State:High
GPIO: 10 State:High
GPIO: 11 State:High
GPIO: 12 State:High
GPIO: 13 State:High
GPIO 13 rising
GPIO: 14 State:High
GPIO: 15 State:High
Error: GPIO 16 'GPIO busy'
Check conflict with GPIO 16 in other programs or in /boot/config.txt
GPIO: 17 State:High
GPIO: 18 State:High
GPIO: 19 State:High
GPIO 19 rising
GPIO: 20 State:Low
GPIO: 21 State:High
GPIO 21 rising
GPIO: 22 State:High
GPIO 22 rising
GPIO: 23 State:High
GPIO: 24 State:High
Error: GPIO 25 'GPIO busy'
Check conflict with GPIO 25 in other programs or in /boot/config.txt
GPIO: 26 State:High
GPIO 26 rising
GPIO: 27 State:High
Waiting for input events:
```

At first it looks like GPIO 16 and 25 have a problem but this is simply because these two GPIOs are being used by the IR Remote Control program (ireventd.py) so can be ignored. Now test the problem Rotary encoder or push button that you are having problems with. In this example when you turn the volume Rotary Encoder you see that GPIO14 and GPIO18 are active.

```
Waiting for input events:
GPIO 15 falling
GPIO 18 falling
GPIO 15 rising
GPIO 15 rising
GPIO 15 rising
GPIO 18 rising
```

However, when you check against the results from the Rotary encoder test in *LCD display problems* There are a number of reasons why your LCD may not be displaying correctly. The most common causes are:

4. For directly connected LCDs, incorrect wiring is most common – check it and check again
5. Incorrect selection of the display type. Reconfigure using the radio-config utility.
6. If using an I2C backpack check that
 - a. you have configured the correct I2C address. Use **i2cdetect -y 1** to display it
 - b. you have correctly set the brightness potentiometer on the I2C backpack

There are a number of LCD diagnostics you can use to check your display depending upon the type:

**lcd_adafruit_class.py, lcd_class.py, lcd_i2c_jhd1313.py, lcd_piface_class.py
lcd_class_gpiozero.py, lcd_i2c_adafruit.py, lcd_i2c_pcf8574.py lcd_vfd.py**

For example, for an LCD directly connected to the GPIO pins (no backpack) run:

```
$ cd /usr/share/radio
$ ./lcd_class.py
```

Testing Rotary encoders on page 221 (See below) you see that it is configured for GPIO14 and GPIO15 and not GPIO15 and GPIO18 and so is incorrectly wired. Correct the wiring to clear the fault.

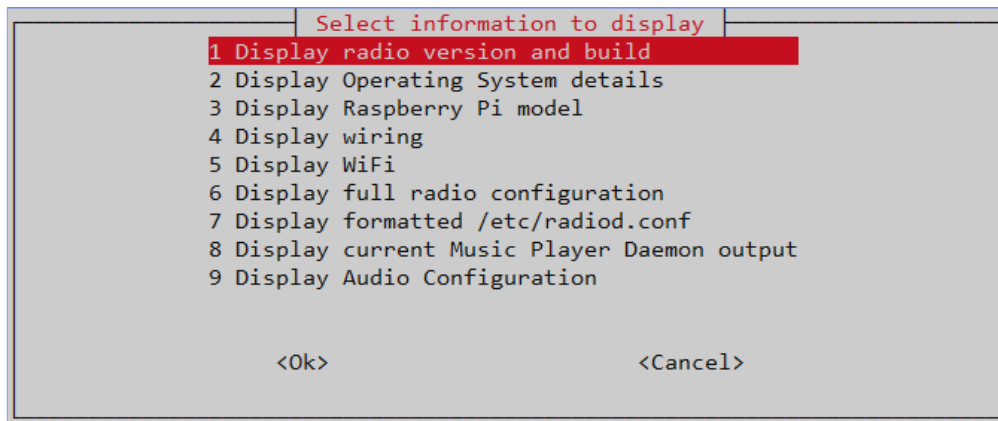
```
Press Ctrl-C to exit diagnostic mode
Test standard rotary encoder Class
Left switch GPIO 14
Right switch GPIO 15
```

The switch settings for Rotary encoders and Push buttons are configured in the **/etc/radiod.conf** configuration file.

```
# Switch settings for Rotary encoders or buttons
menu_switch=17
mute_switch=4
up_switch=24
down_switch=23
left_switch=14
right_switch=15
```

Displaying the Radio and Operating System configuration

- Select option **8 Display Radio and OS configuration** from the diagnostics menu shown in *Figure 259 Diagnostics and Information menu* on page 218.



The following table shows the details of each available report.

Option and description	
1.	Display radio version and build Version number and build, for example Version 8.3 Build 8.3.1
2.	Display Operating System details Displays OS (Bookworm or Trixie) details
3.	Display Raspberry Pi model Raspberry Pi model details eg. b03115: Model 4B, Revision 1.5, RAM: 2GB MB etc.
4.	Display wiring This displays how the wiring for displays, buttons etc. have been configured
5.	Display Wi-Fi Wi-Fi network information such as SSID, channel and signal strength
6.	Display full radio configuration Detailed display of all configuration details (OS and Radio) This creates the file <code>/usr/share/radio/logs/config.log.tar.gz</code> to be sent to support
7.	Display formatted /etc/radiod.conf Easy to read details of the Radio configuration file
8.	Display current Music Player Daemon output This displays the details of the currently playing radio station or track from MPD
9.	Display Audio Configuration Displays all configured sound cards (DACs, onboard jack etc.) and all Audio settings in the relevant system configuration files.

Displaying log files

Log files provide a lot of useful information when it comes to troubleshooting a problem with the Radio or Music Player Daemon (MPD) software. The principal log files are:

- The Radio log file: `/var/log/radiod/radio.log`
- The MPD log file: `/var/log/mpd/mpd.log`
- The daemon log file: `/var/log/daemon.log`

To look at the various log files first log into the Raspberry Pi Radio and then carry out the instructions in the following sections.

Displaying radio program log file

To display the logfile restart the radio and remote software control if running:

```
$ sudo systemctl restart radiod
$ sudo systemctl restart irreventd
```

To display the Radio logfile log into the Raspberry Pi and run the following command:
tail -f /var/log/radiod/radio.log

```
2025-12-05 12:55:01,515 INFO ===== Starting radio =====
2025-12-05 12:55:01,515 INFO Initialising radio pid 1109
2025-12-05 12:55:01,516 INFO Login name pi
2025-12-05 12:55:01,516 INFO User:pi(1000) Group:pi(1000)
2025-12-05 12:55:01,747 INFO Python version 3
2025-12-05 12:55:01,762 INFO Translation code page in radiod.conf = 0
2025-12-05 12:55:02,134 INFO Display code page 0x1
2025-12-05 12:55:02,134 INFO Loaded 'codes.European'
2025-12-05 12:55:02,135 INFO Loaded 'codes.Russian'
2025-12-05 12:55:02,135 INFO Loaded 'codes.English'
2025-12-05 12:55:02,135 INFO Screen LCD Lines=4 Character width=20
2025-12-05 12:55:07,847 INFO Romanize True
2025-12-05 12:55:07,939 INFO Board revision 2
2025-12-05 12:55:08,229 INFO OS release: Debian GNU/Linux 13 (trixie) 64bit
2025-12-05 12:55:08,232 INFO Linux piradio 6.12.47+rpt-rpi-v8 #1 SMP PREEMPT
Debian 1:6.12.47-1+rpt1 (2025-09-16) aarch64 GNU/Linux
2025-12-05 12:55:09,924 INFO Connected to MPD port 6600
2025-12-05 12:55:10,048 INFO UDP Server listening on localhost port 5100
2025-12-05 12:55:10,050 INFO UDP listen:remote 0.0.0.0 port 5100
2025-12-05 12:55:10,676 INFO IP 192.168.1.251
2025-12-05 12:55:13,186 INFO Radio ['/usr/share/radio/radiod.py',
'nodaemon'] Version 8.3
2025-12-05 12:55:13,187 INFO Radio running pid 1109
```

You may be asked to supply a log file in **DEBUG** mode for support purposes. The default logging level is **INFO**. Edit the **/etc/radiod.conf** file and switch on **DEBUG** mode as shown below:

```
# loglevel is CRITICAL,ERROR,WARNING,INFO,DEBUG or NONE
loglevel=DEBUG
```

Restart the radio and run the **tail** command shown above and you will see extra **DEBUG** lines in the log file.

```
2025-12-05 13:17:44,450 INFO ===== Starting radio =====
2025-12-05 13:17:44,451 INFO Initialising radio pid 1114
2025-12-05 13:17:44,451 INFO Login name pi
2025-12-05 13:17:44,451 INFO User:pi(1000) Group:pi(1000)
2025-12-05 13:17:44,691 DEBUG Mounted /dev/sda1 on /media/pi
2025-12-05 13:17:44,700 INFO Python version 3
2025-12-05 13:17:44,708 INFO Translation code page in radiod.conf = 0
2025-12-05 13:17:45,073 INFO Display code page 0x1
2025-12-05 13:17:45,073 INFO Loaded 'codes.European'
2025-12-05 13:17:45,074 INFO Loaded 'codes.Russian'
2025-12-05 13:17:45,074 INFO Loaded 'codes.English'
2025-12-05 13:17:45,075 INFO Screen LCD Lines=4 Character width=20
2025-12-05 13:17:50,846 INFO Romanize True
2025-12-05 13:17:50,926 DEBUG airplay = no
2025-12-05 13:17:50,927 DEBUG audio_config_locked = no
```


Displaying the MPD log entries

In **Debian Trixie**, the traditional syslog system has been replaced with **systemd-journald**. Old log files like **syslog**, **messages**, **auth.log** are no longer used. Instead, all system logs are now stored in a centralized journal, which is accessible using the **journalctl** command.

journalctl | grep -i <daemon name>

To display the MPD log entries, log into the Raspberry Pi and run the following command:

journalctl | grep -i mpd

```
Dec 05 09:13:55 trixie64 sudo[2013]:      root : PWD=/usr/share/radio ;
USER=root ; COMMAND=/usr/bin/ln -s /media/pi /var/lib/mpd/music/media
Dec 05 09:13:55 trixie64 sudo[2017]:      root : PWD=/usr/share/radio ;
USER=root ; COMMAND=/usr/bin/ln -f -s /share /var/lib/mpd/music
Dec 05 13:28:10 trixie64 sudo[2189]:      root : PWD=/usr/share/radio ;
USER=root ; COMMAND=/usr/bin/systemctl start mpd.socket
Dec 05 13:28:11 trixie64 systemd[1]: Listening on mpd.socket.
Dec 05 13:28:11 trixie64 systemd[1]: Starting mpd.service - Music Player
Daemon...
:
```

Displaying other system log entries

Any MPD errors should be displayed. This is usually the failure to load a particular sound driver

Daemons of interest are **mpd**, **librespot** (Spotify) and **bluetoothd**. For example, to look at Bluetooth processes run:

```
$ journalctl | grep -i bluetoothd
Dec 05 09:13:23 trixie64 kernel: Bluetooth: Core ver 2.22
Dec 05 09:13:23 trixie64 kernel: NET: Registered PF_BLUETOOTH protocol
family
Dec 05 09:13:23 trixie64 kernel: Bluetooth: HCI device and connection
manager initialized
Dec 05 09:13:23 trixie64 kernel: Bluetooth: HCI socket layer initialized
Dec 05 09:13:23 trixie64 kernel: Bluetooth: L2CAP socket layer initialized
Dec 05 09:13:23 trixie64 kernel: Bluetooth: SCO socket layer initialized
Dec 05 09:13:23 trixie64 kernel: Bluetooth: HCI UART driver ver 2.3
:
```

Troubleshooting Raspotify

Installation problems

If the curl command to install the Raspotify software fails then carry out a system update and upgrade as shown in *Update to the latest the system packages* on page 98. Retry the curl command.

Raspotify exits with a 101 error code

This is almost certainly an authentication fault. Check your user name and password are correctly set up in **/etc/default/raspotify** and retry. However, it isn't actually necessary to put a username and password in **/etc/default/raspotify** as Raspotify will be using a Raspotify Premium account running on a PC, tablet or mobile phone.

The client connects to Raspotify but no sound heard

Check that the device ID has been added to the **ExecStart** statement in the same **raspotify.service** file.

```
ExecStart=/usr/bin/librespot . . . . . --device=hw:0,0
```



Note: The author does not directly support Raspotify.
Report Raspotify issues to <https://github.com/dtcooper/raspotify/issues>
Raspotify comes with an MIT licence. See <https://opensource.org/licenses/MIT>

Cannot change Raspotify volume

In some cases, this is normal as many cheaper sound cards (DACs) do not have mixer controls. Mixer controls are required to change the volume on the Raspberry Pi when running Spotify or Airplay. In the case of no mixer controls the sound when running Spotify must be controlled from the App on either your mobile device or from the PC application.

However, it is possible to set up a software mixer (SoftVol).

Raspotify initial sound too loud

Edit the `/lib/systemd/system/raspotify.service` file. Locate the following line

```
Environment="VOLUME_ARGS=--enable-volume-normalisation --volume-ctrl linear  
--initial-volume 100"
```

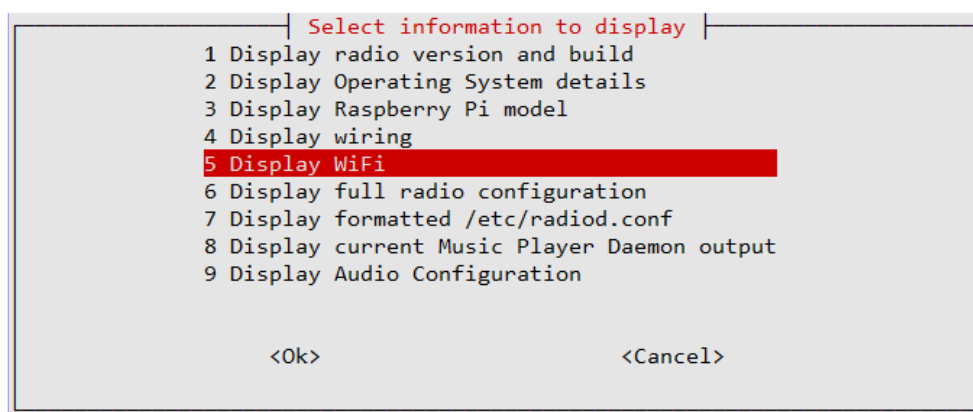
Change to the desired volume setting.

```
--initial-volume 75
```

Network problems

Checking the Wi-Fi network quality

Run `raspi-config` and select option **5 Diagnostics and Information** then select **8 Display Radio and OS configuration**.



Select option **5 Display WiFi**. A screen similar to the following will be displayed along with some notes to explain the displayed output.

```
WiFi configuration details

Wi-Fi network information
-----
Hostname: piradio
Wi-Fi: wlan0 ESSID:"EE-K7987F6"
IP address wlan0: 192.168.1.72
wlan0 Protocol Name:"IEEE 802.11"
Current Frequency:2.462 GHz (Channel 11)
Bit Rate=24 Mb/s Tx-Power=31 dBm
Link Quality=62/70 Signal level=-48 dBm
Power save: on
0: phy0: Wireless LAN
    Soft blocked: no
    Hard blocked: no
```

In the above example the name of the router (ESSID: EE-K7987F6), the Wi-Fi interface (wlan0) IP address, protocol, frequency and channel are displayed. Lastly the really important information is the Signal level in Decibels (dBm). If phy0 is blocked use “rfkill unblock 0” to enable it.

The following table explains the meaning of the Signal level.

Table 20 Wi-Fi signal strength

Signal strength	Signal quality	Notes
-30 dBm	Maximum signal strength, usually attained if the RPi is very close to the Wi-Fi access point	Excellent for all services
-50 dBm	Anything down to this level can be regarded as excellent signal strength	Excellent for all services
-60 dBm	This is still a good, reliable signal strength	Good for all services
-67 dBm	This is the minimum value for all services that require smooth and reliable data traffic.	Minimum required for streaming services including for the radio
-70 dBm	The signal is not very strong, but mostly sufficient. The radio is likely to skip stations.	Only good for Web browsing, email, and the like
-80 dBm	Minimum value required to make a connection. You cannot count on a reliable connection or sufficient signal strength to use services at this level	The radio will not work reliably at this level
-90 dBm	It is very unlikely that you will be able to connect or make use of any services with this signal strength.	The radio cannot operate reliably at this level.

If the signal strength is poor, try re-positioning the radio nearer the router or use a repeater (TP-Link or similar). Position away from radiators or other large metal objects such as fridges and freezers.

Checking response times with ping

Check the response times using the ping command from a PC or another Raspberry Pi. If possible, connect the PC/RPi via an Ethernet cable to the router rather than via Wi-Fi.

```
Command Prompt
C:\Users\bob_r>ping -t 192.168.1.167

Pinging 192.168.1.167 with 32 bytes of data:
Reply from 192.168.1.167: bytes=32 time=5ms TTL=64
Reply from 192.168.1.167: bytes=32 time=7ms TTL=64
Reply from 192.168.1.167: bytes=32 time=5ms TTL=64
Reply from 192.168.1.167: bytes=32 time=9ms TTL=64
Reply from 192.168.1.167: bytes=32 time=8ms TTL=64
Reply from 192.168.1.167: bytes=32 time=140ms TTL=64
Reply from 192.168.1.167: bytes=32 time=7ms TTL=64
Reply from 192.168.1.167: bytes=32 time=17ms TTL=64
Reply from 192.168.1.167: bytes=32 time=4ms TTL=64
Reply from 192.168.1.167: bytes=32 time=161ms TTL=64
Reply from 192.168.1.167: bytes=32 time=4ms TTL=64
Reply from 192.168.1.167: bytes=32 time=6ms TTL=64
Reply from 192.168.1.167: bytes=32 time=75ms TTL=64
Reply from 192.168.1.167: bytes=32 time=5ms TTL=64
Reply from 192.168.1.167: bytes=32 time=5ms TTL=64
Reply from 192.168.1.167: bytes=32 time=7ms TTL=64

Ping statistics for 192.168.1.167:
    Packets: Sent = 16, Received = 16, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 161ms, Average = 29ms
Control-C
```

Figure 263 The ping command

Typical response times are:

Via Ethernet cable - < 1 ms

Via Wi-Fi - 5 to 12 ms with occasional responses < 300 ms

Below is an example of a **bad** network connection.

```
64 bytes from amradio (192.168.1.167): icmp_seq=2923 ttl=64 time=8.62 ms
64 bytes from amradio (192.168.1.167): icmp_seq=2925 ttl=64 time=1127 ms
64 bytes from amradio (192.168.1.167): icmp_seq=2926 ttl=64 time=6.00 ms
64 bytes from amradio (192.168.1.167): icmp_seq=2927 ttl=64 time=14.2 ms
64 bytes from amradio (192.168.1.167): icmp_seq=2927 ttl=64 time=26.6 ms (DUP!)
64 bytes from amradio (192.168.1.167): icmp_seq=2928 ttl=64 time=7.61 ms
64 bytes from amradio (192.168.1.167): icmp_seq=2929 ttl=64 time=6.12 ms
64 bytes from amradio (192.168.1.167): icmp_seq=2945 ttl=64 time=9.00 ms
64 bytes from amradio (192.168.1.167): icmp_seq=2947 ttl=64 time=138 ms
```

A DUP can be seen (packet resent as no response the first one received). Also, the 2nd packet sent took 1.127 seconds to respond.

The above network will give nothing but problems and needs correction. Wired Ethernet connections are very unlikely to give a problem. For a bad Wi-Fi connection consider using a Wi-Fi repeater, such as TP-Link or Netgear, for blind spots in your premises. Put the repeater close to the Raspberry Pi. Search “bad wi-fi” on the Internet for further advice on how to solve this type of problem.

Wi-Fi is currently blocked by rfkill.

No Wi-Fi network is available and the following message is shown during login:

```
Wi-Fi is currently blocked by rfkill.  
Use raspi-config to set the country before use.
```

Running the **raspi-config** network set-up doesn't cure the problem. Run the **rfkill** utility to confirm the problem.

```
$ sudo rfkill list all  
0: phy0: Wireless LAN  
   Soft blocked: yes  
   Hard blocked: no  
1: hci0: Bluetooth  
   Soft blocked: no  
   Hard blocked: no
```

The above display confirms that the Wi-Fi network is blocked. Unblock it with **rfkill**:

```
$ sudo rfkill unblock all
```

Boosting your Wi-Fi network signal



TP-Link dual-band Wi-Fi
range extender

A weak **Wi-Fi** signal can cause a lot of problems such as skipping radio stations or intermittent operation. You can check for this by displaying the **Wi-Fi** details. See *Checking the Wi-Fi network quality* on page 231 and in particular the notes in *Table 20 Wi-Fi signal strength* on page 232. For reliable operation you need a signal strength of between **-30db** (strongest signal) and **-67db** (weakest signal).

To fix a weak **Wi-Fi** signal, try reposition your radio closer to the router and away from large metal objects such as radiators or kitchen equipment such as fridges, freezers or washing machines. Try switching to a **5 GHz** band if not already doing so, or using a **Wi-Fi** range extender as provided by companies such as **TP-Link**.

Radio keeps skipping stations

Cause 1 - Poor WiFi connection

When the Music Player Daemon has problems streaming a particular radio station it skips to the next one. This is a design decision made by the authors of the MPD daemon and is beyond the control of the Bob Rathbone Computer Consultancy. It is extremely irritating when this happens. Either re-positioning your radio or using a Wi-Fi range extender as shown in the previous section may help.

Cause 2 - Internet connectivity problems

Wi-Fi is connectivity at a link level (ISO model) and only means that you have good connectivity to your router. It does not necessarily mean that you have a reliable connection to the Internet. Check this with the ping utility.

```
$ ping -4 google.com  
PING google.com (172.217.76.102) 56(84) bytes of data.  
64 bytes from yulhrp-in-f102.1e100.net (172.217.76.102): icmp_seq=1 ttl=107 time=10.3 ms
```

```
64 bytes from yulhrp-in-f102.1e100.net (172.217.76.102): icmp_seq=2 ttl=107 time=14.8 ms
64 bytes from yulhrp-in-f102.1e100.net (172.217.76.102): icmp_seq=3 ttl=107 time=13.8 ms
64 bytes from yulhrp-in-f102.1e100.net (172.217.76.102): icmp_seq=4 ttl=107 time=15.1 ms
```

The response time should reasonably regular and should have a low response time of around 20 milliseconds or less. If it is irregular or long response times and your **Wi-Fi** connection is OK the try rebooting your router and generally check your incoming connection to your property.

The radio software regularly checks Internet connectivity by pinging **google.com**. In the **/etc/radiod.conf** configuration file you will find the following:

```
# Internet check URL. This must be a reliable URL and port number
# which can be contacted such as google.com
# The port number is normally 80 (HTTP).
# The internet_timeout in in seconds
# Disable by removing the URL from internet_check_url= parameter
internet_check_url=google.com
internet_check_port=80
internet_timeout=10
```

If you It can be disabled by removing the **internet_check_url** URL. There is no evidence at the moment that this will help the problem but it is worth a try.

```
internet_check_url=
```

Cause 3 – Bad station URL

There have been cases where changing the radio station configuration to use a different URL helps prevent skipping. For example for ABC Classic in Australia:

```
http://live-radio01.mediahubaustralia.com/2FMW/mp3/
```

The above URL regularly skips to the next station. Replace the above URL with the following URL which seems to cure this.

```
https://mediaserviceslive.akamaized.net/hls/live/2038316/classicfmnswh/index.m3u8
```

Try searching for an alternative URL for the station you are having trouble with or try contacting the radio station directly to see if they have an alternative URL.

Web interface OM!PD error

When you first run **OM!PD** you are told to that the database is empty and are asked to update it, but you get the following error.

```
Update error!
Failed to open directory:
/etc/chatscripts
  Check media_dir value in Settings -> Config
  Check file permission
```

The above error message is actually wrong. The problem is that the **media_dir** setting has not been set in the **OM!PD** settings. See the section called *Setting up the Music directory* on page 156

Graphical radio shows wrong artwork

This applies only to the `gradio.py` program which displays artwork for the radio station or currently play media track.

In both the case of Radio Station and Media streams, the associated artwork that is pulled from the **Discogs** database depends entirely on the Station name and program details or the Artist, Album and Track name that are submitted as the query to the **Discogs** database. If the submitted query is wrong then the artwork retrieved from the **Discogs** database will be wrong.

If you are finding that a particular radio station or media track then it may be possible to correct this.

To configure radio stations artwork, see section *Configuring radio station artwork selection* on page 188.

To configure media artwork, see section *Configuring media artwork selection* on page 189.

Licences and disclaimer

Licences

The software and documentation for this project is released under the GNU General Public Licence.

The GNU General Public License (GNU GPL or GPL) is the most widely used free software license, which guarantees end users (individuals, organizations, companies) the freedoms to use, study, share (copy), and modify the software. Software that ensures that these rights are retained is called free software. The license was originally written by Richard Stallman of the Free Software Foundation (FSF) for the GNU project.

The GPL grants the recipients of a computer program the rights of the Free Software Definition and uses *copyleft* to ensure the freedoms are preserved whenever the work is distributed, even when the work is changed or added to. The GPL is a *copyleft* license, which means that derived works can only be distributed under the same license terms. This is in distinction to permissive free software licenses, of which the BSD licenses are the standard examples. GPL was the first *copyleft* license for general use. This means that you may modify and distribute the software and documentation subject to the conditions of the licences.

See <http://www.gnu.org/licenses> for further information on the GNU General Public License.

The licences for the source and documentation for this project are:

GNU General Public License.	See http://www.gnu.org/licenses/gpl.html
GNU AFFERO General Public License.	See http://www.gnu.org/licenses/agpl.html
GNU Free Documentation License.	See http://www.gnu.org/licenses/fdl.html

Intellectual Property, Copyright, and Streaming Media

This is an unbelievably complex subject. The author is not a lawyer and cannot offer any legal advice on this subject. If you decide to stream your music content or relay a radio station stream back out to the internet or reproduce such content within a public building or space then you should seek legal advice.

See also: http://en.wikipedia.org/wiki/Copyright_aspects_of_downloading_and_streaming

In general Radio stations are providing a stream to promote their radio station. As media providers they should have arrangements in place to make the content that they provide is legally streamed across the Internet. The question is it legal to listen (or view) such content is a complex one and subject to local and international laws and which vary considerably.

If you implement **Icecast** or any other streaming technology such as **Comcast** to re-stream content within your own home then provided that this is not streamed back out to the Internet or a public location then one would think that you will not encounter any problems (but you never know). If you stream music tracks or relay radio stations back out onto the internet or public space then almost certainly you will be infringing a copyright law or intellectual property rights somewhere. The penalties for such an infringement can be severe.

WARNING: YOU USE THE ICECAST STREAMING IN THIS PROJECT AT YOUR OWN RISK ESPECIALLY IF YOU MAKE THE STREAM CONTENT AVAILABLE ACROSS THE INTERNET OR PUBLIC SPACE, EVEN IF YOU ARE JUST RELAYING AN EXISTING MEDIA STREAM, LEGAL OR OTHERWISE.

Also see the Disclaimer on page 239.

Legality of recording radio streams

Is it legal to record media streams from the Internet?

As usual, when it comes to legal issues, it is not a simple yes/no answer and depends upon the specific laws in your country or region.

Recording Internet radio is generally legal for personal, non-commercial use in many jurisdictions. In many countries, recording internet radio for personal use is often considered permissible under the "fair use" or "fair dealing" doctrines. These exceptions allow individuals to make copies of copyrighted material for personal enjoyment, such as creating offline playlists.

However, this usage must not be for commercial purposes or involve redistribution (See *Intellectual Property, Copyright, and Streaming Media* on page 237). For example, most smart TV's and Internet TV boxes have a recording facility built-in to allow recording of TV programs and other copyrighted material for later personal viewing or listening. There are exceptions in many countries for example, in the United Kingdom you may not record **Police, Taxi or Air Traffic Control** transmissions. Even listening to these may be considered illegal. Strangely possessing the means to listen to such transmissions such as a **radio scanner** or an **Internet Radio** is not itself illegal in the UK. It usually comes down to how you use such devices.

However, using recorded internet radio or other media content for commercial purposes, such as selling or publicly distributing the recordings or playing them on commercial or public premises, is typically prohibited without obtaining proper licenses or permissions from the copyright holders. If you plan to use recorded content for purposes beyond personal enjoyment, it's your responsibility to make sure that you comply with local copyright laws. This usually also applies to sharing recordings even if you do it for free.

For example, in the United Kingdom, you would require either a **PPL** (Phonographic Performance Limited) or **PRS** (Performing Rights Licence) for such usage.

Radio Artwork display legality

Radio station and album artwork is provided by the **Discogs** music and artwork database. See <https://www.discogs.com> Access to the **Discogs** database is provided through the **Discogs API** (Application programmer interface). It isn't perfect and sometimes the incorrect or no artwork will be displayed.



Legal note: Nearly all Artwork is copyrighted and may not be used for any other purpose other than viewing it. The radio program simply displays a reduced size of the artwork to enhance the viewing experience of the radio program. In that respect it is no different from viewing such artwork in a Web Browser. Also, the reduced size to fit the display window loses some of the quality and detail of the original artwork. It is entirely the responsibility of the user to ensure that this feature is used legally. In particular you should not attempt to further distribute any downloaded artwork. Please also see the *Disclaimer* on page 239.

Disclaimer

THIS SOFTWARE AND DOCUMENTATION IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS 'AS IS' AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT OWNER OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS, PENALTIES FOR UNAUTHORISED USE; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE OR DOCUMENTATION, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.



Note: The author does not distribute any recording facility, stream rippers or streaming packages with the **radiod** package. These are exclusively provided by third-party suppliers and require the user to intentionally download and install them separately. It is entirely the responsibility of the user to ensure that these facilities are used legally.



Note: Most artwork displayed in the graphics version of the radio is copyrighted by its authors or copyright holders and is for viewing only. You may not copy or use such artwork for any other purpose other than viewing it in the radio software.

Glossary

ALSA	Advanced Linux Sound Architecture
API	Application Program Interface
CAD	Control and Display (PiFace)
DAC	Digital to Analogue Converter (Digital to audio frequency analogue in this case)
DSI	Display Serial Interface is a specification by the Mobile Industry Processor Interface (MIPI)
DT	Device Tree (Overlay). Device configuration in /boot/firmware/config.txt in Debian
EMI	Electromagnetic Interference (For example fluorescent lighting etc.)
HDMI	High-Definition Multimedia Interface for audio and video plus Ethernet interface.
GPIO	General Purpose IO (On the Raspberry PI)
HDMI	High-Definition Multimedia Interface such as on most TVs and Computer monitors
I2C	Industry standard serial interface (Philips now NXP) using data and clock signals
I2S	Inter-IC Sound (Used in DAC interface) from Philips (Now NXP)
LCD	Liquid Crystal Display, also see OLED
LIRC	Linux Remote Control software

M3U	MPEG3 URL
MPC	Command line client for MPD
MPD	Music Player Daemon
MPEG	Moving Picture Experts Group
MPEG3	Music encoding standard from MPEG
NAS	Network Attached Storage
NFS	Network File System
NVMe	Non-Volatile Memory Express, the new generation of Solid-State Drive (SSD)
OLED	Organic Light Emitting diode. Also, OLED character displays gradually replacing LCDs
OS	Operating system (Debian Trixie or Bookworm in this case)
PCB	Printed Circuit Board
PPL	Phonographic Performance Limited – Licensing authority for copyrighted material
PRS	Performing Rights Licence – Licensing of copyrighted material in own performances
PSSD	Portable SSD disk drive – Also see SSD
RSS	Really Simple Syndication – Web feed usually containing news items
RSSI	Bluetooth Received Signal Strength Indicator in Decibels
SPI	Serial Peripheral Interface (using GPIO pins on the Raspberry Pi)
SSD	Solid State Disk. Also see NVMe
XDG	X Desktop Group. Interoperability software between X-Windows and Wayland windows

Appendix A – Wiring diagrams and lists

The following tables show the wiring for the various versions of the radio. These configurations are normally set up by the **configure_radio.sh** program with the exception of the Vintage radio. See *Configuring the radio* on page 110. It is also necessary to set the **pull_up_down** parameter in **/etc/radiod.conf** depending on wiring.

A.1 Push Button and Rotary Encoder 40-pin wiring

The following table shows the wiring for the 40-pin push-buttons or rotary encoders.

Table 21 40-Pin Push-buttons/Rotary encoder Wiring

Buttons/Encoders	Pin	GPIO	Configuration	Radio function
Menu button	11	17	menu_switch	Menu
Channel down button/Rotary switch A	16	23	down_switch	Channel down
Channel up button/Rotary switch B	18	24	up_switch	Channel up
Mute button	7	4	mute_switch	Mute sound
Volume down button/Rotary switch A	8	14	left_switch	Volume Down
Volume up button/Rotary switch B	10	15	right_switch	Volume Up

A.2 Push Button and Rotary Encoder 26-pin wiring

The following table shows the wiring for the 26-pin push-buttons or rotary encoders.

Table 22 26-Pin Push-buttons/Rotary encoder Wiring

Buttons/Encoders	Pin	GPIO	Configuration	Radio function
Menu button	22	25	menu_switch	Menu
Channel down button/Rotary switch A	19	10	down_switch	Channel down
Channel up button/Rotary switch B	11	17	up_switch	Channel up
Mute button	7	4	mute_switch	Mute sound
Volume down button/Rotary switch A	8	14	left_switch	Volume Down
Volume up button/Rotary switch B	10	15	right_switch	Volume Up

Set **pull_up_down=up** in **/etc/radiod.conf** depending on wiring.

A.3 IQaudIO Cosmic Controller wiring

The following table shows the wiring for the IQaudIO Cosmic controller.

Table 23 IQaudIO Cosmic Controller Wiring

Physical control	Pin	GPIO	Configuration	Radio function
Left hand push button	7	6	down_switch	Channel down
Middle push button	29	5	menu_switch	Menu
Right hand push button	31	4	up_switch	Channel up
Rotary encoder A input	16	23	left_switch	Volume control
Rotary encoder B input	18	24	right_switch	“ “
Rotary encoder Switch	13	27	mute_switch	Mute switch
Left status LED	8	14	rgb_red	Error status
Middle status LED	10	15	rgb_blue	Busy status
Right status LED	36	16	rgb_green	Normal status
IR sensor	22	25	In /boot/firmware/config.txt	Remote control

A.4 Pimoroni Pirate Radio wiring

The following table shows the wiring for the Pimoroni Pirate radio (pHat BEAT). Orientation is with the basic push buttons on the left-hand side. The menu button is on the top left-hand side.

Table 24 Pimoroni Pirate radio (pHat BEAT) Wiring

Pimoroni buttons	Pin	GPIO	Configuration	Radio function
On Off button	32	12	menu_switch	Menu
Channel Up button	29	5	up_switch	Channel up
Mute button	31	6	mute_switch	Mute sound
Channel Down button	16	13	down_switch	Channel Up
Volume Up button	36	16	right_switch	Volume Up
Volume Down Button	37	26	left_switch	Volume Down

The On/Off button used in the Pimoroni Radio software re-assigned as the menu switch with the Rathbone radio software. All LCD outputs are set to zero (No display).

Set `pull_up_down=up` in `/etc/radiod.conf`

A.5 Pimoroni Pirate Audio wiring

The following table shows the wiring for the Pimoroni Pirate radio (pHat BEAT).

Table 25 Pimoroni Pirate radio Audio Wiring

Radio buttons	Pimoroni	Pin	GPIO	Configuration	Radio function
Channel Up button	X	36	16	up_switch	Channel up
Channel Down button	A	29	5	down_switch	Channel Down
Volume Up button	Y	18	20/24 (1)	right_switch	Volume Up
Volume Down Button	B	31	6	left_switch	Volume Down

Note 1: Button Y (Volume up) can be GPIO 24 on earlier versions of the Pirate Audio card. The new settings in `/etc/radiod.conf` are normally set to the following:

```
up_switch=16
down_switch=5
left_switch=6
right_switch=20
```

For earlier versions of the Pirate Audio this is.

```
right_switch=24
```

The Pimoroni Pirate Audio only has four buttons which means there are no buttons available for **Menu** or **Mute**.

For **Mute** press **Volume UP** and **Volume DOWN** together.

For **Menu** press **Channel UP** and **Channel Down** together.

The menu function currently doesn't work that well at the moment but is good enough to get to the Search menu. There is no shutdown function using menu button long press as with other designs.

Set `pull_up_down=up` in `/etc/radiod.conf`

A.6 Vintage Radio Push-button/Rotary Encoder 40-pin wiring

The following table shows the wiring for the 40-pin push-buttons or rotary encoders. This set-up must be manually configured in `/etc/radiod.conf`. It cannot currently be configured by the `configure_radio.sh` script.

Table 26 40-PinPush-buttons/Rotary encoder Wiring

Buttons/Encoders	Pin	GPIO	Configuration	Radio function
Menu button	22	25	menu_switch	Menu
Channel down button/Rotary switch A	19	10	down_switch	Channel down
Channel up button/Rotary switch B	11	17	up_switch	Channel up
Mute button	7	4	mute_switch	Mute sound
Volume down button/Rotary switch A	8	14	left_switch	Volume Down
Volume up button/Rotary switch B	15	10	right_switch	Volume Up

Table 27 Status LED indications

GPIO	Pin	LED	Function
23	16	Red	Error condition, shutdown in progress, IR activity (If configured)
22	15	Blue	Busy condition such as start-up, loading or changing radio stations or tracks.
27	13	Green	Normal operation such as playing stations or tracks.

Table 28 Rotary menu switch

GPIO	Pin	Switch value
24	18	1
8	24	2
7	26	4

Combining the above switch values gives a composite switch value of 0 through 7.

0=Idle, 1=Speak current station/track, 3=Search mode, 4=Source menu, 5=Options Menu 6,7 unused

A.7 SH1106 SPI 1.3" OLED with joystick and 3-buttons

Table 29 SH1106 1.3" OLED with joystick – GPIO pin assignments

Joystick	Buttons	SPI Interface	Radio Function	GPIO Pin	Configuration
Up			Channel Up	6	/etc/radiod.conf
Down			Channel Down	19	" " "
Left			Volume Down	5	" " "
Right			Volume Up	26	" " "
Push			Menu switch	13	" " "
	KEY1		Mute Switch	21	/etc/radiod.conf
	KEY2		Not used	n/a	" " "
	KEY3		Not used	n/a	" " "
		RST_PIN		25	sh1106_config.py
		DC_PIN		24	" "
		CS_PIN		8	" "
		BL_PIN		18	" "



Please note that none of these pin assignments for the SH1106 1.3" OLED display with joystick can be changed by the user and are documented purely for reference.

The GPIO pin assignments for the joystick and button interface are defined in the `/etc/radiod.conf` configuration file and are configured by the `configure_radio.sh` program. The display settings are also configured in `radiod.conf`.

```
menu_switch=13
mute_switch=21
up_switch=6
down_switch=19
left_switch=5
right_switch=26

display_type=SH1106_SPI

display_width=16
display_lines=4

lcd_select=0
lcd_enable=0
lcd_data4=0
lcd_data5=0
lcd_data6=0
lcd_data7=0
```

The GPIO pin assignments for the SPI configuration are defined in the `sh1106_config.py` configuration file in the `/usr/share/radio program` directory.

```
# Pin definition
RST_PIN      = 25
DC_PIN       = 24
CS_PIN       = 8
BL_PIN       = 18
```

Converting the 1.3" OLED pHat to use the I2C interface



Warning: Do not attempt this procedure unless you possess considerable soldering skills. The risk of damaging both the card and the Raspberry Pi is great if you make a mistake.

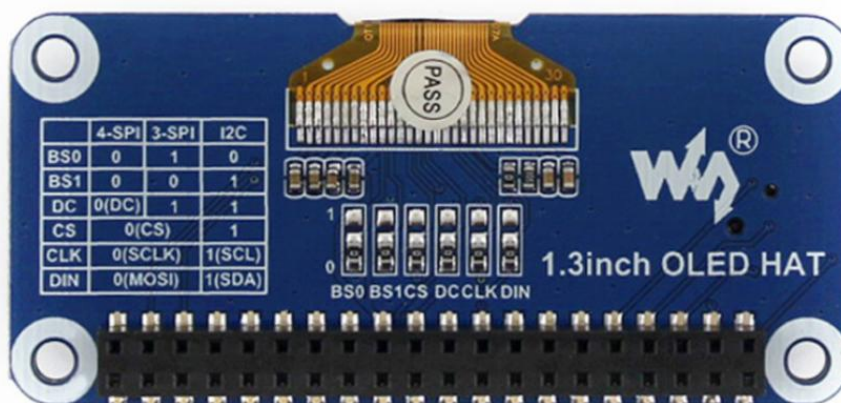


Figure 264 1.3" OLED pHat reverse side

The module uses a 4-wire SPI communication mode by default, that is, BS0, BS1, DC, CS, CLK, and DIN are connected to 0 by default (1 and 0 do not represent the level, but the welding method of connecting or connecting the resistance, the specific hardware link is as follows surface):

Note: The above picture is the welding on the hardware, the following table is the actual hardware connection:

Communication method	BS1/BS0	CS	DC	DIN	CLK
3-wire SPI	0/1	CS	1	MOSI	SCLK
4-wire SPI	0/0	CS	DC	MOSI	SCLK
I2C	1/0	0	1	SDA	SCL

The specific hardware re-wiring for I2C is as follows:

Connect BS0 to 0 to GND, BS1 to 1 to VCC (3.3V), CS to 1 to connect to GND, D/C to 1 to connect to GND, DIN to 1 to connect to Raspberry Pi SDA, CLK to 1 to connect to the tree Raspberry Pi SCL; when using I2C: the high and low states of DC can control the address of the slave device, here is connected to GND, then the 7-bit address of I2C is: 0x3C.

To switch to I2C using the I2C address 0x03 it is necessary to amend the **sh1106_config.py** configuration file. Change the **DeviceSPI** and **Device_I2C** definitions in the configuration file to enable the I2C interface instead of I2C.

```
Device_SPI = 1
Device_I2C = 0
```

To

```
Device_SPI = 0
Device_I2C = 1
```

See https://www.waveshare.com/wiki/1.3inch_OLED_HAT for further information.



The author has not tested the above procedure so if any technical support is required, please contact Waveshare directly at <https://support.waveshare.com>

A.8 Waveshare 1.42" and 1.5" SPI OLED display wiring

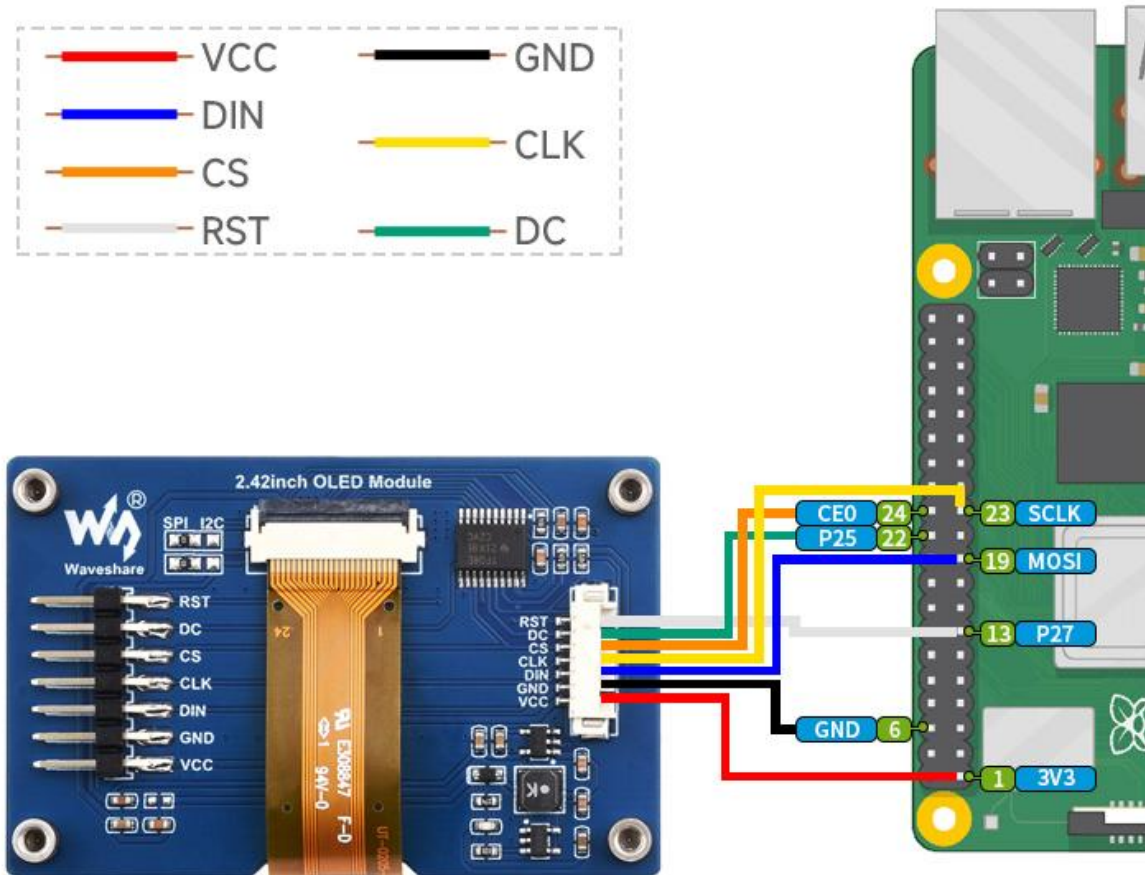


Figure 265 Waveshare 1.42" SPI OLED wiring

Table 30 SPI wiring scheme for OLEDs using the SPI interface

Waveshare	Alt. Name	Description	Colour	RPi Pin	RPi Name
RST	RES	Reset	White	13	GPIO27
DC	D/C	Data/Command	Green	22	GPIO25
CS	CS	Chip Select	Orange	24	CE0
CLK	SCLK	Clock	Yellow	23	SCLK
DIN	SDA	Data In	Blue	19	MOSI
GND	VDD	GND 0V	Black	6	GND
VCC	VSS	+3.3V	Red	1	3V3

RPi Pin numbers in column five are the physical GPIO header pin numbers and not GPIO numbers
The second column are alternative names used by other manufacturers of SPI products

Appendix B – Using a battery pack

It may be that you wish to make your radio portable. This can be done by using a 5V battery pack.



Figure 266 Raspberry Pi 5 Volt battery pack

There are various manufacturers who supply a range of battery 5-Volt packs specifically for the Raspberry Pi. These are charged from any normal 5V charger.

Try to use a battery pack that directly connects to the GPIO header rather than connecting via the micro-USB or USB-C port.

The reason is that if connected by the USB port the internal regulator it will drop the voltage to 4.8 Volts which may give problems with the Wi-Fi connection.

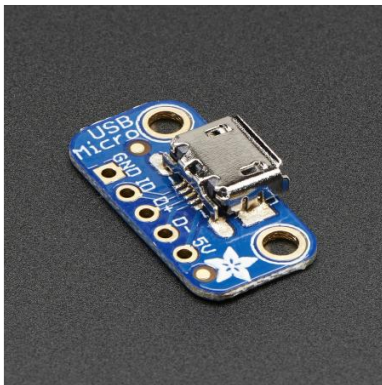


Figure 267 Micro USB breakout board

If the battery pack chosen doesn't connect directly to the GPIO header then purchase a micro-USB breakout board. This allows the 5 volt and GND connections to be connected directly to pins 2 and 6 respectively on the GPIO header. Caution advise as 5V on the wrong pin will destroy your Raspberry Pi.



Note: Connecting a 5V power supply directly to the GPIO pins bypasses the safety polyfuse and the reverse current direction protection diode.

Warning: Under no circumstances should a power source be connected to the GPIO header **3.3V** pins. For more information on applying 5V power see the following article:

<https://github.com/raspberrypi/hats/blob/master/designguide.md#back-powering-the-pi-via-the-gpio-header>

Appendix C – HiFiBerry Device Tree Overlays

While the drivers for the **HiFiBerry** boards are already included in the Raspberry Pi Linux kernel it may be necessary to activate them with additional configurations so-called Device **Tree** (DT) overlays. This requires editing either **/boot/config.txt** (**Bullseye**) or **/boot/firmware/config.txt** (**Bookworm** and **Trixie**) and adding the correct overlay for your **HiFiBerry** device to **config.txt** as shown in the following table.

Table 31 HiFiBerry DAC DT overlays

Device	OS	DT Overlay
DAC for Raspberry Pi 1/DAC+ Light/DAC Zero/MiniAmp/Beocreate/DAC+ DSP/DAC+ RTC	Bullseye	dtoverlay=hifiberry-dac
DAC8x	Bullseye	dtoverlay=hifiberry-dac8x
DAC+ Standard/Amp2/Amp4	Bullseye	dtoverlay=hifiberry-dacplus
“ “ “ “	Bookworm	dtoverlay=hifiberry-dacplus-std
DAC+ Pro/DAC2 Pro	Bullseye	dtoverlay=hifiberry-dacplus
“ “ “ “	Bookworm	dtoverlay=hifiberry-dacplus-pro
DAC2 HD	Bookworm	dtoverlay=hifiberry-dacplushd
DAC+ ADC	Bookworm	dtoverlay=hifiberry-dacplusadc
DAC+ ADC Pro, DAC2 ADC Pro	Bookworm	dtoverlay=hifiberry-dacplusadcpro
Digi+, Digi 2 Standard	Bookworm	dtoverlay=hifiberry-digi
Digi+ Pro, Digi 2 Pro	Bookworm	dtoverlay=hifiberry-digi-pro
Amp+ (not Amp2!)	Bookworm	dtoverlay=hifiberry-amp
Amp3	Bookworm	dtoverlay=hifiberry-amp3
Amp4 Pro	Bookworm	dtoverlay=hifiberry-amp4pro

The above information came from the following article:

<https://www.hifiberry.com/docs/software/configuring-linux-3-18-x/>

A complete and up-to-date of HiFiBerry products will be found at <https://www.hifiberry.com/>

Appendix D - Audio streaming codec notes

A codec (the term is a mashup of the words *code* and *decode*) is computer code that uses compression techniques to shrink video files and to convert between analogue and digital sound.

Radio streaming technologies generally aim for high-quality audio delivery, utilizing various codecs and protocols to balance quality and bandwidth. For live streams, 128kbps AAC is a common standard, other streams offer a higher quality 320kbps AAC-LC stream and even a FLAC option for those with sufficient bandwidth. They also support HLS and MPEG-DASH for newer devices and higher quality streams. The ones supported by the **liquidsoap** recording software are shown below:

D.1 Codecs and Bitrates:

AAC (Advanced Audio Coding):

AAC (Advanced Audio Coding) is an audio coding standard used for *lossy* digital audio compression by losing some of the data. It is designed to be the successor to MP3, offering better sound quality at similar bit rates. AAC supports higher bit rates (up to 320kbps) and can handle 24-bit/44.1kHz audio files, making it a popular choice for streaming and Bluetooth audio. It is widely used in various devices, including those running iOS and Android.

MP3:

Previously used, but being phased out in favour of AAC and other more modern codecs, especially for newer devices. It supports up to 320 kbps at 16 bits.

FLAC (Free Lossless Audio Codec):

Used for higher quality, lossless streaming, particularly on Radio 3, where it can average around 550kbit/s.

128kbps AAC:

The standard for live streams, considered a good balance between quality and bandwidth. Used for Radio 3's high-quality stream, bypassing signal processing and offering what is referred to as "HD Sound".

OPUS:

Opus is a totally open, royalty-free, highly versatile audio codec. OPUS handles bit-rates from 6 kb/s to 510 kb/s. It supports both constant bitrate (CBR) and variable bitrate (VBR).



Note: The above codecs and associated formats are how downloads are stored by the recording facility. The download stream format may be in a different format from the above. The **liquidsoap** program uses **ffmpeg** to convert the download stream to one of the above formats to store it on disc. See the following parameters in **/etc/radiod.conf** for codec and associated format settings.

```
# Codecs list for media playlist creation (Run 'mpd -V' to display others)
CODECS="mp3 ogg flac wav wma aac m4a opus mp4"
:
record_format=mp3
```

Appendix E Radio program and configuration files

A.1 The radio programs

There are three possible radio programs that you can chose from depending upon what you are building as summarised below.

Name	Displays	Started by	Startup file
radiod.py	LCDs or TFTs	systemd	/usr/lib/systemd/system/riod.service
gradio.py	HDMI or touch screens	XDG autostart	/etc/xdg/autostart/radio.desktop
vgradio.py	HDMI or touch screens	XDG autostart	/etc/xdg/autostart/radio.desktop

One of the above radio variants will be selected by installation program or when re-configuring the software with the **radio-config** configuration program. See *Configuring the radio* on page 110

A.2 Radio programs configuration file

All three variants of the radio use the **/etc/radiod.conf** file during operation. Nearly all changes to **radiod.conf** are done by the **radio-config** configuration program during installation and re-configuration although there are exceptions. There is a tutorial about configuring the radio which can be called by running **radio-config** and selecting option 6 followed again by option 6 :
radio-config → 6 Documents and Tutorials → 6 Configuration tutorial (/etc/radiod.conf)

A.3 LCD and TFT versions startup file

The system file for starting and stopping the LCD or TFT versions of the radio is shown below.

```
# Radio systemd script
[Unit]
Description=Radio daemon
After=network.target

[Service]
Type=simple
ExecStart=/usr/share/radio/radiod.py nodaemon
PIDFile=/run/radiod.pid
#Restart=on-failure
#RestartSec=5s

[Install]
WantedBy=multi-user.target
```

A.4 Radio graphic versions startup file

The graphical versions of the radio are started by **XDG (X Desktop Group)** which provides interoperability between the two free-software desktop environments namely **X Window System (X11)** and the newer **Wayland** windowing system on the **Linux** operating system. Using XDG the graphical radio programs are started from **/etc/xdg/autostart/radio.desktop**. See below:

```
[Desktop Entry]
Name=gradio
Comment=Start graphic version of the radio
Exec=sudo /usr/share/radio/gradio.py
Terminal=false
Type=Application
NoDisplay=true
```

The program name in the Exec statement will be either **gradio.py** or **vgradio.py** depending upon the selection made during installation.

A.4.1 Old windows startup files

There are two versions of the Linux Window Desktop. A) **X-Windows** (old) and B) **Wayland**. They each have their own startup files. It is still possible to use these configuration files but they have been replaced with the **XDG** configuration files as previously shown. The problem is that if you are using **/etc/xdg/autostart/radio.desktop** to start the radio program as well as either of the methods described below a second copy of the radio will be started. You need to disable **X-Window** and **Wayland** startup files to prevent this.

X-Windows autostart file

```
$ cat .config/lxsession/LXDE-pi/autostart
@lxpanel --profile LXDE-pi
@pcmanfm --desktop --profile LXDE-pi
@xscreensaver -no-splash
@point-rpi
@sudo /usr/share/radio/gradio.py
```

Disable the last entry by adding '#' at the start of the line.

```
#@sudo /usr/share/radio/gradio.py
```

Wayland autostart file

```
cat .config/labwc/autostart
sudo /usr/share/radio/gradio.py &
```

Disable the last entry by adding '#' at the start of the line.

```
#sudo /usr/share/radio/gradio.py &
```

A.5 Desktop startup icons

The graphical versions of the radio programs can also be started from the Windows Desktop (either **X-Windows** or **Wayland**) instead of at boot time. This is enabled by two files.

/home/pi/Desktop/gradio.desktop and **/home/pi/Desktop/vgradio.desktop**. These are the same for both versions of the desktop.

gradio.desktop

```
[Desktop Entry]
Name=Radio
Comment=Internet radio
Icon=/usr/share/radio/images/radio.png
Exec=sudo /usr/share/radio/gradio.py
Type=Application
Encoding=UTF-8
Terminal=false
Categories=None;
```

If you want to start **vgradio.py** instead of **gradio.py** then amend the **Exec** command above.

Index

- AAC, 203
- activity LED, 10, 44, 45, 71, 72
- Adafruit, 8, 9, 10, 45, 58, 59, 62, 66, 67, 68, 69, 71, 72, 80, 81, 119, 149, 200
- AdaFruit, 8, 45, 182
- AdaFruit industries, 8
- AdaFruit RGB plate, 45, 182
- airflow, 44
- airplay, 185
- Airplay, 162, 163, 190
- Alarm, 178, 181, 199, 200
- Allo Piano**, 32, 134, 135
- alsamixer, 131, 133, 141
- amplifier, 10, 60, 75, 77
- aplay**, 83, 130
- Arduino, 67, 68, 119, 120
- ASX, 203
- AV, 18
- battery, 60, 247
- battery pack, 247
- Bitvise**, 84
- Bluetooth device, 132
- Bookworm, 86, 95, 99, 204
- Bookworm Lite**, 19, 86, 95, 99
- breakout boards, 65
- CAD, 12, 73, 74, 239
- CIFS, 204, 205, 206
- CloudBreak, 190
- colours, 200
- COM-09117 12-step rotary encoder, 50
- configure_radio.sh*, 42, 68, 110, 241
- constructor's gallery, 44
- cooling fans, 78
- Cosmic controller, 42, 72, 73, 241, 243
- Cosmic Controller, 149
- Cyrillic, 6, 21, 143, 175, 176
- Cyrillic character LCD, 175, 176
- DAC, 27, 28, 29, 48, 80, 130, 131, 132, 140, 141, 239
- daemon, 118, 178, 179, 200, 207
- DHCP, 204
- Discogs** database, 187, 188, 236, 238
- dpkg**, 110
- Electromagnetic Interference, 76, 239
- EMI, 76, 239
- equalizer, 193
- espeak, 14, 182
- eSpeak, 6
- ferrite core, 76, 77
- ffmpeg, 249
- Fing, 84
- firmware, 162
- GPIO, 10, 23, 24, 27, 28, 45, 47, 48, 49, 50, 55, 56, 60, 61, 66, 67, 68, 69, 70, 71, 72, 132, 149, 239
- GPIO header, 60, 66, 68, 132
- GPIO pins, 27, 45, 61, 66, 67
- Ground Loop Isolator, 77
- Grove JHD1313 LCD**, 25
- Grove LCD RGB, 25, 116
- HD44870, 17, 20, 22, 42, 57, 63
- HDMI, 19, 83, 130, 131
- heat sink, 78
- HiFiBerry, 26, 27, 28, 48, 75, 130, 132, 140, 141
- hostname, 103
- housing the radio, 43
- I2C, 23, 24, 48, 62, 66, 67, 68, 69, 71, 77, 119, 120, 135, 149, 239
- I2C interface, 23, 42, 43, 66, 67, 69, 72
- Icecast2, 163, 164
- interface board, 63, 64, 67, 69, 75
- iPod 4 pole AV**, 18
- IQaudio, 241
- IQAudio, 19, 28, 42, 48, 49, 58, 62, 71, 72, 73, 75, 76, 130, 132, 149
- IR, 10, 12, 16, 44, 45, 48, 62, 69, 70, 71, 72, 73, 149, 171, 182
- IR sensor, 70, 71
- IR Sensor, 44, 70, 149
- ir-keytable**, 152
- JustBoom, 26, 29, 75, 132
- KY-040 Rotary Encoder, 51
- language file, 170
- LCD, 8, 9, 10, 17, 20, 21, 42, 43, 47, 48, 57, 58, 59, 60, 62, 66, 67, 68, 71, 76, 77, 80, 81, 119, 135, 149, 179, 181, 199, 200, 239
- LED Backlight, 58
- liquidsoap, 3, 5, 166, 167, 249
- LIRC, 239
- Locale, 104
- M3U, 203, 240
- mains filter, 77
- MHS, 40, 126
- micro USB, 19
- MP3, 195, 203

MPC, 240
mpd, 110, 141, 206, 228
 MPD, 110, 118, 179, 200, 206, 207, 240
MPDdroid, 207
MPDroid, 207
 MPEG, 203, 240
 MPEG3, 203, 240
 NAS, 195, 205, 240
 Network Time Protocol, 199, 240
 news feed, 170
 NFS, 204, 205, 240
 NTP, 199
 O!MPD, 155, 198
OLED, 21, 42, 72, 73, 119, 240
 Optical rotary encoders, 57
 Organic Light Emitting Diode, 21
 OS, 205, 240
 Parameters
 codepage, 176
 controller, 176
 language, 175, 176
 rgb_blue, 54
 rgb_green, 54
 rgb_red, 54
 romanize, 176
 streaming_on, 164
 volume_display, 169
 PC, 9, 43, 163, 171, 195, 203, 204
 PCF8574, 67, 68
 PCM, 28
 PCM5102A, 30
 PCM5102A DAC, 30
 pHat BEAT, 12, 73, 242
 Pi Zero, 10, 19
 Pi Zero W, 19
 PiFace, 12, 62, 73, 74, 239
PiFace CAD, 42, 71, 73
 PiFace Control and Display, 12
 Pimoroni, 12, 73, 242
 Pimoroni pHat, 28, 132
 Pimoroni pHAT, 30
 Pirate Audio, 35
 Pirate radio, 12, 35, 73, 242
 PLS, 203
 potentiometer, 68
 power adapter, 60
 power supply switch, 60
 pulseaudio, 31
 Putty, 84
pygame, 99, 192
radiod.conf, 28, 68, 229
 Random, 181
 Raspberry Pi, 1, 4, 8, 9, 10, 17, 18, 44, 45, 60, 66, 69, 70, 71, 76, 83, 110, 118, 120, 141, 149, 153, 195, 200, 239
 Raspberry Pi model 5, 19
Raspbian Bookworm, 86
 remote control, 12, 44, 70, 71, 73, 229
 Revision 1 board, 50
 rfkill, 234
Romanized, 175, 176
 rotary encoder, 9, 10, 42, 44, 50, 51, 67, 69, 71, 149, 182, 199
 RSS, 143, 170, 181, 240
 Russian, 6, 21, 22, 143, 175, 176
 Russian, 175
 screen saver, 194, 195
 Serial Peripheral Bus interface, 12
 service radiod, 118, 170, 229
 SH1106, 24, 42, 115
 SH1106 1.3" OLED, 244
Shoutcast, 154, 200, 201, 202, 203
 smbus2, 119
 SoftVol, 231
 speech, 165
 SPI, 12, 68, 74
 SPI interface, 42, 73, 74
 Spotify, 143, 177, 178, 208, 209
 SSD1306, 23, 42, 115
 SSH, 95
Stretch, 17
 TFT, 8, 39, 42
timesync, 199
 timezone, 100, 101, 102
 tone control, 14, 80
 touch screen, 7, 39, 42, 43
 TSOP38238, 70
 TSOP382xx, 44, 149
 type of radio, 42
 URL, 170, 203, 240
 USB, 10, 19, 60, 67, 76, 77, 195, 196
 USB adaptor, 10
 USB disk drive, 79, 88
 USB stick, 195, 196
 USB to Ethernet adapter, 10, 19
 USB-C, 18, 60
 vintage radio, 13, 14, 43
 Vintage radio, 71, 76
 VNC, 183
 wake-up button, 70
 Waveshare, 38, 39
 Web interface, 153, 161, 204

Wi-Fi, 103
wiring, 23, 24, 50, 57, 68, 81
WM8960, 26, 33, 136
WMA, 195

XDG, 250
xscreensaver, 194
xscreensaver-command, 195
X-Windows, 240